



Econstruct Design & Build Pvt. Ltd.



JAYESH PROJECT

Bengaluru



G+1 Residential



Bengaluru, karnataka 



East Facing



PLOT AREA: 5997.73 SF

TOTAL BUILTUP AREA: 8229.33 SF



- Total Built-up Area

TF – 2766.94SF





Fig.(3) Front Elevation



Fig.(4) Side Elevation



Fig.(5) Ground Floor



Fig.(6) First Floor

SECTION VIEWS



Fig.(7) Section View



Fig.(8) First Floor

STAIRCASE DETAILS

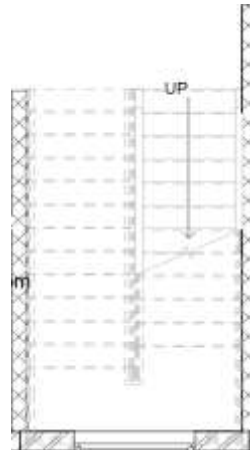
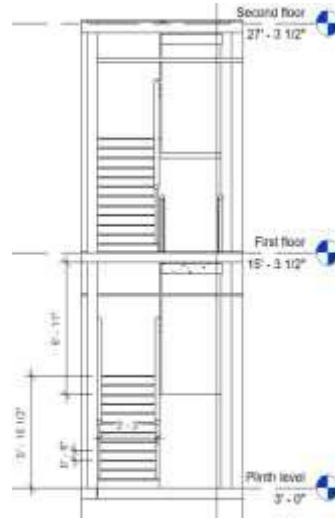
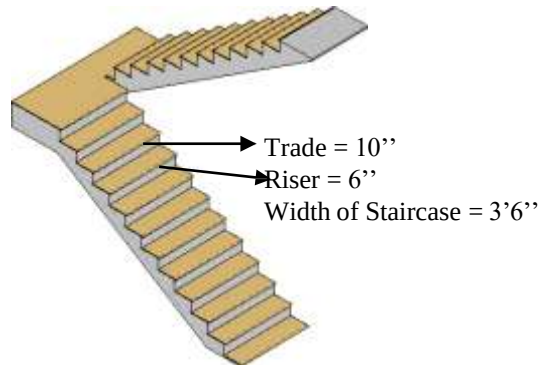
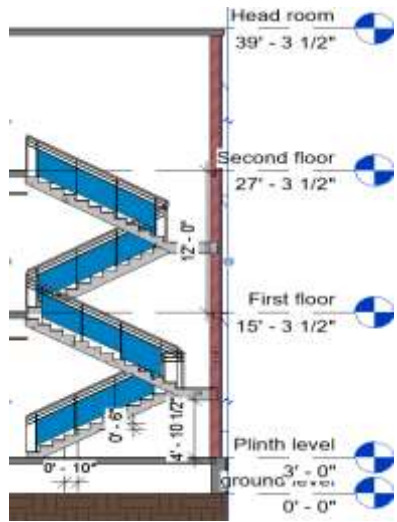


Fig.(10) Isometric view

SCHEDULES

Tile

Total Vitrified Area	2574.65 Sq.Ft
Total Anti skid Area	264.70Sq.Ft
Open Terrace Area	1779.05 Sq.Ft
Staircase Tile	167.96Sq.Ft



False Ceiling

Sr.No	Material	Level	Area
1	Gypsum Board	Ground Floor	2135.78 Sq.Ft
2	Gypsum Board	First Floor	1323.48 Sq.Ft
3	Gypsum Board	Second Floor	213.15 Sq.Ft
		Total Area	3672.41 Sq.Ft



Window

Sr.No	Level	Family and Type	Material	Sill Height	Height	Width	Count
1	GROUND FLOOR	M_Window-Casement-Multi-Sash-Horizontal	Wooden	3'6"	5'	5'	13
2	GROUND FLOOR	M_Window-Double-Hung; washroom windows	Wooden	6'4"	2'	2'	5
3	FIRST FLOOR	M_Window-Casement-Multi-Sash-Horizontal	Wooden	3'6"	5'	5'	4
4	FIRST FLOOR	M_Window-Double-Hung; washroom windows	Wooden	3'0"	5'	5'	5
5	FIRST FLOOR	M_Window-Casement-Multi-Sash-Horizontal	Wooden	6'4"	2'	2'	3
6	SECOND FLOOR	M_Window-Casement-Multi-Sash-Horizontal	Wooden	6'4"	2'	2'	1
7	PLINTH LEVEL	M_Window-Awning-Triple-Vertical: 900 x 1800mm	Wooden	2'	4'	30'	1
						Total Count	32

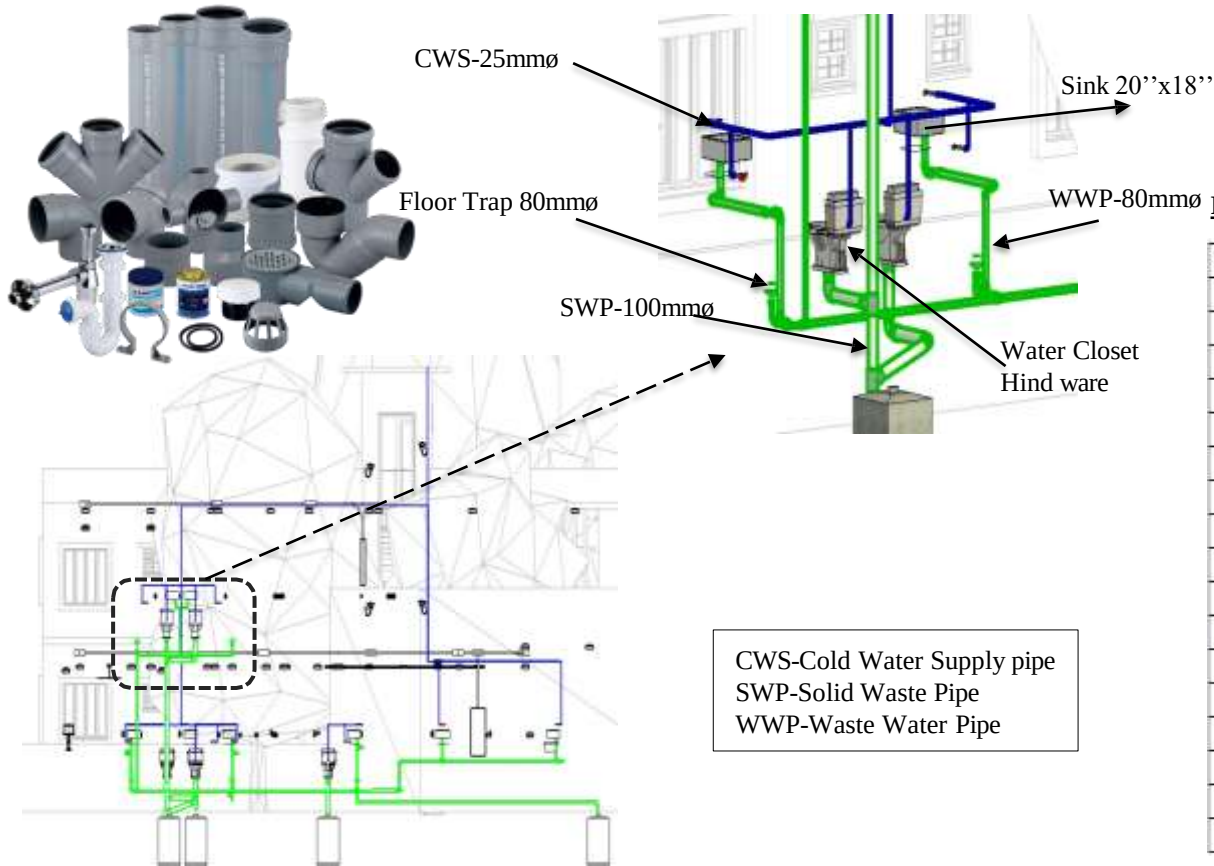


Door

Sr.No	Level	Family and Type	Material	Width	Height	Count
1	GROUND FLOOR	M_Door-Interior-Single-4_Panel-Wood: Bedroom Door	Wooden	3'0"	7'0"	5
2	GROUND FLOOR	M_Door-Interior-Single-Flush_Panel-Wood: interior door	Wooden	2'6"	7'0"	5
3	GROUND FLOOR	M_Door-Exterior-Double-Full Glass-Wood_Clad: pooja door	Wooden-Glass	4'0"	7'0"	1
4	GROUND FLOOR	M_Door-Interior-Double-Full Glass-Wood: main door	Wooden	4'0"	7'0"	1
5	FIRST FLOOR	M_Door-Interior-Single-4_Panel-Wood: Bedroom Door	Wooden	3'0"	7'0"	3
6	FIRST FLOOR	M_Door-Interior-Single-Flush_Panel-Wood: interior door	Wooden	2'6"	7'0"	2
7	Second floor	M_Door-Interior-Single-Flush_Panel-Wood: interior door	Wooden	2'6"	7'0"	1
					Total count	18



Typical Bathroom Section



Pipe Details

Sr.No	DIAMETER OF PIPE	MATERIAL	LENGTHS
1	25mmø	Polyvinyl Chloride-Rigid	631'6"
2	40mmø	Polyvinyl Chloride-Rigid	271'6"
3	80mmø	Polyvinyl Chloride-Rigid	148'8 1/2"
4	100mmø	Polyvinyl Chloride-Rigid	131'6"
		Total	1182'6"

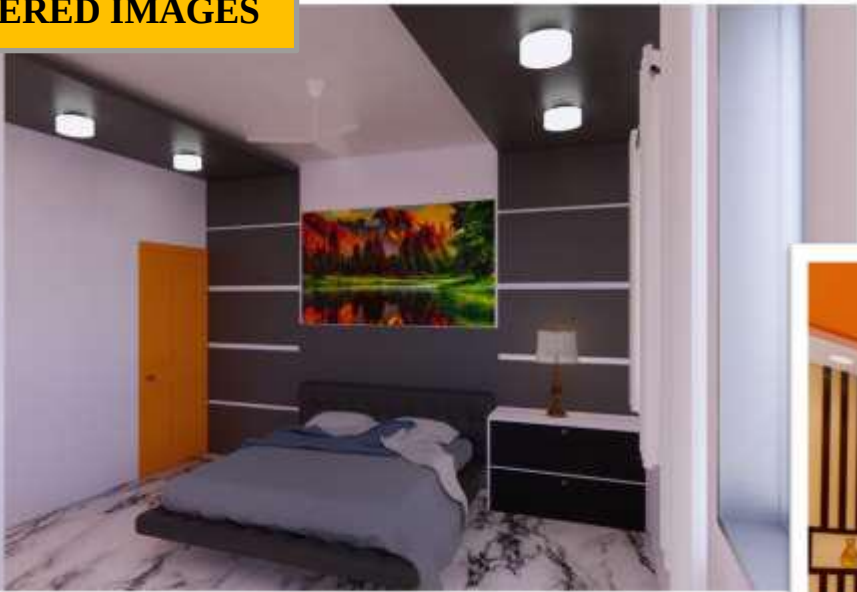
Pipe Accessories Details

Sr.No	Size	Family and Type	Count
1	25mmø-15mmø	Bush Reducer-Standard	35
2	25mmø-25mmø	M_Elbow -Generic Standard	37
3	25mmø-25mmø	M-Tee -Generic Standard	15
4	25mmø	End Cap	2
5	50mmø-40mmø	Bush Reducer-Standard	10
6	40mmø-40mmø	M_Elbow -Generic Standard	9
7	40mmø-40mmø	M-Tee -Generic Standard	8
8	40mmø	End Cap	6
9	80mmø-80mmø	Bush Reducer-Standard	10
10	80mmø-80mmø	M_Elbow -Generic Standard	10
11	80mmø-80mmø	M-Tee -Generic Standard	4
12	80mmø	End Cap	4
13	100mmø	Bush Reducer-Standard	10
14	100mmø-100mmø	M_Elbow -Generic Standard	10
15	100mmø-100mmø	M-Tee -Generic Standard	4
16	100mmø	End Cap	2



Fig.(11) Plumbing section

RENDERED IMAGES



RENDERED IMAGES



INDIRA NAGAR PROJECT

Bengaluru



B+G+5 Residential



Bengaluru, karnataka 



South Facing



PLOT AREA: 6760 SF

TOTAL BUILTUP AREA: 22562.49 SF



SITE PLAN AND PROJECT INFO.

Project Details:

- Total Built-up Area

Basement – 3356.15 SF

Ground Floor – 3061.68 SF

First Floor – 3356.51SF

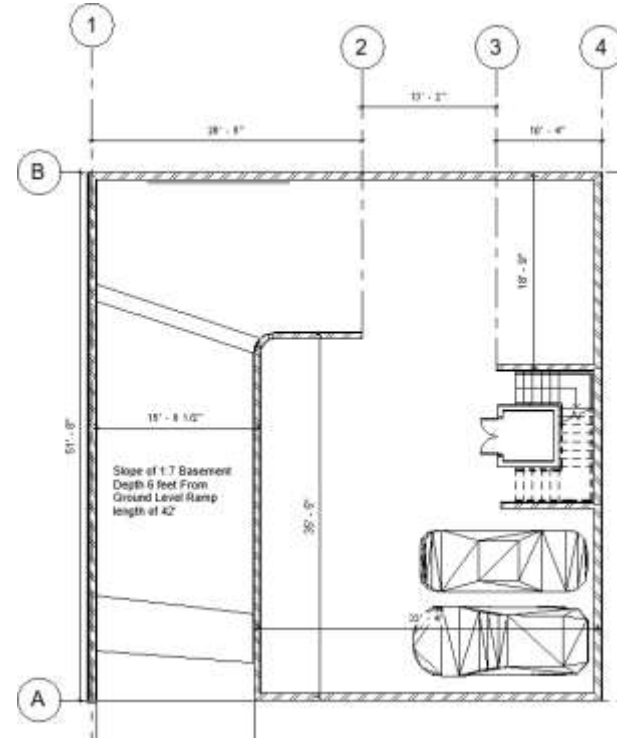


Fig.(1) Basement Floor Plan

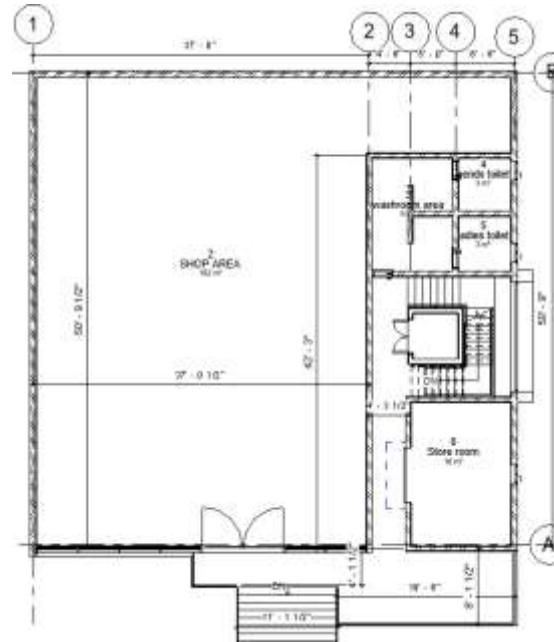


Fig.(2) Ground Floor Plan

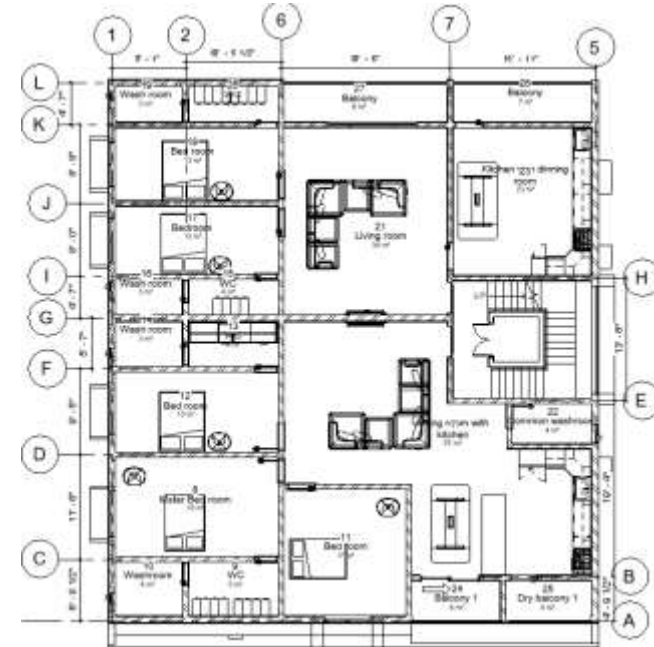


Fig.(3) First Floor Plan

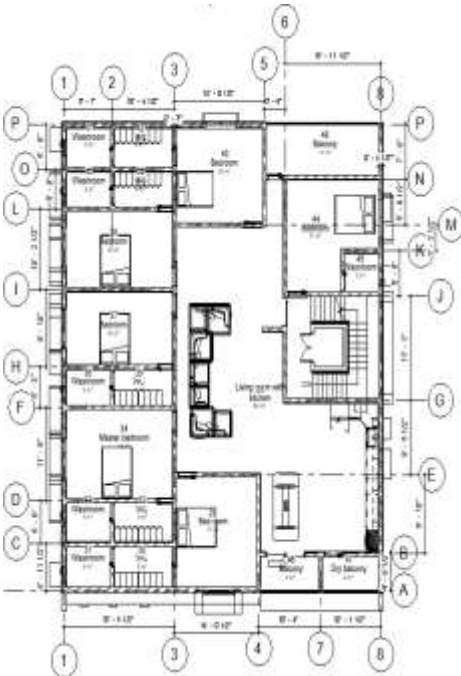


Fig.(4) Second Floor Plan
Second Floor – 3345.28 SF

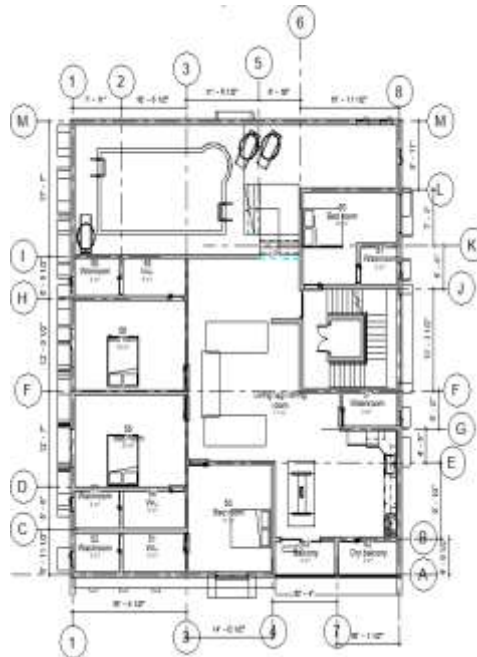


Fig.(5) Third Floor Plan
Third Floor – 3329.78 SF

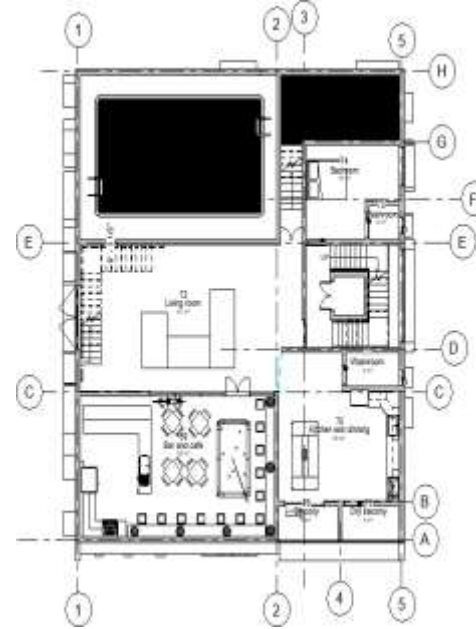


Fig.(6) Fourth Floor Plan
Fourth Floor – 3314.31 SF

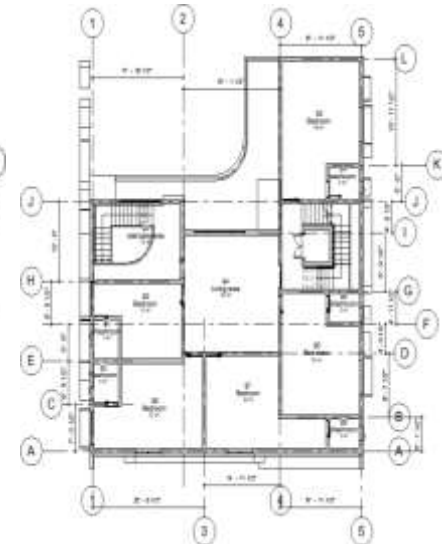


Fig.(7) Fifth Floor Plan
Fifth Floor – 2798.78 SF

ELEVATED VIEWS



Fig.(10) Front Elevation

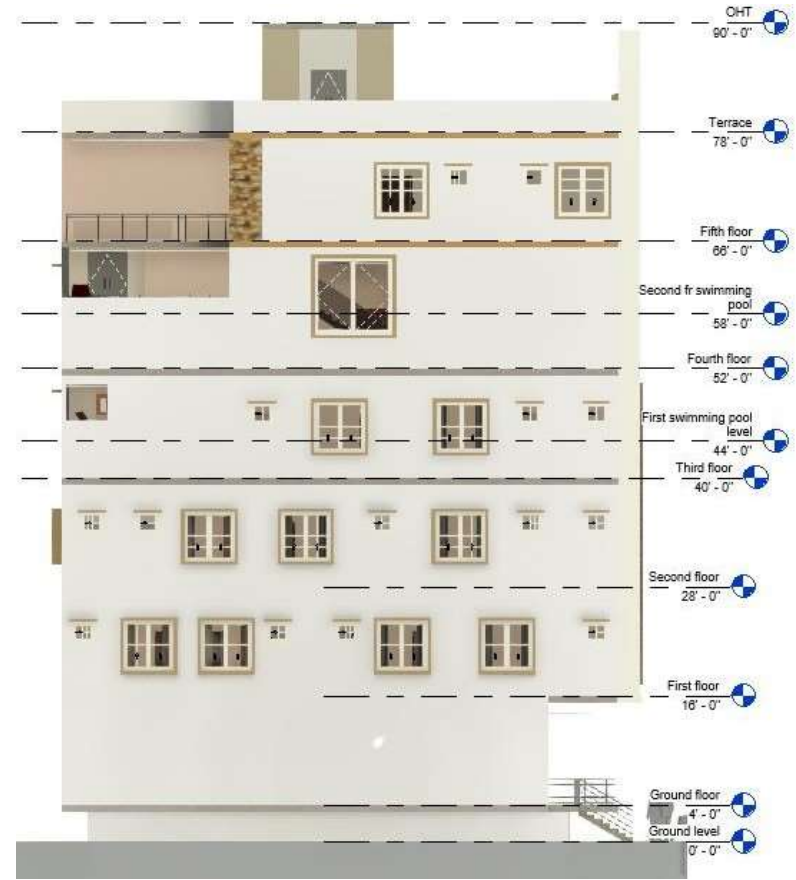


Fig.(11) Side Elevation

ELEVATED VIEWS

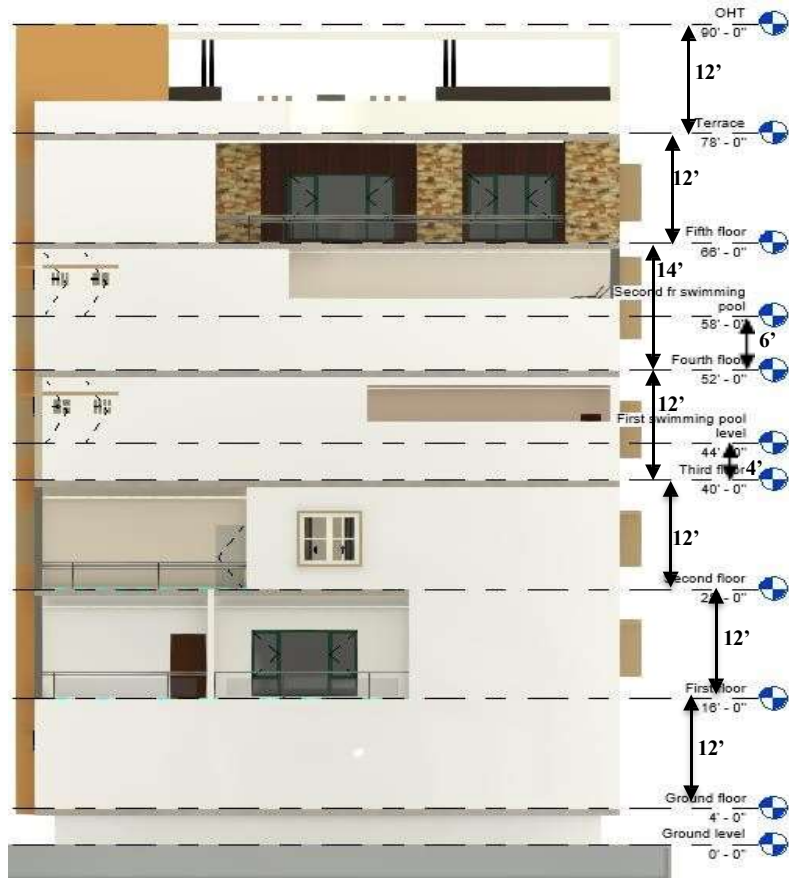


Fig.(10) Front Elevation

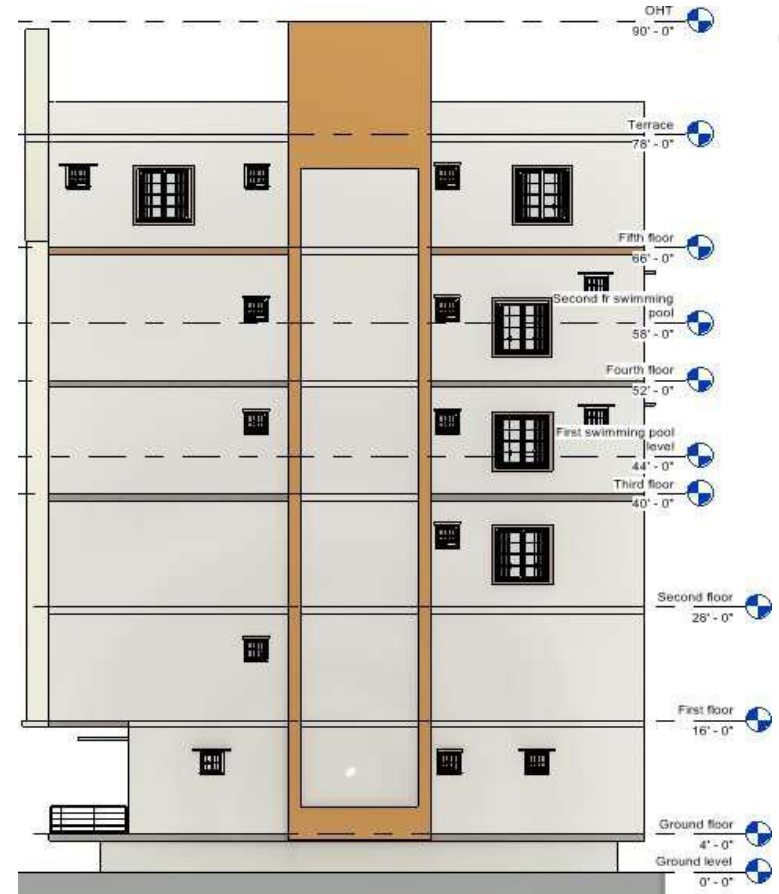


Fig.(11) Side Elevation

3D ISOMETRIC VIEWS

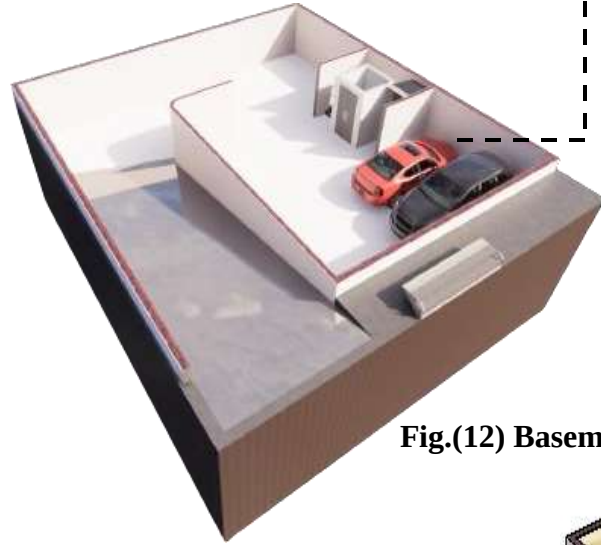


Fig.(12) Basement Floor

Basement Depth 10' (6' From
Ground Level

Commercial area for Shopping
with washroom and 1 store
room attached

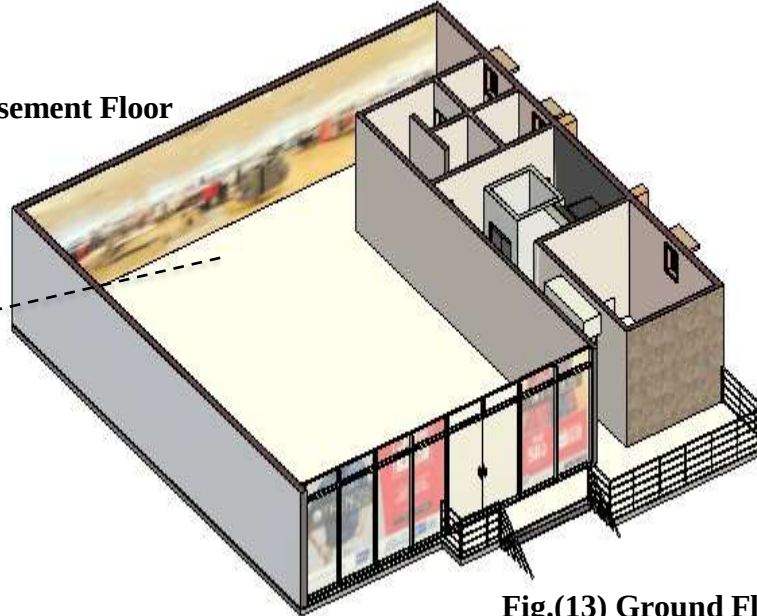


Fig.(13) Ground Floor



First floor:
2 Apartment's
2 BHK & 3BHK

Fig.(14) First Floor

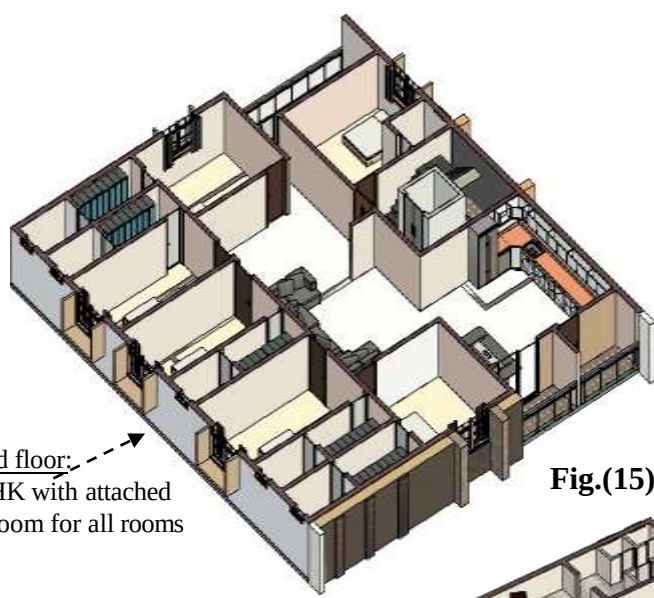


Fig.(15) Second Floor



Fig.(17) Fourth Floor



Fig.(16) Third Floor

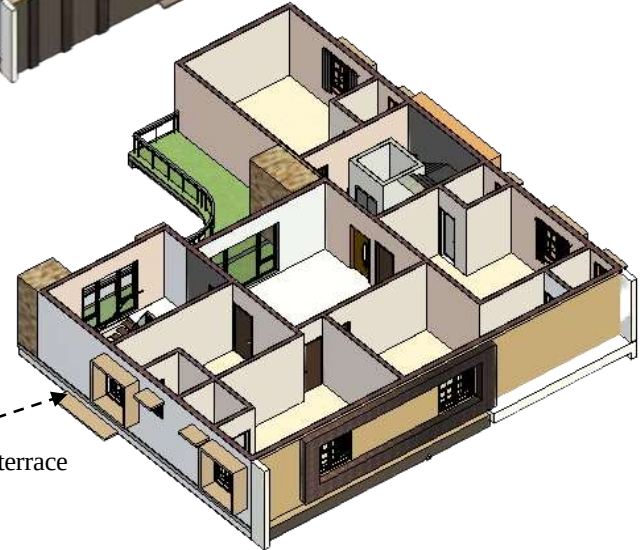


Fig.(18) Fifth Floor

Second floor:
5BHK with attached
bathroom for all rooms

Forth floor:
Duplex villa

Third floor:
3BHK with attached
bathroom for all rooms &
swimming pool.

Fifth floor:
5BHK with open terrace



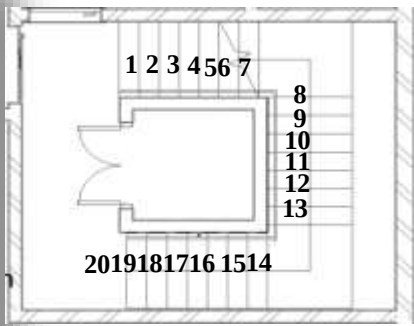
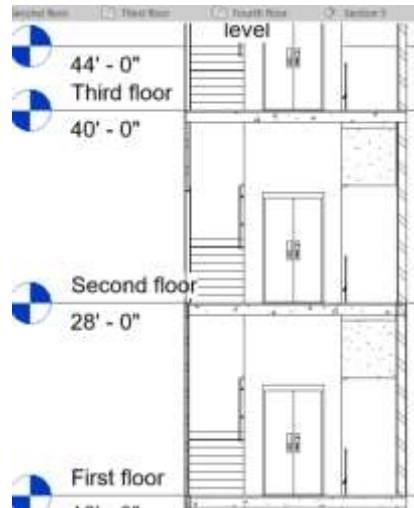
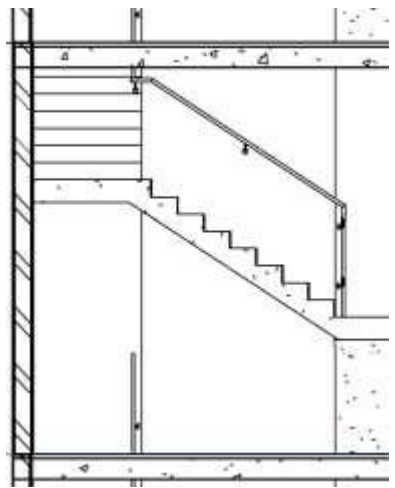
Fig.(19) Section View



Fig.(20) Fifth Floor



STAIRCASE DETAILS



Trade = 10''
Riser = 6''
Width of Staircase = 3'6''

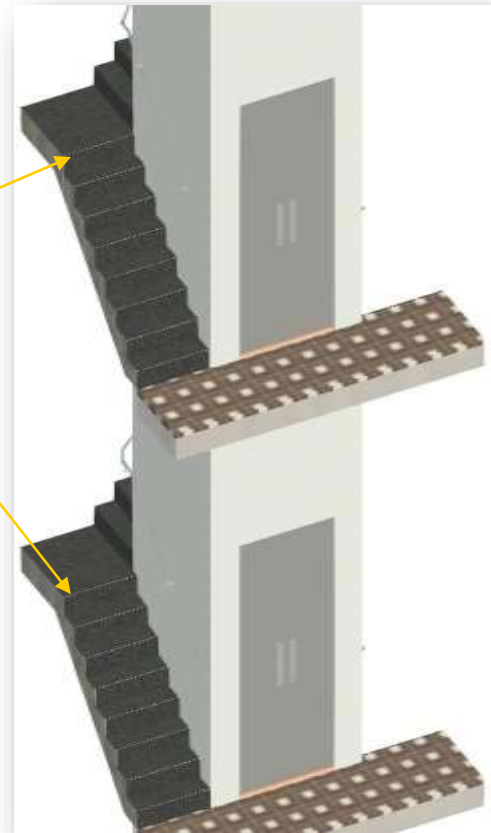


Fig.(21) Staircase Isometric view

SCHEDULES

Tile

Total Vitrified Area	17119.16 Sq.Ft
Total Anti skid Area	9716.28Sq.Ft
Open Terrace Area	2675.47 Sq.Ft
Staircase Tile	697.67Sq.Ft



False Ceiling

Sr.No	Material	Level	Area
1	Gypsum Board	Ground Floor	2679.24 Sq.Ft
2	Gypsum Board	First Floor	2636.2 Sq.Ft
3	Gypsum Board	Second Floor	2636.2 Sq.Ft
4	Gypsum Board	Third Floor	2377.96 Sq.Ft
5	Gypsum Board	Fourth floor	2098.2 Sq.Ft
6	Gypsum Board	Fifth floor	2087.444 Sq.Ft



Window

Sr.No	Level	Family and Type	Material	Sl Height	Height	Width	Count
1	GROUND FLOOR	Window-Sliding-Double	Wooden	6'6"	2'	2'	3
2	FIRST FLOOR	Window-Single-Hung-Double	Wooden	3'	5'	5'	5
3	FIRST FLOOR	Window-Sliding-Double	Wooden	6'6"	2'	2'	5
4	SECOND FLOOR	Window-Single-Hung-Double	Wooden	3'	5'	5'	6
5	SECOND FLOOR	Window-Sliding-Double	Wooden	6'	2'	2'	6
6	THIRD FLOOR	Window-Single-Hung-Double	Wooden	3'	5'	5'	4
7	THIRD FLOOR	Window-Sliding-Double	Wooden	6'	2'	2'	5
8	FIRST FR SWIMMING POOL	Window-Sliding-Double	Wooden	3'	2'	2'	3
9	FOURTH FLOOR	Window-Single-Hung-Double	Wooden	3'	5'	5'	1
10	FOURTH FLOOR	M_Window-Casement-Double: 850 x 1050mm	Wooden	4'	8'	8'	1
11	SECOND FR SWIMMING POOL	Window-Sliding-Double	Wooden	3'	2'	2'	2
12	FIFTH FLOOR	Window-Single-Hung-Double	Wooden	3'	5'	5'	6
13	FIFTH FLOOR	Window-Sliding-Double	Wooden	6'6"	2'	2'	5
						Total Count	52

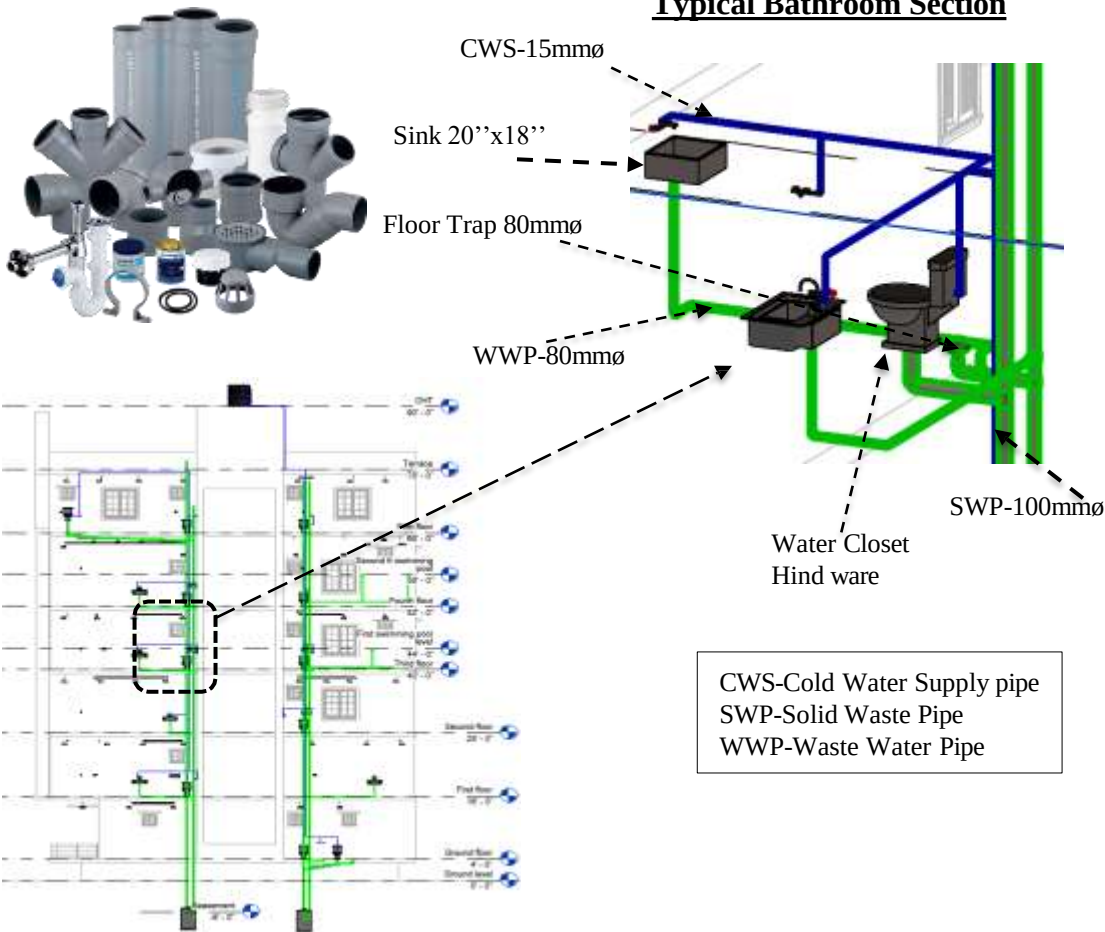


Door

Sr.No	Level	Family and Type	Material	Width	Height	Count
1	GROUND FLOOR	Door-Curtain-Wall-Double-Glass	Glass	9'	7'6"	1
2	GROUND FLOOR	TATA WOOD DOORS	Wooden	2'	7'	3
3	GROUND FLOOR	Door-Overhead-Rolling	Steel	6'	7'6"	1
4	GROUND FLOOR	M_Door-Interior-Double-Full Glass-Wood main door	Wooden	4'6"	7'6"	1
5	FIRST FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door PS-002 Bed room door	Wooden	7'6"	7'6"	7
6	FIRST FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door@61	Wooden	2'6"	7'6"	10
7	SECOND FLOOR	M_Door-Interior-Single-Finish_Panel-Wood interior door	Wooden	2'6"	7'6"	1
8	SECOND FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door	Wooden	3'6"	7'6"	8
9	SECOND FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door@61	Wooden	2'6"	7'6"	13
10	SECOND FLOOR	Door-Sliding-Weather_Shield-Contemporary-3_Panel	Glazed Door	6'6"	7'6"	1
11	THIRD FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door@61	Wooden	3'	7'6"	4
12	THIRD FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door	Wooden	2'6"	7'6"	8
13	FOURTH FLOOR	Door-Curtain-Wall-Double-Glass: Door-Curtain-Wall-Double-Glass	Glazed Door	4'6"	7'6"	2
14	FOURTH FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door	Wooden	3'6"	7'6"	2
15	FOURTH FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door@61	Wooden	2'6"	7'6"	2
16	FOURTH FLOOR	Door-Sliding-Weather_Shield-Contemporary-3_Panel: Default lead type catlao	Glazed Door	6'6"	7'6"	1
17	FIFTH FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door	Wooden	3'6"	7'6"	4
18	FIFTH FLOOR	Doors-Door sets_TATA Doors_TATA Wooden Door@61	Wooden	2'6"	7'6"	8
					TOTAL COUNT	76



Typical Bathroom Section



Pipe Details

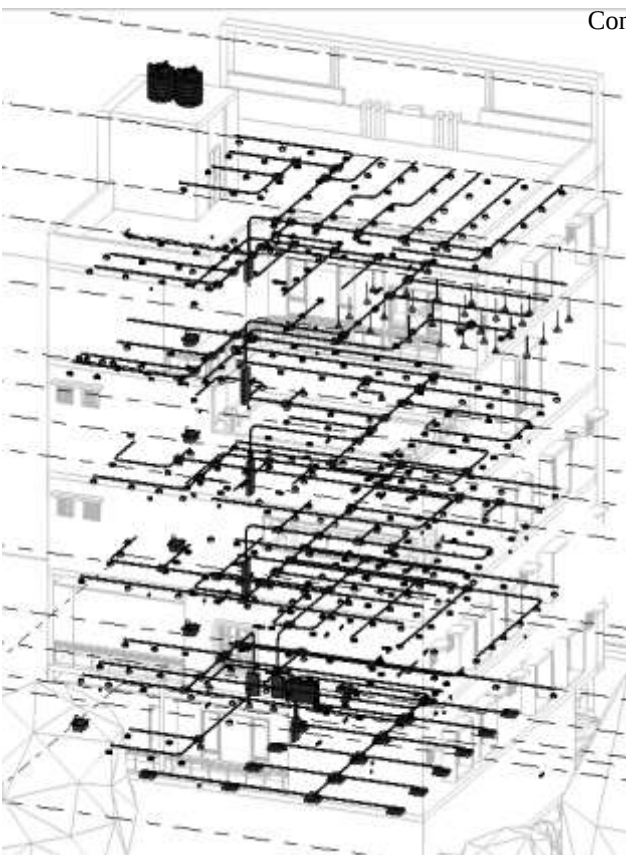
Sr.No	DIAMETER OF PIPE	MATERIAL	LENGTHS
1	15mmø	Polyvinyl Chloride-Rigid	841'8"
2	40mmø	Polyvinyl Chloride-Rigid	307'7"
3	80mmø	Polyvinyl Chloride-Rigid	489'2"
4	100mmø	Polyvinyl Chloride-Rigid	464'7"
		Total	2103'4"

Pipe Accessories

Sr.No	Size	Family and Type	Count
1	15mmø-15mmø	M Transition-Standard	2
2	15mmø-15mmø	M Elbow -Generic Standard	156
3	15mmø-15mmø	M-Tee -Generic Standard	68
4	15mmø	End Cap	40
5	40mmø-40mmø	M Elbow -Generic Standard	78
6	80mmø-40mmø	M Transition-Standard	24
7	80mmø-80mmø	M Transition-Standard	5
8	80mmø-80mmø	M Elbow -Generic Standard	52
9	80mmø-80mmø	M-Tee -Generic Standard	50
10	80mmø-80mmø	M_Cross - Generic: Standard	2
11	80mmø	End Cap	2
12	100mmø-100mmø	M Elbow -Generic Standard	44
13	100mmø-100mmø	M-Tee -Generic Standard	24
14	100mmø-100mmø	M Transition-Standard	3
15	100mmø	End Cap	2

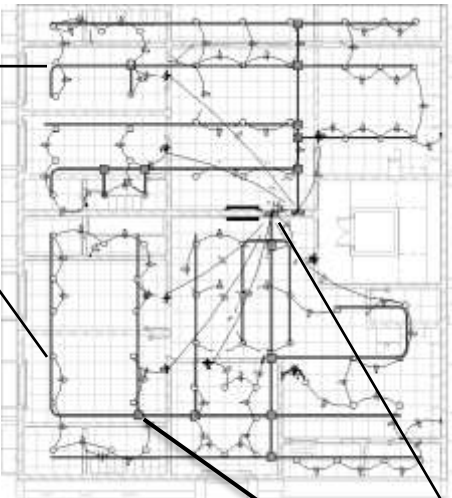


ELECTRICAL DETAILS



Elbow

Conduit 27mm



Branch Panel: SF PANEL

Location: Supermarket Building (Ground Floor) Area: Shop 1

Panel: 1 (Main) Area: 3

4 AC, 10 Amp Break Type: Mini Circuit Breaker (MCB) Rating: 10 A

Sl. No.	Conduit Description	Qty	Rate	Rs.	Sl. No.	Qty	Rate	Rs.	Conduit Description	Sl. No.
1	Lighting - Ceiling Light	204	1	204	1	204	1	204	Lighting - Ceiling Light	1
2	Lighting - Ceiling Light	204	1	204	2	204	1	204	Lighting - Ceiling Light	2
3	Lighting - Ceiling Light	204	1	204	3	204	1	204	Lighting - Ceiling Light	3
4	Lighting - Ceiling Light	204	1	204	4	204	1	204	Lighting - Ceiling Light	4

Sl. No.	Conduit Description	Connected Load	Demand Factor	Estimated Demand	Panel Rating
1	Lighting - Ceiling Light	500 W	1.25%	625 W	
2	Other	100 W	0.50%	50 W	
3	Accessories	500 W	100.00%	500 W	
					Total Connected Load: 1125 W
					Total Est. Demand: 1175 W
					Rating: 125 A
					Panel Rating: 125 A

Electrical fixtures and Equipment's

Sr.No	FAMILY AND TYPE	COUNT
1	M Plain Recessed Lighting Fixture: 600x600 - 120	19
2	Ceiling Light - Flat Round: 60W - 120V	340
3	Ceiling Light - Flat Round Two Switch Light	48
4	M Lighting Switches	203
5	M Duplex Receptacle-Standard	83
	Total	693



Switch with Receptacle

Conduit Junction box

Conduit and its Accessories Details

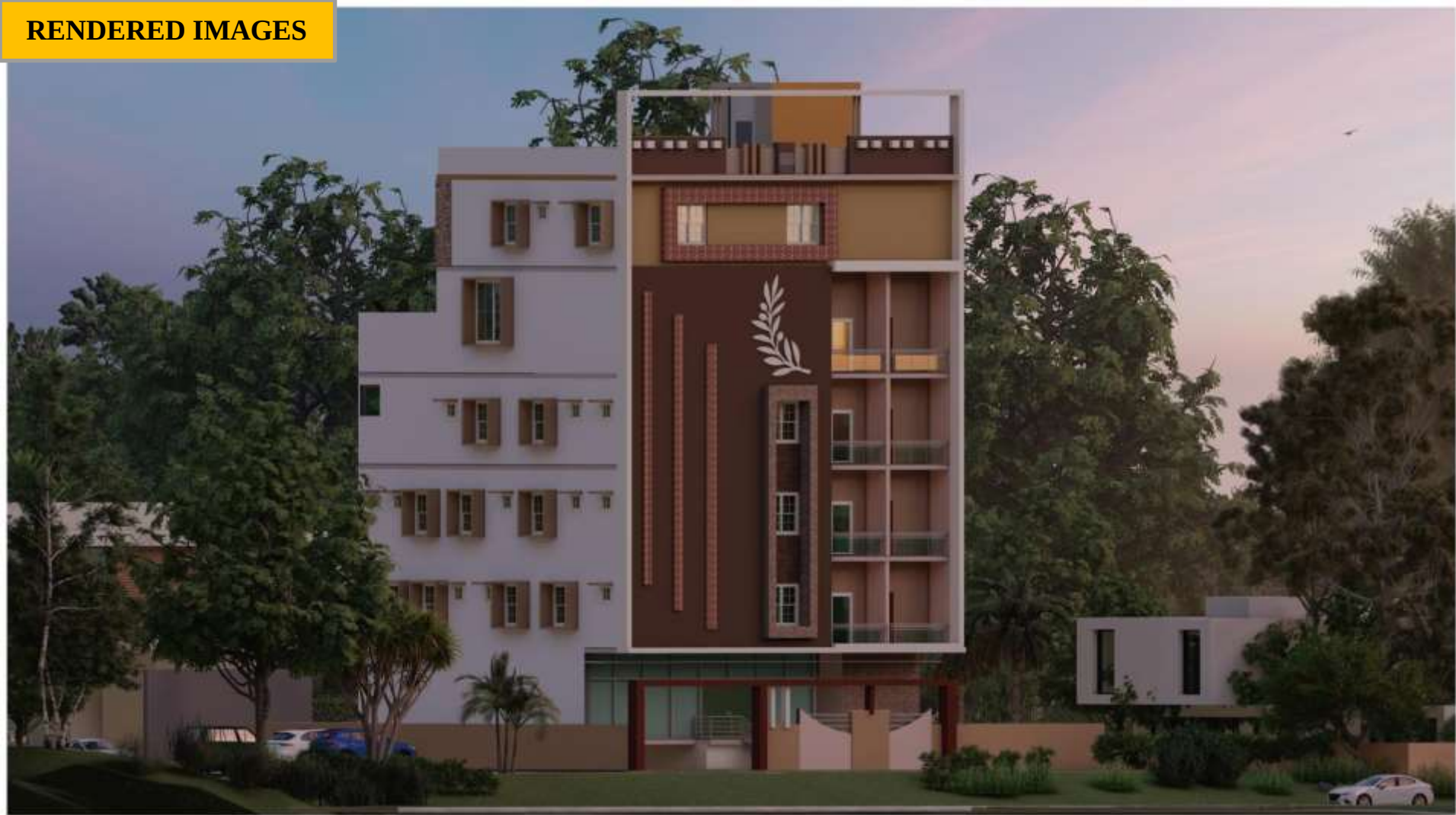
Sr.No	SIZE	FAMILY AND TYPE	UNIT	COUNT
1	27mm	Conduit without Fittings Rigid Nonmetallic Conduit (RNC Sch 80)	Length	780.84'
2	27mm-27mm	M Conduit Elbow - without Fittings - RNC: Standard	Nos	56
3	27mm-27mm	M Conduit Junction Box - Cross - PVC: Standard	Nos	79
4	27mm-27mm	M Conduit Junction Box - Tee - PVC: Standard	Nos	74



RENDERED IMAGES



RENDERED IMAGES



RENDERED IMAGES



RENDERED IMAGES



RENDERED IMAGES



LIGHTING ANALYSIS

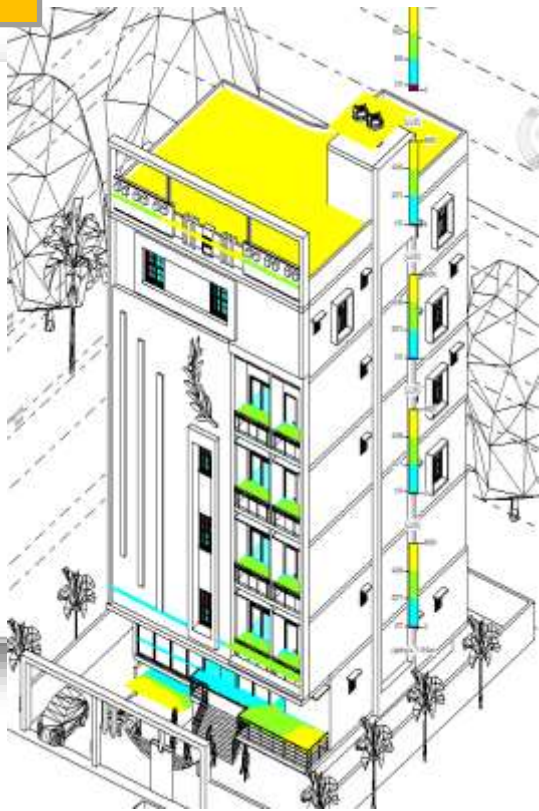
Lighting Analysis - Results Summary

Custom Analysis
For all Rooms Included in Daylighting

Total Both - 4% Passing
87% either time below threshold
10% either time above threshold

9:00 am - 19% Passing
July 15
GHI: 464, DNI: 473, DHI: 49
75% below threshold
6% above threshold w/o shades

3:00 pm - 8% Passing
July 15
GHI: 96, DNI: 147, DHI: 19
85% below threshold
7% above threshold w/o shades



Fig(22) Lighting Analysis Model

SOLAR ANALYSIS

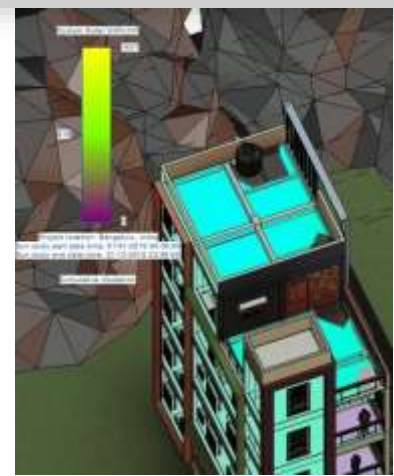
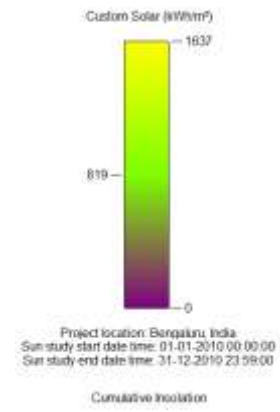


Fig (23) Solar Analysis Model

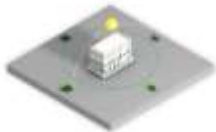
ENERGY ANALYSIS

Building Orientation:

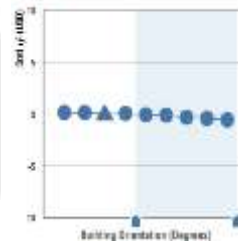
Building Orientation

Rotates a building clockwise from 0 degrees, e.g. 90 degrees rotates the North side of the building to face East.

Current Setting:
135 - 90



Building Orientation



Wall Window Ratio:

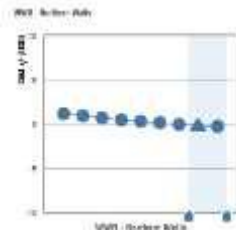
WWR - Northern Walls

Window-Wall-Ratio (glazing area / gross wall area) interacts with window properties to impact daylighting, heating & cooling.

Current Setting:
BIM (8%) - 0%



Editing: WWR - Northern Wall



Payback Limit:

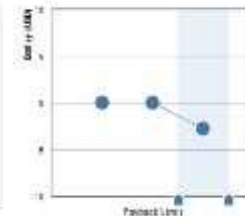
PV - Payback Limit

Use the payback period to define which surfaces will be used for the PV system. Surfaces with shading or poor solar orientation may be excluded.

Current Setting:
30 yr



Editing: PV - Payback Limit



Surface Coverage:



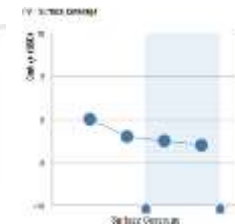
PV - Surface Coverage

Defines how much roof area can be used for PV panels, assuming area for maintenance access, rooftop equipment and system infrastructure.

Current Setting:
75 % - 90 %



Editing: PV - Surface Coverage



SUN PATH

Cobie Sheet

[illegible]

SR.No	NAME OF WORK	COST (.)
1.	Site cleaning & preliminary works	4,36,064/-
2.	Concreting work	83,61,106/-
3.	Fe-500 TMT Rebar	82,61,107/-
4.	Mason Work	26,92,224/-
5.	Tile Work	35,98,935/-
6.	Door and Window works	9,46,549/-
7.	Plastering work	1,25,035/-
8.	False ceiling	11,73,036/-
9.	Painting work	97,028/-
10.	Plumbing and Electric work	8,64,170/-
11.	Water proofing work	1,89,280/-
12.	Landscape work	1,92,000/-
	Total Cost (A)	2,74,90,041/-

SR.No	ADDITIONAL CHARGES	COST (Rs.)
1.	Add 5% Overhead Charges	15,53,187/-
2.	Add 1% General Consultant charge	2,74,900/-
3.	Add 1% Water Charges	50,000/-
4.	Contractors Profit 10%	27,49,004/-
5.	Add 5% Contingency Charges	15,53,187/-
6.	Adding 12% GST	39,14,032/-
	Total Cost (B)	90,40,925/-
	Total Cost (A)+(B)	RS. 3,65,30,966/-

MSP GANTT CHART

G+5 Storey Building	527 days	Wed 22-06-22	Thu 27-06-24
Paper work	26 days	Wed 22-06-22	Wed 27-07-22
Preparation of Drawing/Design stage	17 days	Thu 28-07-22	Fri 19-08-22
Labour supply and vendor selection activity	18 days	Wed 17-08-22	Fri 09-09-22
Pre construction phase	10 days	Sat 10-09-22	Fri 23-09-22
Construction phase	459 days	Fri 23-09-22	Thu 27-06-24
Sub structure	28 days	Fri 23-09-22	Wed 02-11-22
Start of construction phase	0 days	Fri 23-09-22	Fri 23-09-22
Excavation	4 days	Sat 24-09-22	Wed 28-09-22
PCC	3 days	Thu 29-09-22	Sat 01-10-22
G+5 Storey Building	527 days	Wed 22-06-22	Thu 27-06-24
Paper work	26 days	Wed 22-06-22	Wed 27-07-22
Purchase of land/Registration	15 days	Wed 22-06-22	Tue 12-07-22
Contract signing between Client and Builder	7 days	Wed 13-07-22	Thu 21-07-22
Selection of Engineer/Architect	5 days	Fri 22-07-22	Wed 27-07-22
Preparation of Drawing/Design stage	17 days	Thu 28-07-22	Fri 19-08-22
Conceptualization	2 days	Thu 28-07-22	Fri 29-07-22
Design Drawing	6 days	Fri 29-07-22	Thu 04-08-22
Structural Consultant meeting	1 day	Fri 05-08-22	Fri 05-08-22
Modification as per structural consultant	2 days	Mon 08-08-22	Tue 09-08-22
Working drawing formation	2 days	Wed 10-08-22	Thu 11-08-22
GFC Drawing preparation	1 day	Fri 12-08-22	Fri 12-08-22
Drawing approvals from authority	3 days	Sat 13-08-22	Tue 16-08-22
Internal Authority approvals	2 days	Wed 17-08-22	Thu 18-08-22
Clients approval	1 day	Fri 19-08-22	Fri 19-08-22
Labour supply and vendor selection activity	18 days	Wed 17-08-22	Fri 09-09-22
Labour contractor	2 days	Wed 17-08-22	Thu 18-08-22
Material supply vendor selection procedure	4 days	Fri 19-08-22	Wed 24-08-22
RFI for materials	2 days	Wed 24-08-22	Thu 25-08-22
Evaluation of price and quality parameter	2 days	Fri 26-08-22	Mon 29-08-22
Negotiation of price and selection of vendor	1 day	Tue 30-08-22	Tue 30-08-22
Equipments procurements(Renting)	2 days	Wed 31-08-22	Thu 01-09-22

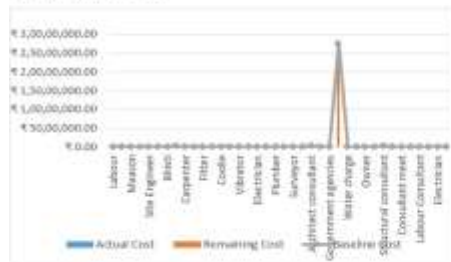


REPORTS

RESOURCE COST OVERVIEW

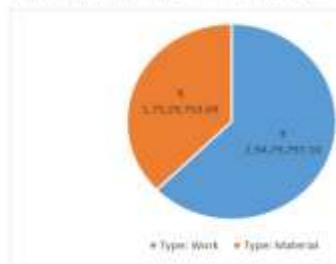
COST STATUS

Cost status for work resources.



COST DISTRIBUTION

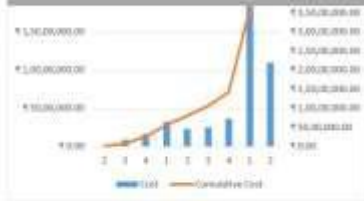
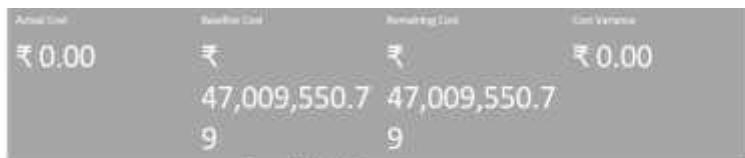
How costs are spread out amongst different resource types.



COST DETAILS

Cost details for all work resources.

CASH FLOW



Resource	Remaining Cost	Actual Cost	Cost	ACWP	BCWP	BCWS
G+5 Storey Building	₹ 47,009,550.79	₹ 0.00	₹ 47,009,550.79	₹ 0.00	₹ 0.00	₹ 15,000.00

RESOURCE SHEET OVERVIEW:

Resource	Actual Work	Actual Cost	Standard Rate
Labour	0 hrs	₹ 0.00	₹ 534.00/day
Helper	0 hrs	₹ 0.00	₹ 500.00/day
Mason	0 hrs	₹ 0.00	₹ 745.00/day
Supervisor	0 hrs	₹ 0.00	₹ 1,000.00/day
Site Engineer	0 hrs	₹ 0.00	₹ 1,800.00/day
Painter	0 hrs	₹ 0.00	₹ 714.00/day
Brick	0 hrs	₹ 0.00	₹ 714.00/day
Blocker	0 hrs	₹ 0.00	₹ 645.00/day
Carpenter	0 hrs	₹ 0.00	₹ 784.00/day
Mason (Brick Layer)	0 hrs	₹ 0.00	₹ 784.00/day
Fitter	0 hrs	₹ 0.00	₹ 784.00/day
Blacksmith	0 hrs	₹ 0.00	₹ 714.00/day
Conke	0 hrs	₹ 0.00	₹ 645.00/day
JCB	0 hrs	₹ 0.00	₹ 6,000.00/day
Vibrator	0 hrs	₹ 0.00	₹ 350.00/day
Dumpers	0 hrs	₹ 0.00	₹ 3,300.00/day
Electrician	0 hrs	₹ 0.00	₹ 1,000.00/day
Electrical Helper	0 hrs	₹ 0.00	₹ 500.00/day
Plumber	0 hrs	₹ 0.00	₹ 1,000.00/day
Watchman	0 hrs	₹ 0.00	₹ 1,000.00/day

Resource	0 hrs	₹ 0.00	₹ 1,000.00/day
Resource	0 hrs	₹ 0.00	₹ 1,000.00/day
Resource	0 hrs	₹ 0.00	₹ 1,000.00/day

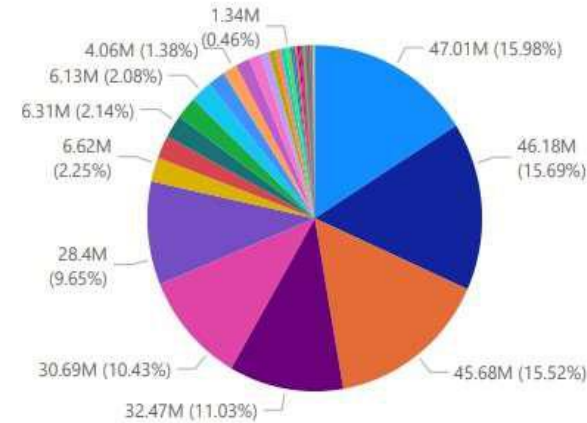
Architect	0 hrs	₹ 0.00	₹ 1,000.00/day
Electrical Helper	0 hrs	₹ 0.00	₹ 500.00/day
Plumber	0 hrs	₹ 0.00	₹ 1,000.00/day
Watchman	0 hrs	₹ 0.00	₹ 1,000.00/day
Surveyor	0 hrs	₹ 0.00	₹ 2,000.00/day
Draftsman	0 hrs	₹ 0.00	₹ 2,000.00/day
Architect consultant	0 hrs	₹ 0.00	₹ 84,256.00/day
Contractors charge	0 hrs	₹ 0.00	₹ 0.00/hr
Government agencies	0 hrs	₹ 0.00	₹ 500.00/day
MSP Consultant	0 hrs	₹ 0.00	₹ 980.00/day
Water charge	0 hrs	₹ 0.00	₹ 0.00/hr
Temporary electric supply	0 hrs	₹ 0.00	₹ 0.00/hr
Over	0 hrs	₹ 0.00	₹ 0.00/hr
Legal work consultant	0 hrs	₹ 0.00	₹ 5,000.00/day
Structural consultant	0 hrs	₹ 0.00	₹ 81,536.00/day
Structural designer	0 hrs	₹ 0.00	₹ 1,000.00/day
Consultant meet	0 hrs	₹ 0.00	₹ 5,000.00/day
Nagara palike approval	0 hrs	₹ 0.00	₹ 2,000.00/day
Labour Consultant	0 hrs	₹ 0.00	₹ 2,000.00/day
Project manager	0 hrs	₹ 0.00	₹ 5,000.00/day
Electrician	0 hrs	₹ 0.00	₹ 1,000.00/day

Resource	0 hrs	₹ 0.00	₹ 1,000.00/day
Resource	0 hrs	₹ 0.00	₹ 1,000.00/day
Resource	0 hrs	₹ 0.00	₹ 1,000.00/day

Name	Resource_Names
Anti termite treatment for floor PCC	Anti termite treatment[200],Coolie,Labour
Architectural phase	
Basement	Electrical conduit[1],Electrician,Helper,Labour
Basement	Tile joint sealent[1],Tiles[92.52],Tiles cleaner[1],Tiling work[1]
Basement floor	
Basement floor	Plaster mortar[8.63],Beldar[200%],Coolie[200%],Labour[400%
Basement paint	Painting work[60]
BBS for Column work	Beldar,Coolie,Helper,Labour,Steel[557.14]
BBS for Column work	Blacksmith,Carpenter,Fitter
BBS for slab and beam	Beldar,Carpenter,Steel[84.25]
Bore well	Coolie,Electrician,Fitter,Helper
Casting of concrete	Beldar,Bhisti,Concrete[1],Labour,Site Engineer
Casting for slab and beam	Beldar,Concrete[106],Contractor,Fitter,Helper
Casting for slab and beam	Beldar,Concrete[114],Contractor,Fitter,Helper
Casting for slab and beam	Beldar,Concrete[42],Contractor,Fitter,Helper
Casting for slab and beam	Beldar,Concrete[62],Contractor,Fitter,Helper
Casting of brick work	Beldar,Bricks[100],Carpenter,Coolie,Labour
Casting of brick work	Beldar,Bricks[62],Carpenter,Coolie,Labour
Casting of brick work	Beldar,Bricks[75],Carpenter,Coolie,Labour
Casting of brick work from lintel level	Beldar,Bricks[20],Coolie,Helper,Mason(Brick Layer)
Casting of concrete for column	Beldar,Concrete[114],Coolie,Helper,Labour,Owner
Casting of concrete for column	Beldar,Concrete[181.88],Coolie,Helper,Labour,Owner
Casting of concrete for column	Concrete[1],Labour,Maason,Site Engineer,Supervisor
Clients approval	Owner
Column marking	Coolie[400%],Labour[500%],Site Engineer,Supervisor,Surveyor
Compacting the filled material	Back filling[1]
Completion of Basement floor	
Completion of fifth floor	
Completion of first floor	
Completion of fourth floor	
completion of plastering work	
Completion of pre construction work	

Cost by Name

POWER BI REPORTS



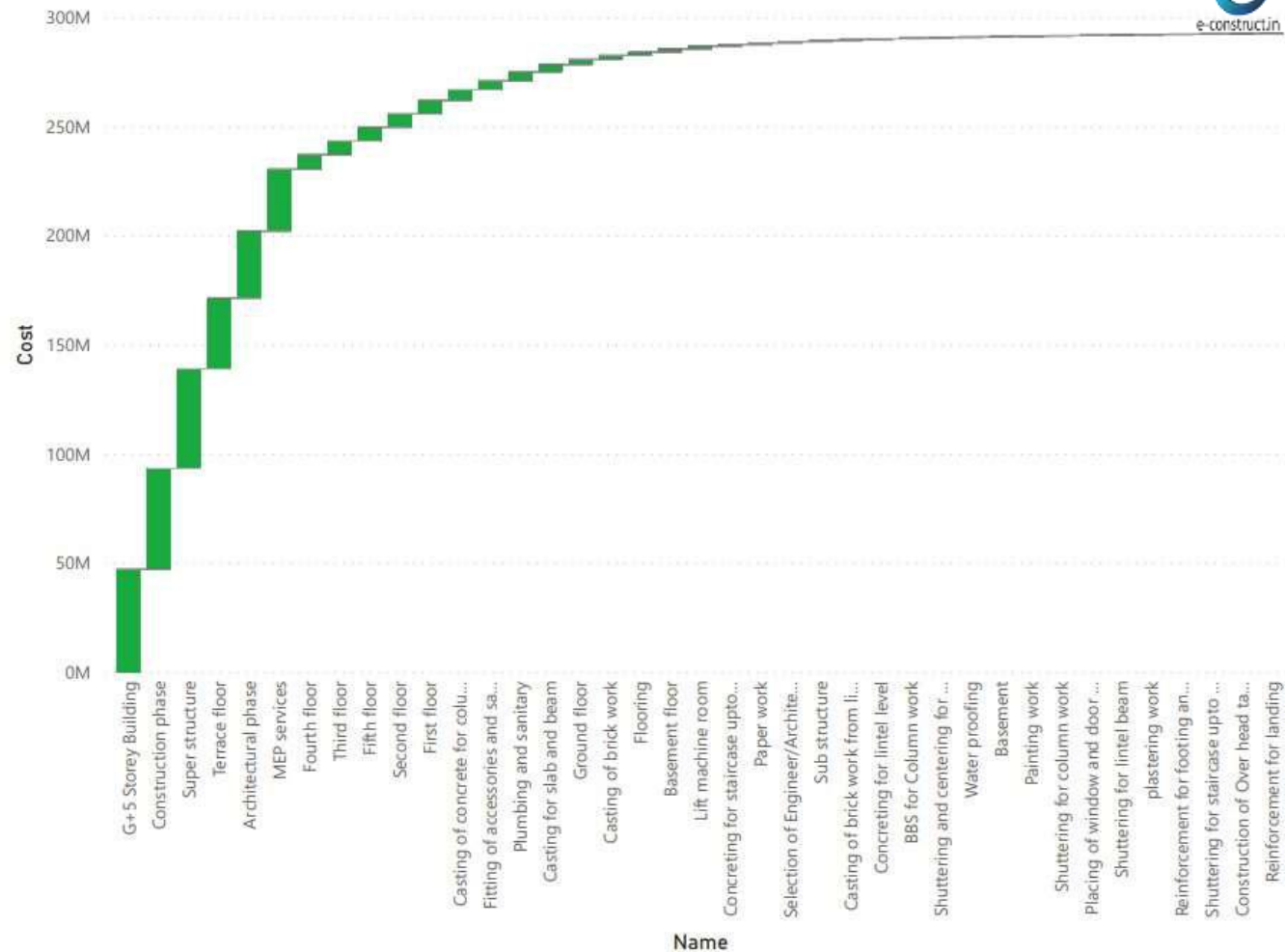
Name

- G+5 Storey Building
- Construction phase
- Super structure
- Terrace floor
- Architectural phase
- MEP services
- Fourth floor
- Third floor
- Fifth floor
- Second floor
- First floor
- Casting of concrete for c...
- Fitting of accessories an...
- Plumbing and sanitary
- Casting for slab and beam
- Ground floor
- Casting of brick work
- Flooring
- Basement floor
- Lift machine room
- Concreting for staircase ...
- Paper work
- Selection of Engineer/Ar...
- Sub structure
- Casting of brick work fro...

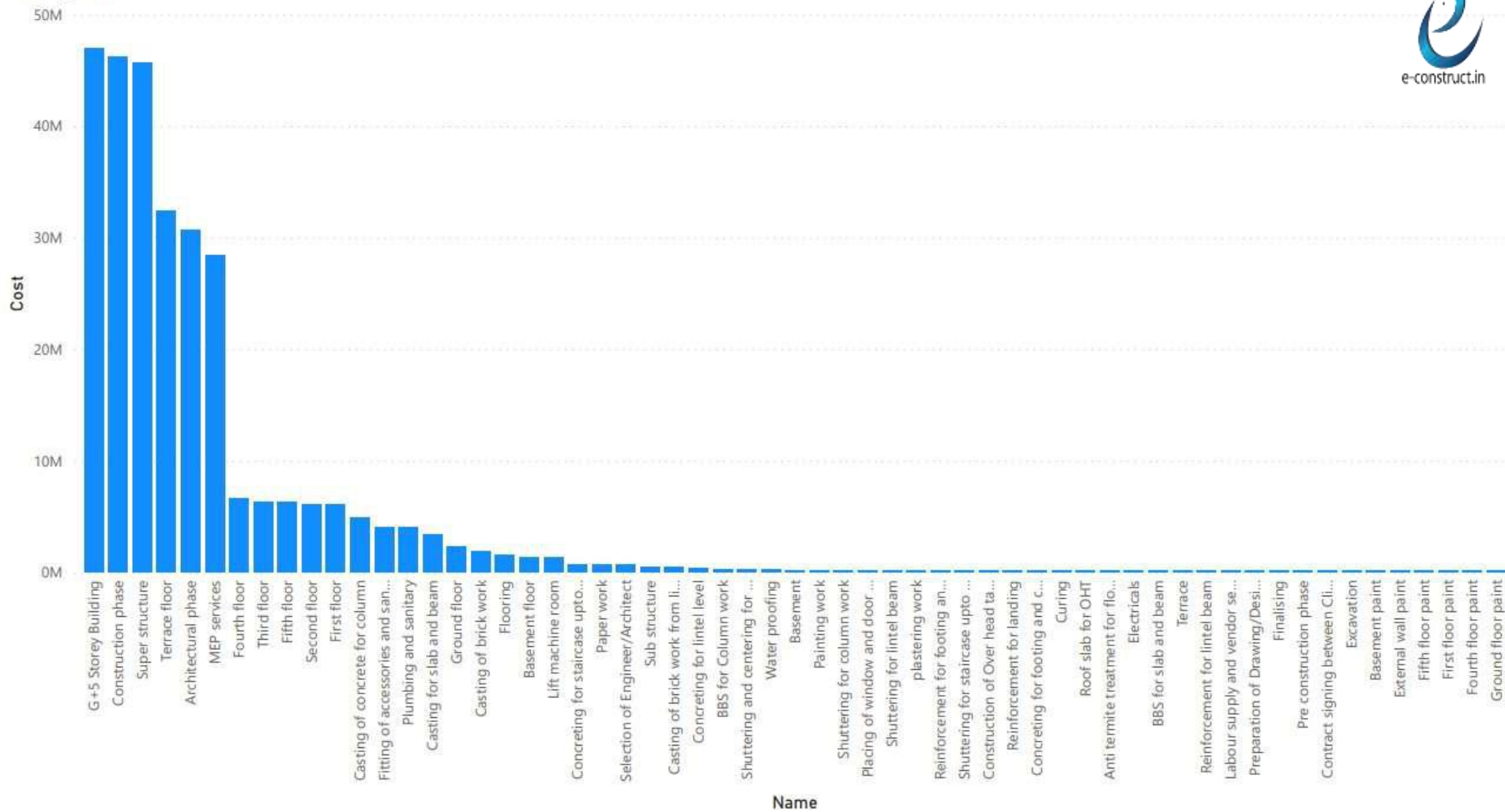
Name	Duration
Anti termite treatment for floor PCC	1 day
Architectural phase	104 days
Basement	1 day
Basement	4 days
Basement floor	1.5 days
Basement floor	49 days
Basement paint	4 days
BBS for Column work	2 days
BBS for slab and beam	4 days
Bore well	2 days
Casting of concrete	1 day
Casting for slab and beam	2 days
Casting for slab and beam	3 days
Casting of brick work	6 days
Casting of brick work from lintel level	2 days
Casting of concrete for column	1 day
Clients approval	1 day
Column marking	1 day
Compacting the filled material	2 days
Completion of Basement floor	0 days
Completion of fifth floor	0 days
Completion of first floor	0 days
Completion of fourth floor	0 days
completion of plastering work	0 days
Completion of pre construction work	0 days
Completion of Second floor	0 days
Completion of sub structure phase	0 days
Completion of Third floor	0 days
Conceptualization	2 days
Concreting for footing and column starter	1 day
Concreting for lintel level	1 day

Cost by Name

● Increase ● Decrease ● Total



Cost by Name



RAVIRAJ PROJECT

Bengaluru



G+4 Residential



Bengaluru, karnataka



South Facing



PLOT AREA: 800 SF

TOTAL BUILTUP AREA: 2852 SF



SITE PLAN AND PROJECT INFO.

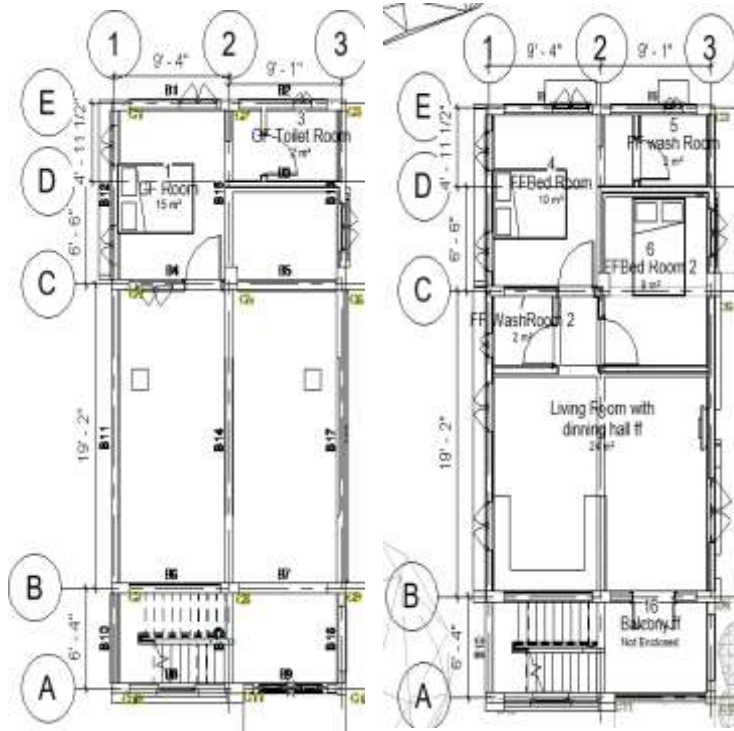


Fig.(1) Ground Floor Plan



Fig.(2) First Floor Plan

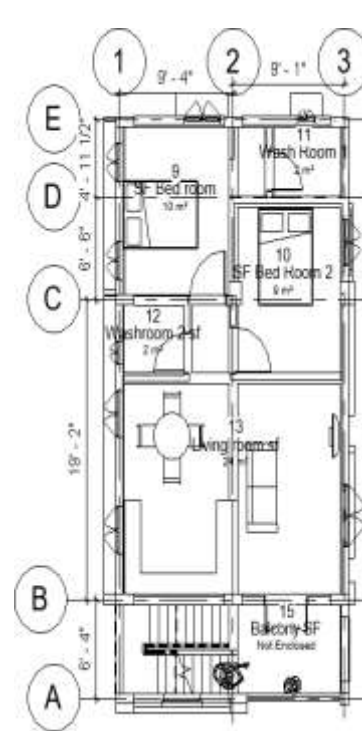


Fig.(3) Second Floor Plan

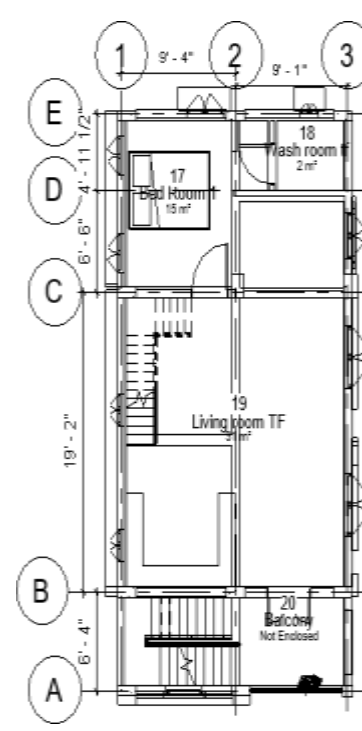


Fig.(4) Third Floor Plan

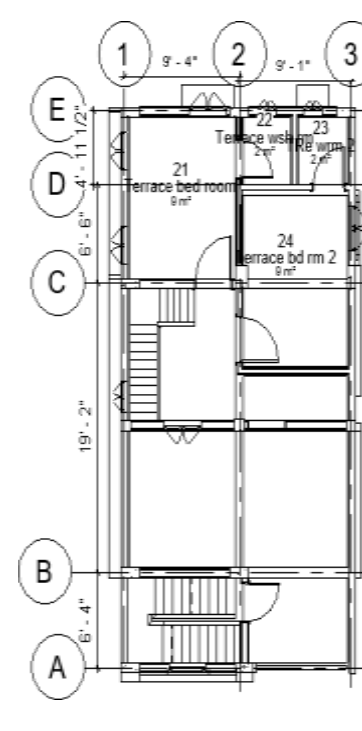


Fig.(5) Terrace Floor Plan

ELEVATED VIEWS

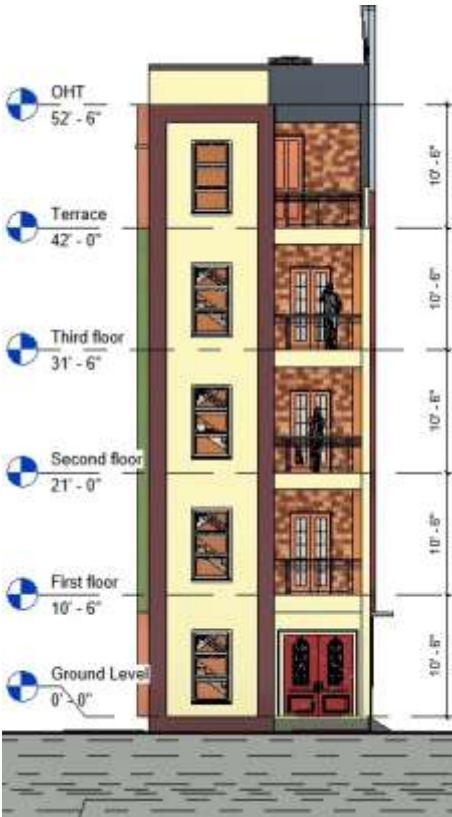


Fig.(6) Front Elevation

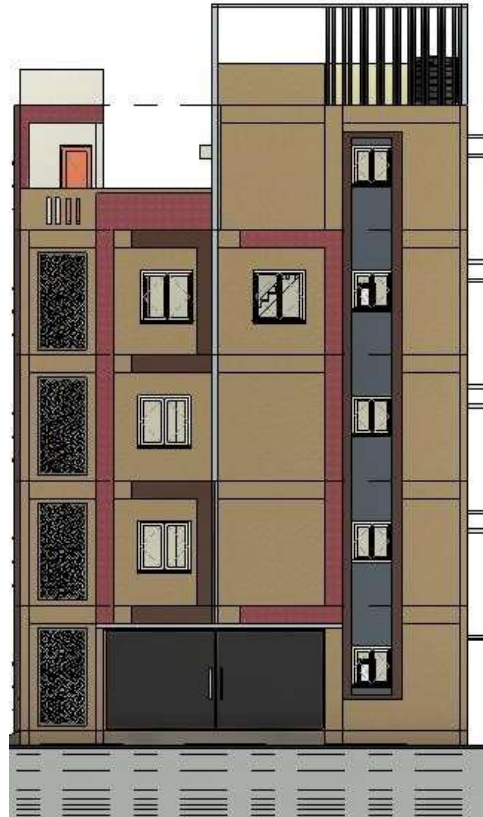


Fig.(7) Side Elevation



Fig.(8) Side Elevation

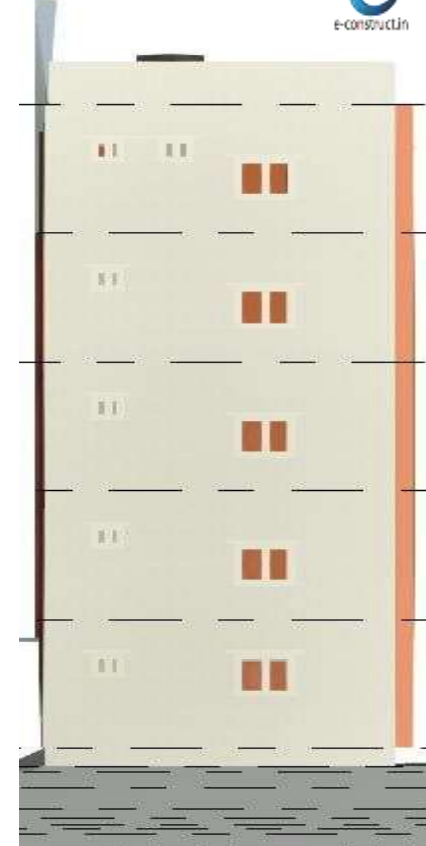


Fig.(9) Back Elevation

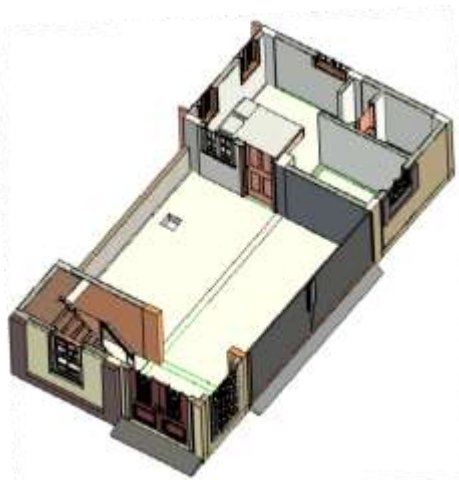


Fig.(10) Ground Floor

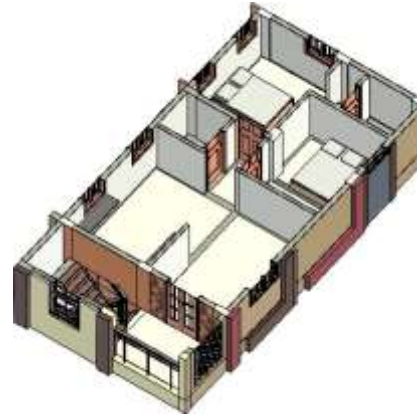


Fig.(11) First Floor



Fig.(12) Second Floor

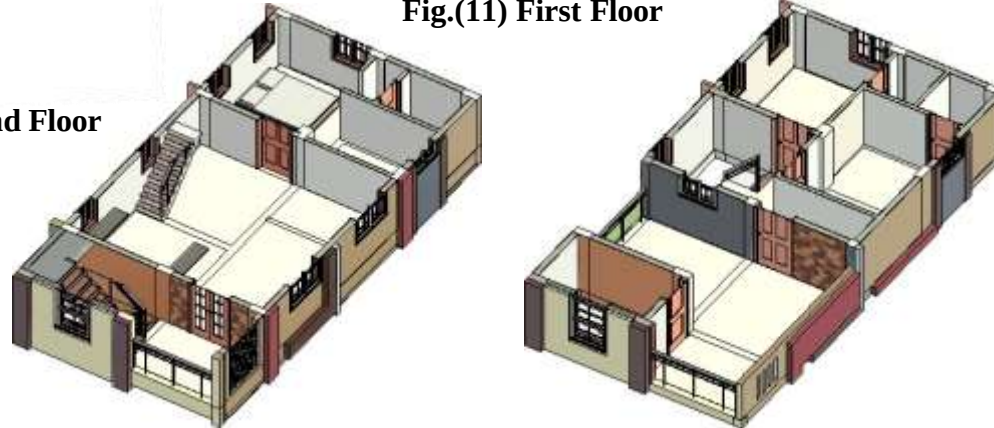


Fig.(13) Third Floor

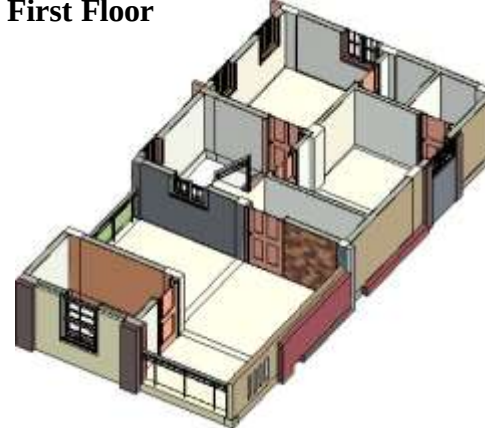


Fig.(14) Terrace Floor

Project Details:

- Total Built-up Area :-
 - Ground Floor – 228.16 SF
 - First Floor – 713.17 SF
 - Second Floor – 713.17 SF
 - Third Floor – 713.17 SF
 - Terrace Floor – 484.33 SF

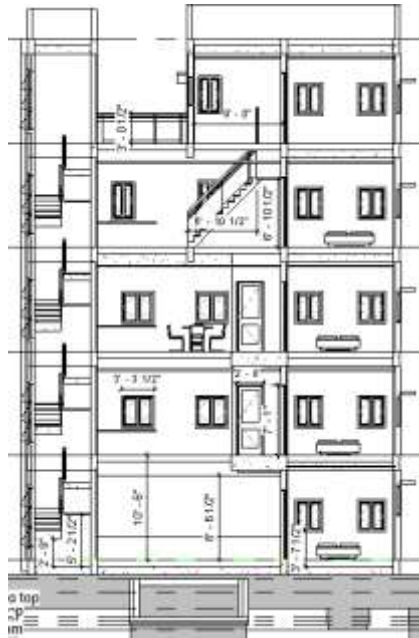
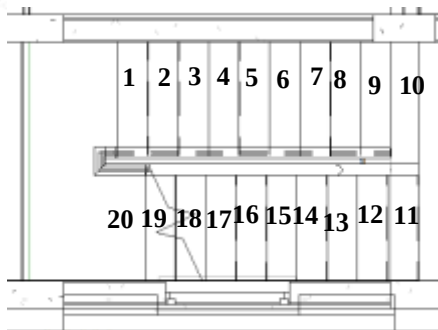
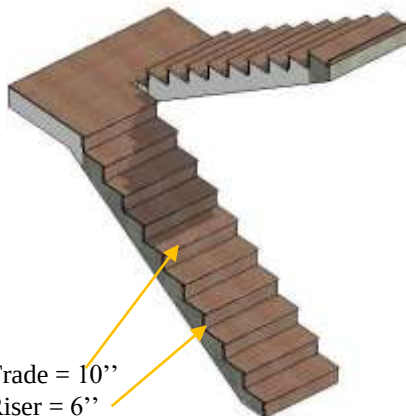
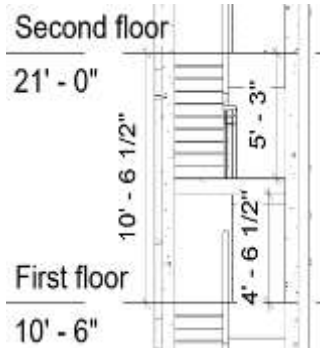
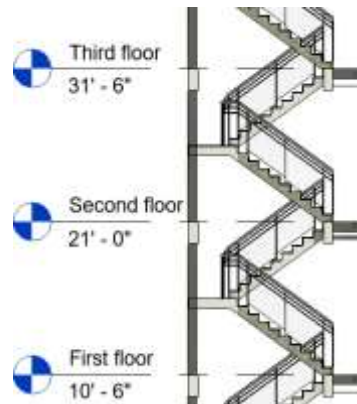


Fig.(15) Section View



Fig.(16) First Floor

STAIRCASE DETAILS



Trade = 10"
Riser = 6"
Width of Staircase = 3'6"

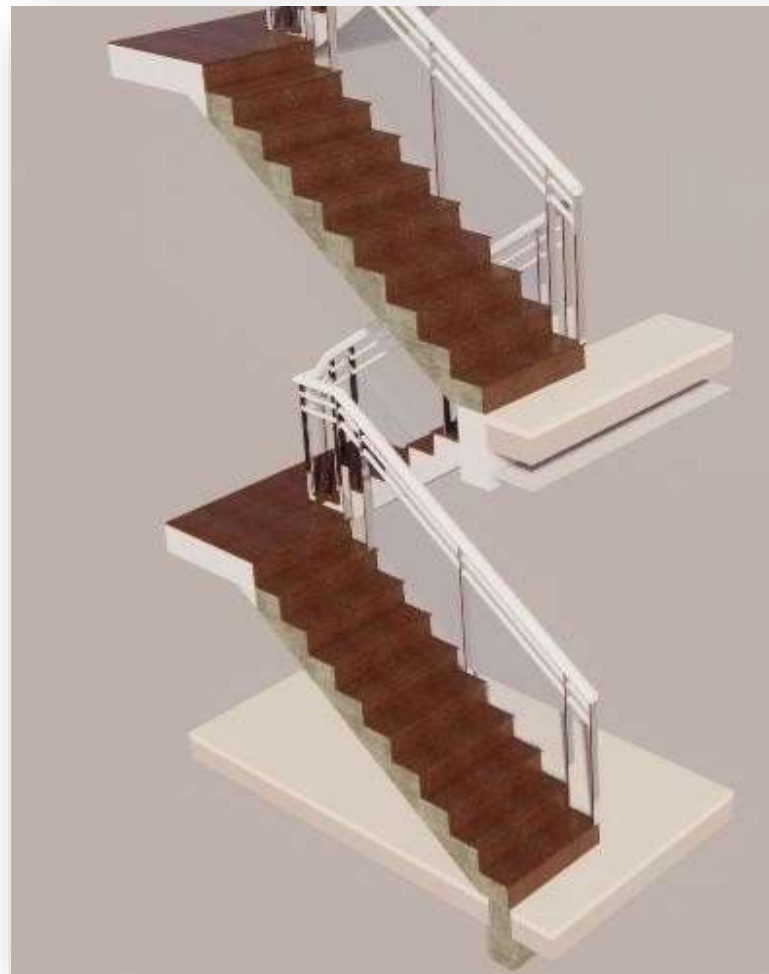


Fig.(17) Isometric View

SCHEDULES

Tile work

Sr.No	MATERIAL	AREA
1	Total Vitrified Area	1775.4 Sq.Ft
2	Anti Skid Tile for Toilets	172.16Sq.Ft
3	Anti Skid Tile for Passage and Parking	519.38 Sq.Ft
4	Staircase Tile	262.0Sq.Ft
	Total	2728.94 Sq.Ft



False Ceiling

Sr.No	Material	Level	Area
1	Gypsum Board	Ground Floor	581.04 Sq.Ft
2	Gypsum Board	First Floor	602.56 Sq.Ft
3	Gypsum Board	Second Floor	548.76 Sq.Ft
4	Gypsum Board	Third Floor	527.24 Sq.Ft
5	Gypsum Board	Terrace floor	344.32 Sq.Ft
	Total		2593.92 Sq.Ft



Window

Sr.No	Level	Material	Total No of Windows
1	GROUND FLOOR	Wooden	7
2	FIRST FLOOR	Wooden	10
3	SECOND FLOOR	Wooden	10
4	THIRD FLOOR	Wooden	10
5	TERRACE FLOOR	Wooden	9
	Total		46



Room Schedule

Sr.No	LEVEL	AREA
1	GROUND LEVEL	228.16 SF
2	FIRST FLOOR	713.17 SF
3	SECOND FLOOR	713.17 SF
4	THIRD FLOOR	713.17 SF
5	TERRACE	484.33 SF
6	OHT	489.34 SF
	Total	3374.01 SF



Room Legend

- Balcony SF
- Living room sf
- SF Bed room
- SF Bed Room 2
- Wash Room 1
- Washroom 2 sf

Door

Sr.No	Level	Family and Type	Material	Width	Height	Count
1	GROUND FLOOR	M_Door-Interior-Single-4_Panel-Wood	Wooden	30"	7'6"	1
2	GROUND FLOOR	M_Door-Interior-Single-Flush_Panel-Wood: interior door	Wooden	26"	7'6"	1
3	GROUND FLOOR	Victorian_front_door_1824": Victorian_front_door	Wooden-Glass	60"	7'6"	1
4	FIRST FLOOR	M_Door-Interior-Single-4_Panel-Wood	Wooden	30"	7'6"	2
5	FIRST FLOOR	M_Door-Interior-Single-Flush_Panel-Wood: interior door	Wooden	26"	7'6"	2
6	FIRST FLOOR	M_Door-Interior-Double-Full Glass-Wood: Double door	Wooden-Glass	30"	7'6"	1
7	SECOND FLOOR	M_Door-Interior-Single-4_Panel-Wood	Wooden	30"	7'6"	2
8	SECOND FLOOR	M_Door-Interior-Single-Flush_Panel-Wood: interior door	Wooden	26"	7'6"	2
9	SECOND FLOOR	M_Door-Interior-Double-Full Glass-Wood: Double door	Wooden-Glass	30"	7'6"	1
10	THIRD FLOOR	M_Door-Interior-Single-4_Panel-Wood	Wooden	30"	7'6"	1
11	THIRD FLOOR	M_Door-Interior-Single-2_Panel-Wood:Interior door	Wooden	26"	7'6"	1
12	THIRD FLOOR	M_Door-Interior-Double-Full Glass-Wood: Double door	Wooden-Glass	30"	7'6"	1
13	TERRACE	M_Door-Interior-Single-4_Panel-Wood	Wooden	26"	7'6"	3
14	TERRACE	M_Door-Interior-Single-2_Panel-Wood: washroom door	Wooden	30"	7'6"	3
					Total	22



STRUCTURAL DETAILS

FOOTING SCHEDULE (M25,Fe500)

FOOTING NUMBERS	COLUMN NUMBERS	FOOTING TYPE	FOOTING DIMENSION					FOOTING REINFORCEMENT				REMARKS
			STEP	L	B	D1	D2	BOTTOM		TOP		
								ALONG B	ALONG L	ALONG B	ALONG L	
FC1	C1,C2,C3	COMBINED	-	5900	1200	500	-	T 12@125c/c	T 12@125c/c	T 12@100c/c	T 12@100c/c	-
FC2	C4,C5,C6	COMBINED	-	5900	2450	500	-	T 12@125c/c	T 12@125c/c	T 12@100c/c	T 12@100c/c	-
FC3	C7,C8,C9,C10,C11,C12	COMBINED	-	5900	3480	500	-	T 12@125c/c	T 12@125c/c	T 12@100c/c	T 12@100c/c	-

Fig.(17) Structural Model:

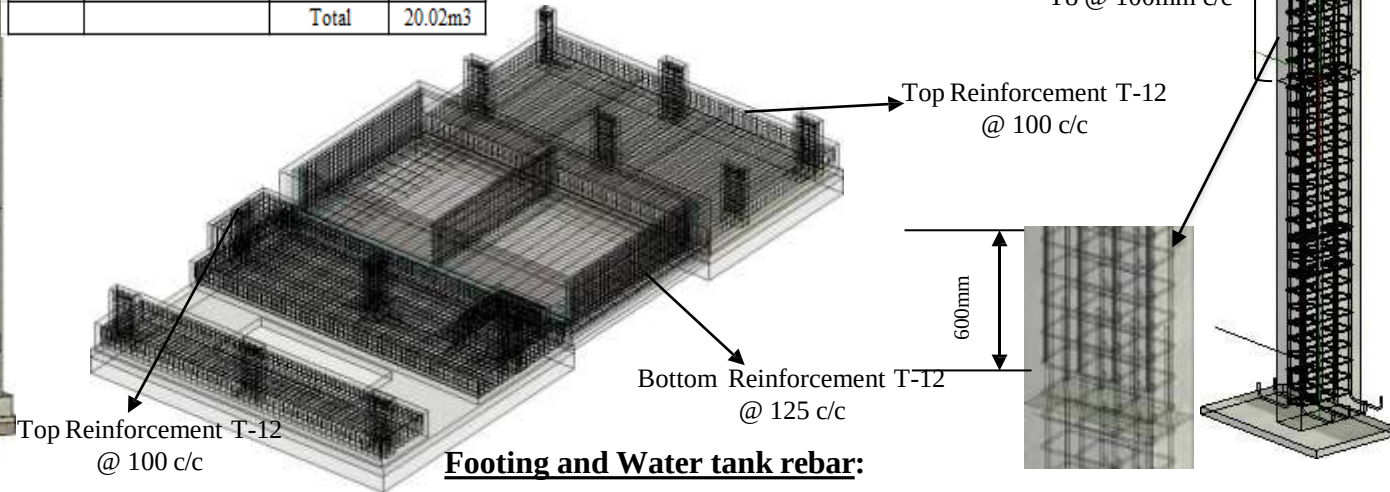


Concrete for Column:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Concrete-Rectangular-Column: 200x450mm	M-30	16.3m3
2	Concrete-Rectangular-Column: 300 x 450mm	M-30	2.42m3
3	Concrete-Rectangular-Column: 230x230mm	M-30	1.3m3
	Total		20.02m3

Concrete for Footing:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Concrete-Cast in situ (Footing)	M30	21.06m3
2	Pedestal beam Concrete	M30	1.21m3
	Total		22.27m3



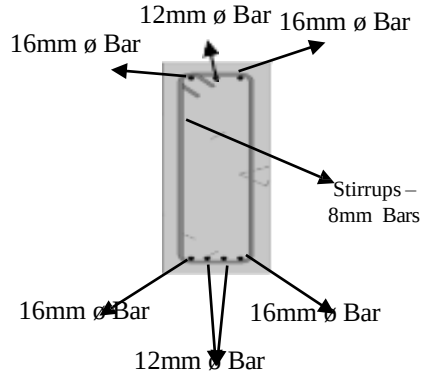
Main bars:8 Nos of 12mmØ Bars

T8 @ 100mm c/c

T8 @ 250 mm c/c

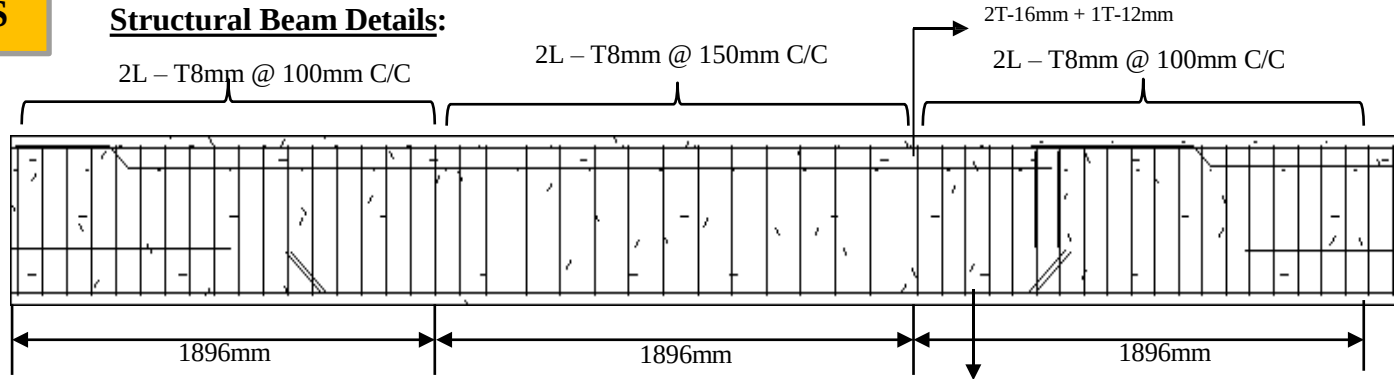
T8 @ 100mm c/c

STRUCTURAL DETAILS



Beam Section:

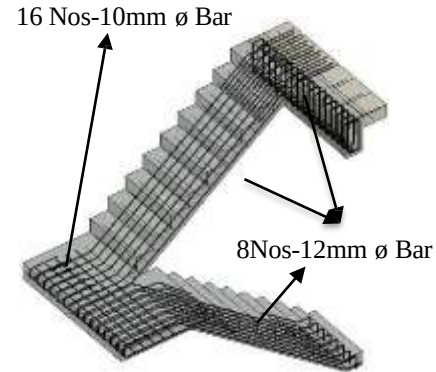
Structural Beam Details:



Concrete Consumption by Beam:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Concrete-Rectangular Beam: 200x450mm	M25	17.73m3
2	Concrete-Rectangular Beam: 200x600mm	M25	14.06m3
	Total		31.80m3

Structural Staircase Details:

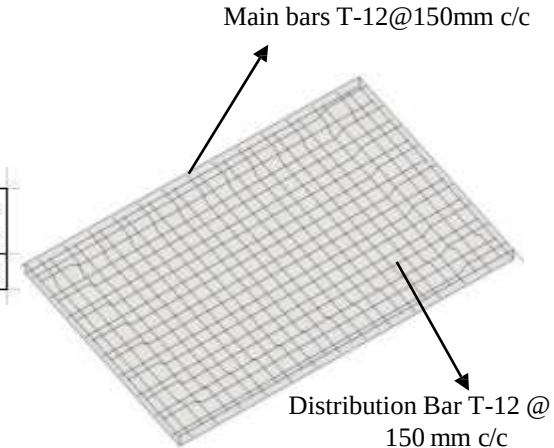


Concrete for Staircase:

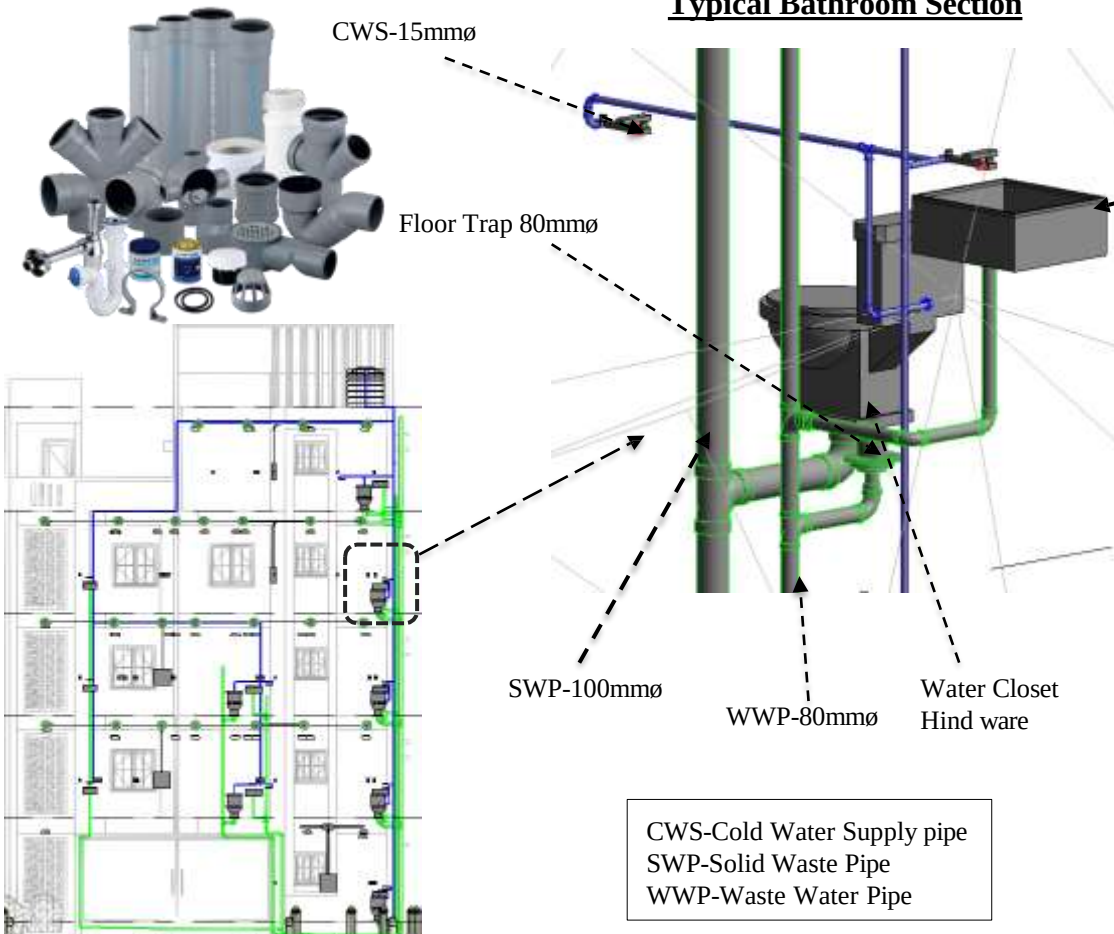
Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Concrete Cast in Situ	M25	9.6m3

Structural Model:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Floor 6 th Slab	M25	46.13m3



Typical Bathroom Section



Pipe Details

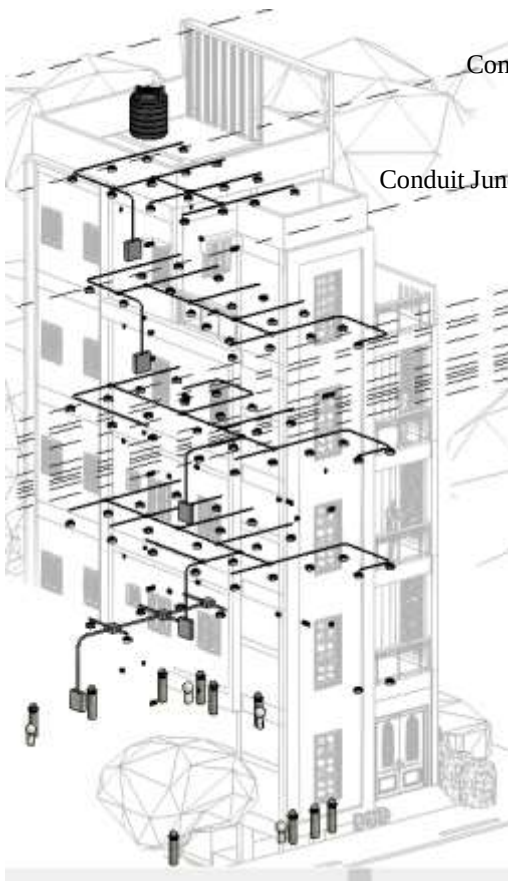
Sr.No	DIAMETER OF PIPE	MATERIAL	LENGTHS
1	15mmø	Polyvinyl Chloride-Rigid	289'2"
2	40mmø	Polyvinyl Chloride-Rigid	187'9"
3	80mmø	Polyvinyl Chloride-Rigid	65'2"
4	100mmø	Polyvinyl Chloride-Rigid	54'1"
		Total	595'2"

Sink 20"x18"

Pipe Accessories Details

Sr.No	Size	Family and Type	Count
1	15mmø-15mmø	M Plug PVC Standard	19
2	15mmø-15mmø	M Elbow -Generic Standard	52
3	15mmø-15mmø	M-Tee -Generic Standard	26
4	40mmø	M Plug PVC Standard	1
5	40mmø-40mmø	M Elbow -Generic Standard	22
6	40mmø-40mmø	M-Tee -Generic Standard	3
7	50mmø	M Plug PVC Standard	2
8	50mmø-40mmø	M Reducer-Standard	9
9	50mmø-50mmø	M Elbow -Generic Standard	13
10	80mmø-80mmø	M-Tee -Generic Standard	18
11	80mmø-80mmø	M Plug PVC Standard	1
12	80mmø-80mmø	M Elbow -Generic Standard	2
13	80mmø-80mmø	M Plug PVC Standard	44
14	100mmø	M-Tee -Generic Standard	1
15	100mmø-80mmø	M Reducer-Standard	3
16	100mmø	M Elbow -Generic Standard	4
17	100mmø-80mmø	M-Tee -Generic Standard	7

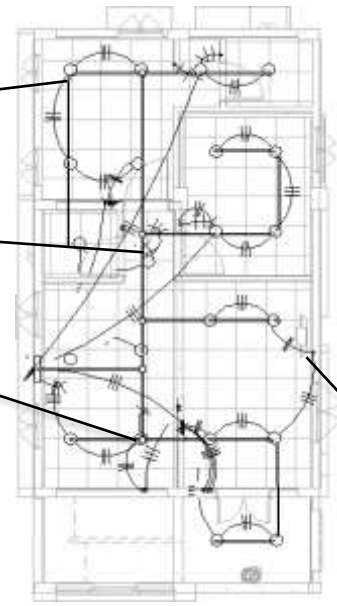




Elbow

Conduit 27mm

Conduit Junction box



Electrical fixtures and Equipment's

Sr.No	FAMILY AND TYPE	COUNT
1	M_Bollard Light: Incandescent - 120V	13
2	M_Ceiling Light - Flat Round: 60W - 120V	86
3	Ceiling Light - Flat Round:Two Switch Light	26
4	M_Spot Light - Exterior: 300W Incandescent - 120V	4
5	M_Street Light - Standard: 400W Halogen - 120V	4
Total		133



Switch with Receptacle

Conduit and its Accessories Details

Sr.No	SIZE	FAMILY AND TYPE	UNIT	COUNT
1	27mm	Conduit without Fittings: Rigid Nonmetallic Conduit (RNC) Standard	Length	423'1"
2	27mm-27mm	M_Conduit Elbow - without Fittings - RNC: Standard	Nos	16
3	27mm-27mm	M_Conduit Junction Box - Cross - PVC: Standard	Nos	4
4	27mm-27mm	M_Conduit Junction Box - Tee - PVC: Standard	Nos	21

Branch Panel: FF PANEL 2

Location:
Scale: 1/8"=1'-0"
Drawing: (blank)
Sheet: 1

Sheet: 1
Date: 10/10/10
Rev: 1
Scale: 1/8"=1'-0"
Drawing: (blank)

Notes:

Panel	Conduit Description	Qty	Panel	Qty	Panel	Qty	Panel	Qty	Panel	Qty
1	27mm RNC	1	2	1	3	1	4	1	5	1
2	27mm RNC	1	3	1	4	1	5	1	6	1
3	27mm RNC	1	4	1	5	1	6	1	7	1
4	27mm RNC	1	5	1	6	1	7	1	8	1
5	27mm RNC	1	6	1	7	1	8	1	9	1
6	27mm RNC	1	7	1	8	1	9	1	10	1
Total										
Total Length		117'00"	Total Qty		100'00"	Total Cost		100'00"		

Panel Schedule

Panel	Conduit Description	Qty	Panel	Qty	Panel	Qty	Panel	Qty	Panel	Qty
1	27mm RNC	1	2	1	3	1	4	1	5	1
2	27mm RNC	1	3	1	4	1	5	1	6	1
3	27mm RNC	1	4	1	5	1	6	1	7	1
4	27mm RNC	1	5	1	6	1	7	1	8	1
5	27mm RNC	1	6	1	7	1	8	1	9	1
6	27mm RNC	1	7	1	8	1	9	1	10	1
Total										
Total Length		117'00"	Total Qty		100'00"	Total Cost		100'00"		

Notes:

RENDERED IMAGES







RENDERED IMAGES



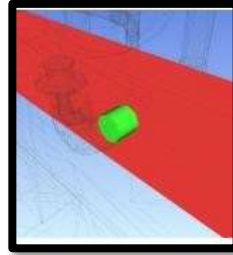
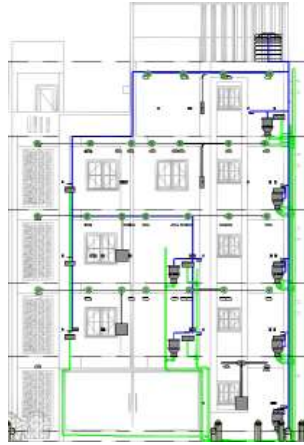
RENDERED IMAGES



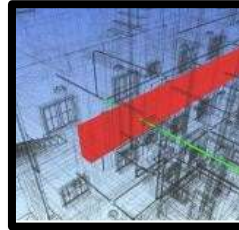
CLASH DETECTION



V/S



1. Element ID:358353
2. Element ID:1336556



1. Element ID:358873
2. Element ID:1380366

Plumbing Model

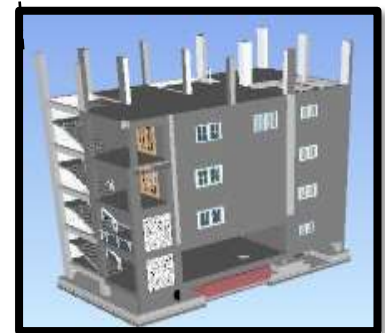
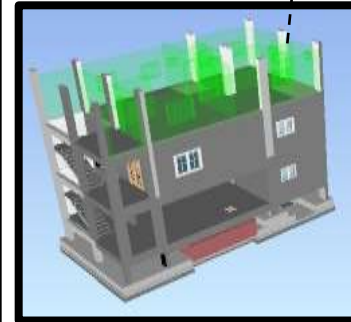
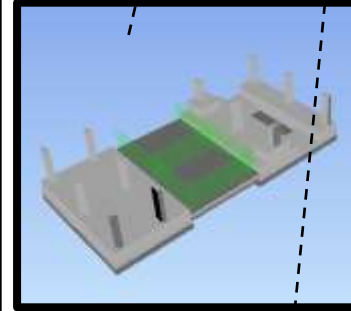
Structural Model

Clash Detection Analysis

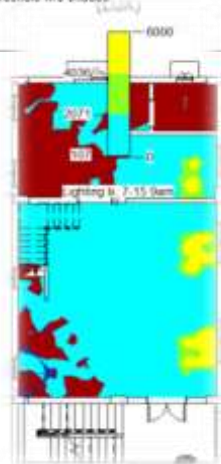
Name	Status	Clashes	New	Active	Reviewed	Approved	Resolved
Architecture vs Plumbing	Done	379	372	0	5	2	0

TIME LINER

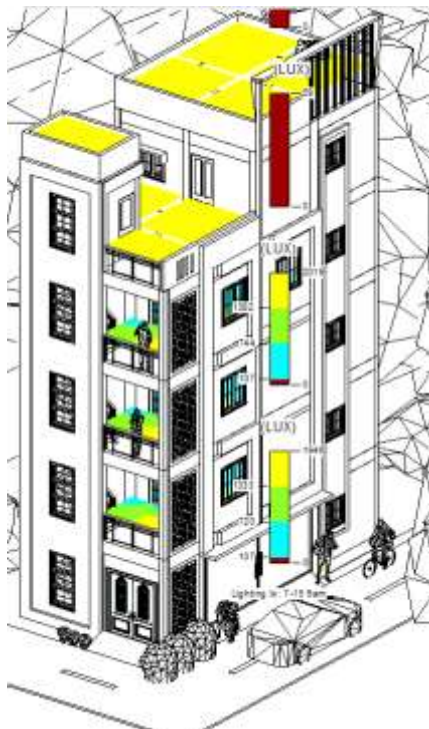
PROJECT START DATE 04-JULY- 2022	04-JULY-2022 To 07-01-2023	07-01-2023 To 14-05-2023	14-05-2023 To 28-08-2023
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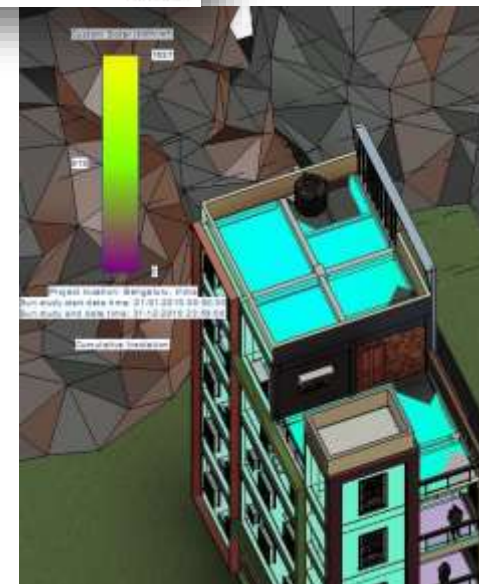
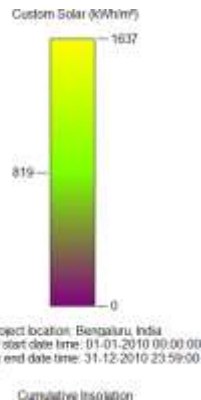
LIGHTING ANALYSIS(6D)



Lighting Analysis Model



SOLAR ANALYSIS(6D)



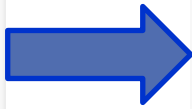
Solar Analysis Model

Solar Analysis Result

ENERGY ANALYSIS(6D)



Energy Model



Setting the building standard according to standard value

ENERGY ANALYSIS (6D)

Building Orientation:

Building Orientation

Rotates a building clockwise from 0 degrees, e.g. 90 degrees rotates the North side of the building to face East.

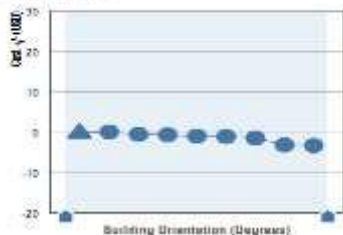
Current Setting:
BIM - 90



Energy Cost
USD / m² / yr



Building Orientation



Daylighting & Occupancy cont:

Daylighting & Occupancy Controls

Represents typical daylight dimming and occupancy sensor systems.

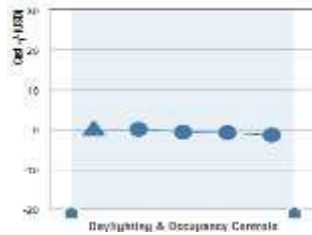
Current Setting:
BIM - Daylighting &
Occupancy Controls



Energy Cost
USD / m² / yr



Daylighting & Occupancy Controls



Payback Limit:

PV - Payback Limit

Use the payback period to define which surfaces will be used for the PV system. Surfaces with shading or poor solar orientation may be excluded.

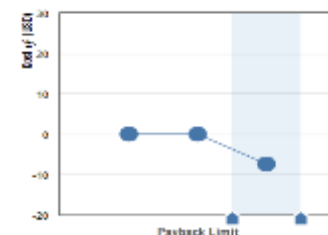
Current Setting:
30 yr



Energy Cost
USD / m² / yr



PV - Payback Limit



Surface Coverage:

PV - Surface Coverage

Defines how much roof area can be used for PV panels, assuming area for maintenance access, rooftop equipment and system infrastructure.

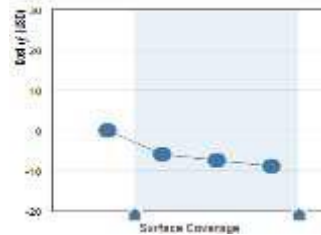
Current Setting:
60 % - 90 %



Energy Cost
USD / m² / yr

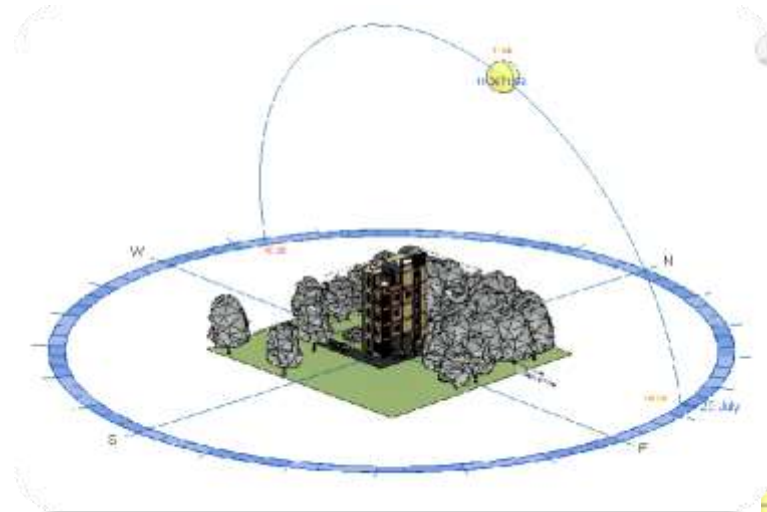


PV - Surface Coverage



LIGHTING ANALYSIS

Sun Path



Cobie Sheet



File	CODEA	This CODEA spreadsheet is an example file that comes with the CODEA Extension 1.0. Individual worksheets are organized by project phase as shown below.
Version	1	
Release	1	
System	IFC2x1	
Region	an IBC	
Purpose		
Details		
Pre-Phase	Sheet	Contents
	Content	People and Companies
Early Design Worksheets	Sheet	Contents
	Project, Site, and Facility	
	Floor	Vertical levels and exterior areas
	Space	Rooms
	Zone	Sets of spaces sharing a specific attribute
	Type	Types of equipment, products, and materials
Detailed Design Worksheets	Sheet	Contents
	Component	Individually named or to be named items
	System	Sets of components providing a service
	Assembly	Assemblies for Types, Component and others
	Connection	Logical connections between components
	Impact	Economic, Environmental and Social impacts of various stages in the life cycle
Construction Worksheets	Sheet	Contents
		NOTE: Submittals and approvals added on 'Documents' worksheet
		NOTE: Manufacturers and model added on 'Type' worksheet
		NOTE: Serial and tag added on 'Component' worksheet

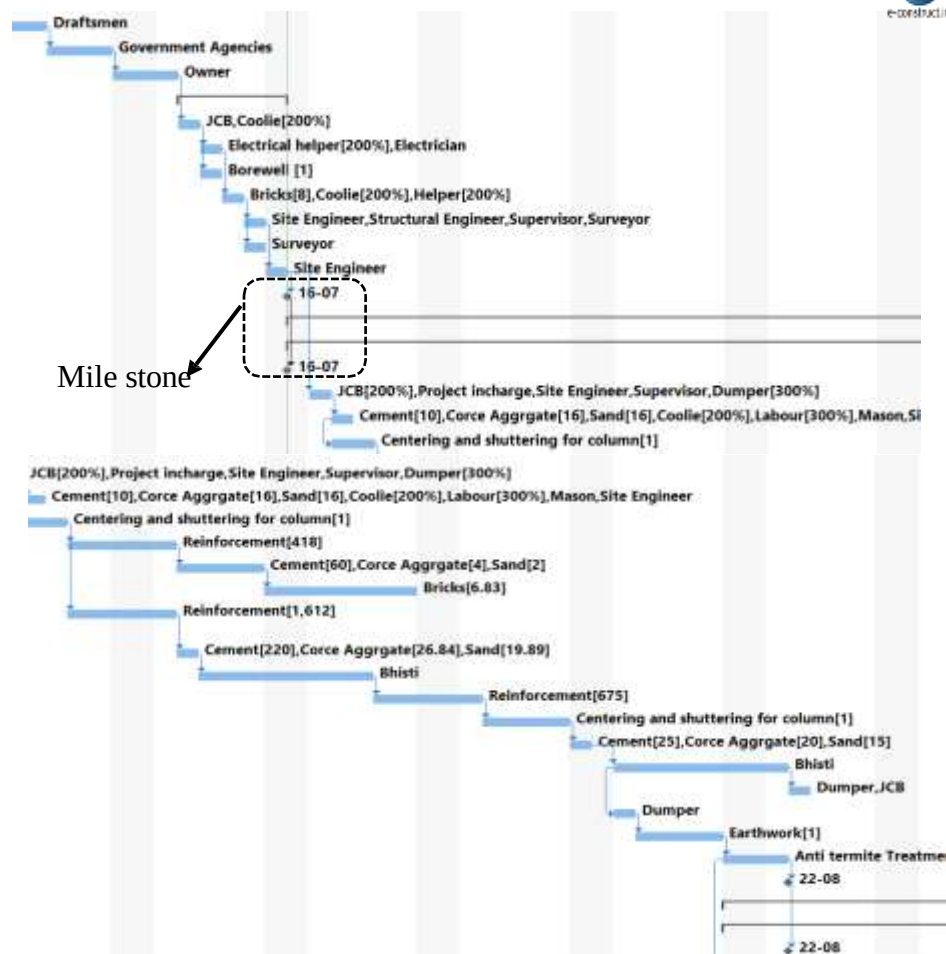
[illegible]

SR.No	NAME OF WORK	COST (.)
1.	Site cleaning & preliminary works	2,68,844/-
2.	Concreting work	11,15,916/-
3.	Fe-500 TMT Rebar	9,15,916/-
4.	Mason Work	3,36,768/-
5.	Tile Work	1,86,000/-
6.	Door and Window works	3,80,200/-
7.	Plastering work	43,958/-
8.	False ceiling	2,09,536/-
9.	Painting work	2,24,881/-
10.	Plumbing and Electric work	3,27,694/-
11.	Water proofing work	25,468/-
12.	Landscape work	1,94,000/-
	Total Cost (A)	52,59,360/-

SR.No	ADDITIONAL CHARGES	COST (Rs.)
1.	Add 5% Overhead Charges	15,53,187/-
2.	Add 1% General Consultant charge	52,101/-
3.	Add 1% Water Charges	52101/-
4.	Contractors Profit 10%	5,21,019/-
5.	Add 5% Contingency Charges	2,96,834/-
6.	Adding 12% GST	7,48,022/-
	Total Cost (B)	17,22,183/-
	Total Cost (A)+(B)	RS. 69,81,543/-

MSP GANTT CHART

Ravira Project	360 days	Mon 04-07-22	Mon 28-08-23			₹ 6,757,574.84
Paper work	27 days	Mon 04-07-22	Wed 03-08-22			₹ 120,000.00
Purchase of land/Registration	15 days	Mon 04-07-22	Wed 20-07-22		Owner	₹ 0.00
Contract signing between Client and Builder	7 days	Thu 21-07-22	Thu 28-07-22	3	Contractor[1],Ow	₹ 100,000.00
Selection of Engineer/Architect	5 days	Fri 29-07-22	Wed 03-08-22	4	Architect consulta	₹ 20,000.00
Preparation of Drawing/Design stage	39 days	Mon 04-07-22	Wed 17-08-22			₹ 72,200.00
Contract drafting	2 days	Tue 09-08-22	Wed 10-08-22	5FS+4 days	Draftsmen,Quantit	₹ 4,800.00
Design and drawing preparation	6 days	Thu 11-08-22	Wed 17-08-22	7	Architect consulta	₹ 62,000.00
GFC Drawing preparation	2 days	Mon 04-07-22	Tue 05-07-22		Draftsmen	₹ 2,400.00
Drawing approvals from authority	3 days	Wed 06-07-22	Fri 08-07-22	9	Government Agenci	₹ 3,000.00
Clients approval	2 days	Sat 09-07-22	Mon 11-07-22	10	Owner	₹ 0.00
Pre construction phase	5 days	Tue 12-07-22	Sat 16-07-22			₹ 104,600.00
Site Clearance	1 day	Tue 12-07-22	Tue 12-07-22	11	JCB,Coolie[200%]	₹ 5,600.00
Temporary power	1 day	Wed 13-07-22	Wed 13-07-22	13	Electrical helper[2]	₹ 1,700.00
Bore well	1 day	Wed 13-07-22	Wed 13-07-22	13	Borewell [1]	₹ 60,000.00
Temporary Shed material procurement	1 day	Thu 14-07-22	Thu 14-07-22	14	Bricks[8],Coolie[2]	₹ 27,800.00
Study of drawing and assigning the work	1 day	Fri 15-07-22	Fri 15-07-22	16	Site Engineer,Struc	₹ 6,500.00
Fixing of Benchmark	1 day	Fri 15-07-22	Fri 15-07-22	16	Surveyor	₹ 2,000.00
Column marking	1 day	Sat 16-07-22	Sat 16-07-22	17	Site Engineer	₹ 1,000.00
Completion of pre construction work	0 days	Sat 16-07-22	Sat 16-07-22	19		₹ 0.00
Construction phase	348 days	Sat 16-07-22	Mon 28-08-23			₹ 6,460,774.84
Sub structure	32 days	Sat 16-07-22	Tue 23-08-22			₹ 573,421.80
Start of construction phase	0 days	Sat 16-07-22	Sat 16-07-22	20		₹ 0.00
Excavation	1 day	Mon 18-07-22	Mon 18-07-22	19	JCB[200%],Project	₹ 20,500.00
PCC	1 day	Tue 19-07-22	Tue 19-07-22	24	Cement[10],Corce	₹ 48,950.00
Formwork for footing and column starting	2 days	Tue 19-07-22	Wed 20-07-22	25SS	Centering and shut	₹ 2,976.25
Reinforcement for footer task	4 days	Thu 21-07-22	Mon 25-07-22	26	Reinforcement[4]	₹ 21,350.00



REPORTS

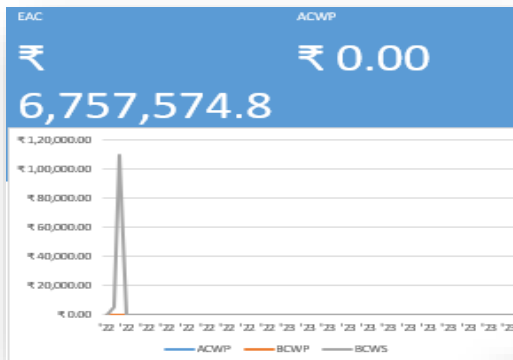
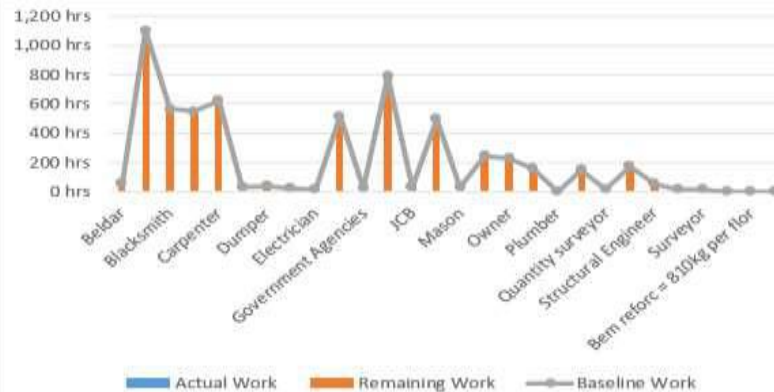
CASH FLOW:



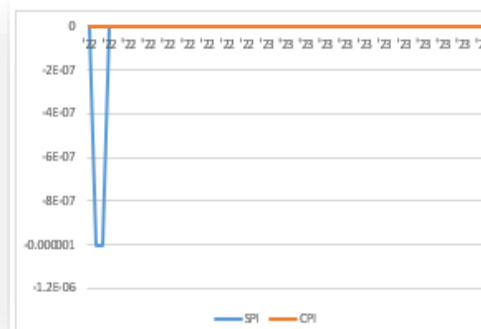
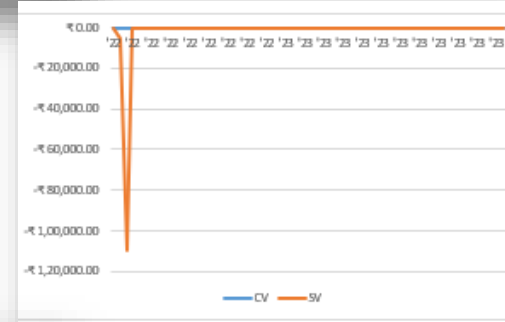
RESOURCE SHEET OVERVIEW:

RESOURCE STATS

Work status for all work resources.



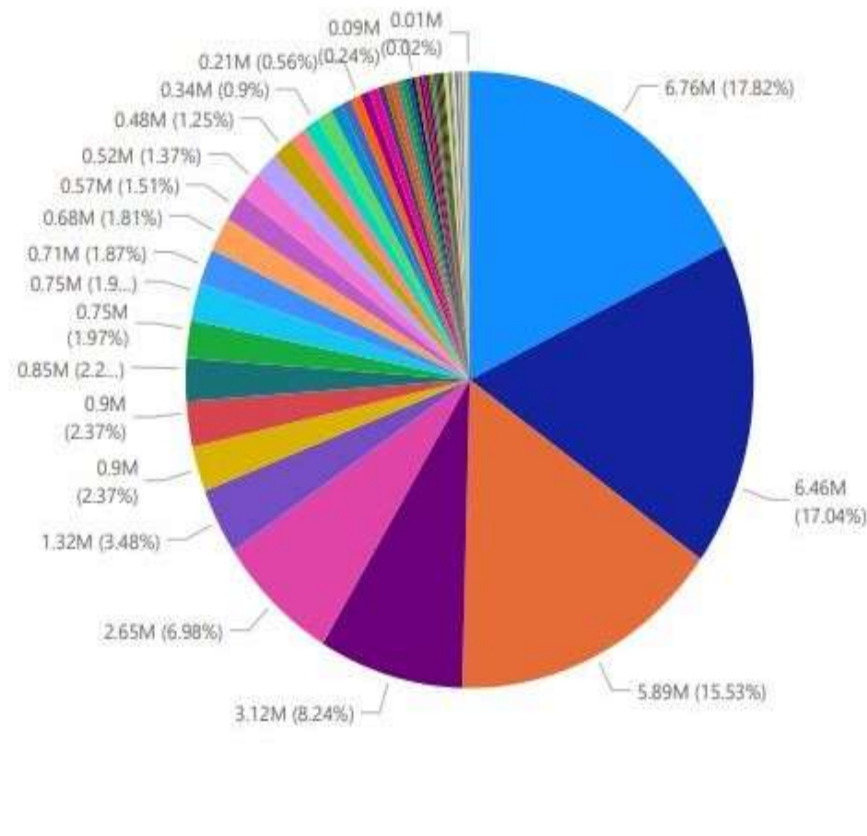
EARNED VALUE:



Name	Baseline_Cost
Concreting for footing and column starter	1,52,801.50
Concreting for staircase upto landing	20,407.00
Concreting for water tank	32,900.00
Construction of super structure	0.00
Construction phase	64,60,774.84
Contract drafting	4,800.00
Contract signing between Client and Builder	1,00,000.00
Curing	74,970.00
Curing for footing and column starter	4,998.00
Design and drawing preparation	62,000.00
Drawing approvals from authority	3,000.00
Electricals	1,38,475.00
Excavation	20,500.00
Filling the excavated material	2,000.00
Final outdoor pipe	31,535.00
Finalising	13,21,133.49
Finishing of First floor	0.00
First floor	8,99,081.60
First floor paint	47,832.20
Fixing of Benchmark	2,000.00
Flooring	6,84,588.00
Formwork for footing and column starting	2,976.25
GFC Drawing preparation	2,400.00
Grass and landscaping in the area	5,21,010.00
Ground floor	2,05,547.60
Ground floor paint	47,832.20
Ground Level	7,45,854.00
GST charge	7,45,854.00

Baseline_Cost by Name

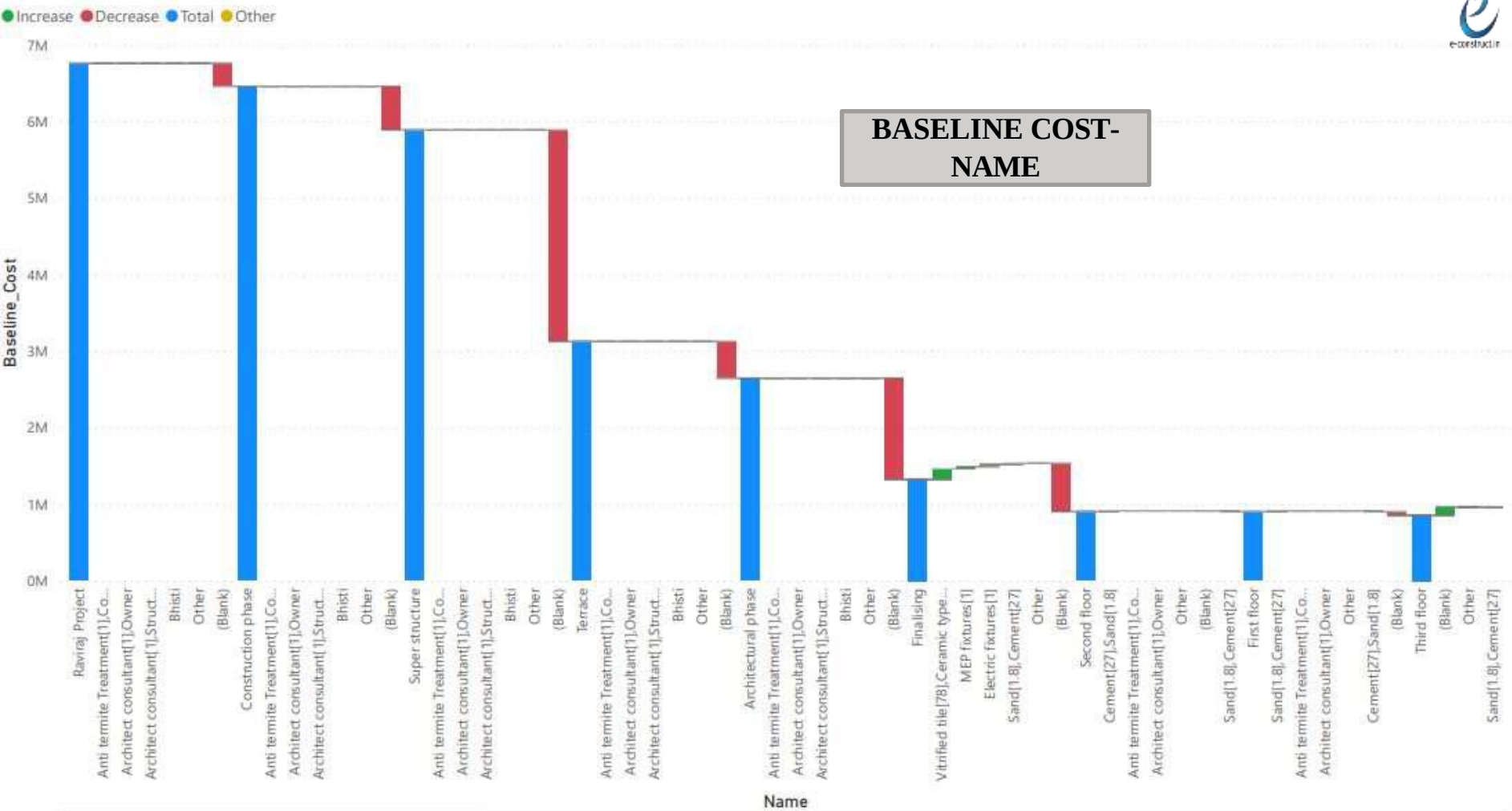
BASELINE COST-NAME



Name

- Raviraj Project
- Construction phase
- Super structure
- Terrace
- Architectural phase
- Finalising
- Second floor
- First floor
- Third floor
- GST charge
- Ground Level
- BBS for slab and beam
- Flooring
- Sub structure
- Grass and landscaping in the area
- Casting of concrete for column
- Placing of window and door frame
- BBS for Column work
- Casting for slab and beam
- Casting of brick work
- Painting work
- Shuttering and centering for slab an..

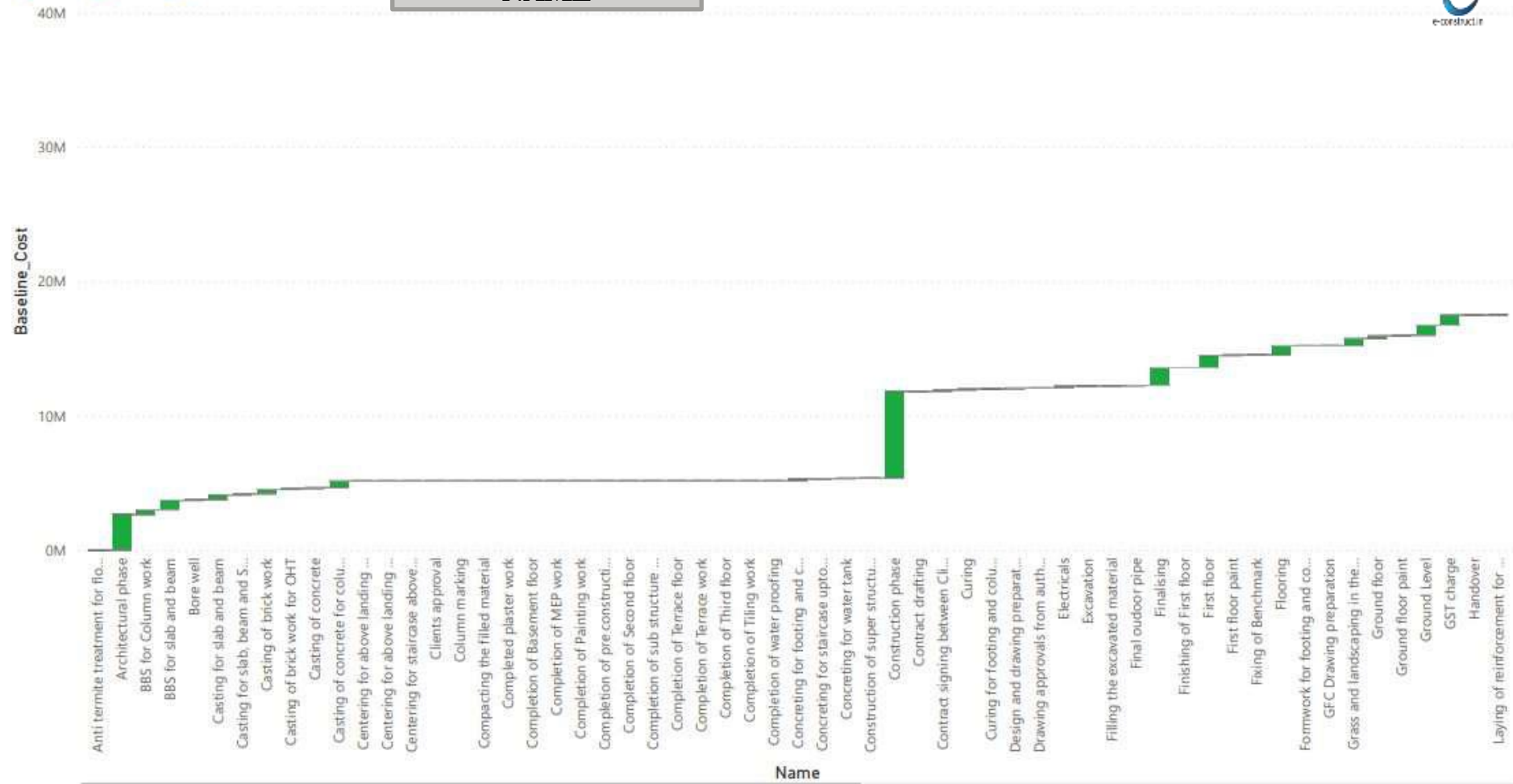
Baseline_Cost by Name and Resource_Names



Baseline_Cost by Name

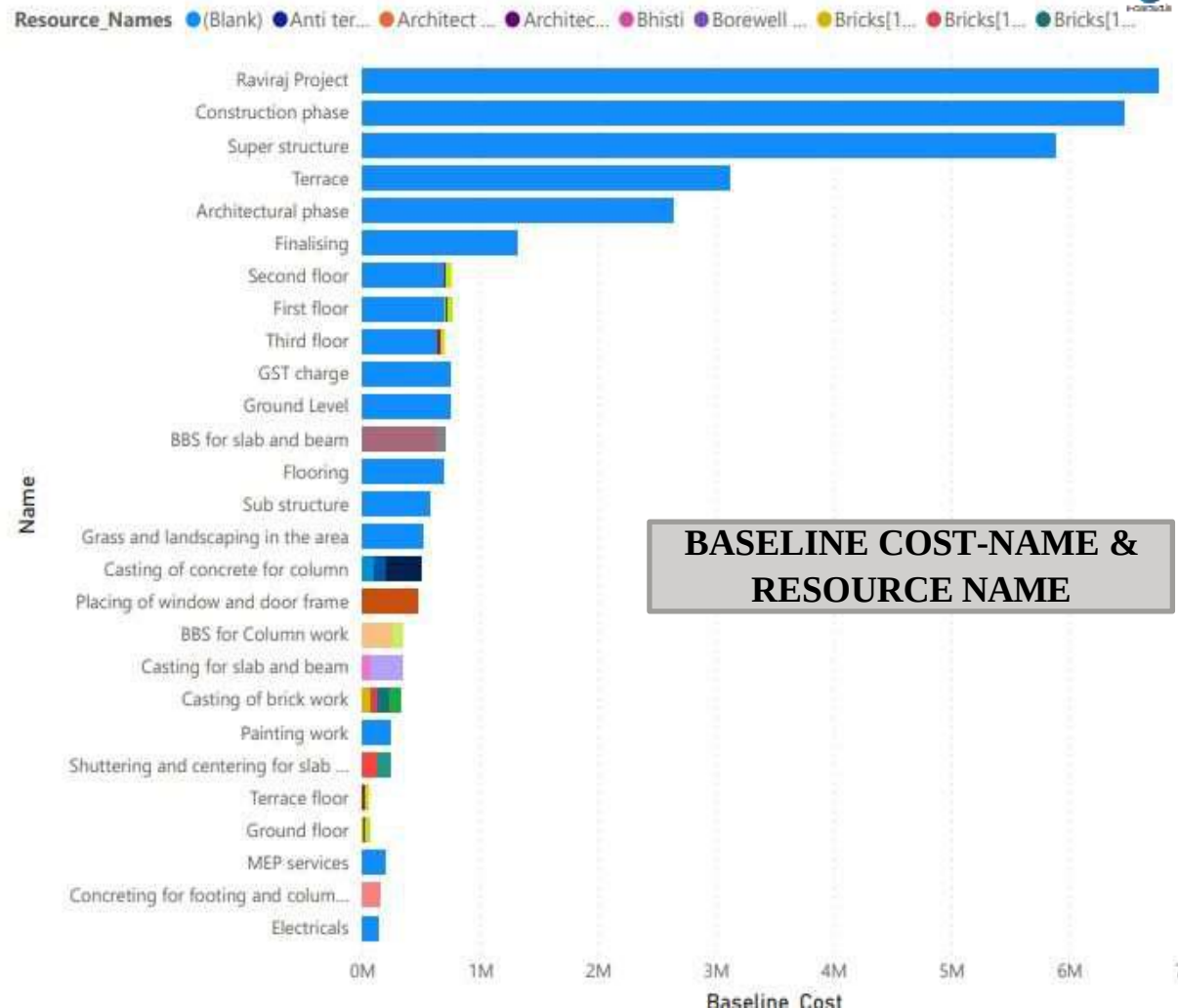
● Increase ● Decrease ● Total

BASELINE COST- NAME



Name	Resource_Names
Anti termite treatment for floor PCC	Anti termite Treatment[1],Co
Architectural phase	
BBS for Column work	Reinforcement[737],Blacksm
BBS for Column work	Reinforcement[737],Blacksm 0%]
BBS for slab and beam	Reinforcement[1,956],Blacksm
BBS for slab and beam	Reinforcement[800],Blacksm
Bore well	Borewell [1]
Casting for slab and beam	Cement[102],Corce Aggrgate
Casting for slab and beam	Cement[134],Corce Aggrgate
Casting for slab, beam and Staircase	Cement[134],Corce Aggrgate
Casting of brick work	Bricks[11.64],Bhisti,Helper,M Engineer
Casting of brick work	Bricks[15.96]
Casting of brick work	Bricks[18.39],Bhisti,Helper,La
Casting of brick work	Bricks[18.65],Carpenter,Cool
Casting of brick work for OHT	Bricks[22]
Casting of concrete	Cement[25],Corce Aggrgate]
Casting of concrete for column	Cement[34],Corce Aggrgate[41],Sand[23],Carp helper[200%],Electrician,Fitte
Casting of concrete for column	Corce Aggrgate[41],Sand[23]
Casting of concrete for column	Corce Aggrgate[41],Sand[23],Ceme Mason(Brick Layer)
Centering for above landing staircase	
Centering for above landing staircasse	
Centering for staircase above landing	
Clients approval	Owner
Column marking	Site Engineer
Total	

Baseline_Cost by Name and Resource_Names

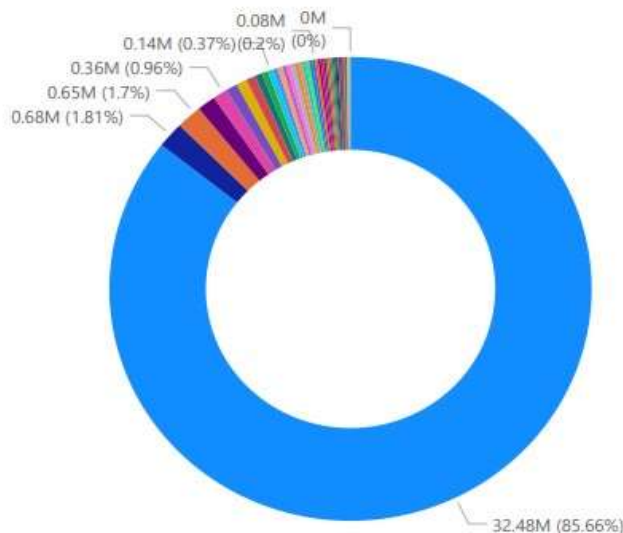


Resource Names

Anti termite Treatment[1],Coolie[600%],Helper[400%]
Architect consultant[1],Owner
Architect consultant[1],Structural Engineer,MEP Engineer[1]
Bhisti
Borewell [1]
Bricks[11.64],Bhisti,Helper,Mason(Brick Layer),Project incharge,Site Engineer
Bricks[15.96]
Bricks[18.39],Bhisti,Helper,Labour,Mason(Brick Layer),Project incharge
Bricks[18.65],Carpenter,Coolie,Helper,Mason(Brick Layer),Site Engineer
Bricks[22]
Bricks[6.83]
Bricks[8],Coolie[200%],Helper[200%]
Cement[10],Corce
Aggrgate[16],Sand[16],Coolie[200%],Labour[300%],Mason,Site Engineer
Cement[102],Corce Aggrgate[12],Sand[8]
Cement[134],Corce Aggrgate[16],Sand[9],Beldar,Coolie,Fitter,Helper
Cement[17],Sand[1.2]
Cement[220],Corce Aggrgate[26.84],Sand[19.89]
Cement[25],Corce Aggrgate[20],Sand[15]
Cement[27],Sand[1.8]
Cement[34],Corce
Aggrgate[41],Sand[23],Carpenter[200%],Coolie[400%],Electrical helper[200%],Electrician,Fitter[200%]
Cement[6],Corce Aggrgate[0.47],Sand[0.45]
Cement[6],Corce Aggrgate[0.47],Sand[0.45],Coolie,Helper,Mason
Cement[60],Corce Aggrgate[4],Sand[2]
Centering and shuttering for column[1]
Centering and shuttering for column[1],Bhisti,Carpenter,Helper
Centering and shuttering for column[1],Fitter,Labour
Centering and shuttering for slab-beam and staircase[1]

Total

Baseline_Cost by Resource_Names

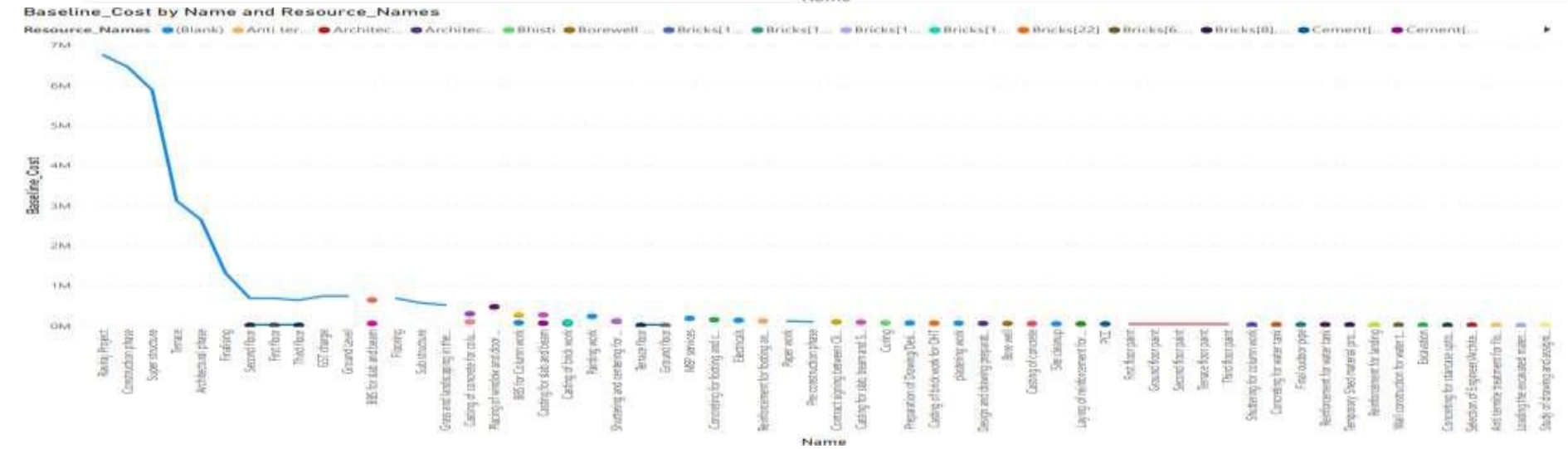
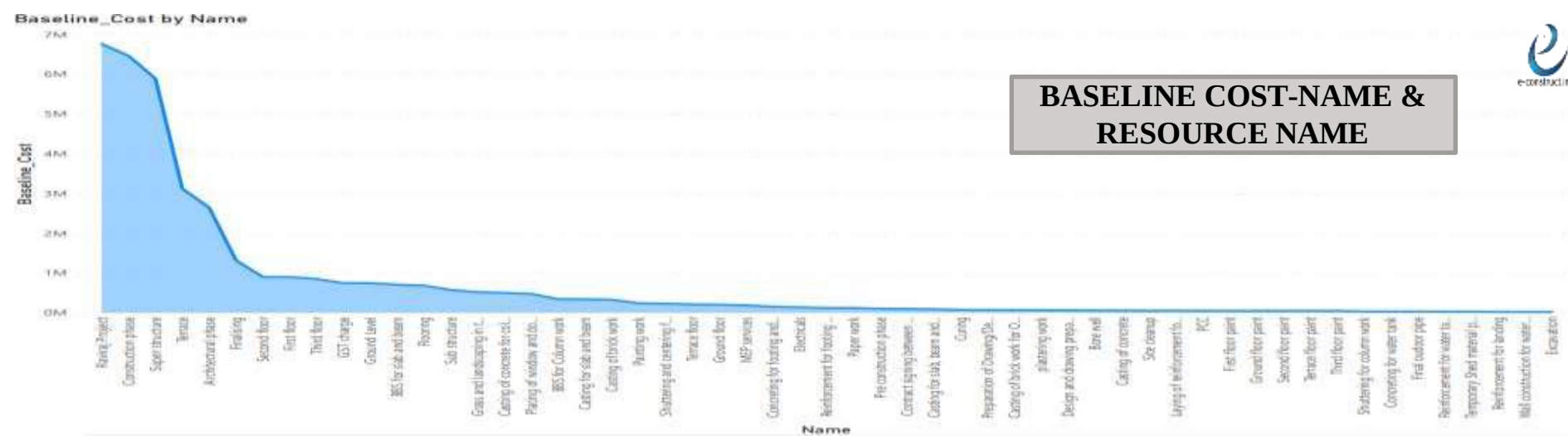


**BASELINE COST-
RESOURCE NAME**

Resource Names

- (Blank)
- Vitrified tile[78], Ceramic type[5.2], S...
- Reinforcement[1,956], Blacksmith, C...
- Doors[1], Windows[1]
- Cement[134], Corce Aggrgate[16], S...
- Corce Aggrgate[41], Sand[23], Ceme...
- Reinforcement[737], Blacksmith, Car...
- Painter, Painting[1]
- MEP fixtures[1]
- Cement[220], Corce Aggrgate[26.84...
- Electric fixtures[1]
- Centering and shuttering for slab-...
- Reinforcement[1,612]
- Centering and shuttering for slab-...
- Cement[34], Corce Aggrgate[41], Sa...
- Bricks[18.39], Bhisti, Helper, Labour, ...
- Contractor[1], Owner
- Corce Aggrgate[41], Sand[23], Ceme...
- Bricks[18.65], Carpenter, Coolie, Help...
- Bhisti
- Reinforcement[737], Blacksmith[400...
- Bricks[11.64], Bhisti, Helper, Mason (B...
- Bricks[22]
- Cement[102], Corce Aggrgate[12], S...
- Reinforcement[800], Blacksmith, Car...

BASELINE COST-NAME & RESOURCE NAME



HSR PROJECT

Bengaluru



B+G+4 Commercial



Bengaluru, karnataka 



East Facing



PLOT AREA: 3998.30 SF

TOTAL BUILTUP AREA: 1395.23 SF



SITE PLAN AND PROJECT INFO.

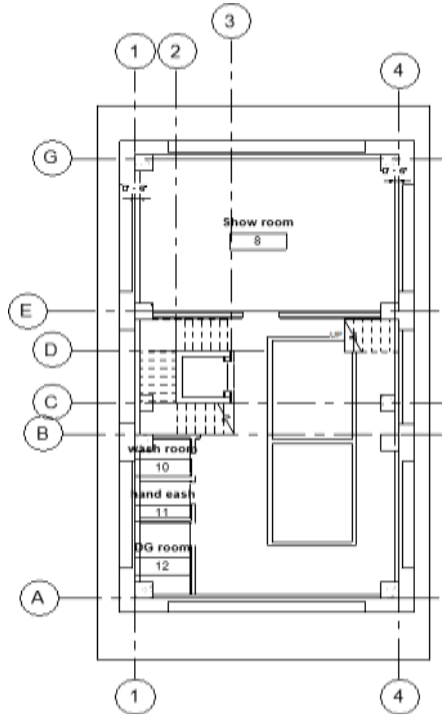


Fig.(1) Basement Floor Plan

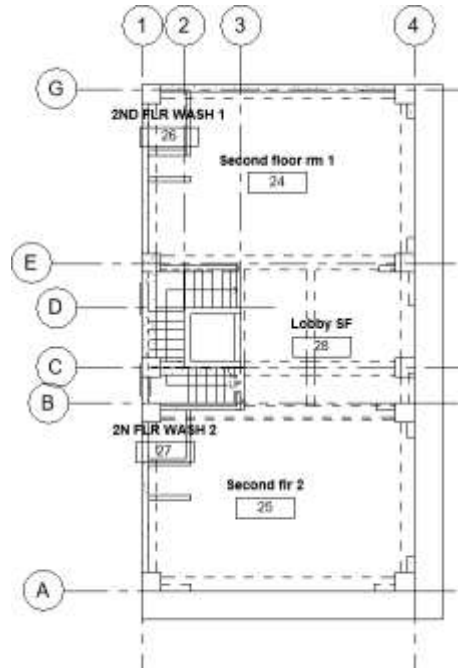


Fig.(2) Typical Floor Plan

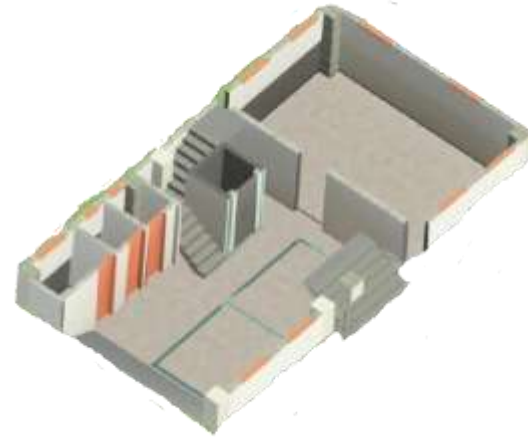


Fig.(3) Basement Floor Plan

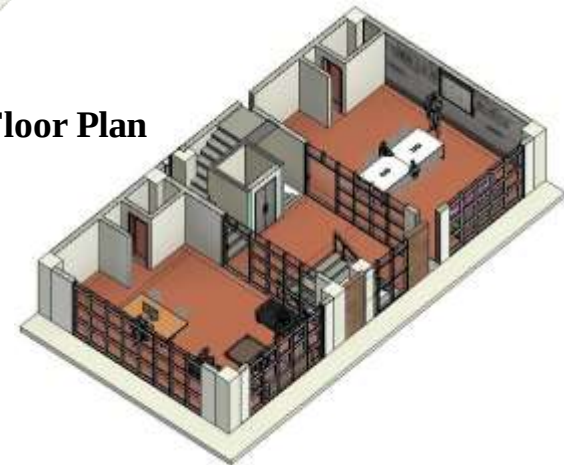


Fig.(4) Typical Floor Plan

SCHEDULES

Door Schedule

Sr.No	Level	Family and Type	Material	Width	Height	Count
1	BASEMENT	M_Door-Interior Single-I_Panal Wood 20" Washroom door	Wooden	20"	70"	2
2	BASEMENT	M_Door-Interior Single-I_Panal Wood Greenhouse door 20"	Wooden	20"	70"	1
3	GROUND FLOOR	M_Door-Interior Single-I_Panal Wood Washroom doors for typical floor	Wooden	20"	70"	2
4	GROUND FLOOR	Door-Curtain-Wall Single-Glass Door-Curtain Wall Single-Glass	Glass	40"	70"	2
5	FIRST FLOOR	M_Door-Interior Single-I_Panal Wood Washroom doors for typical floor	Wooden	20"	70"	2
6	FIRST FLOOR	Door-Curtain-Wall Single-Glass Door-Curtain Wall Single-Glass	Glass	40"	70"	2
7	SECOND FLOOR	M_Door-Interior Single-I_Panal Wood Washroom doors for typical floor	Wooden	20"	70"	2
8	SECOND FLOOR	Door-Curtain-Wall Single-Glass Door-Curtain Wall Single-Glass	Glass	40"	70"	2
9	THIRD FLOOR	M_Door-Interior Single-I_Panal Wood Washroom doors for typical floor	Wooden	20"	70"	2
10	THIRD FLOOR	Door-Curtain-Wall Single-Glass Door-Curtain Wall Single-Glass	Glass	40"	70"	2
11	FOURTH FLOOR	M_Door-Interior Single-I_Panal Wood Washroom doors for typical floor	Wooden	20"	70"	2
12	FOURTH FLOOR	Door-Curtain-Wall Single-Glass Door-Curtain Wall Single-Glass	Glass	40"	70"	2
13	TERRACE	Basement door	Wooden	20"	70"	1
14	LEFT DOORS	Door-Exterior Double-80" door	Aluminium	77"	70"	6
		Total				30



Room Schedule

Sr.No	LEVEL	AREA
1	BASEMENT	1853.08SF
2	GROUND FLOOR	1565.43SF
3	FIRST FLOOR	1565.43SF
4	SECOND FLOOR	1565.43SF
5	THIRD FLOOR	1565.45SF
6	FOURTH FLOOR	1565.45SF
7	TERRACE	1566.44SF
8	OHT	512.82SF
	Total	11758.54SF



Tile work

Sr.No	MATERIAL	AREA
1	Total Vitrified Area(Office)	5596.37 Sq.Ft
2	Anti Skid Tile for Toilets	225.42Sq.Ft
3	Anti Skid Tile for Passage and Parking	664.43 Sq.Ft
	Total Vitrified Area(Office)	6486.22 Sq.Ft



False Ceiling

Sr.No	Material	Level	Area
1	Custom Based Ceiling	Ground Floor	290.52 Sq.Ft
2	Custom Based Ceiling	First Floor	1172.84 Sq.Ft
3	Custom Based Ceiling	Second Floor	1162.08 Sq.Ft
4	Custom Based Ceiling	Third Floor	1162.08 Sq.Ft
5	Custom Based Ceiling	Fourth floor	1162.08 Sq.Ft
		Total	4949.6 Sq.Ft



Window Schedule

Sr.No	Level	Family and Type	Material	Sill Height	Height	Width	Count
1	GROUND FLOOR	Window-Awning-Double-Horizontal: basement ventilators	Wooden	3'0"	2'	4'6"	9
2	GROUND FLOOR	Window-Awning-Double-Horizontal: Washroom window basement	Wooden	3'0"	2'0"	2'0"	2
3	GROUND FLOOR	Window-Fixed: 36" x 65"	Wooden	4'3"	5'5"	3'0"	2
4	FIRST FLOOR	Window-Fixed: 36" x 65"	Wooden	4'3"	5'5"	3'0"	2
5	SECOND FLOOR	Window-Fixed: 36" x 65"	Wooden	4'3"	5'5"	3'0"	2
6	THIRD FLOOR	Window-Fixed: 36" x 65"	Wooden	4'3"	5'5"	3'0"	2
7	FOURTH FLOOR	Window-Fixed: 36" x 65"	Wooden	4'3"	5'5"	3'0"	2
8	TERRACE	Window-Fixed: 36" x 65"	Wooden	4'3"	5'5"	3'0"	2
						Total Count	23



Curtain Wall

Sr.No	Material	Level	Area
1	System Panel Glazed	All levels	2625.65 Sq.Ft



Structural Model:



Concrete for Footing:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Concrete for footing	M-30	65.51m ³
	Total		65.51m ³

Concrete for Column:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Concrete-Rectangular-Column: 2'x2'	M-30	61.01m ³
	Total		61.01m ³

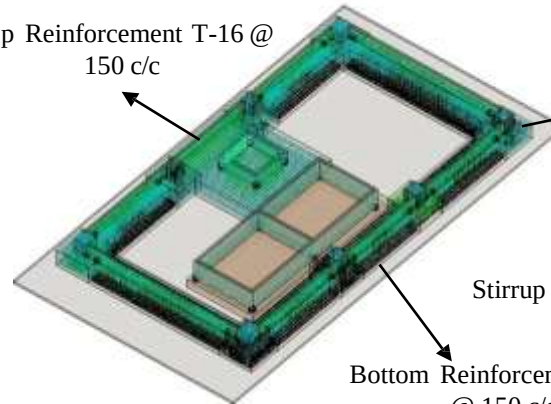
Concrete for UGT:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Basic Wall 6" wall	M-30	5.754m ³
2	Above slab for UGT	M-30	2.962m ³
	Total		8.716m ³

Concrete for Shear wall:

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Basic Wall 8" wall	M-30	27.64m ³

Top Reinforcement T-16 @ 150 c/c



Top Reinforcement T-16 @ 150 c/c

Stirrup bar T-8 @ 150 c/c

Bottom Reinforcement T-16 @ 150 c/c

Footing and Water tank rebar:

Main bars:4 Nos of 20 mm ϕ Bars

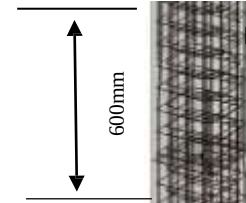
T8 @ 200 mm c/c

4L+4L 8mm ϕ bars 200mm c/c

T8 @ 200 mm c/c

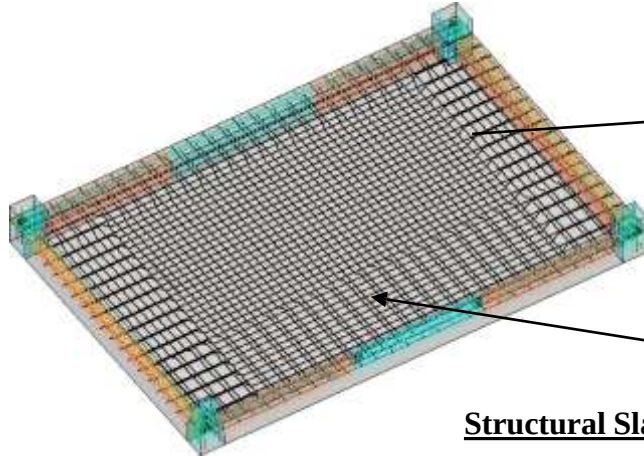
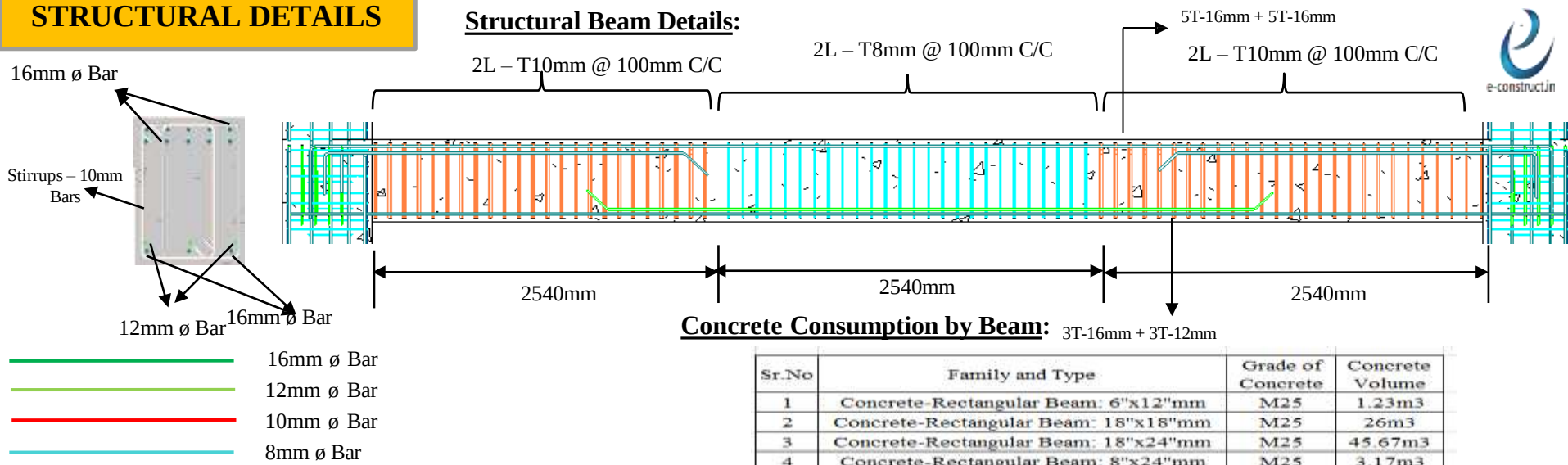
Main bars:12 Nos of 16 mm ϕ Bars

T8 @ 200mm c/c



STRUCTURAL DETAILS

Structural Beam Details:



Structural Slab Details:

Main bars T-10@150mm c/c

Main bar Bar T-10 @ 150 mm c/c

Sr.No	Family and Type	Grade of Concrete	Concrete Volume
1	Basic Wall 8" wall	M-25	103.42m3

PLUMBING DETAILS

Typical Bathroom Section

CWS-15mmø

Floor Trap 80mmø

SWP-100mmø

WWP-80mmø

Water Closet
Hind ware

CWS-Cold Water Supply pipe
SWP-Solid Waste Pipe
WWP-Waste Water Pipe

Pipe Details

Sr.No	DIAMETER OF PIPE	MATERIAL	LENGTHS
1	15mmø	Polyvinyl Chloride-Rigid	358'5"
2	40mmø	Polyvinyl Chloride-Rigid	50'9"
3	80mmø	Polyvinyl Chloride-Rigid	148'4"
4	100mmø	Polyvinyl Chloride-Rigid	145'1"
		Total	702'2"

Sink 20''x18''

Pipe Accessories Details

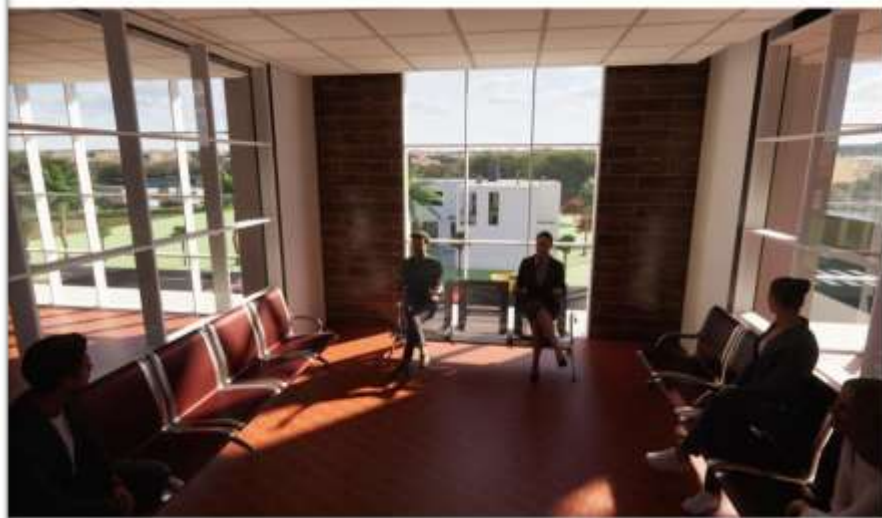
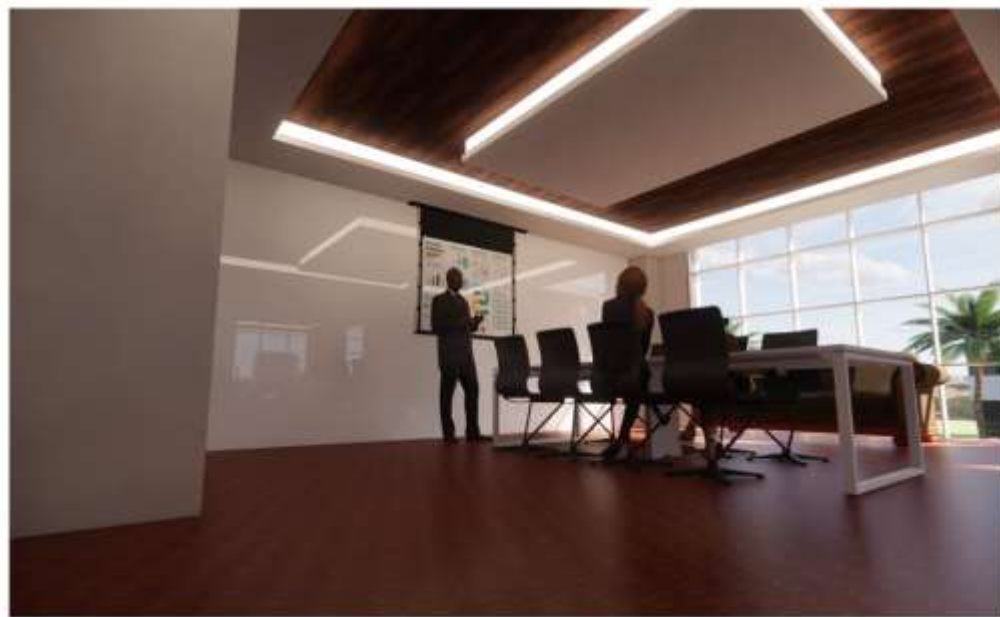
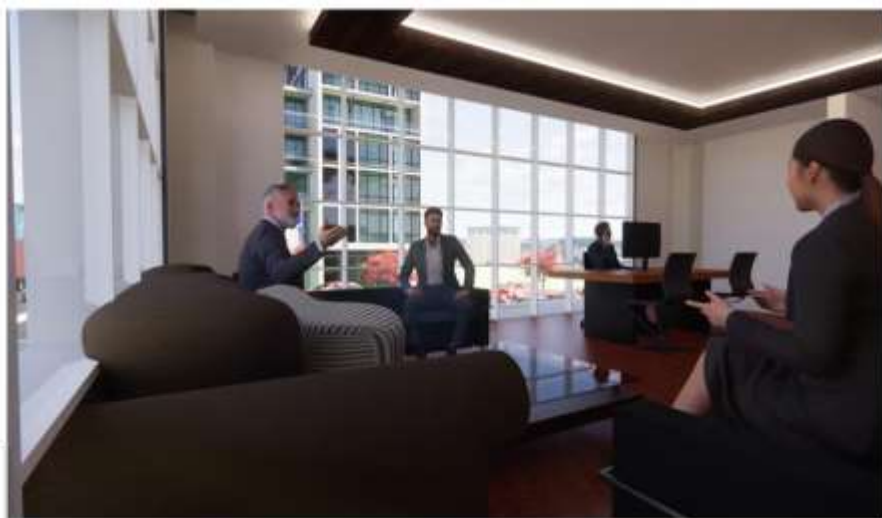
Sr.No	Size	Family and Type	Count
1	15mmø-13mmø	Bush Reduer-Standard	27
2	15mmø-15mmø	M_Elbow -Generic Standard	87
3	15mmø-15mmø	M-Tee -Generic Standard	39
4	15mmø	M_Cap	31
5	80mmø-40mmø	Bush Reducer-Standard	7
6	40mmø-40mmø	M_Elbow -Generic Standard	24
7	40mmø-40mmø	M-Tee -Generic Standard	5
8	80mmø-80mmø	M_Transition-Standard	1
9	80mmø-80mmø	M_Elbow -Generic Standard	17
10	80mmø-80mmø	M-Tee -Generic Standard	20
11	80mmø	M_Cap	2
12	100mmø-100mmø	M_Elbow -Generic Standard	19
13	100mmø-100mmø	M-Tee -Generic Standard	12
14	100mmø	M_Cap	2





RENDERED IMAGES



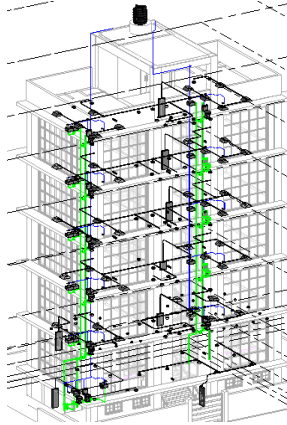


CLASH DETECTION (4D)

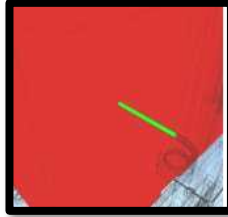


Structural Model

V/S



Plumbing Model



1. Element ID:362821
2. Element ID:1384409

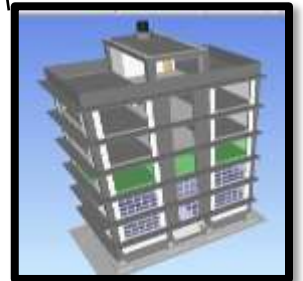
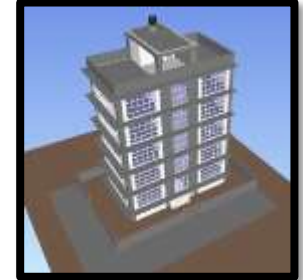
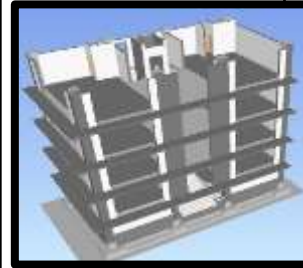
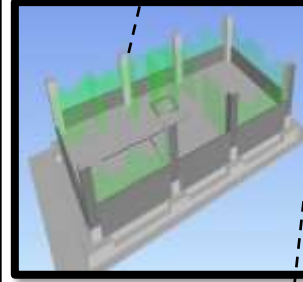
1. Element ID:359755
2. Element ID:1385341

Clash Detection Analysis

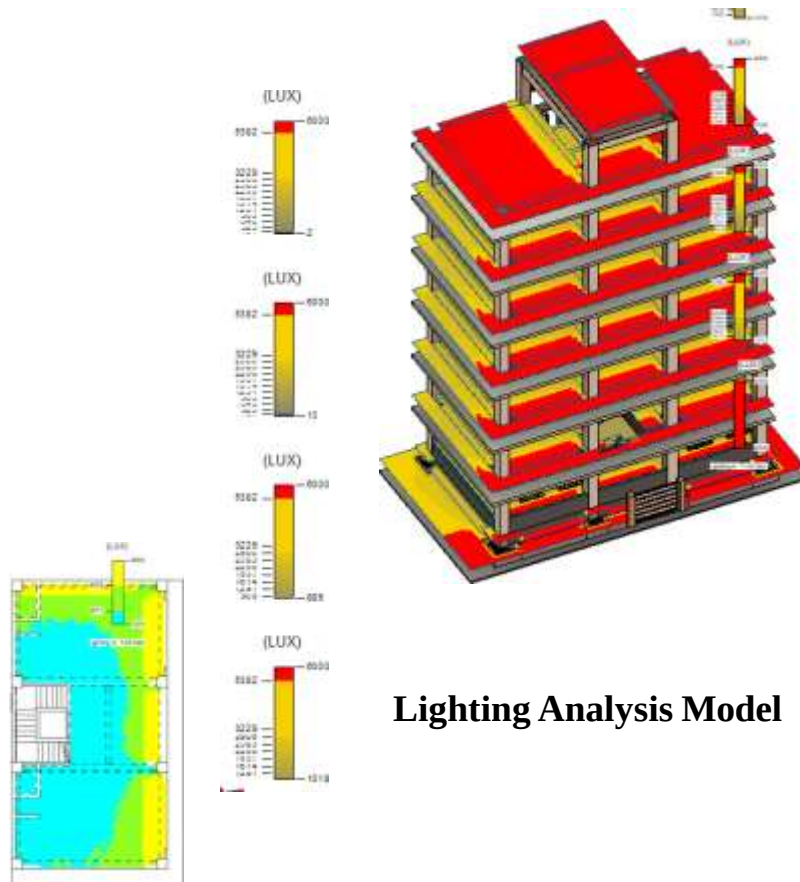
Name	Status	Clashes	New	Active	Reviewed	Approved	Resolved
Architecture vs Plumbing	Done	248	236	1	6	2	1

TIME LINER

PROJECT START DATE 04-JULY- 2022	04-JULY-2022 To 07-01-2023	07-01-2023 To 14-05-2023	14-05-2023 To 28-08-2023
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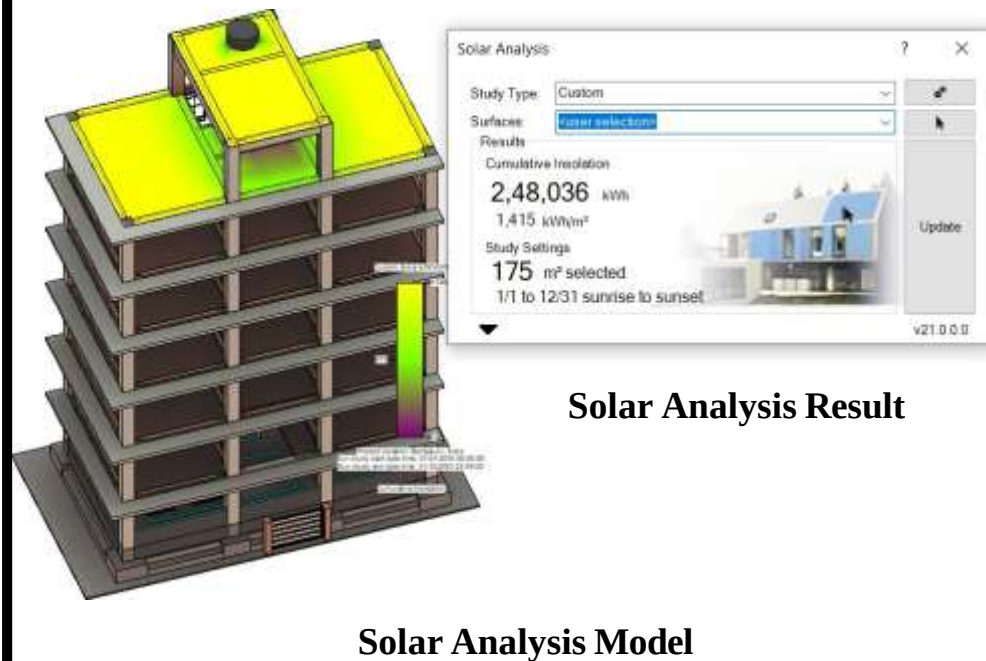


LIGHTING ANALYSIS(6D)



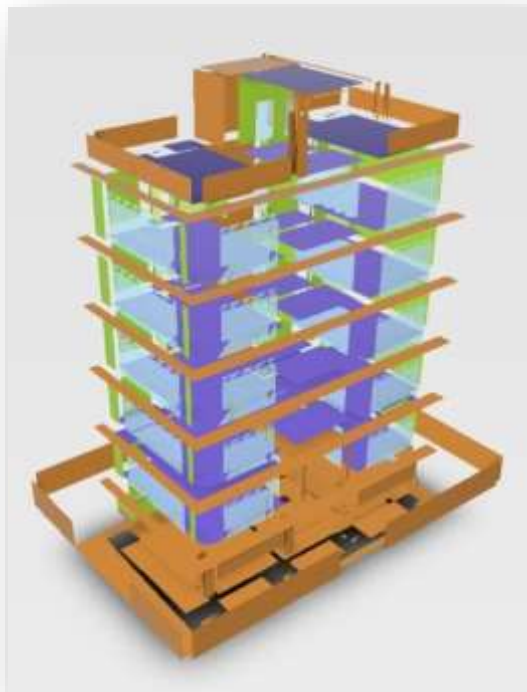
Lighting Analysis Model

SOLAR ANALYSIS(6D)

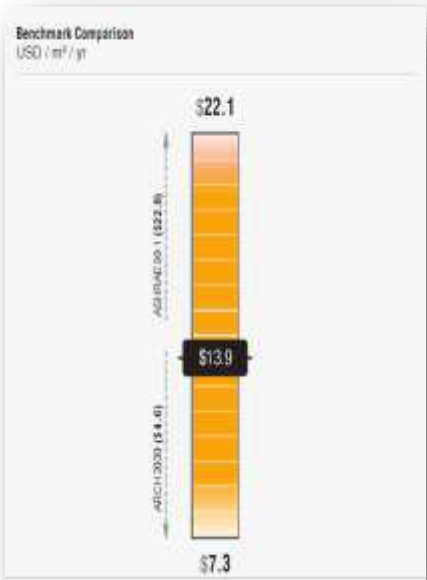
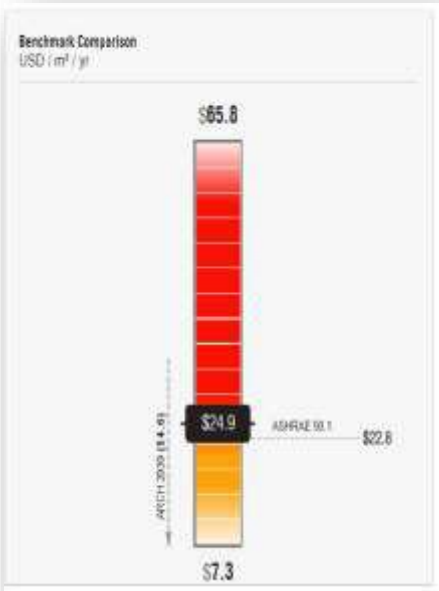


Solar Analysis Result

Solar Analysis Model



Energy Model



Setting the building standard according to standard value

ENERGY ANALYSIS (6D)

Building Orientation:

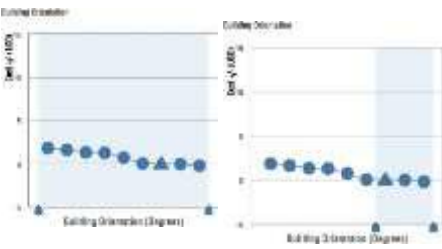
Building Orientation

Rotates a building clockwise from 0 degrees, e.g. 90 degrees rotates the North side of the building to face East.

Current Setting:
135 - 270



Editing: Building Orientation



Wall Window Ratio:

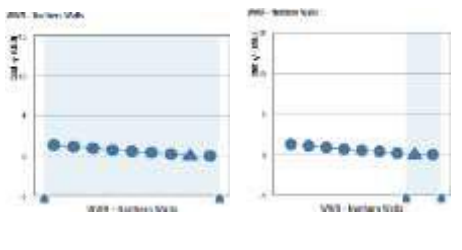
WWR - Northern Walls

Window-Wall-Ratio (glazing area / gross wall area) interacts with window properties to impact daylighting, heating & cooling.

Current Setting:
95% - 0%



Editing: WWR - Northern Walls



Lighting Efficiency:

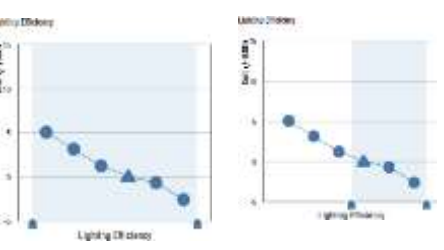
Lighting Efficiency

Represents the average internal heat gain and power consumption of electric lighting per unit floor area.

Current Setting:
20.45 W/m² - 3.23 W/m²



Editing: Lighting Efficiency



Payback Limit:

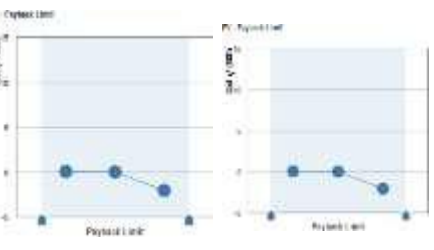
PV - Payback Limit

Use the payback period to define which surfaces will be used for the PV system. Surfaces with shading or poor solar orientation may be excluded.

Current Setting:
10 yr - 30 yr



Editing: PV - Payback Limit



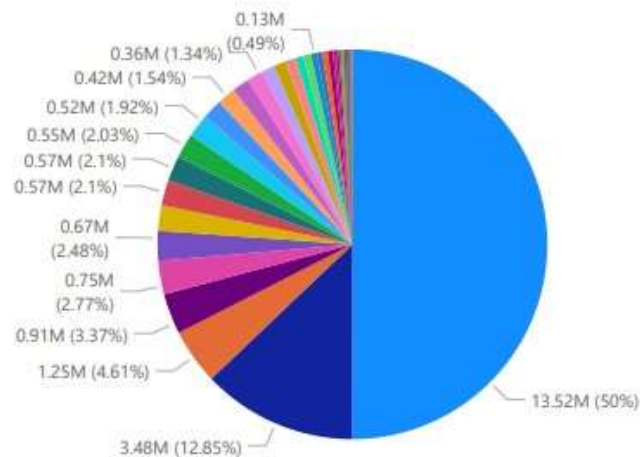
SR.No	NAME OF WORK	COST (.)
1.	Site cleaning & preliminary works	3,89,244/-
2.	Concreting work	40,08,131/-
3.	Fe-500 TMT Rebar	36,08,131/-
4.	Mason Work	8,25,600/-
5.	Tile Work	1,86,000/-
6.	Door and Window works	8,12,457/-
7.	Plastering work	1,18,100/-
8.	False ceiling	2,09,536/-
9.	Painting work	5,28,331/-
10.	Plumbing and Electric work	5,33,436/-
11.	Water proofing work	25,693/-
12.	Landscape work	1,44,000/-
	Total Cost (A)	1,17,12,690/-

SR.No	ADDITIONAL CHARGES	COST (Rs.)
1.	Add 5% Overhead Charges	15,53,187/-
2.	Add 1% General Consultant charge	1,15,028/-
3.	Add 1% Water Charges	1,15,028/-
4.	Contractors Profit 10%	11,50,287/-
5.	Add 5% Contingency Charges	6,60,403/-
6.	Adding 18% GST	24,96,323/-
	Total Cost (B)	46,52,100/-
	Total Cost (A)+(B)	RS. 1,63,64,790/-

Name	Baseline_Cost	Resource_Names
HSR PROJECT	1,35,20,197.67	
BBS for slab and beam	34,75,577.00	Carpenter[200%],Coolie[500%]
Casting of slab ,beam and staircase	12,45,545.00	Cement[240],Corce aggregate[
BBS for column work	8,63,658.00	Coolie,Fitter,Helper,Reinforcem
Casting of brick wok	7,47,852.00	Bricks[27.05],Coolie,Labour,Ma
Reinforcement for footing and column starting	6,71,475.00	Reinforcement[8,953]
Concreting for footing and column starting	3,85,180.00	Cement[544],Corce aggregate[
Casting of concrete for column	3,29,142.00	Cement[76],Corce aggregate[1
Shuttering and centering for slab and beam	2,87,308.00	Centering and shuttering for sl
Placing of window and door frames	2,73,000.00	Doors[1]
Concreting for staircase upto landing	2,36,940.00	Cement[42],Corce aggregate[1
Basement	1,95,388.50	Painting charge[1]
First floor	1,95,388.50	Painting charge[1]
Fourth floor	1,95,388.50	Painting charge[1]
Ground floor	1,95,388.50	Painting charge[1]
Second floor	1,95,388.50	Painting charge[1]
Terrace	1,95,388.50	Painting charge[1]
Third floor	1,95,388.50	Painting charge[1]
Laying of reinforcement for plinth level	1,76,325.00	Reinforcement[2,351]
Reinforcement for watertank	1,74,246.00	Reinforcement[2,323.28]
PCC	1,32,314.00	Cement[184],Corce aggregate[
Basement	1,31,399.81	Tile work[1]
First floor	1,31,399.81	Tile work[1]
Fourth floor	1,31,399.81	Tile work[1]
Ground floor	1,31,399.81	Tile work[1]
Second floor	1,31,399.81	Tile work[1]
Terrace	1,31,399.81	Tile work[1]
Third floor	1,31,399.81	Tile work[1]
First floor	1,23,861.52	Curtain glass [1],Coolie,Helper,
Fourth floor	1,23,861.52	Curtain glass [1],Coolie,Helper,
Ground floor	1,23,861.52	Curtain glass [1],Coolie,Helper,
Total	2,70,40,395.34	

Baseline_Cost by Name

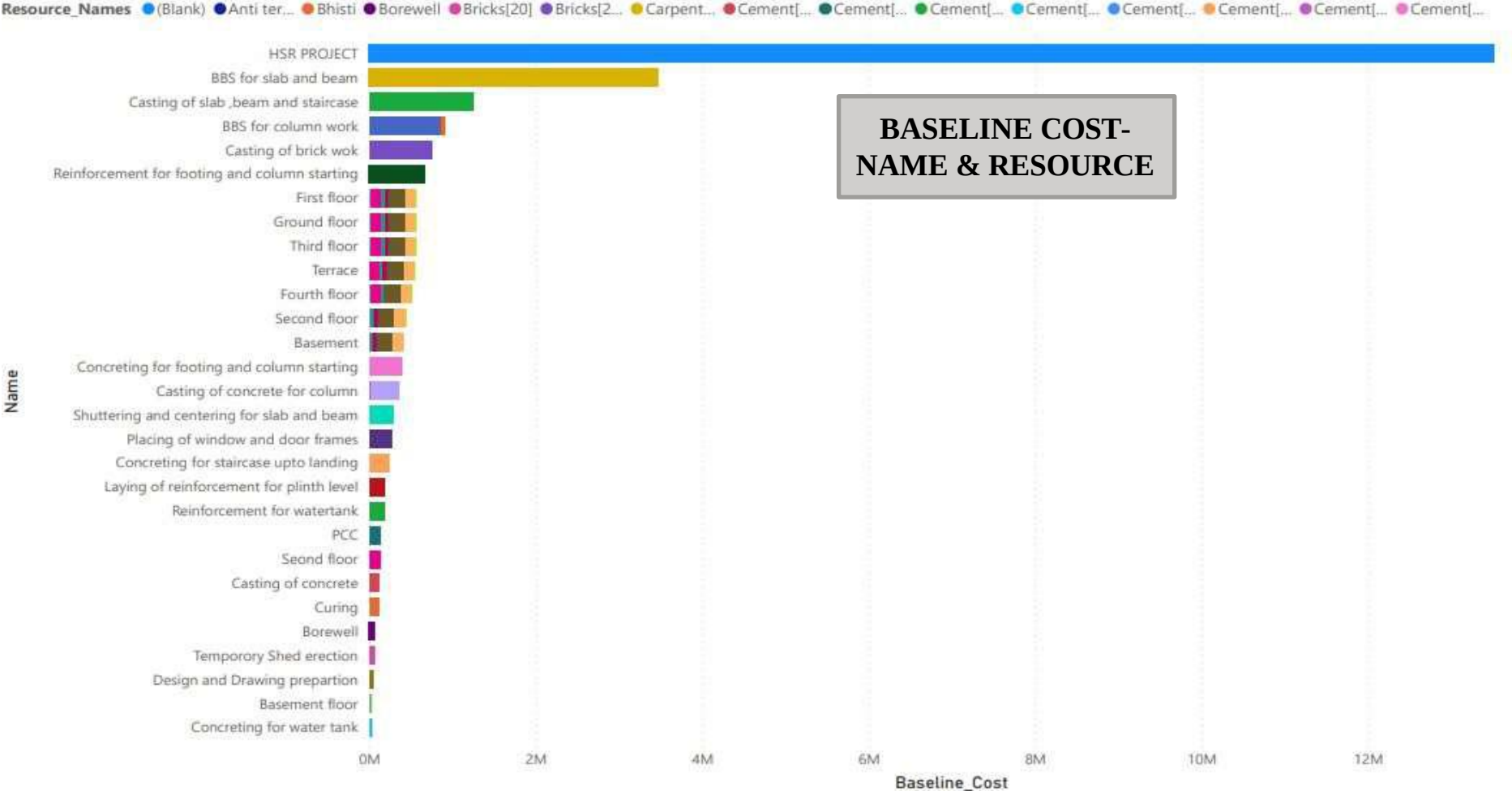
BASELINE COST-NAME



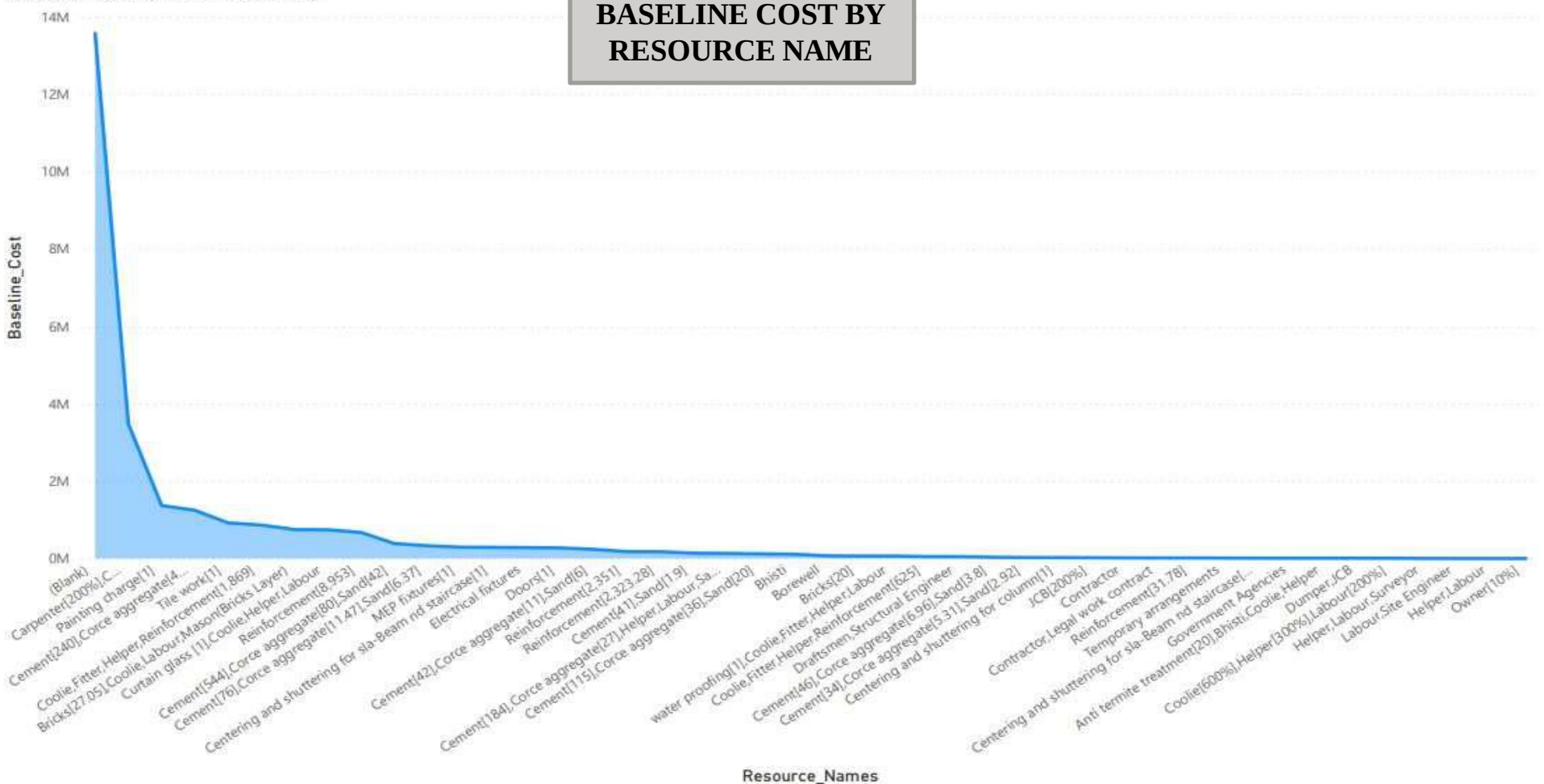
Name

- HSR PROJECT
- BBS for slab and beam
- Casting of slab ,beam and staircase
- BBS for column work
- Casting of brick wok
- Reinforcement for footing and column starting
- First floor
- Ground floor
- Third floor
- Terrace
- Fourth floor
- Second floor
- Basement
- Concreting for footing and column starting
- Casting of concrete for column
- Shuttering and centering for slab and beam
- Placing of window and door frames
- Concreting for staircase upto landing
- Laying of reinforcement for plinth level
- Reinforcement for watertank
- PCC
- Seond floor
- Casting of concrete
- Curing
- Borewell

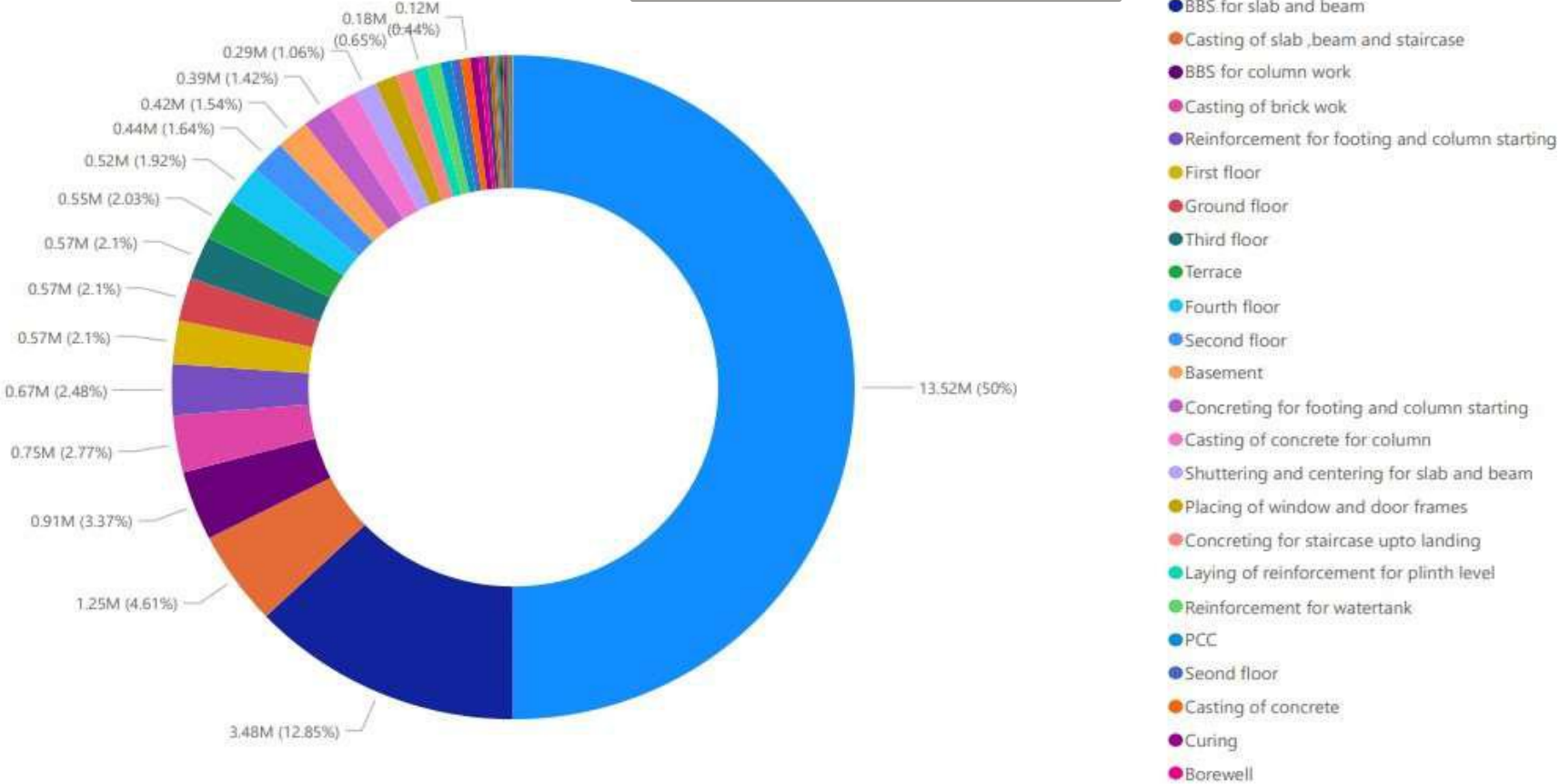
Baseline_Cost by Name and Resource_Names



Baseline_Cost by Resource_Names



BASELINE COST- BY
NAME





PROJECT INFORMATION

PROJECT LOCATION: MANGALORE,INDIA

SITE AREA : 12642 Sq. mts.

TOTAL BUILT-UP AREA: 41383 Sq. mts.

NO. OF FLOORS : 2B+LG+4

HEIGHT Of BUILDING : 36.1m

FLOOR TO FLOOR HT. :4.5m (L.W.G To Third floor) 3.

WIDTH OF BUILDING :74 m

LENGTH OF BUILDING : 102 m

TOTAL GATES : 2No.s (One on the East & South-east vehicular entry)

TOTAL NO. OF OPEN CAR-PARK: 118 No. s

TOTAL NO. OF PARKING SPACES : 365No.s

TOTAL NO. OF 2-WHEELER: 32No.s



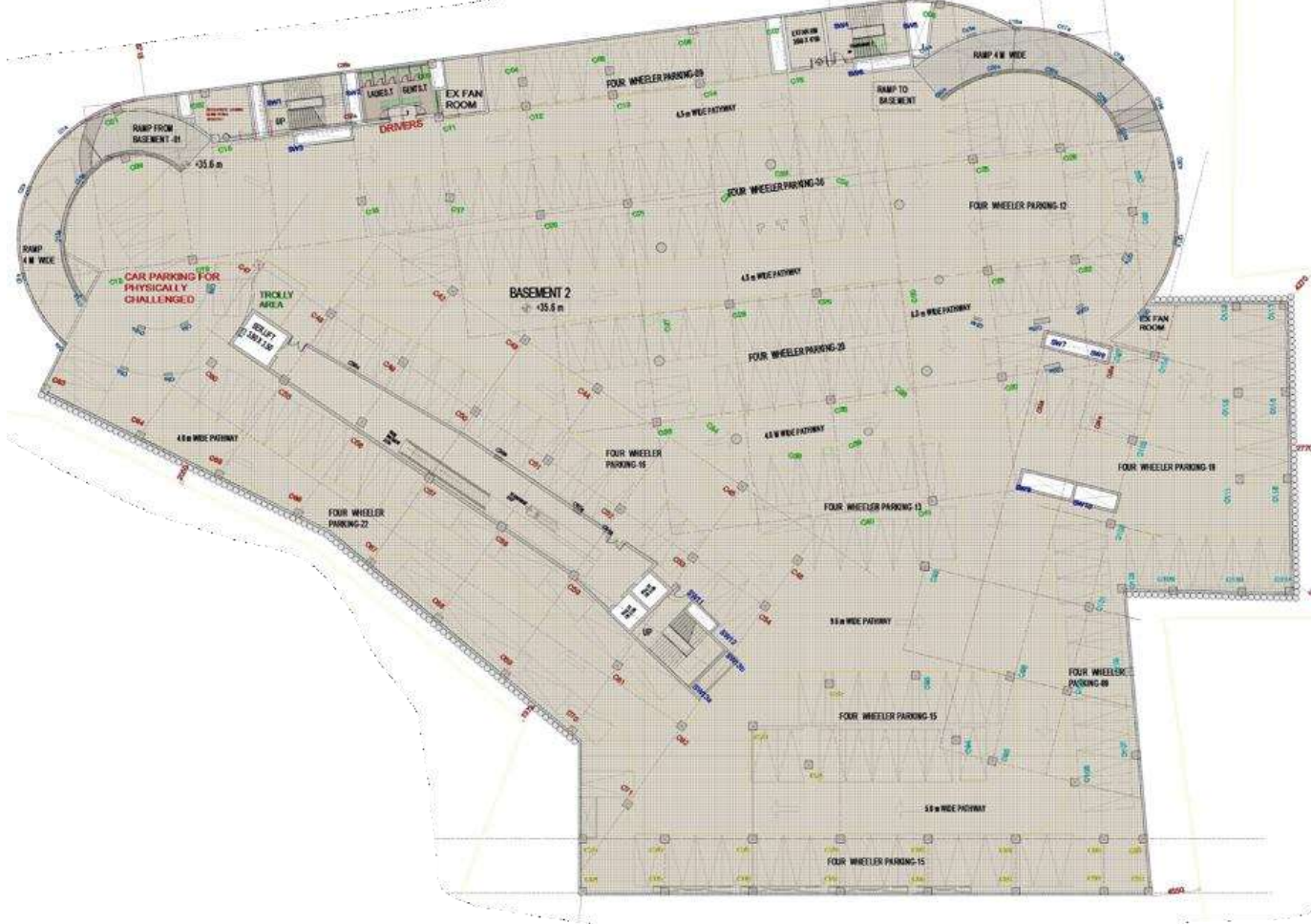
BASEMENT LEVEL 2 PLAN :

FEATURES

Total Built-Up Area = 7095 Sq. mts.
Car Parks = 189 No.s
2 ramps for Vehicular movement
3-staircases
Washrooms



BASEMENT LEVEL 1 PLAN :



FEATURE

Total Built Up Area = 7080 Sq. mts.

Car Parks = 176 No.s

2-Wheelers = 32 No.s

2 ramps for Vehicular movement

3-staircases

Washrooms



LOWER GROUND FLOOR LEVEL PLAN :

FEATURES

Total Built-Up Area =5500 Sq. mts.

2 ramps for Vehicular movement

3-staircases

1 Escalator

Washrooms

A.H.U Rooms



GROUND FLOOR LEVEL PLAN :

FEATURES

Total Built-Up Area =4800 Sq. mts.

2-Wheeler Parking :57 No.s

3- Wheeler Parking :17 No.s

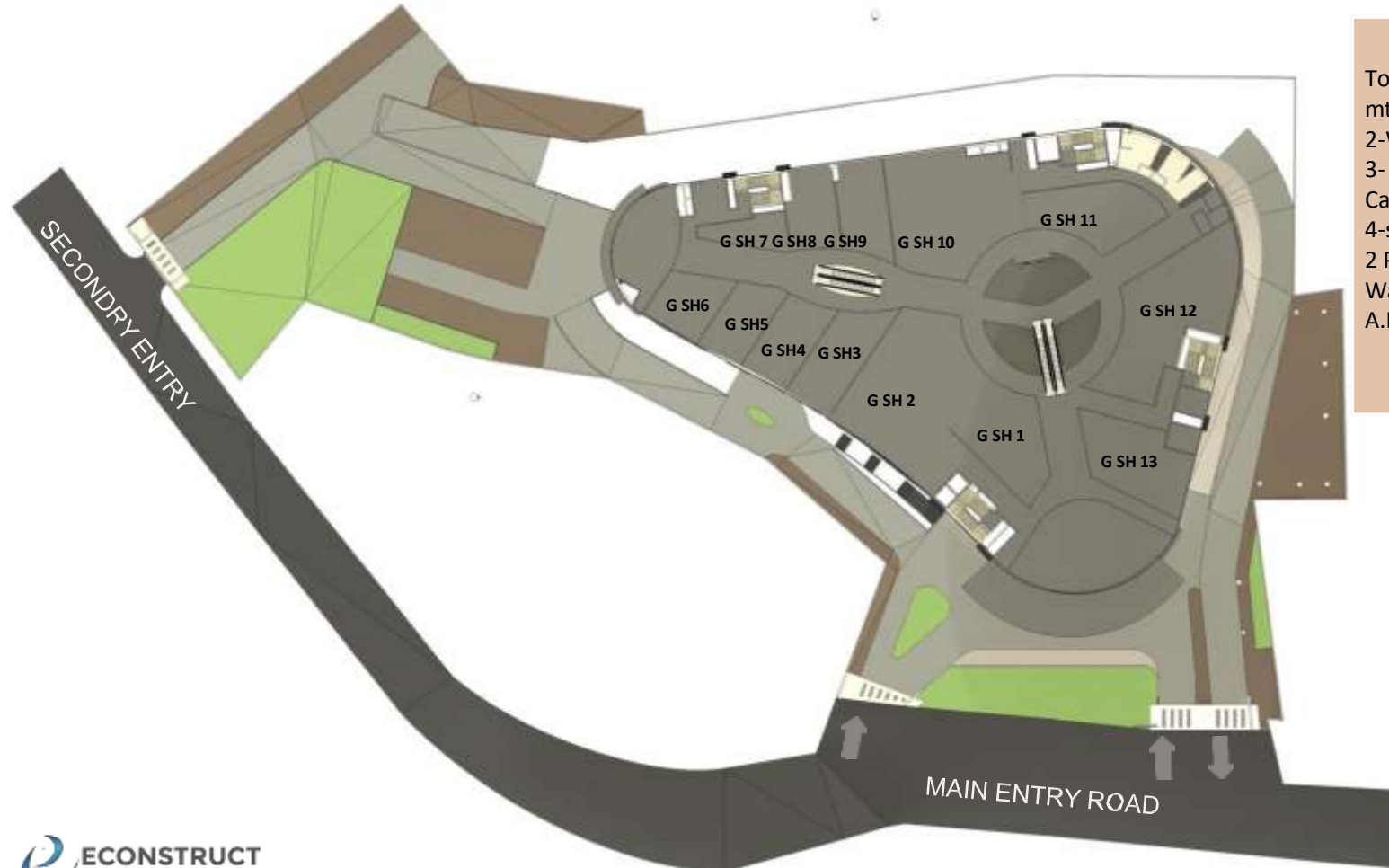
Car –Parking : 29 No.s

4-staircases

2 Pairs Of Escalator

Washrooms

A.H.U Rooms



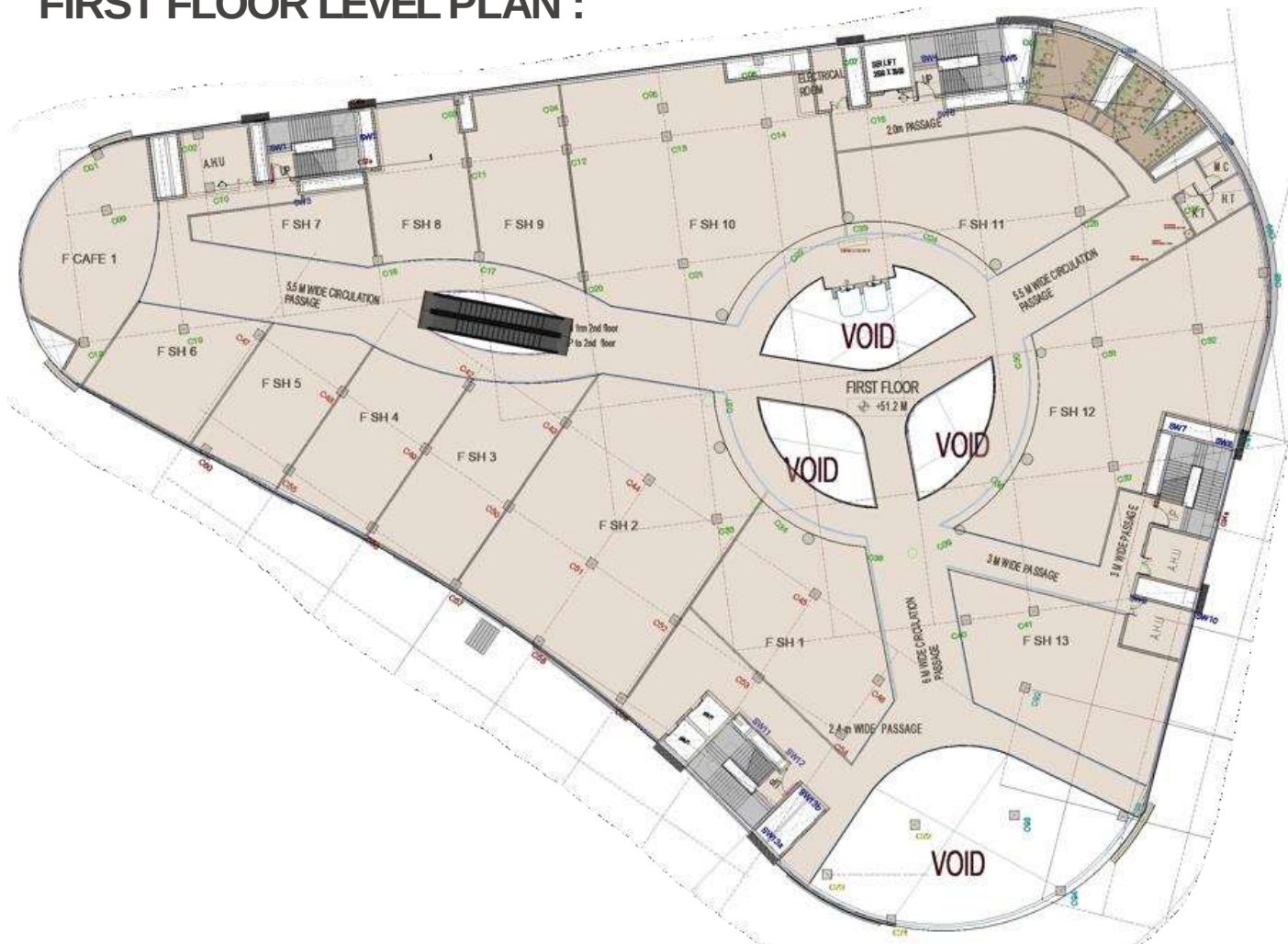
FIRST FLOOR LEVEL PLAN :



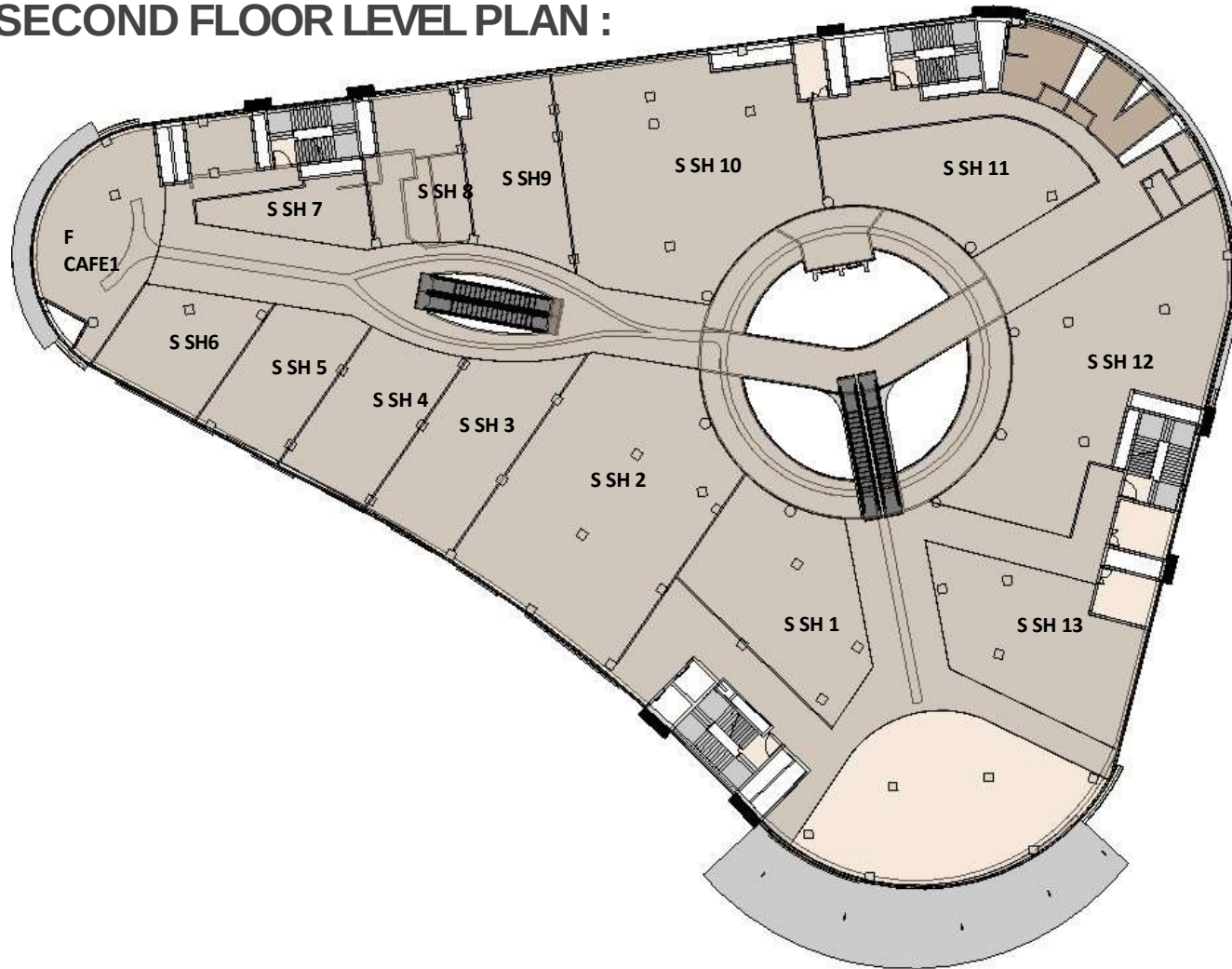
FEATURES

Total Built-Up Area =4764Sq. mts.

- 1 Service lift
- 2 capsule Lifts
- 2 Lifts
- 4-staircases
- 1 Escalator
- Washrooms
- A.H.U Rooms



SECOND FLOOR LEVEL PLAN :



LEGENDS

Total Built-Up Area = 4825 Sq. mts.

4-staircases

2 Pair Of Escalator

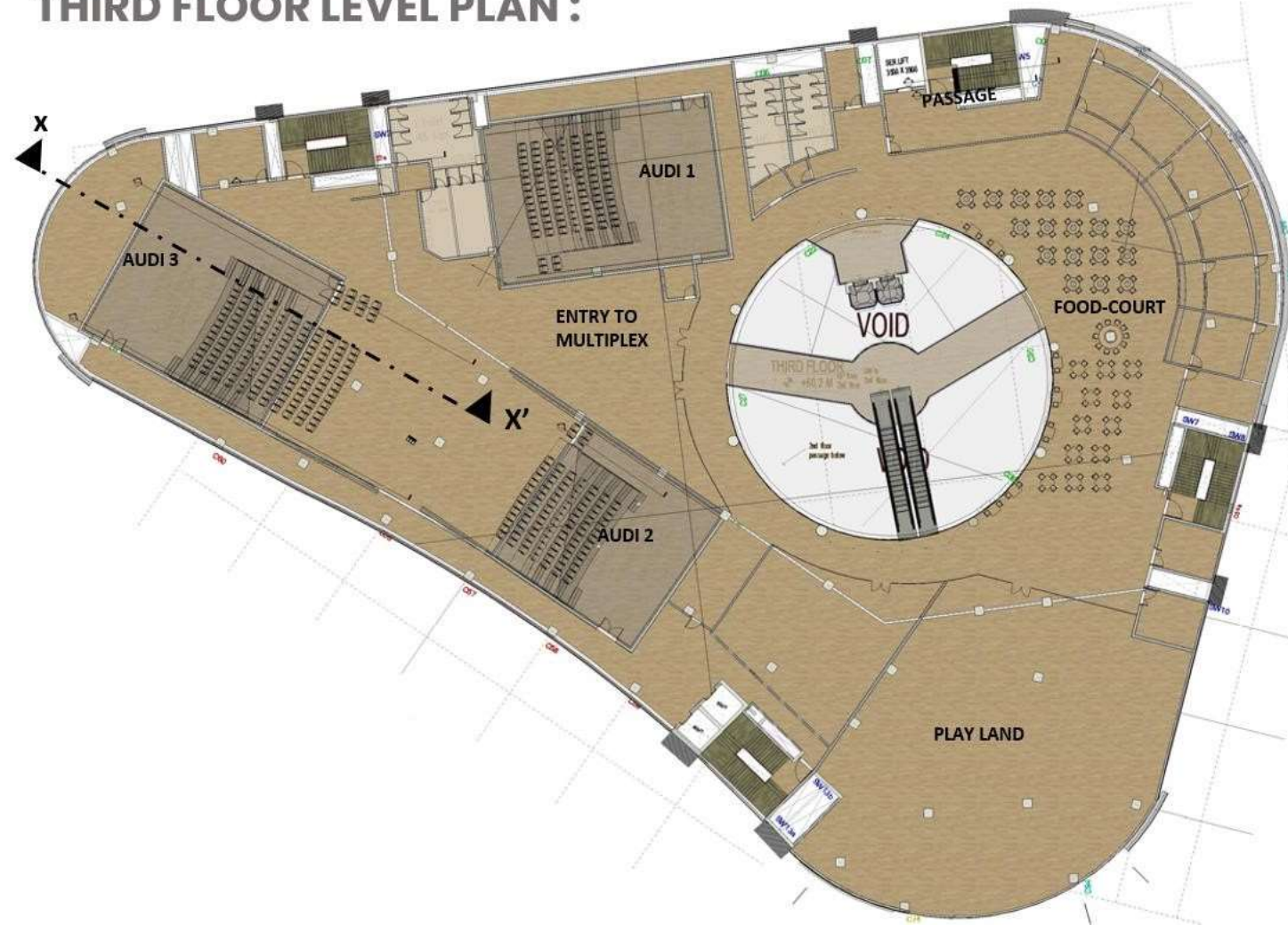
Washrooms

A.H.U Rooms

Shops : 13 No. s + 1 Café



THIRD FLOOR LEVEL PLAN :



KEY- FEATURES

Total Built-Up Area = 4417Sq. mts.

4-staircases

3-Theatres:

AUDI -1 =150 Seating Capacity

AUDI -2 =150 Seating Capacity

AUDI -3 =150 Seating Capacity

Food -Court 205 No.s seating

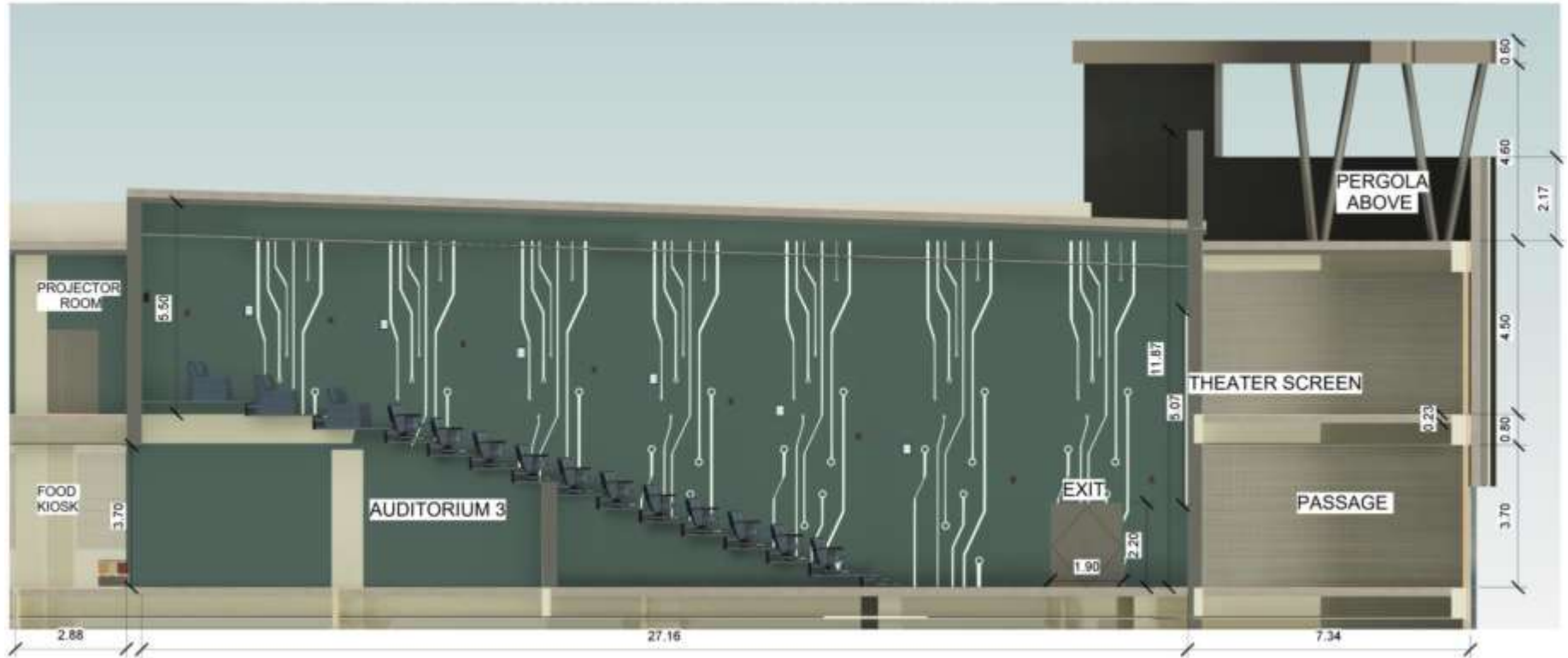
1 Pair of Escalator

Washrooms

A.H.U Rooms



AUDITORIUM SECTION :



3D Views Of Reliance Mall



3D Views Of Reliance Mall



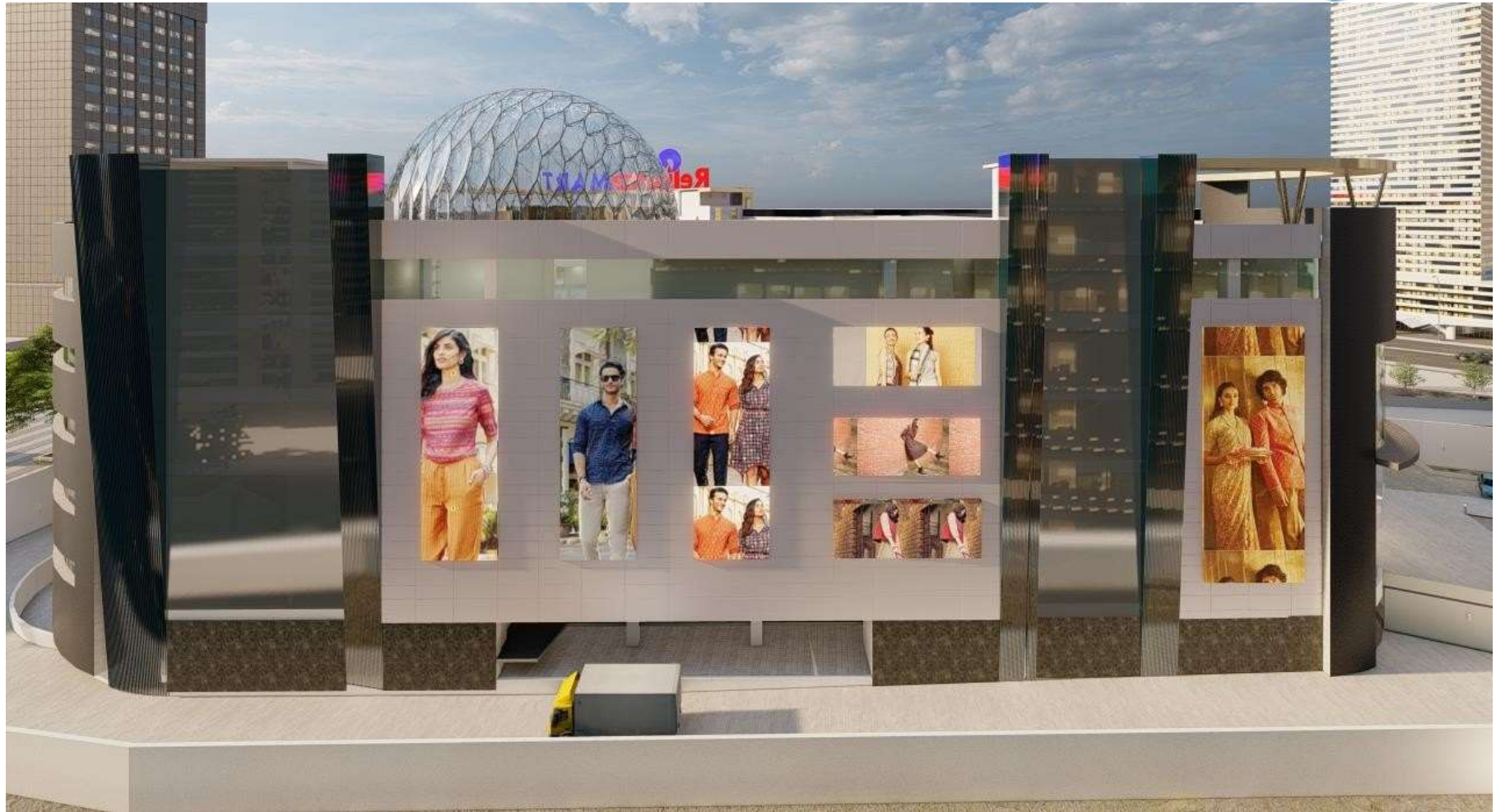
3D Views Of Reliance Mall



3D Views Of Reliance Mall



3D Views Of Reliance Mall



3D Views Of Reliance Mall



3D Views Of Reliance



3D Interior Views Of Reliance Mall



AREA OF THE MALL AT THE GR.FLOOR



SHOP AREA OF THE MALL



SHOP AREA OF THE MALL



CAFE INTERIOR OF THE MALL AT FIRST FLOOR

3D Interior Views Of Reliance Mall



TICKET – COUNTER & FOOD COUNTER FOR MULTIPLEX



FOOD – COURT AREA OF THE MALL AT THE 3RD FLOOR

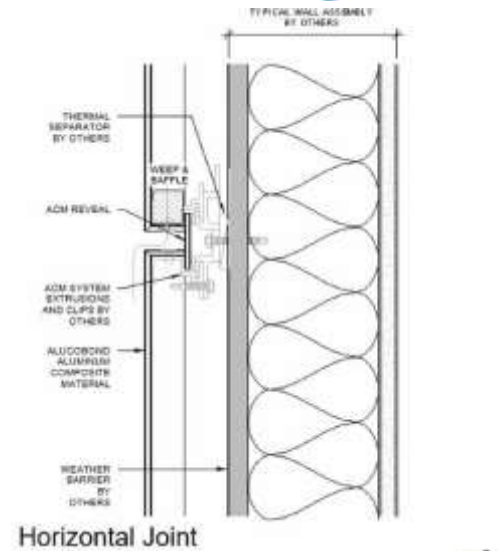
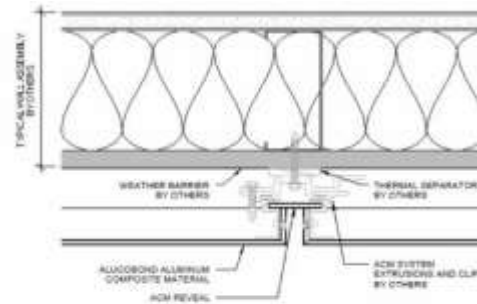
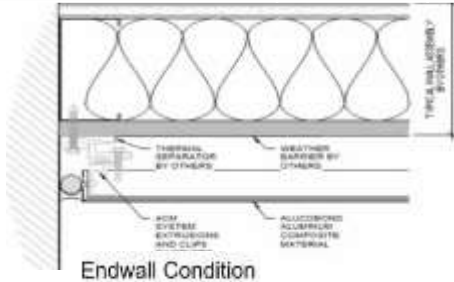


MOVIE THEATRE OF THE MALL AT THE 3RD FLOOR

Exterior Facade details Of Reliance Mall

SCHEDULE FOR WALL TREATMENTS

WALL TREATMENTS	QUANTITY	UNITS	RATE/UNIT	UNITS	COST
EXTERIOR WALL FAÇADE METALLIC COLOUR ALUCOBOND	29600.75	Sq. ft.	280.00	Sq. ft.	82,88,210.00
EXTERIOR WALL FAÇADE SOLID COLOUR ALUCOBOND LAYER	19073.65	Sq. ft.	500.00	Sq. ft.	95,36,825.00

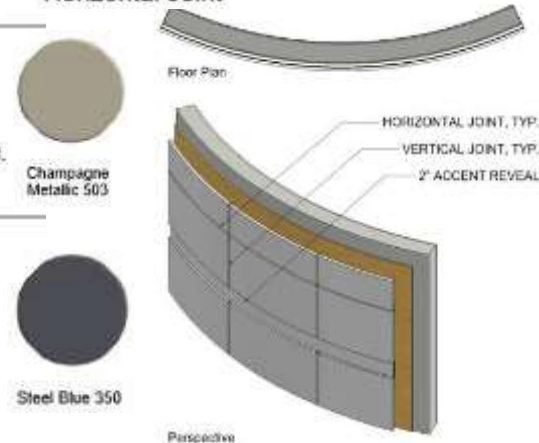


ALUCOBOND® metallic colours

Changing light conditions and perspectives endow these timeless elegant colours with a glowing, lively appearance.

ALUCOBOND® solid colours

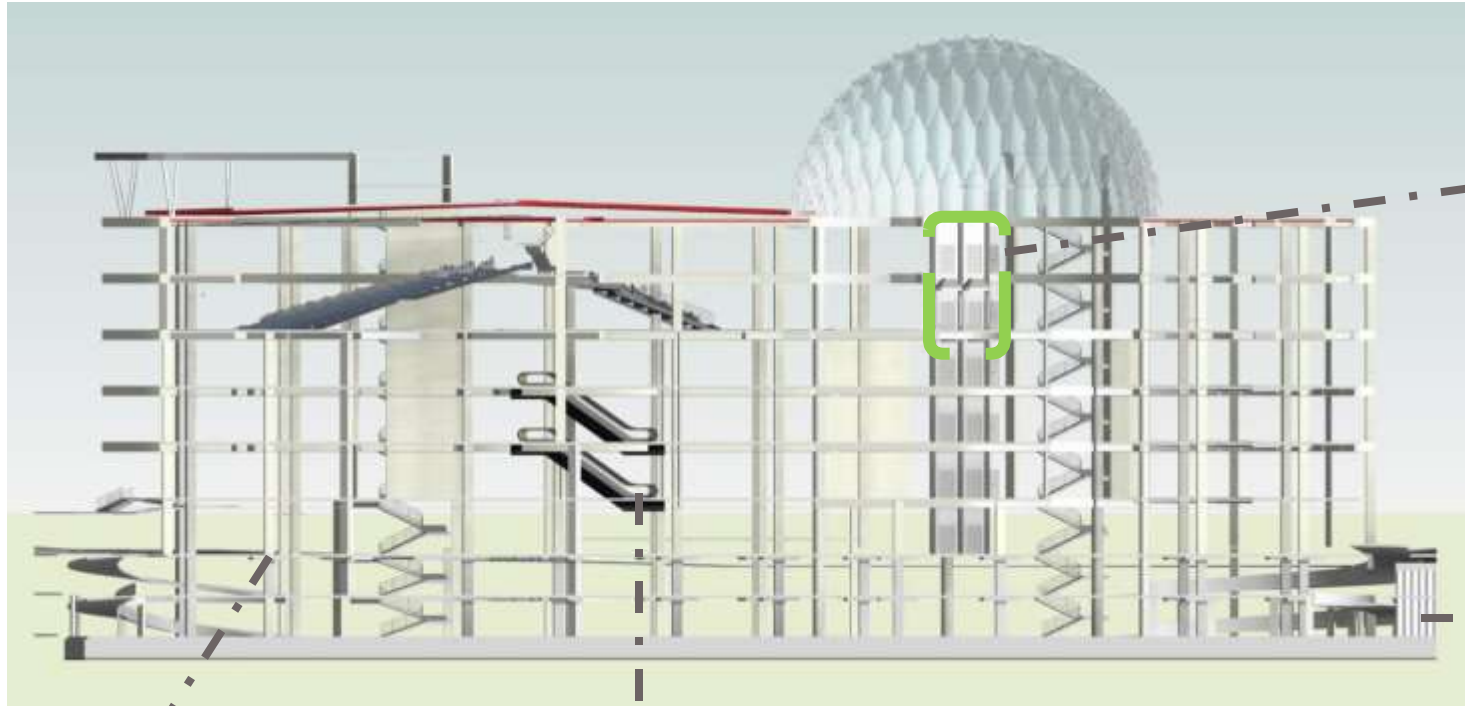
From delicate white to strong red: solid colours generate a unified appearance without any special effects.



3D Structure Of Reliance Mall



Section Of Reliance Mall



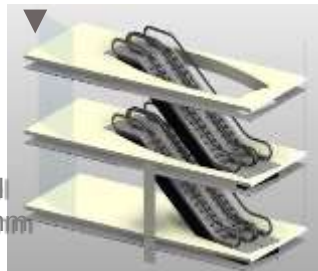
Capsule Lifts : 2.46 x
1.855
No. Of lift : 2 No.s
Cost of one lift : 20
Lakhs



Shoring Pile
Pile cap : 1000x1000x900mm
Pile Dia. : 600mm
No. Of piles : 198 No. s



Columns with drop panel
Dimensions : 750 x 750 mm
No. Of lift : 108 No. s



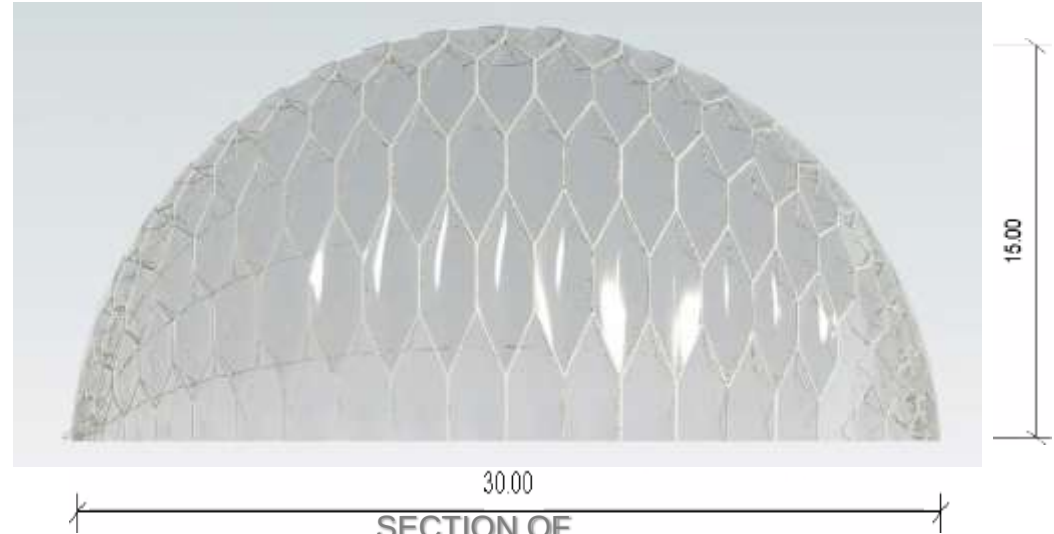
Escalators : 12
x 1.53
No. Of lift : 7 No. s
Cost of one
escalator : 28



3D Structure Of Reliance Mall



**3-D VIEW OF
DOME**



**SECTION OF
DOME**

SCHEDULE FOR DOME TREATMENTS :

DOMES ELEMENTS	QUANTITY	UNITS	RATE/UNIT	UNITS	COST
STEEL SPACE FRAME	37129.45	Tons	60,000.00	Tons	-
GLASS PANELS	29213.25	Sq. ft.	120.00	Sq. ft.	34,95,90.00



Structural Details Of Reliance Mall

A) STRUCTURAL WALL

Sr. No.	Description	Dimensions	Grade
1	Retaining Wall	230mm Thickness	M30
2	Shear Wall	300mm Thickness	M40
3	Shear Wall	230mm Thickness	M40

C) COLUMN

Sr. No.	Description	Dimensions (length X width) &(Diameter)	Grade
1	Column 1	750mm x 750mm	M40
2	Column 2	230mm x 750mm	M40
3	Circular Column	800 mm Dia.	M40
4	Circular Column	1000 mm Dia.	M40

B) SLAB

Sr. No.	Description	Dimensions	Grade
1	General Slab	150mm Thickness	M30
2	Stair Slab	175mm Thickness	M30
3	OHWT and LMR	200mm Thickness	M30
4	Sunken Slab	150mm Thickness	M30
5	Flat Slab	200mm Thickness	M30
6	Flat Slab Drop	350mm Thickness	M30

D) BEAM

Sr. No.	Description	Dimensions (Width x depth)	Grade
1	Beam	800mmx230mm	M30
2	Beam	750mmx230mm	M30
3	Beam	600mmx300mm	M30



Water up

Face

L5-L7 Ph1

L5-L7 Ph2

L5-L7 Ph3

L5-L7 Ph4

L5-L7 Ph5

L5-L7 Ph6

L5-L7 Ph7

L5-L7 Ph8

Sliding Plan (D.O.F. x3)

SHORING PILES DETAILS

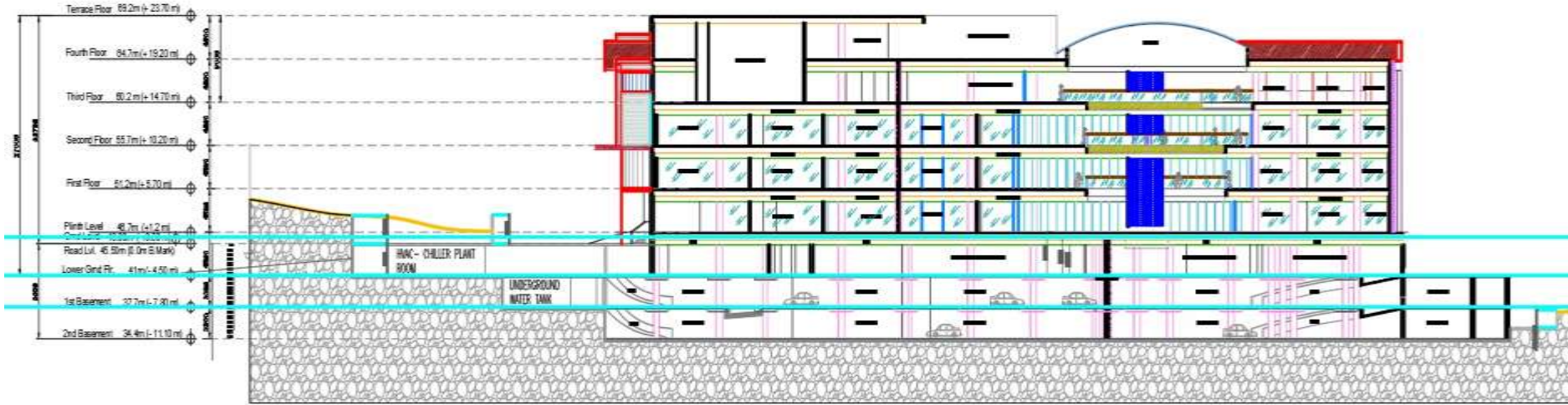
Description	Diameter	Depth	Grade	No. of Piles
Shoring Pile	600mm	7500mm	M30	198

PILE BEAMS DETAILS

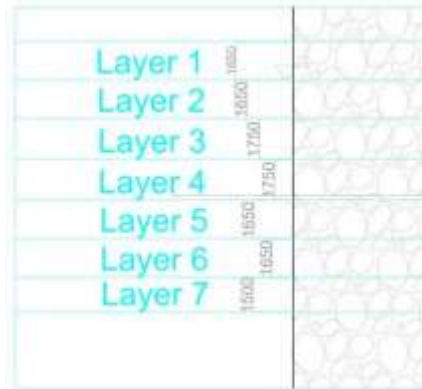
Description	Width	Depth	Grade	Length
Pile Beam	750mm	750mm	M30	118800mm

The process of excavation has been divided into 9 –Phases respectively . Where each phase consists of two layers at a time .

Excavation Sequence Of Reliance Mall



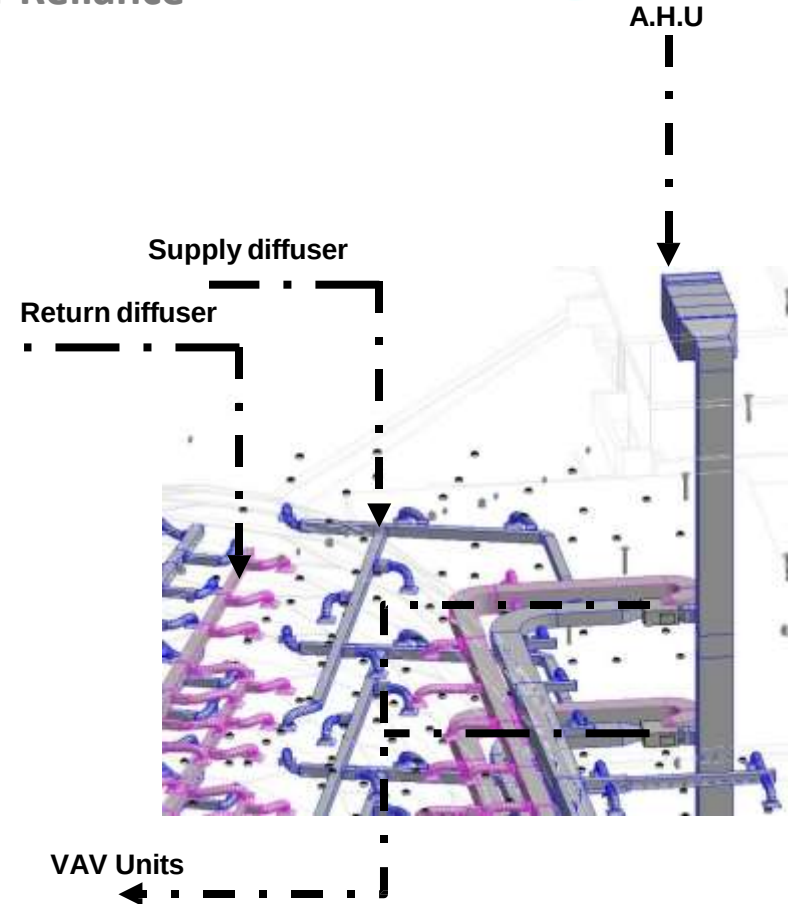
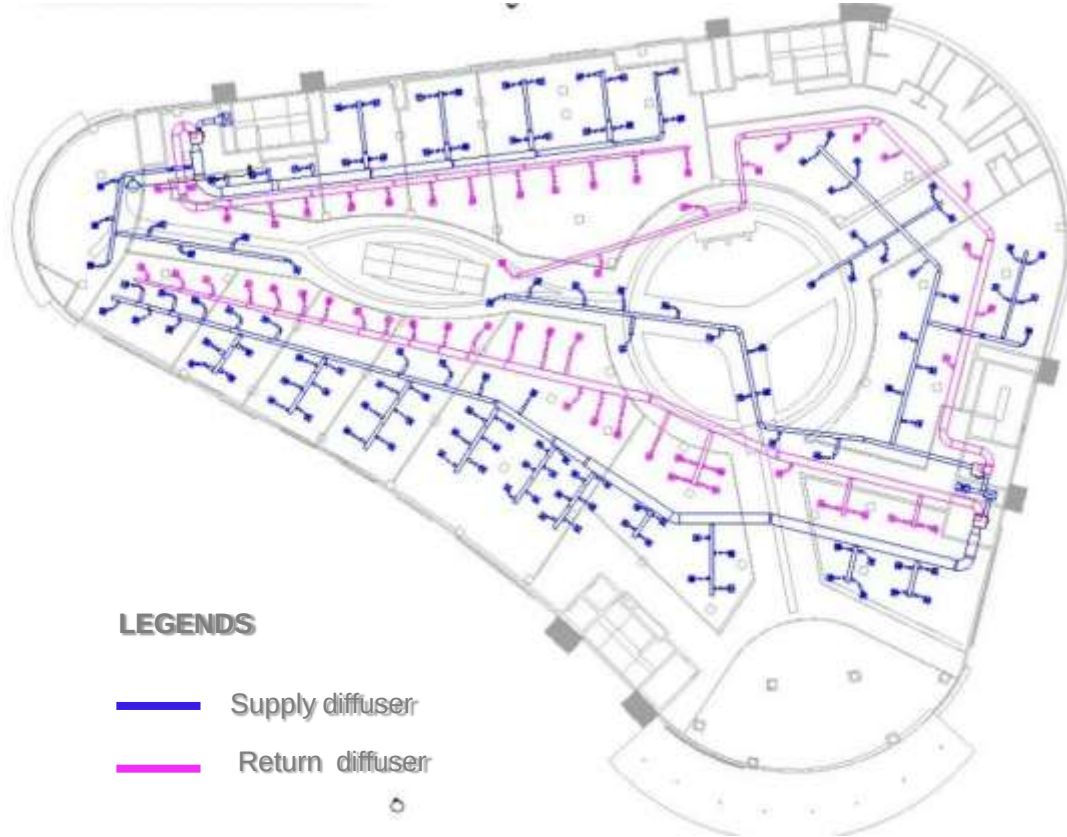
SEQUENCE OF EXCAVATION :



PHASE -1

PHASE NAME	AREA EXCAVATED IN Sq. mts.
PHASE -1	1943.84
PHASE -2	1931.97
PHASE -3	1065.58
PHASE -4	907.69
PHASE -5	859.67
PHASE -6	914.58
PHASE -7	1668.75
PHASE -8	953.32
PHASE -9	775.53

2D & 3D HVAC Of Ground floor for Reliance Mall



3D VIEW SHOWING HVAC

2D Electrical Of Typical floor Reliance Mall



LEGENDS

-  MCB
-  Switch
-  15A socket with switch
-  5/15A Universal socket with switch
-  5A socket with switch
-  Ceiling Fixture
-  1x 4' LED tube light
-  Exhaust Fan
-  Wall-mounted fixture
-  Meter box

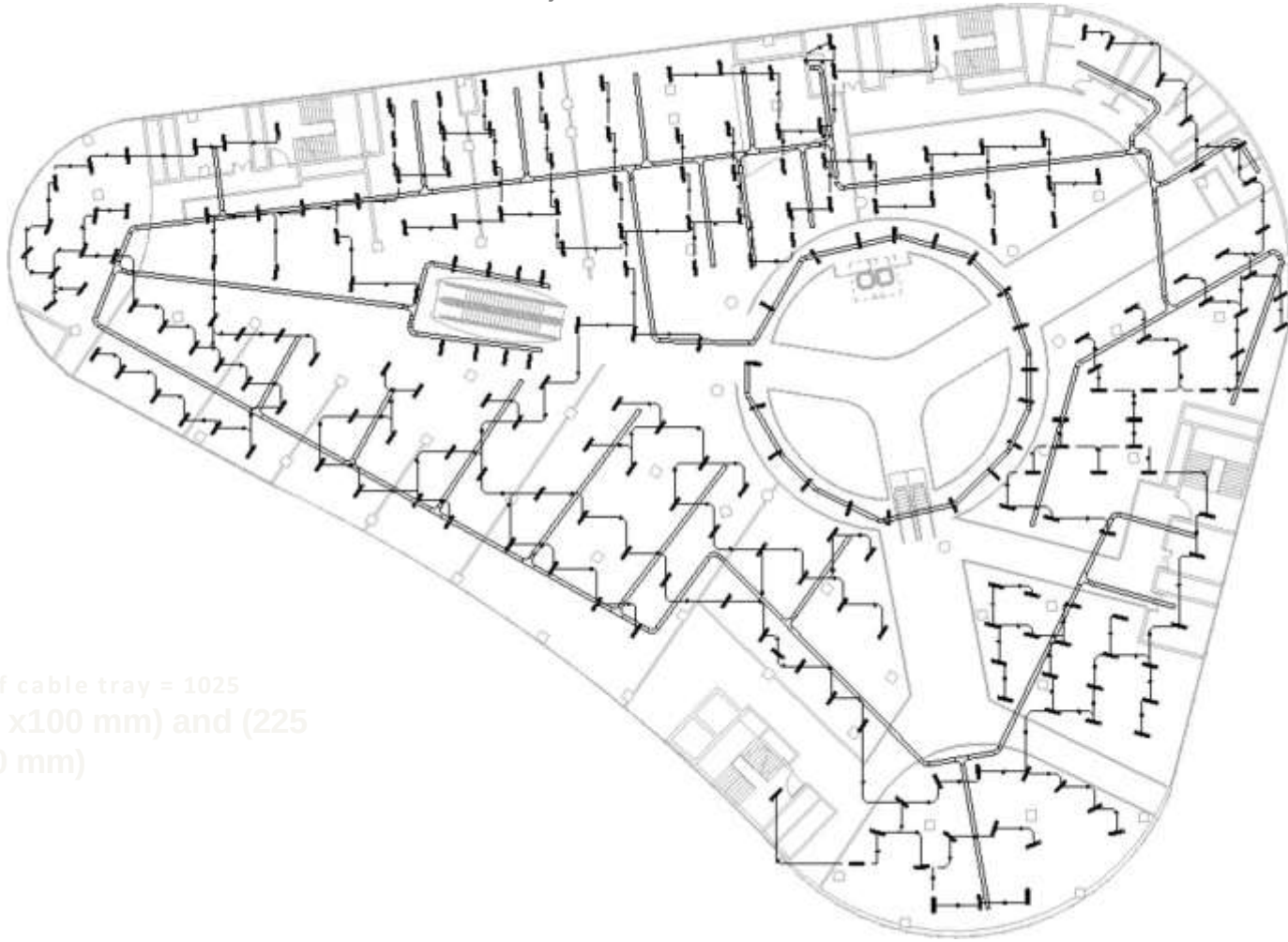


3D Electrical Of Reliance Mall at the Basement -1 &-2 Levels



LED 20W 1197
CEILING FIX 15W
3464
SWITCHES 6A 1180

3D Electrical Of Reliance Mall For GF, First & Second Floor



Total no of cable tray = 1025
(300 mm x100 mm) and (225
mm x100 mm)

Electrical Details Of Reliance Mall



Electrical Panel Schedule Of Reliance Mall

SL. NO.	FLOORS	TOTAL CONNECTED LOAD	TOTAL ESTIMATED DEMAND
1.	FIRST BASEMENT	6144VA	6144VA
2.	SECOND BASEMENT	6144VA	6144VA
3.	GROUND FLOOR	5616VA	5616VA
4.	FIRST FLOOR	10080VA	10080VA
5.	SECOND FLOOR	10080VA	10080VA
6.	THIRD FLOOR	8496VA	8496VA

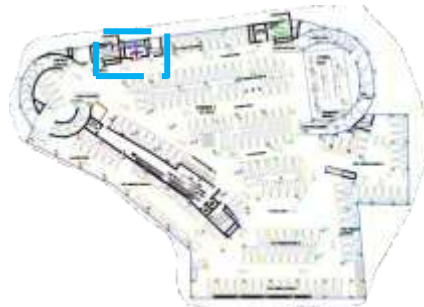
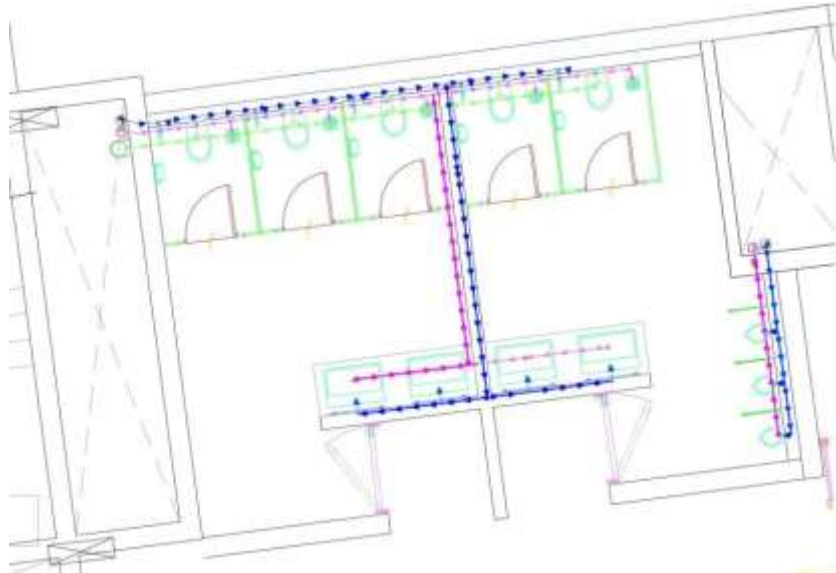
Electrical Equipment Schedule Of Reliance Mall

SL. NO.	EQUIPMENT	CONSUMPTION (W)	COUNT
1.	Fluorescent Tubelight	20	1197
2.	Ceiling Fixtures	15	3464
3.	Compound Wall Light	15	600
4.	Spot Light	18	3200
5.	Solar Path Light	2	63
6.	Mast Light	250	6

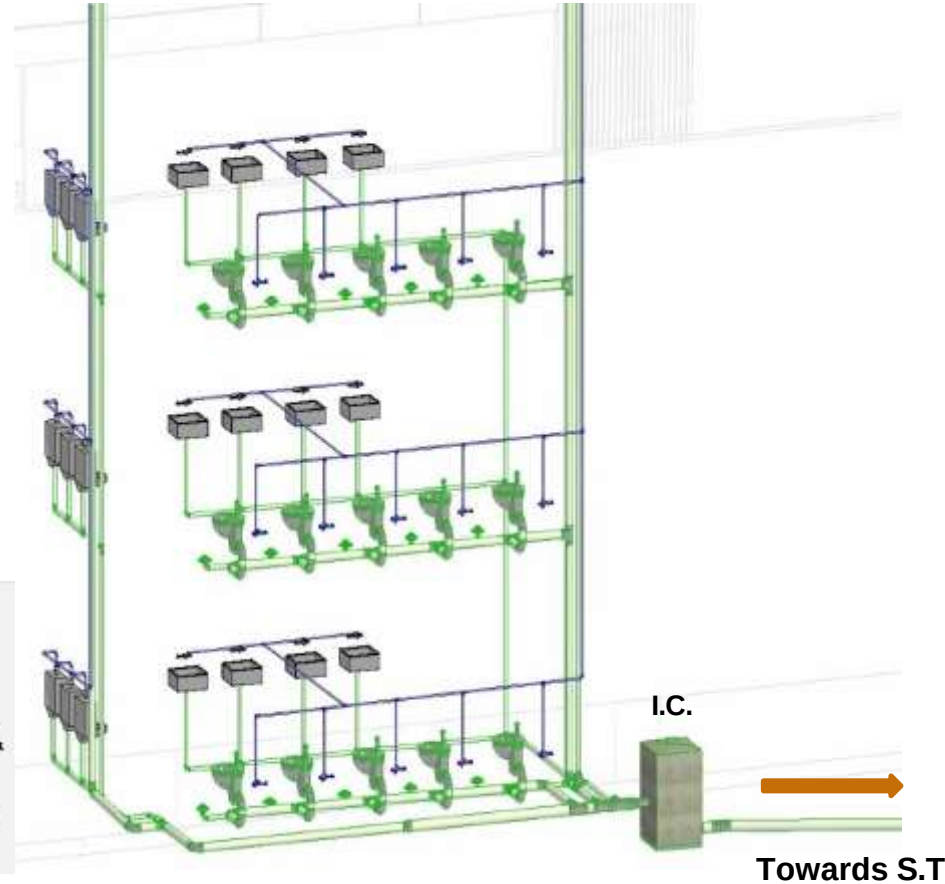
Electrical Devices Schedule Of Reliance Mall

SL. NO.	EQUIPMENT	COUNT
1.	1 Way Switch 16/20A 230V	10500
2.	2Way Switch 16/20A,230V	2500
3.	Universal socket, 6pins, 10/20A ,230V	4200
4.	Universal switch socket, 6pins, 10/20A ,230V (combine)	2500
8.	D.P Switch, 32A ,230V,	550

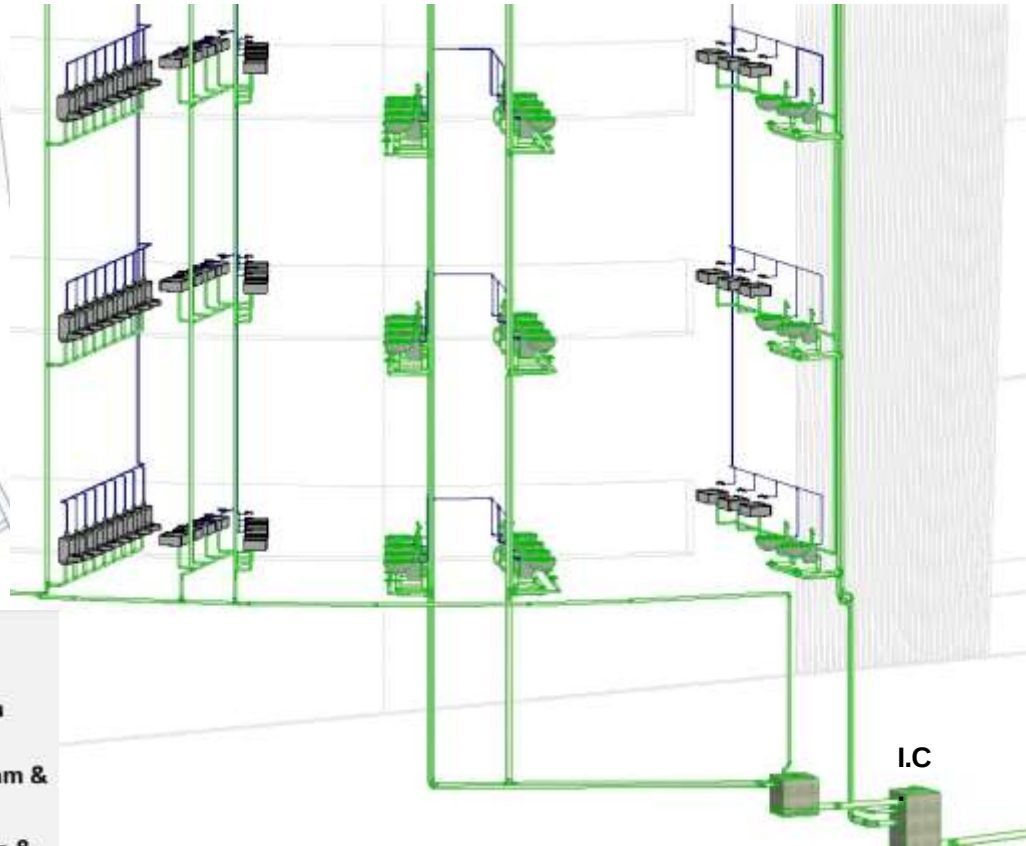
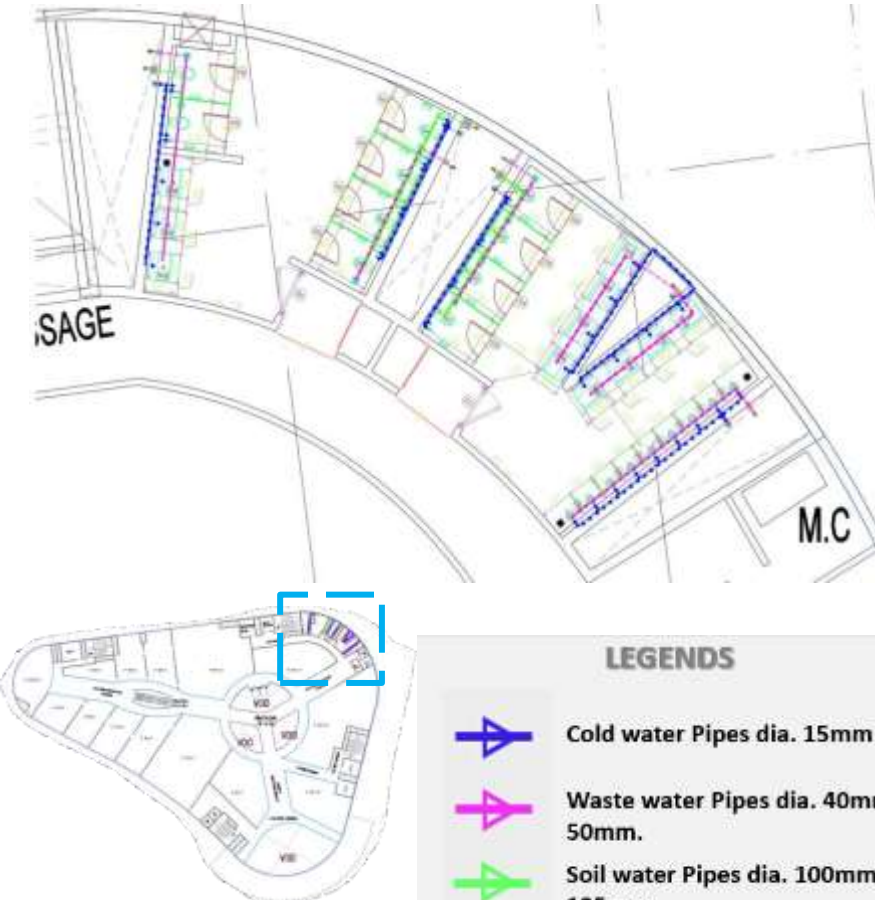
2D& 3D Plumbing Of Washrooms at Basement -2 to Lower Ground Level



LEGENDS	
	Cold water Pipes dia. 15mm
	Waste water Pipes dia. 40mm & 50mm.
	Soil water Pipes dia. 100mm & 125mm.



2D & 3D Plumbing Of Washrooms for GR. to 2nd Floor



LEGENDS



Cold water Pipes dia. 15mm

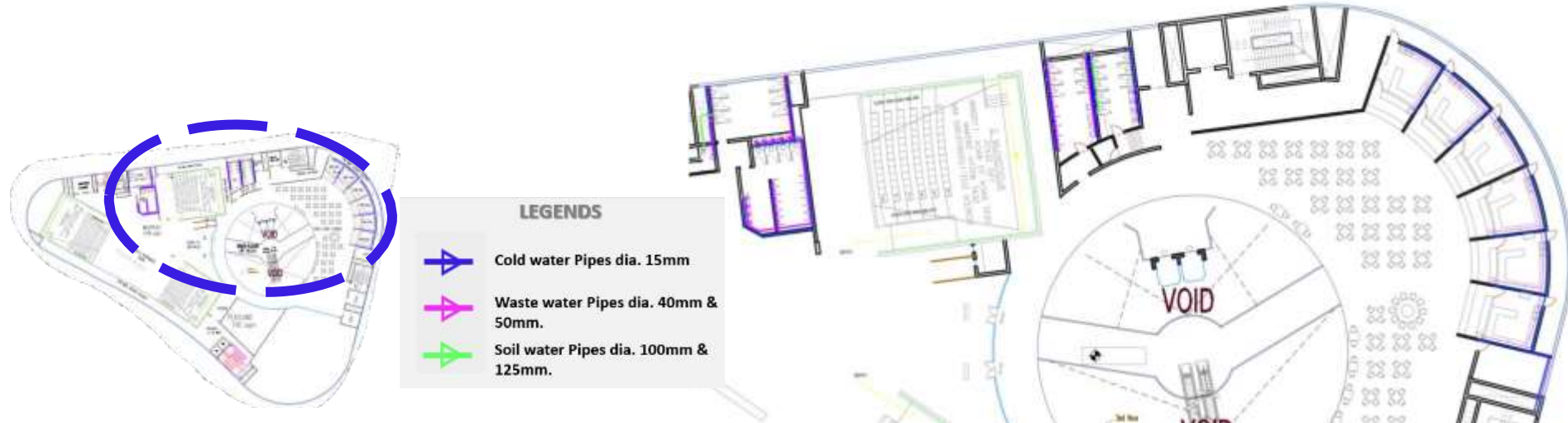
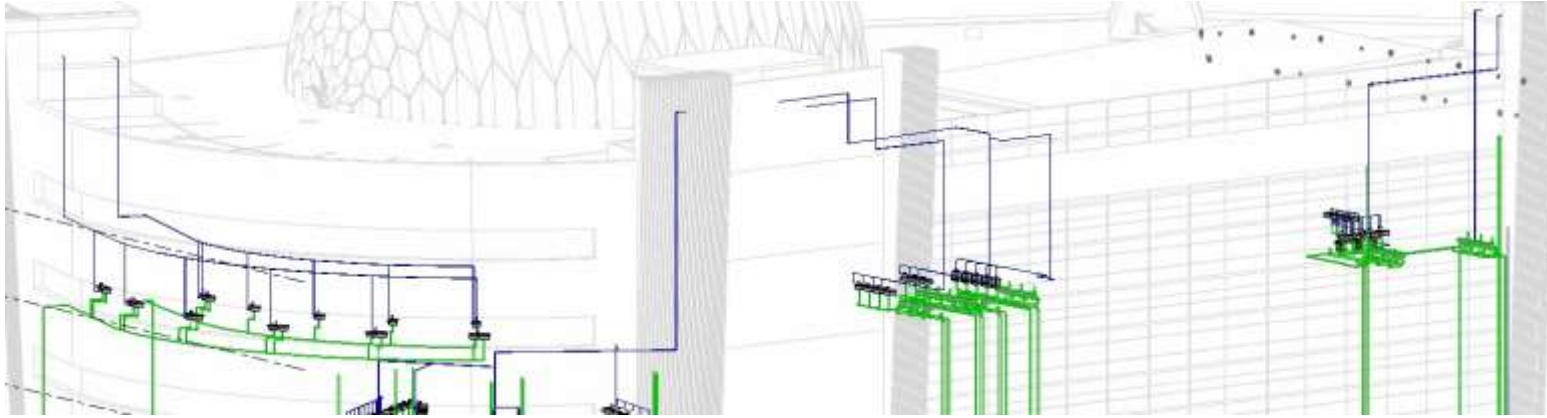


Waste water Pipes dia. 40mm & 50mm.



Soil water Pipes dia. 100mm & 125mm.

2D & 3D Plumbing Of Third Floor



Plumbing Details Of Reliance Mall



For W.C. =2000L/D, For H/W=9000 L/D &
For Cleaning = 13750 L/D
Total water demand =43,750 Litres Per day

Plumbing Fixtures Details:

SL. NO.	FAMILY & TYPE	COUNT
1.	Wash Basin	154
2.	Double –Sided Sink	6
3.	Single Sink	6
4.	Taps	140
5.	Urinals	140
6.	Wall Mounted W.C's	137
7.	Flush valves	140
8.	Health faucet	168
9.	Paper roll holder	154
10.	Floor traps	168
11.	P-traps	140
12.	Inspection chambers	3

Pipe Details:

SL. NO.	DAIMETE R	FAMILY	TYPE
1.	15mm	Pipe Types : PVC DWV	Cold water
2.	40mm	Pipe Types : PVC DWV	Waste water
3.	50mm	Pipe Types : PVC DWV	Waste water
4.	100mm	Pipe Types : PVC DWV	Solid waste
5.	125 mm	Pipe Types : PVC DWV	Solid waste

2D Fire- Fighting Of Reliance Mall



LEGENDS

-  Main Water Supply Line
-  Sprinkler - Line
-  Ceiling Sprinkler Systems

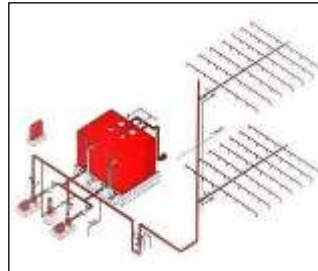


MAIN WATER SUPPLY -01 Total sprinklers= 155

MAIN WATER SUPPLY -02 Total sprinklers= 113

Area covered by single sprinkler = 12m^2

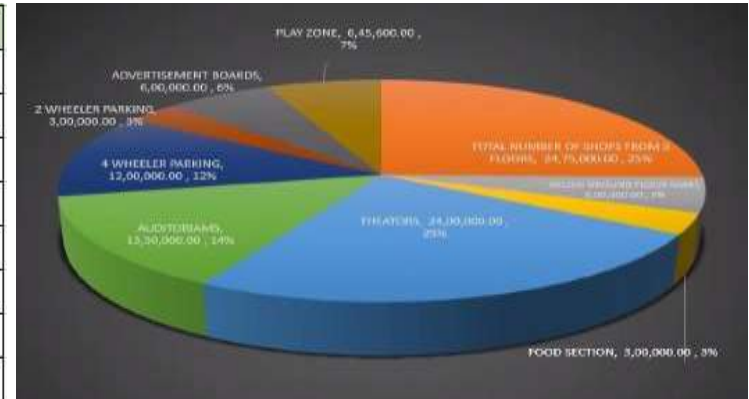
- 146 ceiling Sprinkler Systems per Floor has been Provided.
- As per the norms 125 mm dia. main supply pipe has been provided 132.17 running mts. Per floor.
- Sprinkler – Line 40mm, 32mm, 25mm dia. Pipe has been given 756 running mts. Per floor.
- Fire fighting Pumps: Jockey pumps No. 4, Main Pumps 2 No.s & 2 No.s Sprinkler Pumps.
- Tank Capacity 2,20,000 Liters .



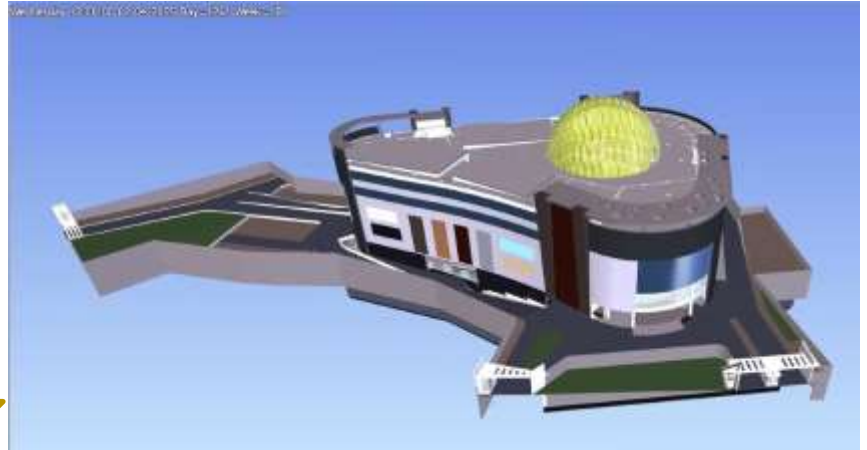
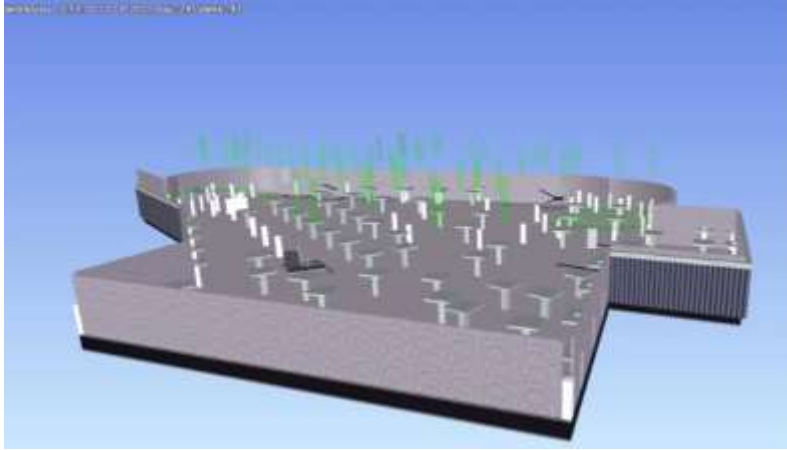
Finance Management Of Reliance Mall

REVENUE GENERATION	Qty.	Unit	Rate (in Rs.)	Amount (in Rs.)	
			Avg. Rent		
TOTAL NUMBER OF SHOPS FROM 3 FLOORS	45	per month	₹ 55,000.00	24,75,000.00	
BELOW GROUND FLOOR MART	1	per month	₹ 5,00,000.00	5,00,000.00	
FOOD SECTION	6	per month	₹ 50,000.00	3,00,000.00	
THEATORS	3	per month	₹ 8,00,000.00	24,00,000.00	
AUDITORIAMS	3	per month	₹ 4,50,000.00	13,50,000.00	
4 WHEELER PARKING	30000	per month	₹ 40.00	12,00,000.00	1000/day 4 wheeler vehicles
2 WHEELER PARKING	15000	per month	₹ 20.00	3,00,000.00	500/day 2 wheeler vehicles
ADVERTISEMENT BOARDS	20	per month	₹ 30,000.00	6,00,000.00	
PLAY ZONE	1	per month	₹ 6,45,600.00	6,45,600.00	
			TOTAL	97,70,600.00	
1 YEAR OF RETURNS FROM THE MALL				11,72,47,200.00	
THEREFORE IT TAKES 19 YEARS TO GET THE RETURN OF INVESTMENT					
				19.6004328358418	years

INFLATION RATE CALCULATION		
FUTURE VALUE = PRESENT VALUE $(1+R/100)^n$		
PRESENT VALUE	2,29,80,95,868.79	Cr
INFLATION RATE AS PER 2040 IS 3.9%	2.068689945	$(1+R/100)^{19}$
FUTURE VALUE OF RELIANCE MALL AFTER 19 YEARS	4754047817	Cr.
	475,40,47,817Cr.	



4D Time liner Simulation

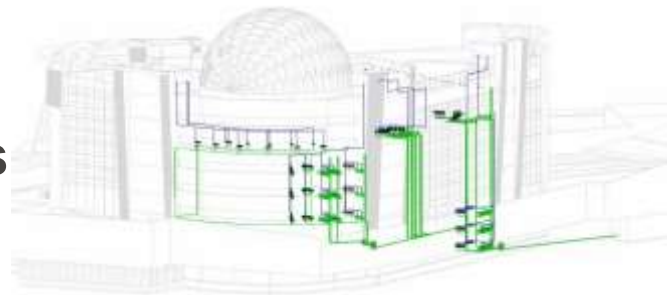


4d Clash-Detection



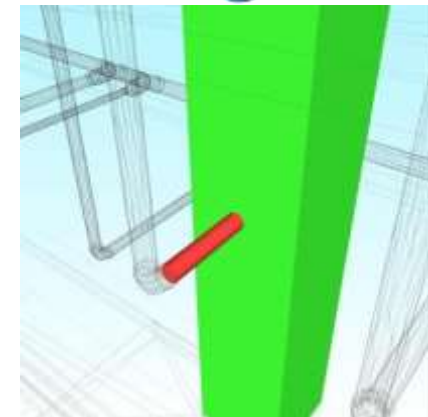
STRUCTURE MODEL

V/s



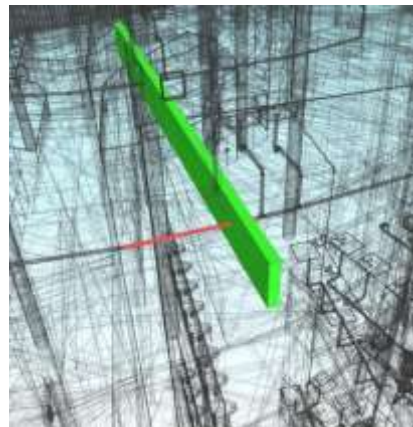
PLUMBING MODEL

Name	Status	Clashes	New	Active	Reviewed	Approved	Resolved
Plumbing Vs S	Done	737	521	0	0	1	215



Element ID 1520503

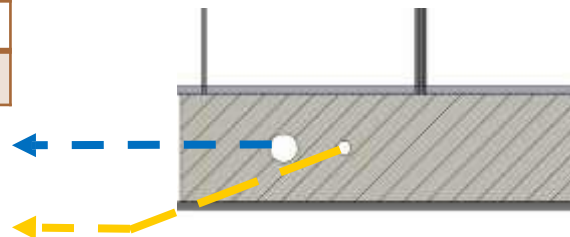
Element ID 934184



Element ID 1458312

Element ID 337763

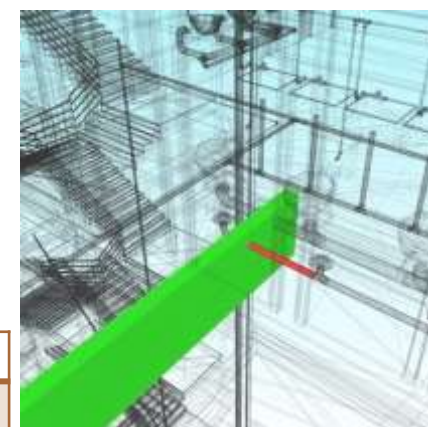
Clash resolved by core cutting in the beam



After doing multiple iterations the clashes has been rectified and reduced to zero.

Element ID 1479036

Element ID 334944



TENDER DOCUMENT

E-Construct Design & Build Pvt. Ltd



Model E-Tender Document

Tender No:- 01/2021-2022

National Competitive Bid through Percentage Rate tender.

For

"Construction of Reliance mall (2B + LG + GF + 3) in Mangaluru,
Karnataka, India".

October 2021

Head office:-

2nd floor, venkatesh heights, parappa aghare main road,
opp. Sai poorna premier apartment, Bangalore-56006

TENDER DOCUMENT

FOR
CONSTRUCTION OF RELIANCE MALL
AT
MANGALORE, KARNATAKA, INDIA



OCTOBER-2021

Reliance Industries Limited (RIL)

3rd Floor, Maker chambers-IV, Nariman point, Mumbai-400 021.

E-Construct Design & Build Pvt. Ltd

Notice Inviting E-Tender

NTT No: Tender/001/2021-2022

Date: 23.10.2021

E-Tender-notice

1. Name of Work:

Tenders are invited on the website <http://www.ril.com> the eligible bidders having Digital signature certificate (DSC) issued from any agency authorized by Controller of Certifying Authority(CCA), Govt. of India and which can be traced upto the chain of trust to the root certificate of CCA, for the following work.

Name of work:	Construction of Reliance mall at Mangalore
Total Estimated cost	Rs. 156.87.54,728.00
Earnest money	Rs. 35,00,000.00
Period of completion	2 years

Tender Inviting Authority	Contact Person(s)/Tender Dealing Officer(s)
Reliance Industries Limited 3C, 3005, Maker chambers-iv, 08/0080 Point, Mumbai-400-021.	08/13881-3005, (General Manager), Bangalore, Karnataka. Contact No: 0713-2842593, 08421895355

2. Time Schedule of Activities:

Sl. No.	Particulars	Time schedule
a.	Tender e-Publication Date	Data & Time as mentioned in the website: http://www.ril.com
b.	Document download/Sale Start Date	
c.	Document download & Bid submission end Date	
d.	Bid Submission Start Date	
e.	Start Date for seeking Clarification on-line	
f.	Last Date for seeking Clarification on-line	
h.	Date of Pre-Bid Meeting (if required)	
i.	Bid Opening Date	

CONTRACT DOCUMENT

CONTRACT DOCUMENT

FOR
CONSTRUCTION OF RELIANCE MALL
AT
MANGALORE, KARNATAKA, INDIA
CONTRACT NUMBER: 01/2021-2022

CONTRACT BETWEEN

Reliance Industries Limited



&

E-Construct Design & Build Pvt. Ltd



"CONTRACT AGREEMENT BETWEEN THE OWNER AND THE CONTRACTOR"

CONSTRUCTION OF RELIANCE MALL

This CONTRACT is made on 1st September 2021 between M/s. Reliance Industries Limited (RIL) having its corporate office at office address: 3rd Floor, Maker Chambers-IV, ~~Reliance point,~~ Mumbai-400 021. (hereinafter referred to as the OWNER) which expression shall, unless repugnant to the context or meaning hereof, be deemed to mean and include its successors and assignees) of the one part.

and

M/s. E-Construct Design & Build Pvt. Ltd. having its principal place of business at 2nd Floor, ~~Yerkaudhar~~ Heights, Parasanna Agrahara Main Road, Opp. Sai Poorna Premier Apartment, Kudla, Karnataka-560-068. (hereinafter called the "CONTRACTOR" (which expression shall, unless repugnant to the context or meaning hereof, be deemed to mean and include its successors and assignees) of the other part. Capitalized terms not otherwise defined in this Contract shall have the meanings given such terms in the General Conditions of Contract, attached hereto and incorporated by reference into this Contract.

WHEREAS

A. M/s. Reliance Industries Limited (RIL) having its registered office at 3rd Floor, Maker Chambers-IV, ~~Reliance point,~~ Mumbai-400 021. (hereinafter called "CLIENT/OWNER") has awarded a Main Contract (as hereinafter defined) to E-Construct Design & Build Pvt. Ltd. (hereinafter called "CONTRACTOR") for perform works.

B. The Owner now wishes to utilize the resources of CONTRACTOR to perform the Services (as hereinafter defined), which services comprise a vital portion of the work and is desirous of being provided and having executed certain works relating to Civil mentioned work, or referred to in the Tender Documents including General Conditions of Contract, Special Conditions of contract, ~~Annexure~~, Specifications, Drawings, Schedule of Rates, various Minutes of Meetings and other documents as called for in the tender.

C. All contractors performing services for the ultimate benefit of Owner are to be bound by terms and conditions consistent with the terms and conditions of the Main Contract.

Ref No: RIL/OMA/101/21-22/001

Date: 15.10.2021

To,

E Construct Design & Build Pvt. Ltd.

2nd Floor, ~~Yerkaudhar~~ Heights,

~~Kudla, Karnataka-560-068.~~

Kind Attn: Mr. ~~Abhishek J. Modi~~

Subject: Construction of Reliance Mall

Reference: Our Tender Document floated Dated 01.09.2021.

Your Quotation submitted dated 10.09.2021.

Subsequent discussion with you in meeting held at our corporate office on 11.09.2021

Letter of Acceptance

Further to your proposals referred above and subsequent meetings we had, we are pleased to inform you that your firm has been selected for the works mentioned above for the estimated cost of work **Rs.156,87,54,726/-** (One Fifty Six ~~Crores~~ Eighty Seven ~~Lakhs~~ Fifty Four Thousand Seven Hundred Twenty Eight only). The rates are inclusive of all Taxes.

1. The scope of work includes material (except steel reinforcement) and cement, to be supplied free-of-cost by the client at site and ~~labor~~ for Civil Structural works, finishing works and allied services works comprising of single building block with single basement and four stories, all as per tender drawings and documents complete from all aspect in all respects.
2. The project commencement date will be reckoned as the 30th day from issuance of this document. Accordingly, the commencement date shall be 15.10.2021 and completion date shall be 15.10.2024.

This LOA shall be treated as order to commence mobilization works at site and work order and other contractual documents shall be signed within 15 days from date of this letter with inclusion of mutually agreed terms as finalized between both the parties.

Thanking you.

Yours faithfully,

For Owner

(Head - Contract)

1.0 – BIM EXECUTION PLAN (BEP) AGREEMENT

By signature below, this BIM Execution Plan, dated 2021-10-22 is herewith approved and will be incorporated as an amendment to the Design Professional Agreement, as a change order to the Contractor Agreement, and as an amendment to other separate consulting and commissioning agent agreements as they may apply to the list of parties co-signing this document.

The BEP shall be updated and amended at each major project phase deliverable, as new key parties or entities are brought on board the project and incorporated into the BIM Execution Plan.

Owner Representatives:

Director	Md Akram
Addl Director	Shaishita Afreen

Design Team

Design professional	Aditya
Structural engineer	Sai
MEP engineer	Hemanshu
Civil engineer	Faheem
Other	

Construction team

General contractor	Ramcharan
Mechanical contractor	Maresh babu
Electrical contractor	Vijay
Plumbing contractor	Dhanush
Structural contractor	Yash
Other	

Consultant	Shivam
-------------------	--------

2.0 – OVERVIEW

The intent of this BIM Execution plan is to provide a framework that will let the Owner, Design Professional, engineers, and Contractor deploy building information modeling (BIM) technology and best practices on this project. This plan delineates roles and responsibilities of each party at each phase, the scope and level of detail of information to be developed, data to be managed and shared, processes defined, and software to be utilized.

3.0 – PROJECT INITIATION

This section defines the Core Collaboration Team, the project objectives, project phases, and overall communication plan throughout the project's phases.

3.1 – PROJECT INFORMATION

RELIANCE MART LOCATED AT MANGALORE ,KARNATAKA. THIS IS G+4 RCC PROJECT TOTAL AREA 7095 Sqm. COMMERCIAL PROJECT HAVING TWO BASEMENT PARKING RETAIL SHOPS INCLUDING MOVIE THEATER,SHOPPING COMPLEX ETC.

Project Name:	RELIANCE MART G-4
Project Number:	RM20210922
Project Address:	MANGLURU ,KARNATAK
Project Description:	COMMERCIAL

3.2 – PROJECT GOALS AND OBJECTIVES

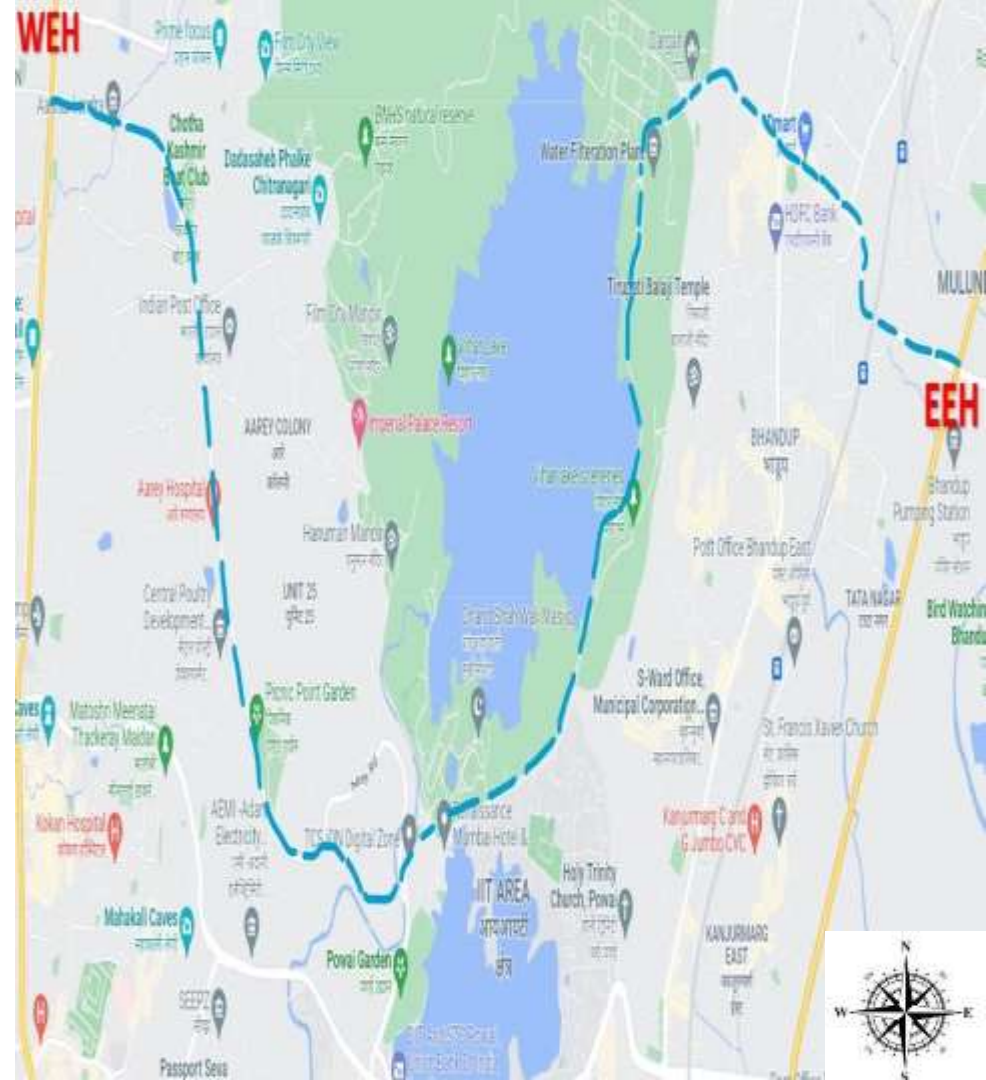
Project Goal(s)	Achieved if	Project Timeframe
Provide UGA OUA with a useful BIM model that can assist in future renovations, additions and space management.	Model is accurate and contains all major components and systems in Revit format, updated to reflect as-built conditions of construction	Completion of Project

GOREGAON MULUND LINK ROAD (GMLR)-TEAM 4



PROJECT DETAILS

Project Name	GMLR Project
Location:	Mumbai(Goregaon to Mulund)
Length of Road	15.86Km
Width of Road	20m (Carriage 15m+2m divider + 1.5*2 Shoulder width)
No. of Junctions	2 No's
No. of Toll Plaza	5 No's



[illegible]

BACKGROUND STUDY

- ❑ Mumbai is known as commercial capital of India which has grown rapidly over past decade.
- ❑ The population of Mumbai is approximately around 40 Lakhs.
- ❑ Everyday registration of Vehicles is 500 nos.
- ❑ Infrastructure activity is also growing rapidly in order to provide accommodation to all their citizens.
- ❑ Present road network is under Saturated level.



EASTERN & WESTERN EXPRESS HWY



- ☐ Roads of both the highways gets blocked regularly.
- ☐ Speed 11 to 20 km/h
- ☐ Congestion
- ☐ Loss of Time & Energy

TRAFFIC CONGESTION IN JVLR & GHODBUNDER

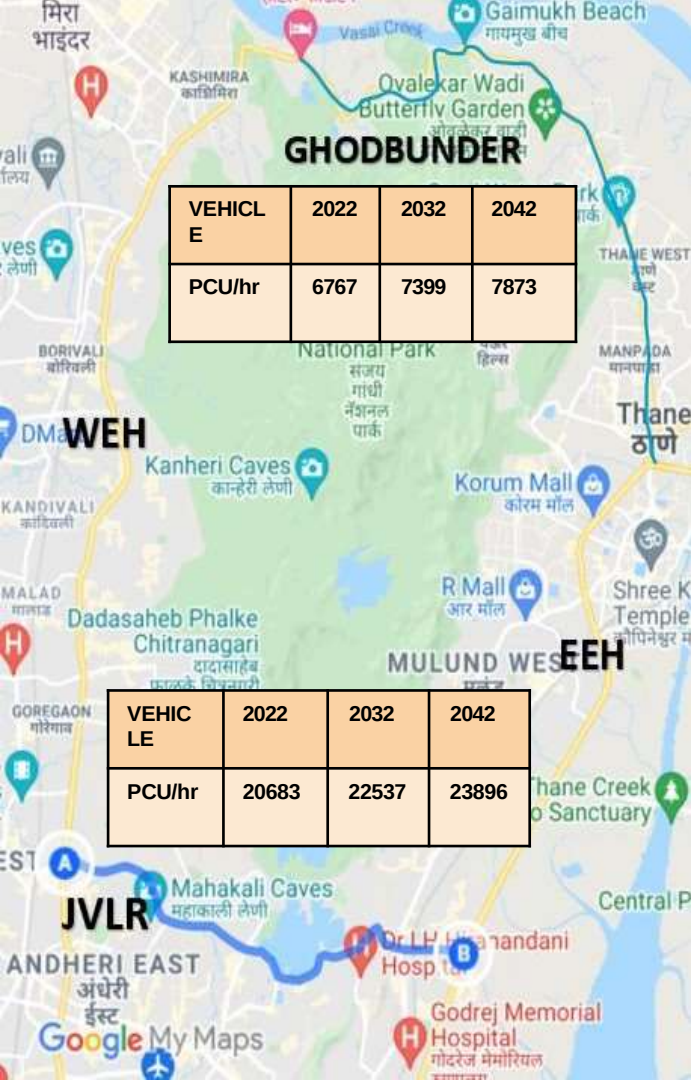


JVLR



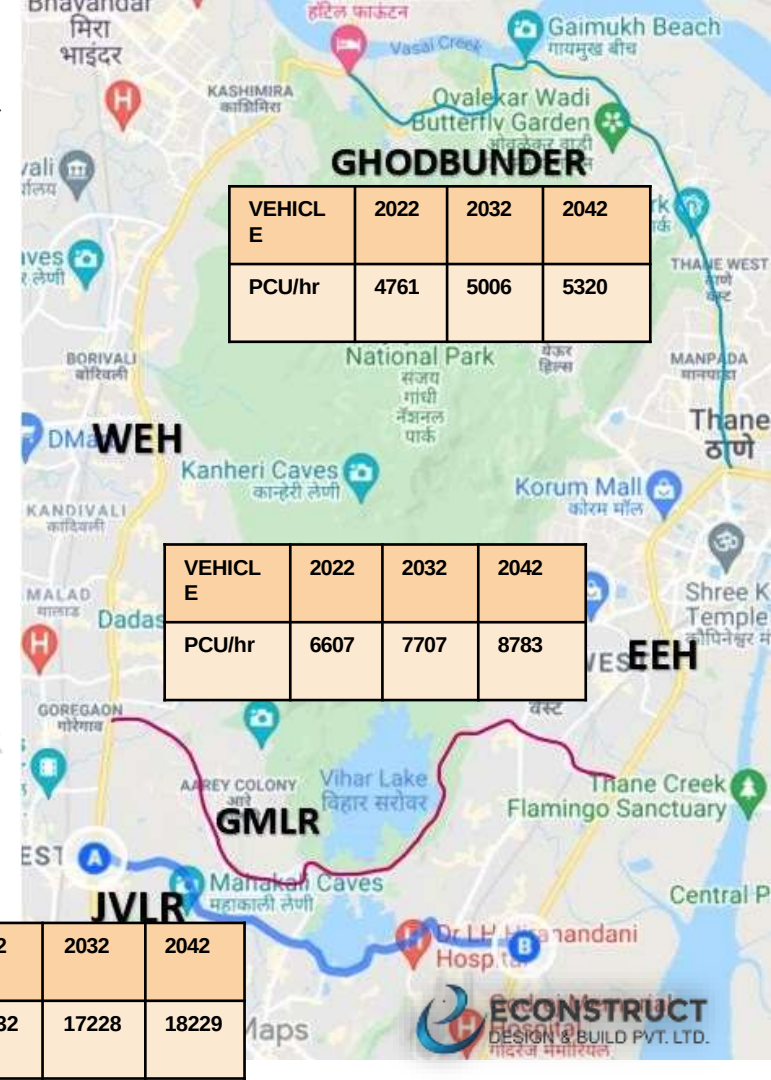
GHODBUNDER

- ☐ Roads have reached to the Saturation level
- ☐ Speed 11 to 20 km/h
- ☐ Congestion
- ☐ Loss of Time & Energy
- ☐ Environmental Impact



TRAFFIC STUDY

BEFORE



AFTER

VEHICLE	2022	2032	2042
PCU/hr	16132	17228	18229

GMLR SWOT ANALYSIS



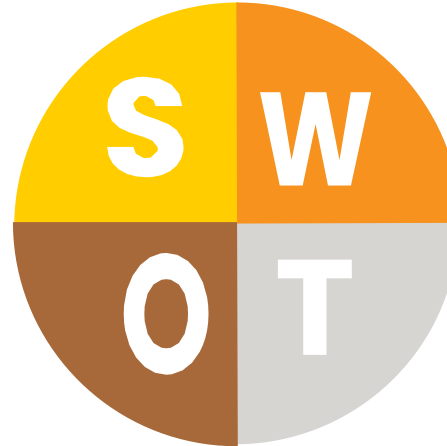
STRENGTHS

- Reduce traffic Congestion and Increase the travel time



- Employment opportunity
- Revenue Generation

OPPORTUNITIES



WEAKNESSES

- Cutting down the Trees
- Relocation of high tension wires, Underground water and gas pipelines

- Land acquisition
- Demolition of structures

THREATS

STARTING POINT OF THE PROJECT



KEY PLAN



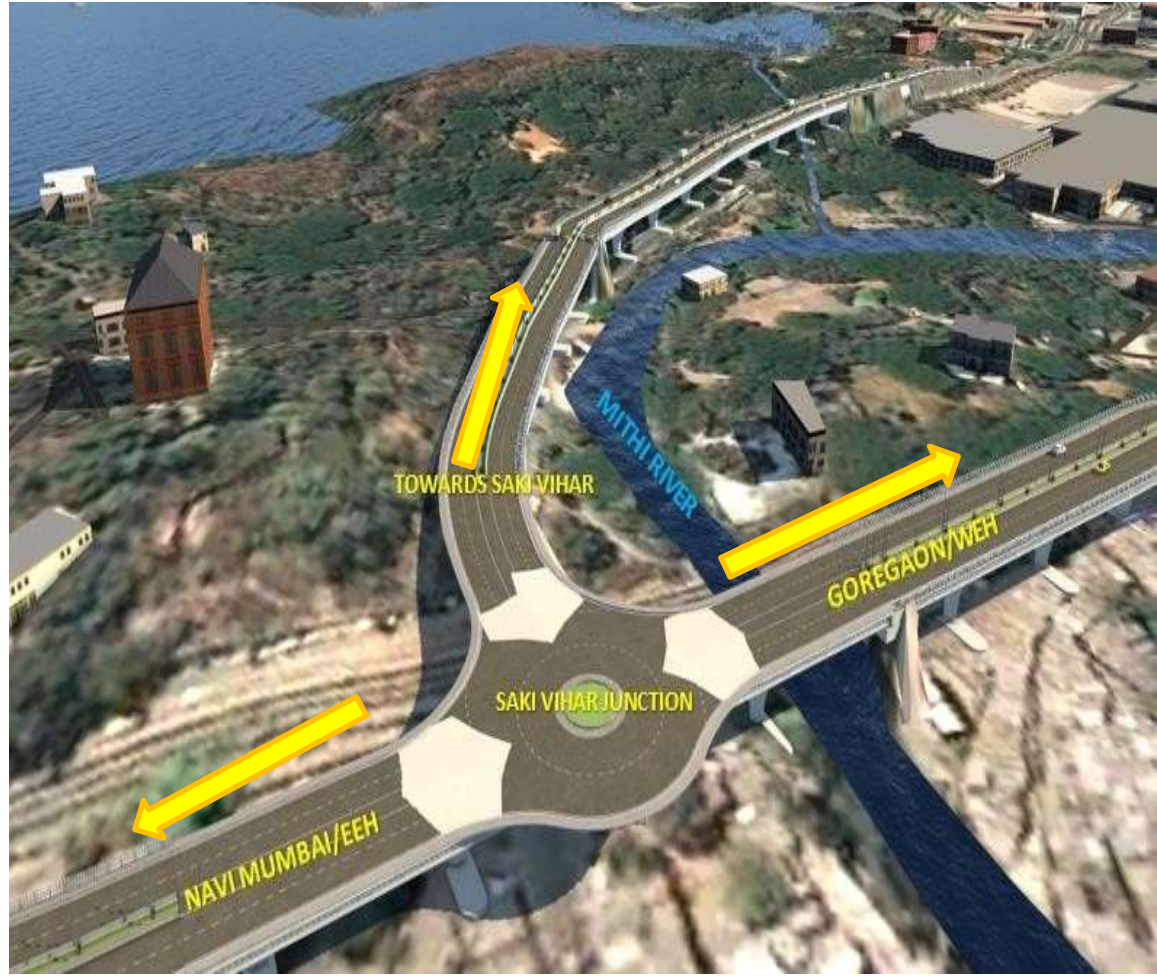
VIEW FROM EASTERN EXPRESS HIGHWAY ENTRY



SAKI VIHAR JUNCTION POINT



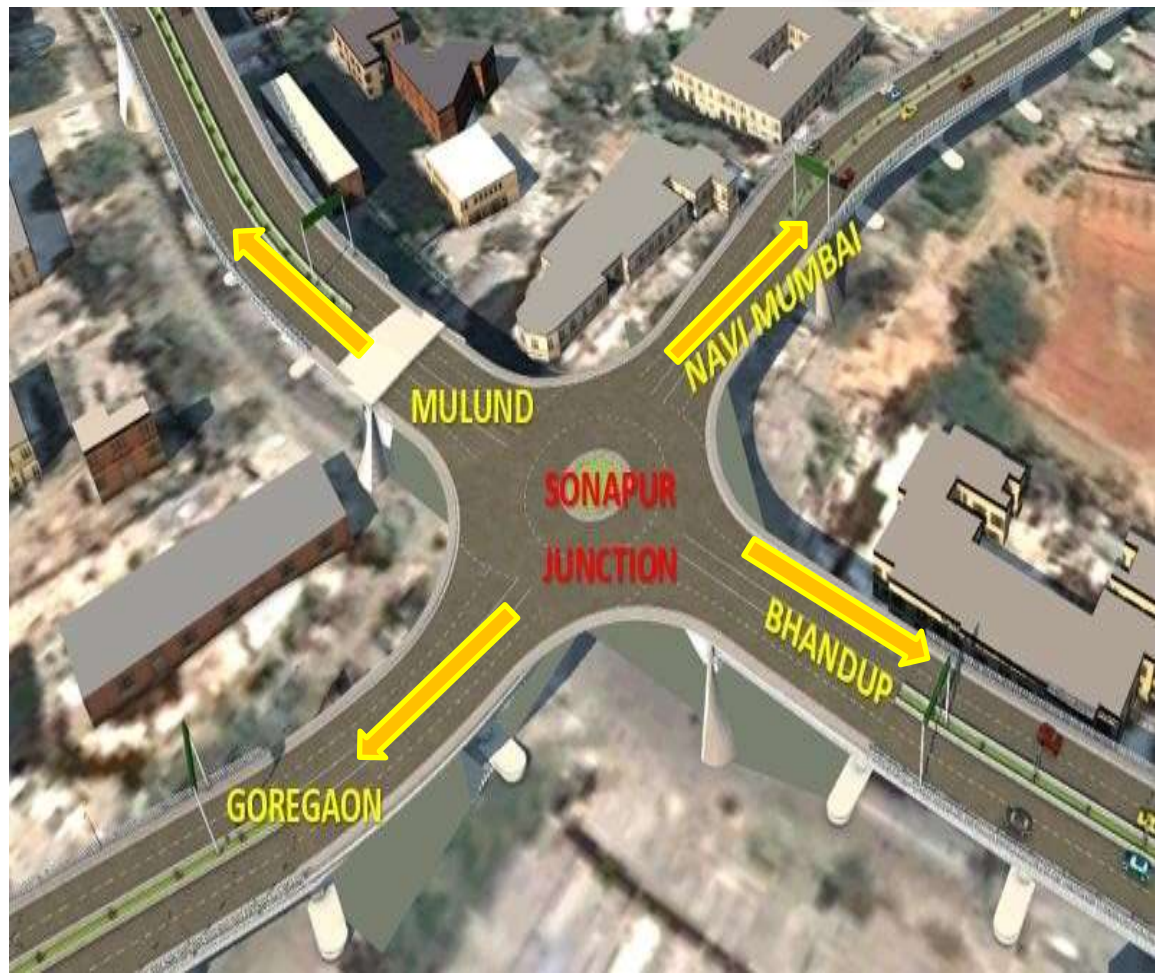
KEY PLAN



SONAPUR JUNCTION POINT



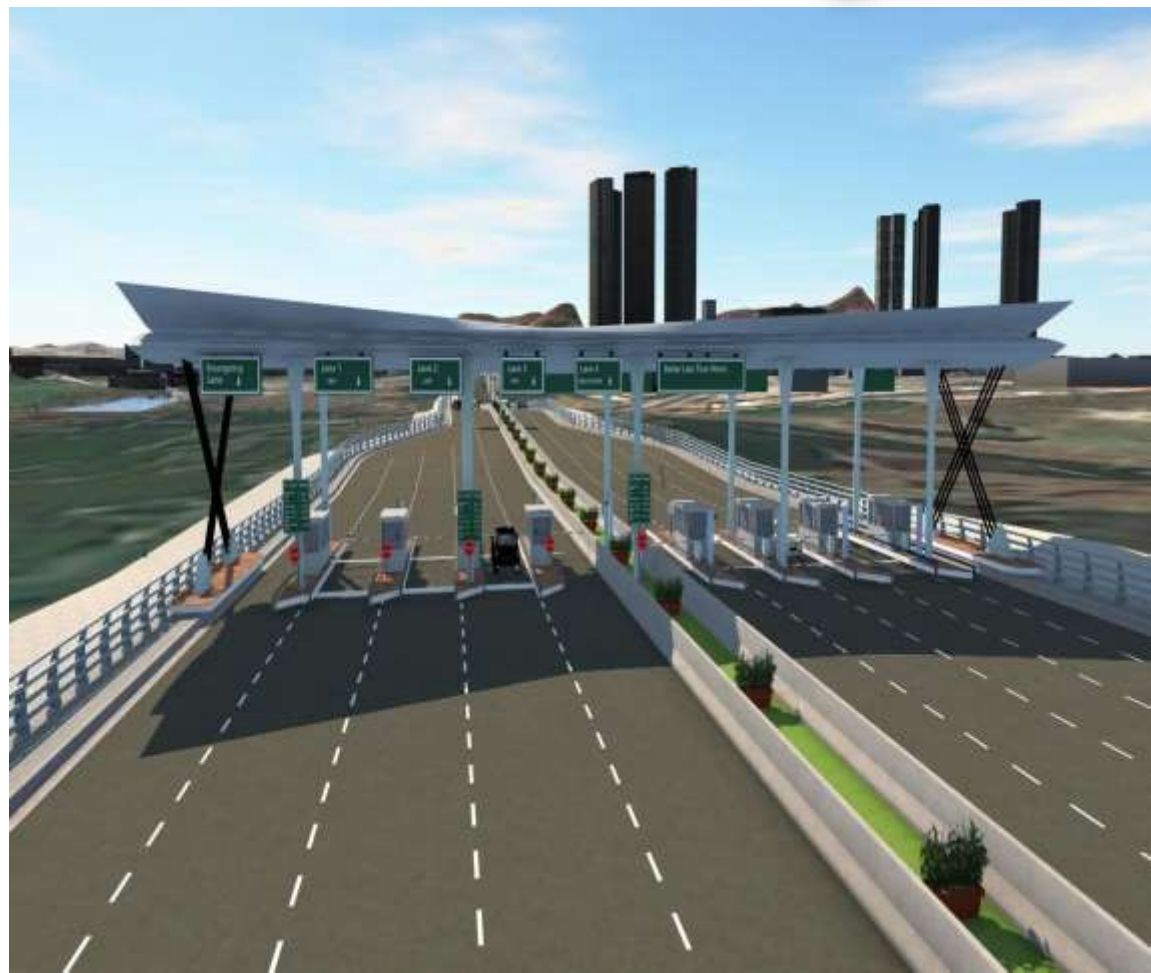
KEY PLAN



END POINT OF THE PROJECT



KEY PLAN



BRIDGE DETAILS

Pile Cap

L=5m

W=22m

D=2m

Longitudinal Overhang = 1.5m

Transverse Overhang = 1.5m

Pile

Longitudinal row = 4m

Transverse row = 6m

No. of piles = 18

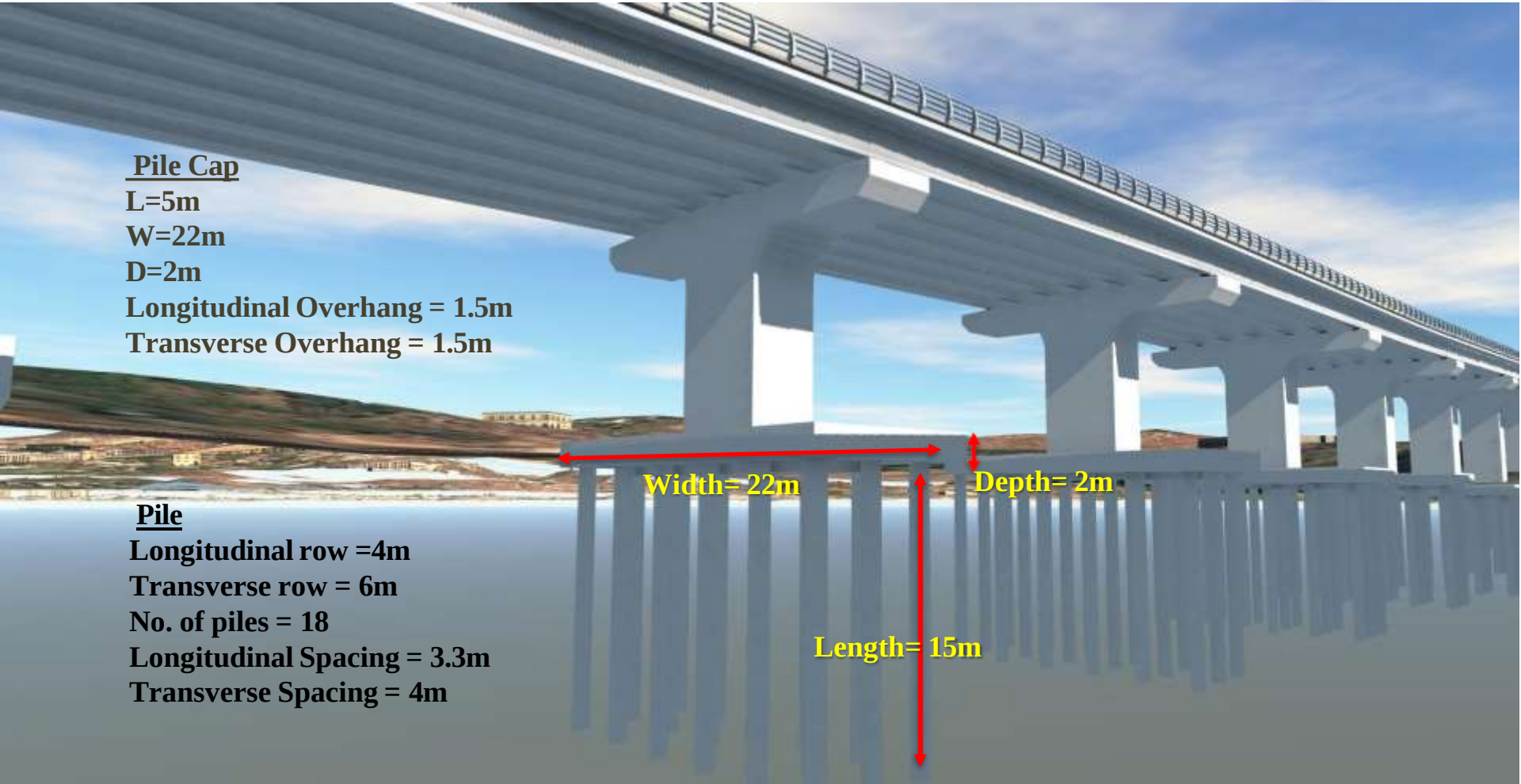
Longitudinal Spacing = 3.3m

Transverse Spacing = 4m

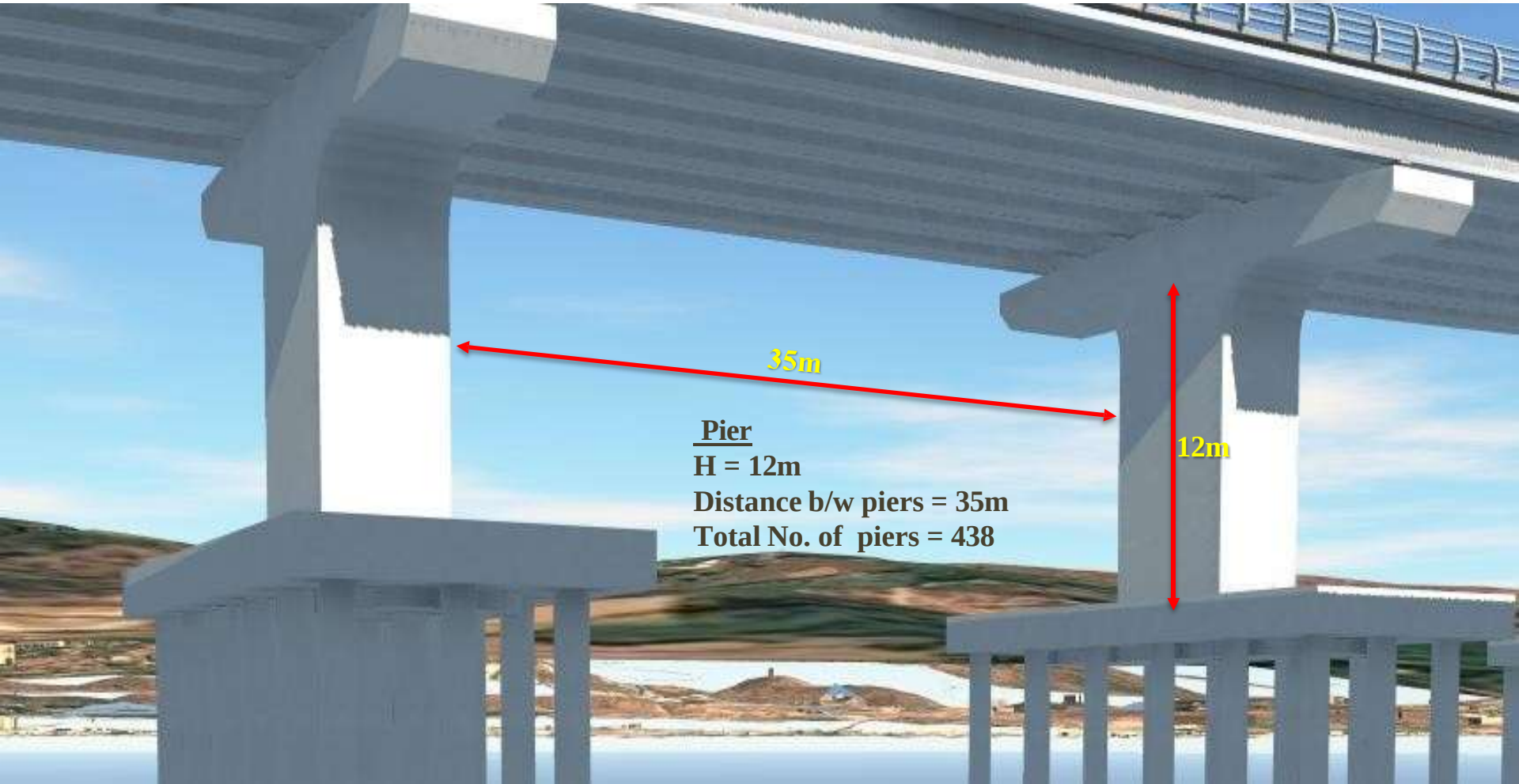
Width= 22m

Depth= 2m

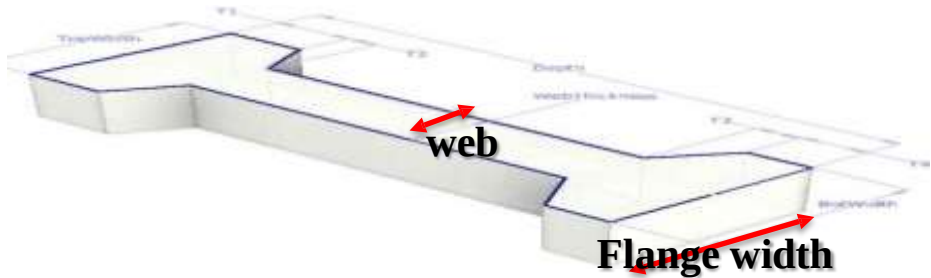
Length= 15m



Pier detail:



Girder details:



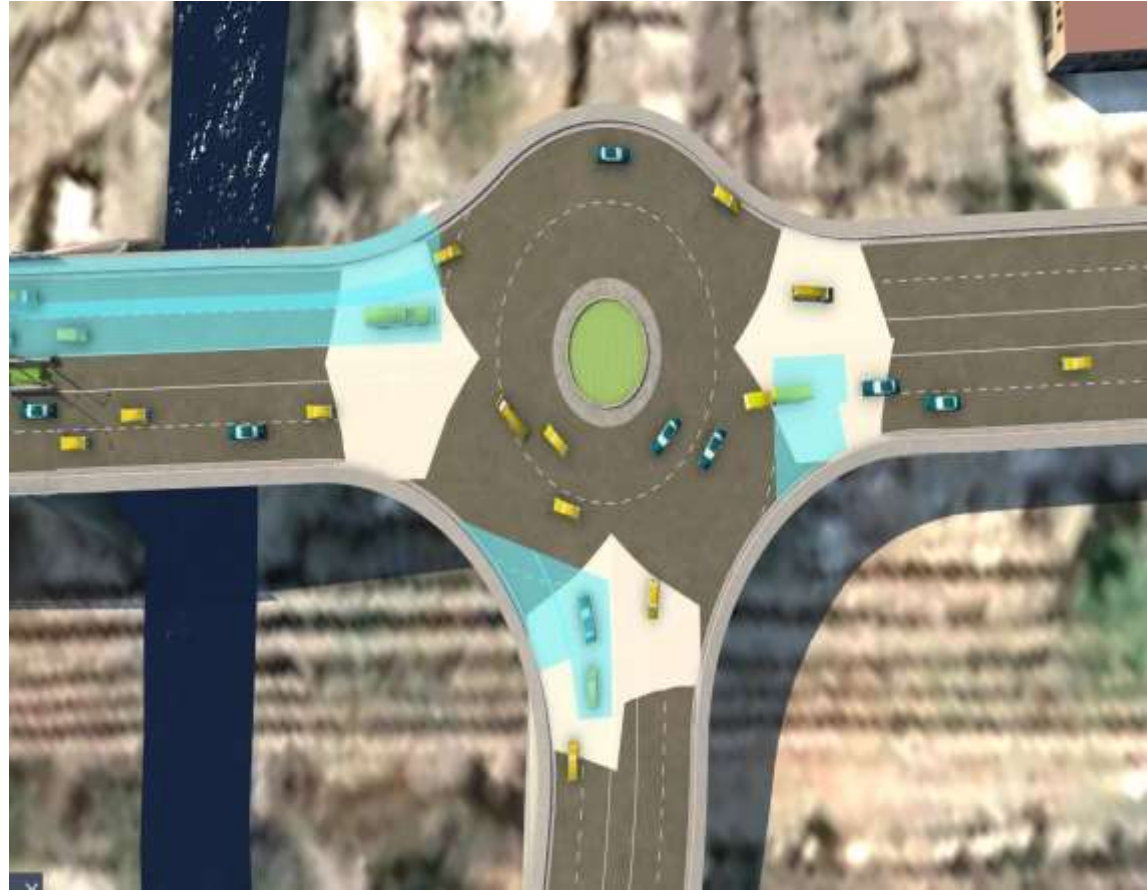
- ☐ No of Girders = 8
- ☐ Spacing b/w two Girders = 2.61m
- ☐ Nominal Length = 35m
- ☐ Material : Pre Cast Concrete
- ☐ I section girder

TRAFFIC MANUVERING STUDY ON ROTARY

Analysis	Per lane
Delay threshold	60 Sec.
Result Volume	Per Lane
Simulation time	Peak Hrs.
Delay time	36sec

AT SAKI VIHAR JUNCTION

Year of Study : 2022

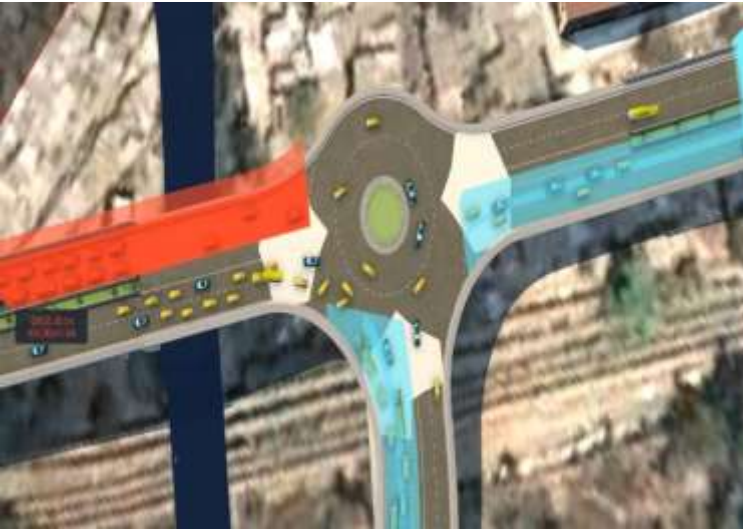


TRAFFIC MANUVERING STUDY ON ROTARY

Analysis	Per lane
Delay threshold	60 Sec.
Result Volume	Per Lane
Simulation time	Peak Hrs.
Delay time	63sec

AT SAKI VIHAR JUNCTION

Year of Study : 2042



TRAFFIC MANUVERING STUDY ON ROTARY

Analysis	Per lane
Delay threshold	60 Sec.
Result Volume	Per Lane
Simulation time	Peak Hrs.
Delay time	18sec

AT SONAPUR JUNCTION

Year of Study : 2022



TRAFFIC MANUVERING STUDY ON ROTARY

Analysis	Per lane
Delay threshold	60 Sec.
Result Volume	Per Lane
Simulation time	Peak Hrs.
Delay time	79sec

AT SONAPUR JUNCTION

Year of Study : 2042



STOPPING SIGHT DISTANCE STUDY FOR DESIGN SPEED OF 60Km/hr NEAR VIHAR LAKE

FOR FORWARD PASS



FOR BACKWARD PASS



STOPPING SIGHT DISTANCE STUDY FOR DESIGN SPEED OF 80Km/hr NEAR VIHAR LAKE

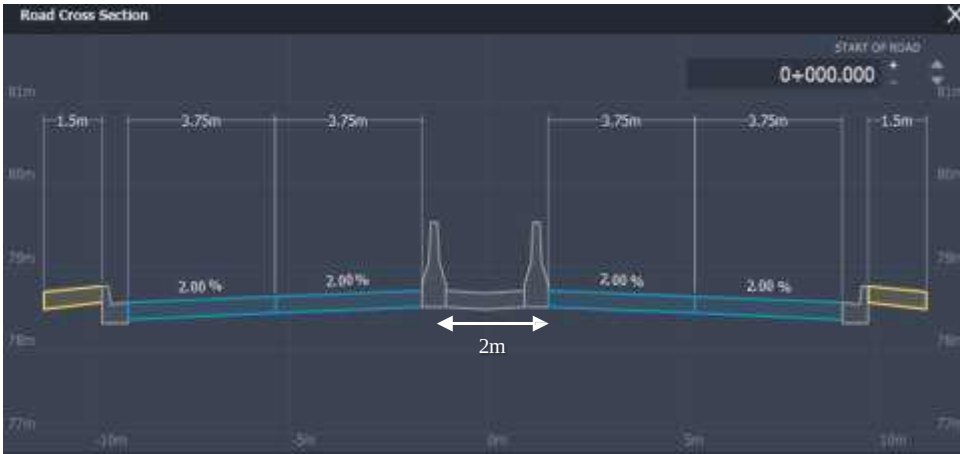
FOR FORWARD PASS



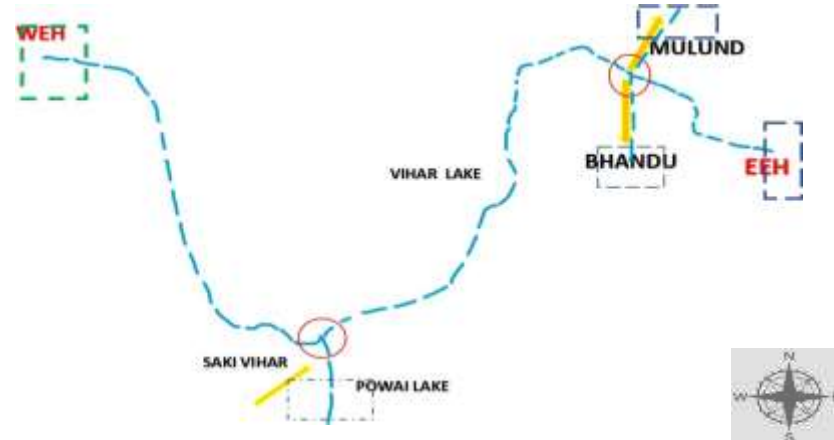
FOR BACKWARD PASS



CROSS SECTION OF ROAD AND TOLL PLAZA



TOLL PLAZA



DISTANCE BETWEEN TOLL PLAZAS

1.	Aarey RD to EEH	15.6Km
2.	Aarey RD to Mulund	14.6Km
3.	Aarey RD to Bhandup	14.4Km
4.	Aarey RD to Saki vihar	6.3Km



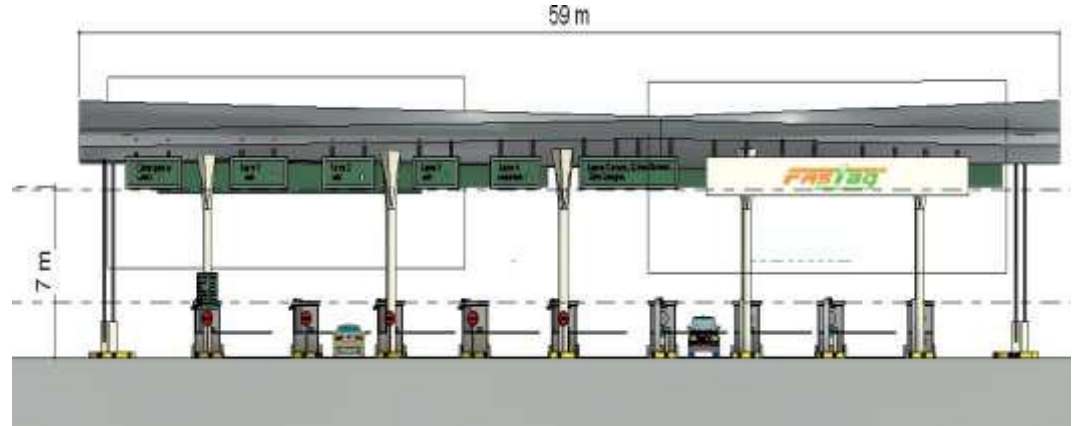
TOLL PLAZA DIFFERENT VIEWS

TOLL NO	LOCATION	NUMBER OF BOOTHS
1.	WEH(ARREY ROAD)	9
2.	SAKI VIHAR	9
3.	MULUND	9
4.	BHANDUP	9
5.	EEH	9



SIDE ELEVATED VIEW

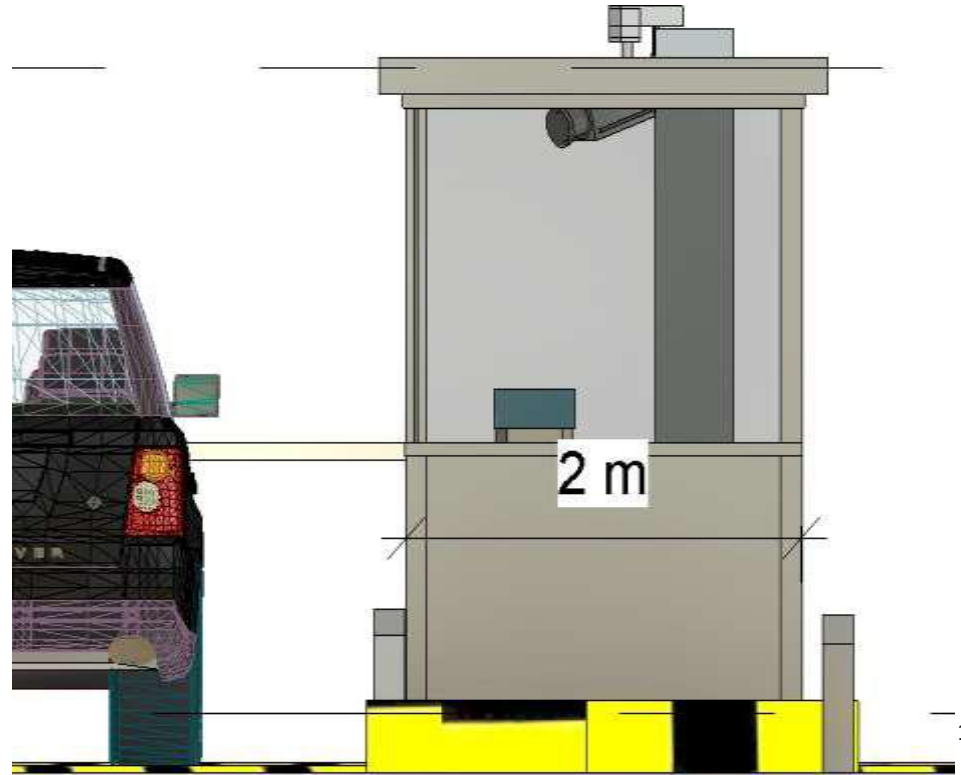
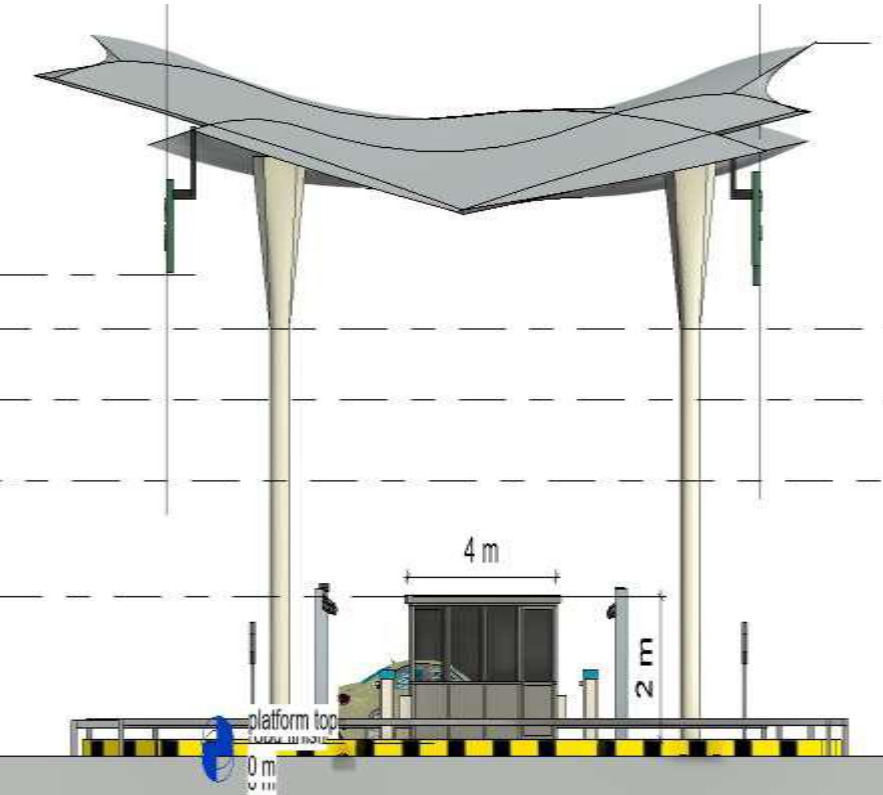
FRONT ELEVATION



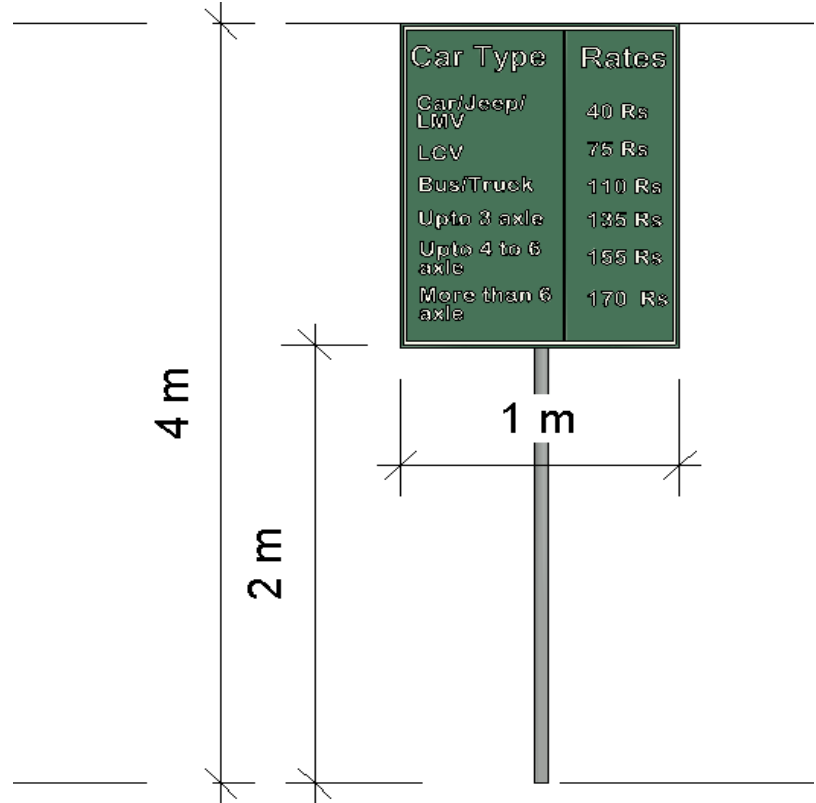
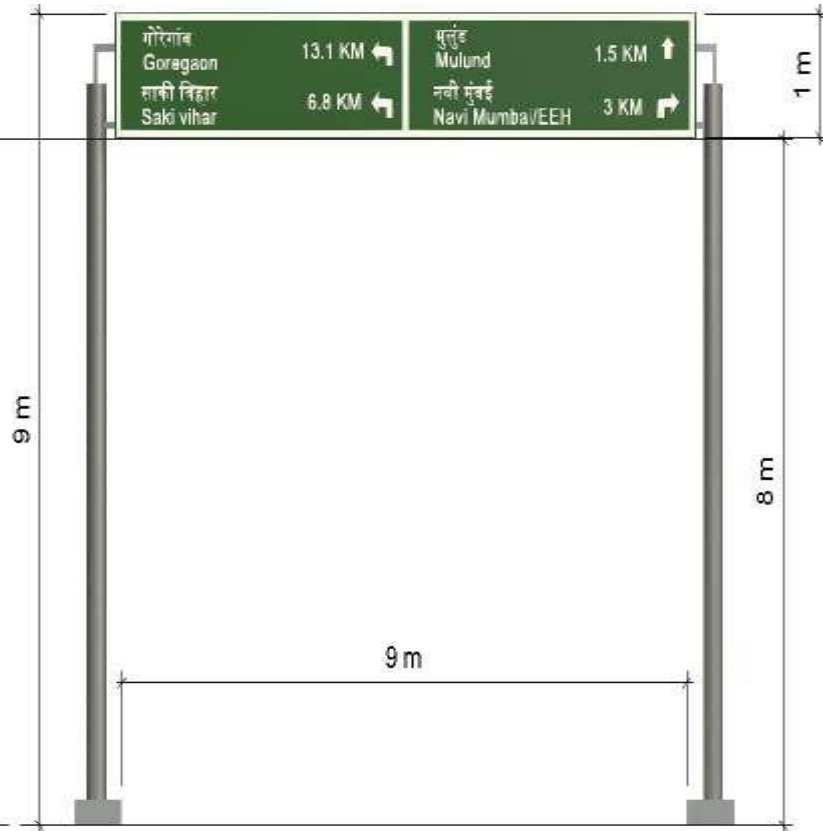
3D VIEW

TOLL PLAZA DIFFERENT VIEWS

BOOTH DIMENSION



Board Details



TOLL PLAZA RENDER VIEWS



TOLL PLAZA RATES



TOLL FARE TO BE CHARGED ON WESTERN HIGHWAY TO EASTERN HIGHWAY & VICE- VERSA

TOLL FARE CHARGED ON SAKI VIHAR PAVAI LAKE TOLL ROAD BOOTH

Type of Vehicle	Single Journey	Return Journey	Monthly pass	Commercial vehicles registered within the district plaza	Type of Vehicle	Single Journey	Return Journey	Monthly pass	Commercial vehicles registered within the district plaza
Car/jeep/van	40	70	2000	35	Car/jeep/van	30	50	1800	30
LCV	70	120	3300	60	LCV	40	70	2000	35
BUS/TRUCK	100	180	4500	150	BUS/TRUCK	70	120	3300	60
Up to 3 Axles	110	200	5400	160	Up to 3 Axles	110	200	5400	160
4 to 6 Axles	135	250	7200	220	4 to 6 Axles	135	250	7200	220
HCM/EME	155	290	8500	250	HCM/EME	155	290	8500	250
7 or more axles	170	320	9400	280	7 or more axles	170	320	9400	280

TOLL PLAZA RATES

TOLL FARE TO BE CHARGED ON SONAPUR JUNCTION TOLL ROAD BOOTH

Type of Vehicle	Single Journey	Return Journey	Monthly pass	Commercial vehicles registered within the district plaza
Car/jeep/van	30	50	2000	35
LCV	50	90	3300	60
BUS/TRUCK	80	150	5400	150
Up to 3 Axles	115	230	7200	220
4 to 6 Axles	130	260	7500	250
HCM/EME	150	290	8500	250
7 or more axles	160	320	9400	280



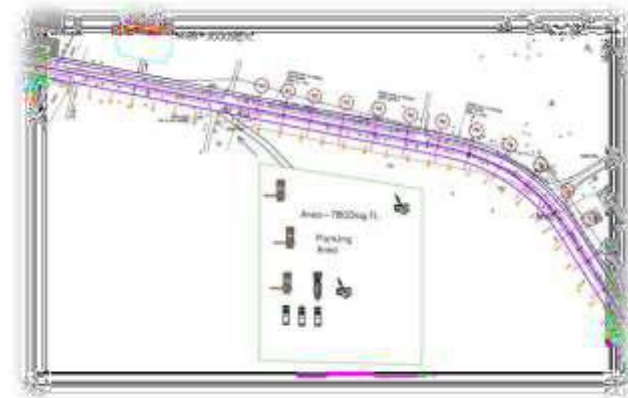
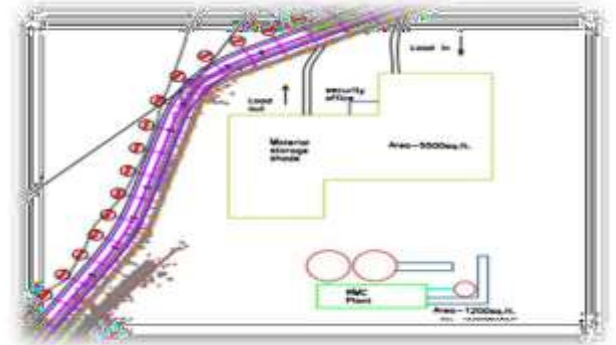
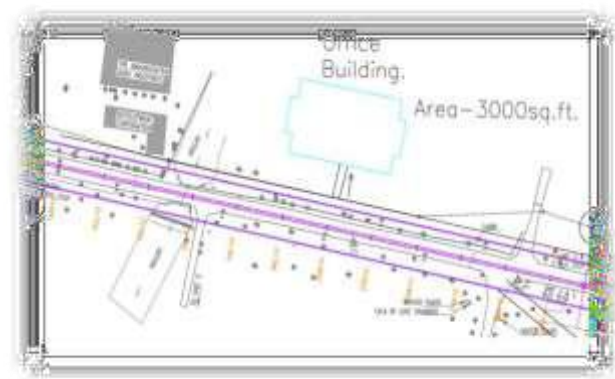
Toll Plaza Collection

TOLL FARE TO BE CHARGED ON WESTERN HIGHWAY TO EASTERN HIGHWAY & VICE- VERSA

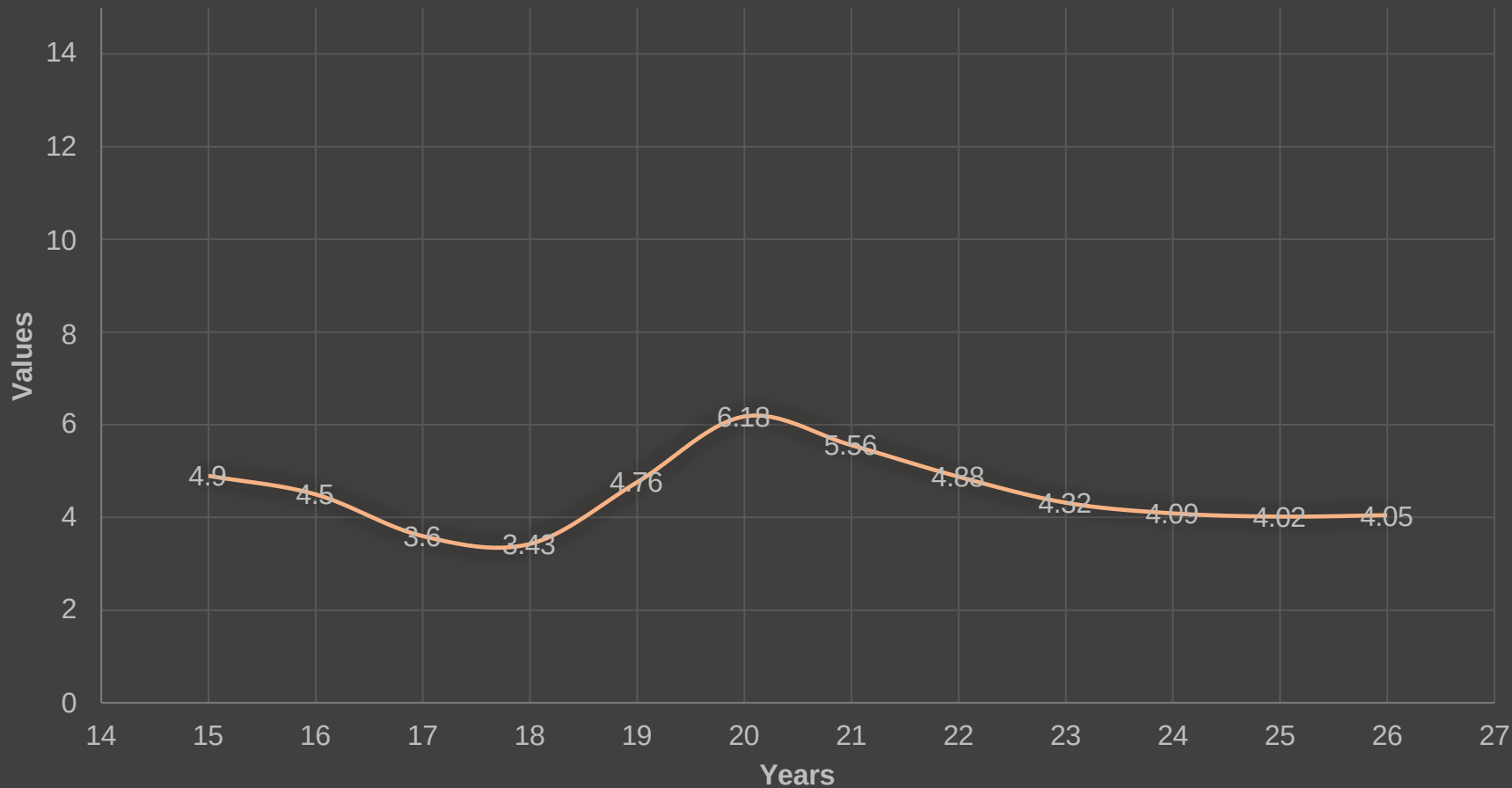
Car/Jeep/Van	Toll rate in Rs.	No. of vehicles per day	Toll collection per day by 1 toll plaza
LCV	35	15000	507000
BUS/TRUCK	65	4500	283500
Up to 3 Axles	95	4500	418500
4 to 6 Axles	110	2040	216240
HCM/EME	145	1980	277200
7 or more Axles	160	1980	306900

Total toll collection	Per day	20,09,340	
	Per month	6,02,80,200	
	Per year	73,34,09,100	
Total project cost	Rs.	2549,82,84,685	2549.82Cr
Project cost recovery	In years	7 years	

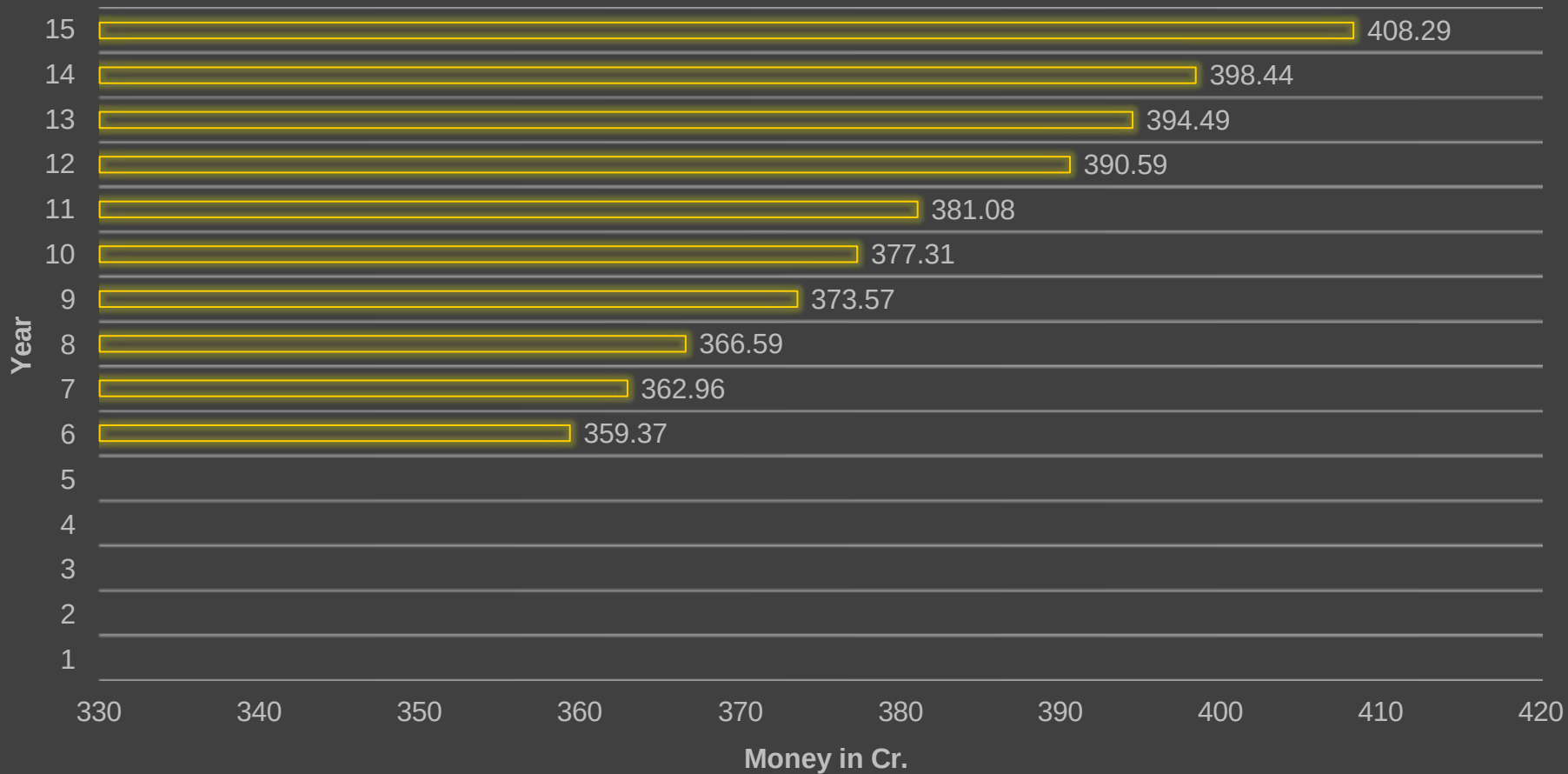
Layout Plan



Inflation rate (2015-2026)



Earnings Per Year (2015-2029)



Total cost recovery period of the project is 7 years.

and we are taking profits from next 3 years.

**Total profit =(Total 10 years Revenue generated by toll plaza) – (total cost
of the project)**

=(3812,73,08,559) - (2549,82,84,685)

Total profit = 1262,90,23,874 /-

BILL OF QUANTITY :-

SR No	NAME OF WORK	COST (Rs.)
1	Site Clearance & preliminary Works	9,74,35,326/-
2	Tests on Pile	69,85,470/-
3	Bridge Substructure And Foundation(CIP)	416,27,64,454/-
4	Abutment(Retaining Wall)	36,94,11,563/-
5	Superstructure	501,33,50,193/-
6	Asphalt	81,97,89,000/-
7	Road Component	34,87,59,861/-
8	Steel Fe 500D Substructure	455,41,60,000/-
9	Road Furniture	4,53,23,210/-
10	Toll Plaza (5 No's.)	3,86,11,925/-
	Total Cost (A)	Rs.1649,52,02,927/-

❖ Continue.....

Sr. No.	Additional Charges	Cost(Rs.)
1	Add 5% Overhead Charges	Rs. 82,47,60,146/-
2	Add 1% Water Charges	Rs. 16,49,52,029/-
3	Add 5% Contingency Charges	Rs.82,47,60,146/-
4	Contractor Profit 15%	Rs.247,42,80,439/-
5	General Consultant 5%	Rs.82,47,60,146/-
	Total Cost (B)	Rs. 511,35,12,906/-

Total Cost (A)	Rs. 1649,52,02,927/-
Total Cost (B)	Rs. 511,35,12,906/-
Total Cost (A)+(B)	Rs. 2160,87,15,834/-

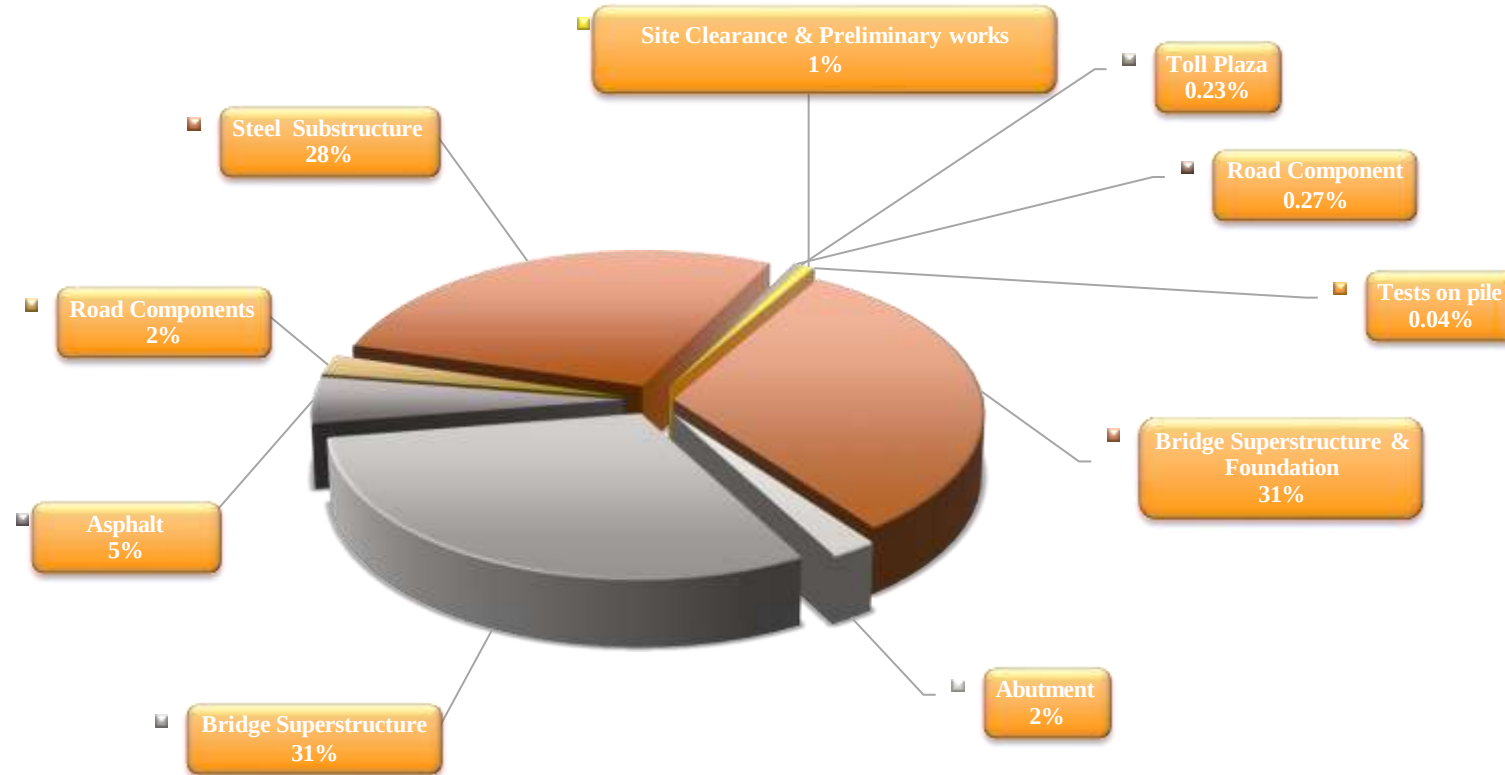
❖ Continue.....

Total Cost (A)+(B)	Rs. 2160,87,15,834/-
Add 9% CGST	Rs. 194,47,84,425/-
Add 9% SGST	Rs. 194,47,84,425/-
Grand Total	Rs. 2549,82,84,685/-

IN WORDS –

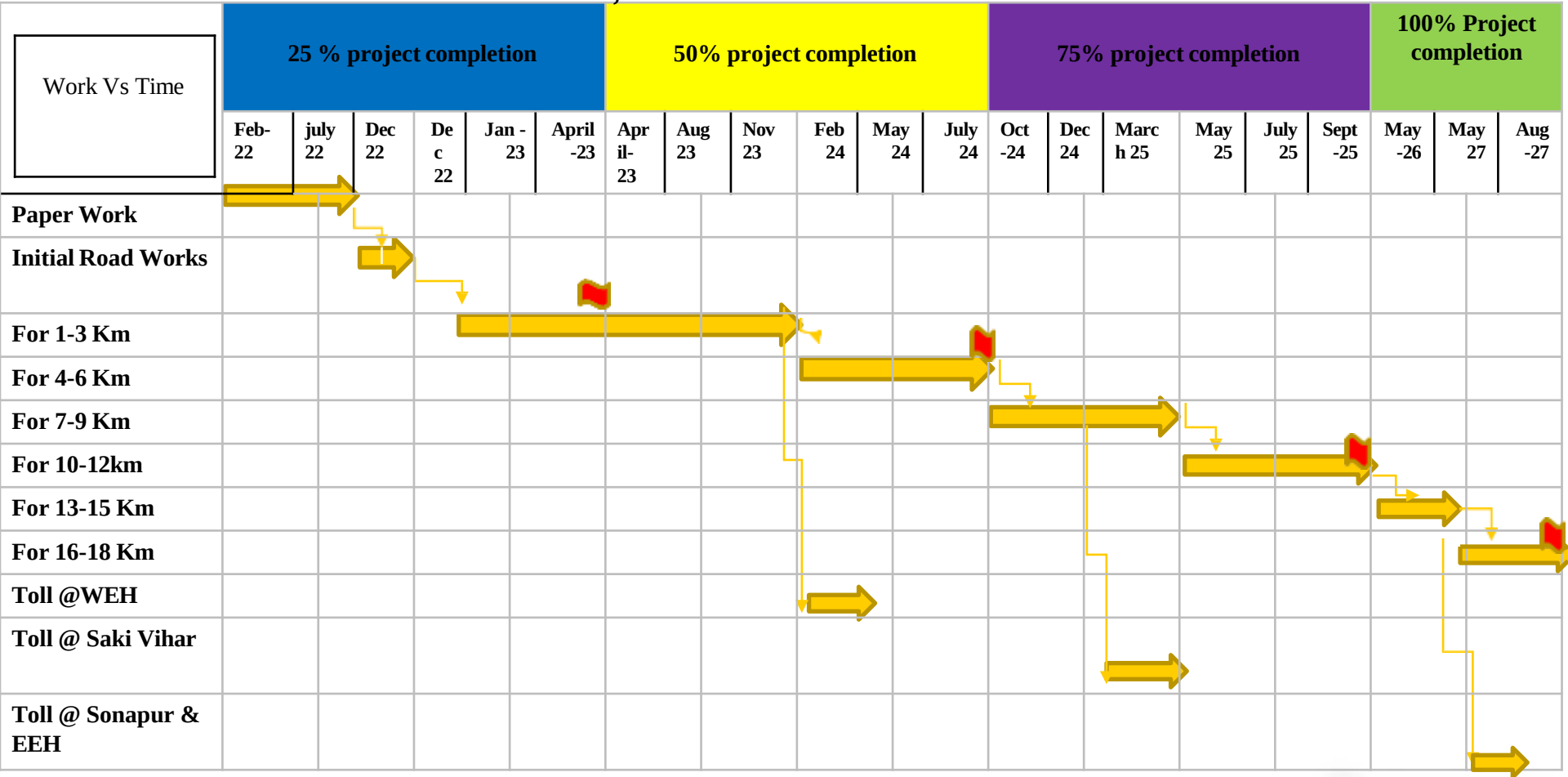
**Rupees Three Thousand Three Hundred Seventy Two Crore Fifty Five Lakh
Sixty Thousand One hundred Fifteen only GMLR PROJECT-FINAL
BOQ.xlsx**

Percentage Of Expenses



- Site Clearance & Preliminary works
- Tests on pile
- Bridge Substructure & Foundation (Cast In Place)
- ABUTMENT
- SUPERSTRUCTURE
- ASPHALT
- ROAD COMPONENTS
- STEEL FE 500D STEEL SUBSTRUCTURE
- ROAD FURNITURE
- TOLL PLAZA

Project Schedule of GMLR



Milestone

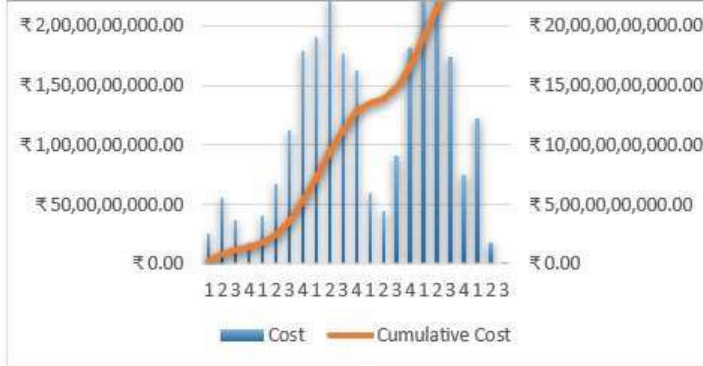


Task

CASH FLOW

Cash flow of GMLR

Actual Cost	Baseline Cost	Remaining Cost	Cost Variance
₹ 94,500.00	₹ 0.00	₹ 25,498,190,185.00	₹ 25,498,284,685.00



Name	Remaining Cost	Actual Cost	Cost	ACWP	BCWP	BCWS
GMLR Project	₹ 25,498,190,185.00	₹ 94,500.00	₹ 25,498,284,685.00	₹ 0.00	₹ 0.00	₹ 0.00

Concession agreement

- In this project, the agreement was done on basis of BOT contract, between **municipal corporation of greater Mumbai (MCGM)** and **Econstruct design and build pvt ltd.** to design, finance, build an infrastructure and operate a project for a period of 15 years including 5 year of construction.
- BOT contract is a form of leasing, where the government (project sponsor) allows a private entrepreneur (project promoter) to design, finance, and build an infrastructure facility.
- **“Concessionaire”** means Econstruct Build and Design Private Limited in this project.
- The **Concession Period** shall commence on the Appointed Date and shall end on the Termination Date.



- In return, the project promoter is permitted to collect tolls and operate the facility for a specified period(called as concession period), during which he is expected to recover all of his costs and earn a reasonable profit.
- After that period, the project is returned to the public entity originally granted the concession.



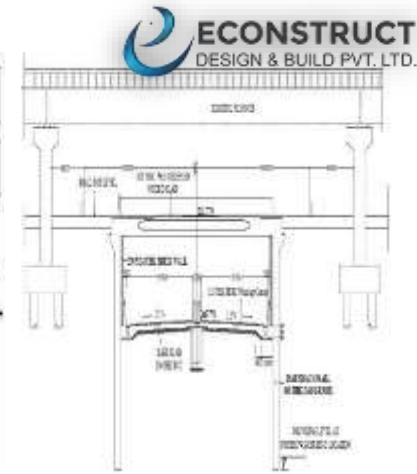
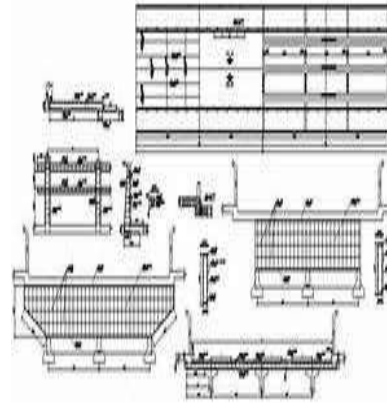
- Elevated corridor connecting WEH near Goregaon Flyover and EEH near Mulund running along Aarey milk colony, Powai Lake and Bhandup Complex area along Trunk Main water line.
- It is about approx. 15.38km of an elevated road with 1.7km connector and 0.9km at grade road at entry/exit at WEH & EEH.
- The concessionaire is not operating the project through the main company, so they are established the separate company and they are operate with new company & that is known as SPV entity.
- The Concessionaire shall for due and faithful performance of its obligations during the Construction Period provide to MMRDA a bank guarantee from any Bank in the form set forth in Schedule 'F' ("Performance Security") for a sum equivalent to Rs.100 million Rupees (one hundred million) within 120 (one hundred twenty) days of the date of this Agreement as a performance security.



- The Concessionaire shall within 60 days from the date of this agreement open and establish the Escrow Account with a Bank (The “Escrow Bank”).
- In which all inflows and outflows of cash on account of capital and revenue receipts and expenditures shall be credited and debited.
- **“Exempted Vehicles”** means vehicles that shall be exempted from payment of Fees which includes Ambulances, Fire Tenders, Ministers of the GOM, Judges of the Supreme Court of India, Governors, Chief Ministers, Leaders of opposition in Lok Sabha, Rajya Sabha, forces in uniform including armed forces, Para military forces and police, etc...
- Additional Toll way shall not be opened to traffic before expiry of 8 (eight) years from the appointed date, if additional Toll way required.



- Three copies each of all Drawings to MMRDA and the Independent Consultant for review and comments.
- The Concessionaire shall Appoint Supervisor's for Monitor and control the activities of Contractors under their respective Project Agreements as may be necessary.
- Remove all surplus construction machinery and materials, Waste materials (Including, without limitation, Hazardous materials and waste water), rubbish and other debris (Including without limitation accident debris) and shall keep the Site in a neat and clean condition before go into operate.
- The Independent Consultant shall report to MMRDA about their work, services, and activities of the this project.
- During the Construction Period, the Concessionaire shall be responsible for maintaining the Site including the existing lanes of the Project Highway at its own cost and expense.



- The Concessionaire shall modify, repair or otherwise make improvements to the existing lanes before expiry of six months from the Appointed Date so that the existing lanes comply with requirements provided in Schedule and At least 2 (two) lanes of the Project Highway are ordinarily open to traffic at all times.
- The Concessionaire shall bear all the expenses relating to Tests under this Agreement.
- The Project shall be deemed to be complete and open to traffic, only when the Completion Certificate or the Provisional Certificate is issued by MMRDA.
- If any dispute between the Parties then they can resolve the problem by two method **Amicable resolution and Arbitration resolution.**



FACILITY MANAGEMENT

It is a professional management discipline focused on the efficient and effective delivery to the road users like to ensure functionality, comfort, safety and efficiency of the built the environment

It includes health & safety, recruitment & training, empowerment, social housing & support, maintenance & technology



List of the facility management provided in project

- Ambulance and fire engine are provided near toll plaza
- For security purpose electronic surveillance are provided
- Welfare arrangements such as toilets and drinking water are installed



- Electrical poles and reflector are placed at regular interval
- Small plants are grown to absorb or minimize the pollution
- Proper sign board are provided



- Cleaning and repair of the road periodically.
- Proper drainage are provided.
- Barrier are provided at both side of the road and etc.
- Dustbin are provided at regular intervals.

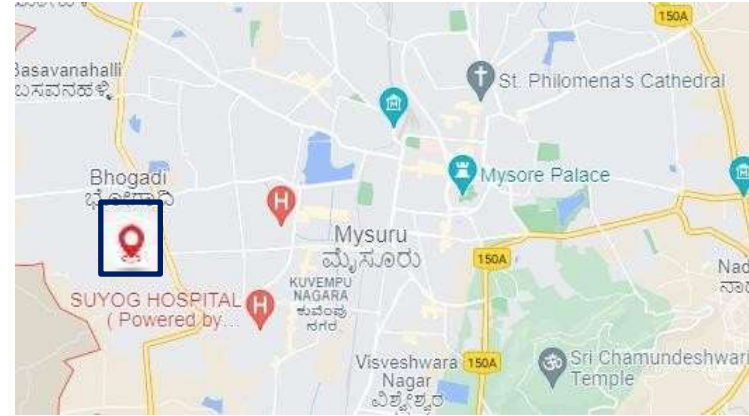


THE EMPIRE PROJECT MYSURU



PROJECT DETAILS

- Project Title : The Empire project Mysuru
- Project Location : Bogadi main road Mysuru Karnataka
- Project Type : Residential Building
- Site Details : 1. Plot area – 7097 Sq.m (76391.50 sq. Ft)
2. Total Built up area – 10050 Sq.m (108177.3 sq. Ft)
- No of floors : 2B + G + 9.
- Height of Building : 36.60m (120').
- Floor to floor height : 2.9m
- Total no of parking spaces : 25 nos.
- Approach road width : 6m (Including entry & exit)



Mysuru airport 18km



Mysuru Bus stand
6.8km



Bengaluru International airport
210km



Project Location

AMENITIES PROVIDED



Club House



Assigned parking spaces



Solar Panels



CCTV Surveillance



Security Guards

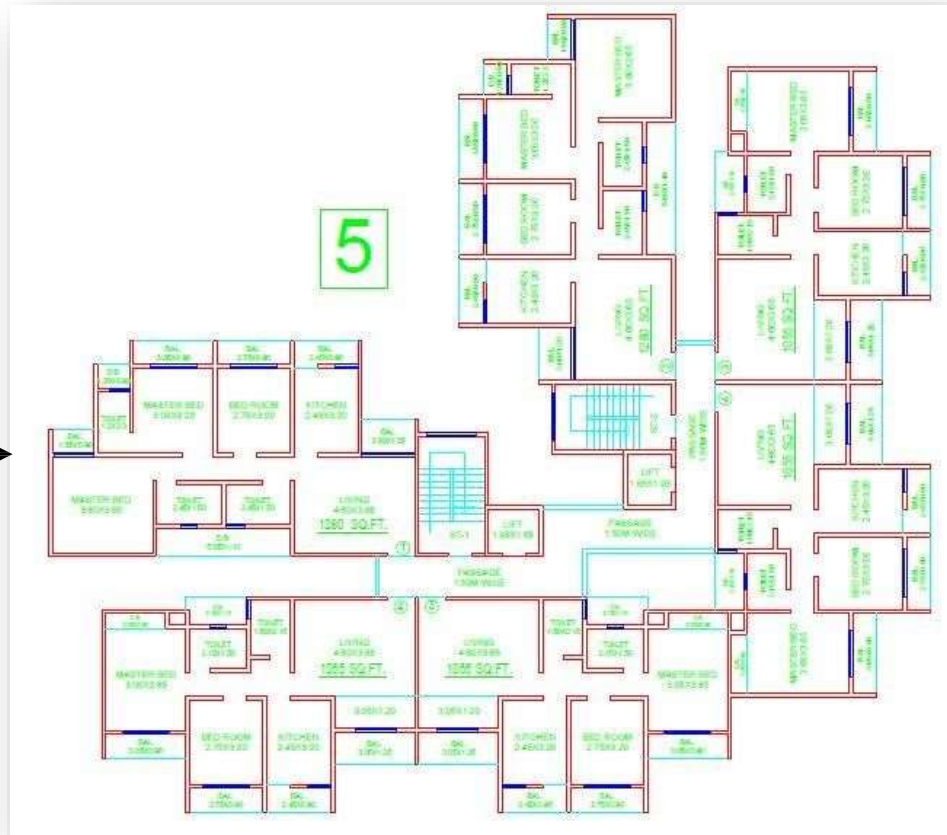


Swimming Pool

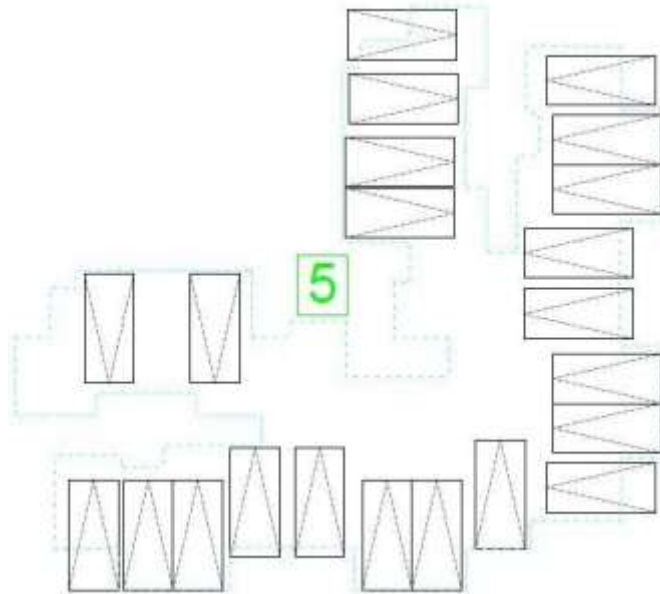


Garden

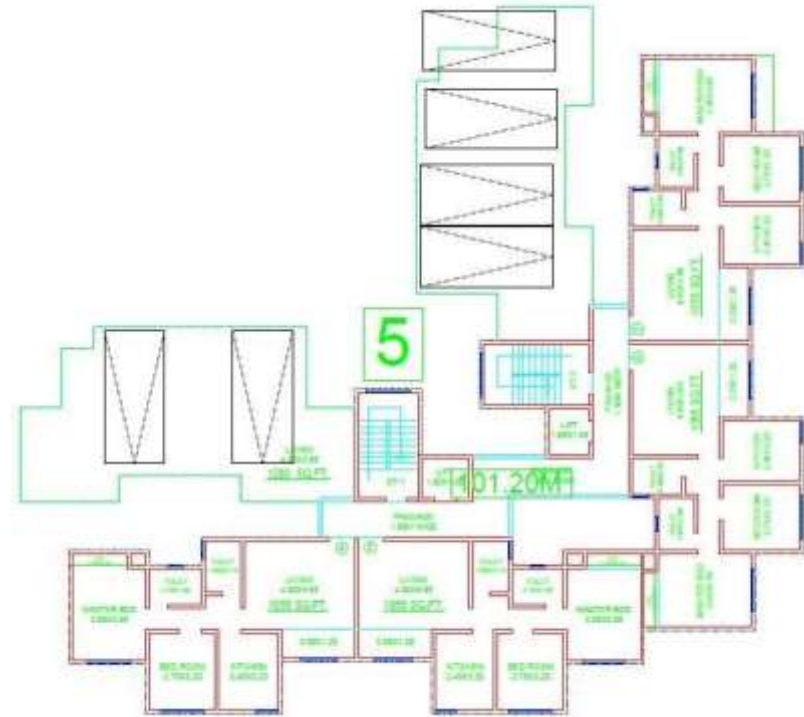
BUILDING LOCATION IN SITE LAYOUT



FLOOR PLANS



Basement Plan



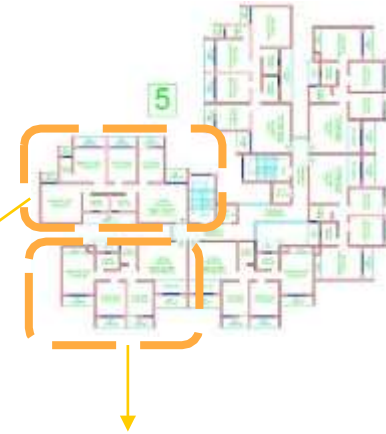
Ground Floor Plan

TYPICAL FLOOR PLAN (FIRST TO NINETH FLOOR)

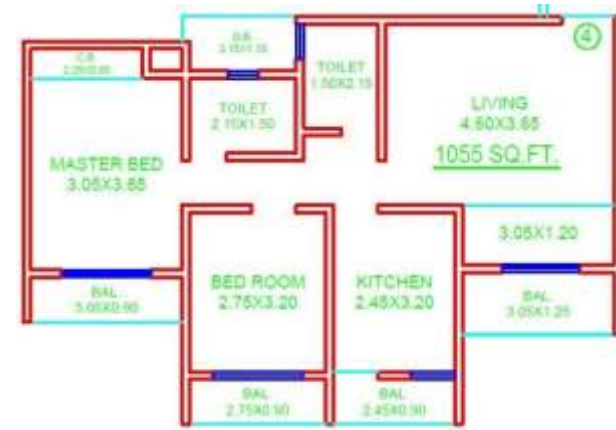


BUILDING DETAILS

- In Ground floor there are totally 4 flats each flat is of 1055 Sq. Ft of 2BHK.
- In Typical floors there are totally 6 flats. 4 flats are of 1055 Sq. Ft of 2BHK & 2 flats are of 1280 Sq. Ft consisting of 3BHK.
- There are 2 staircases & 2 lifts to access the buildings.



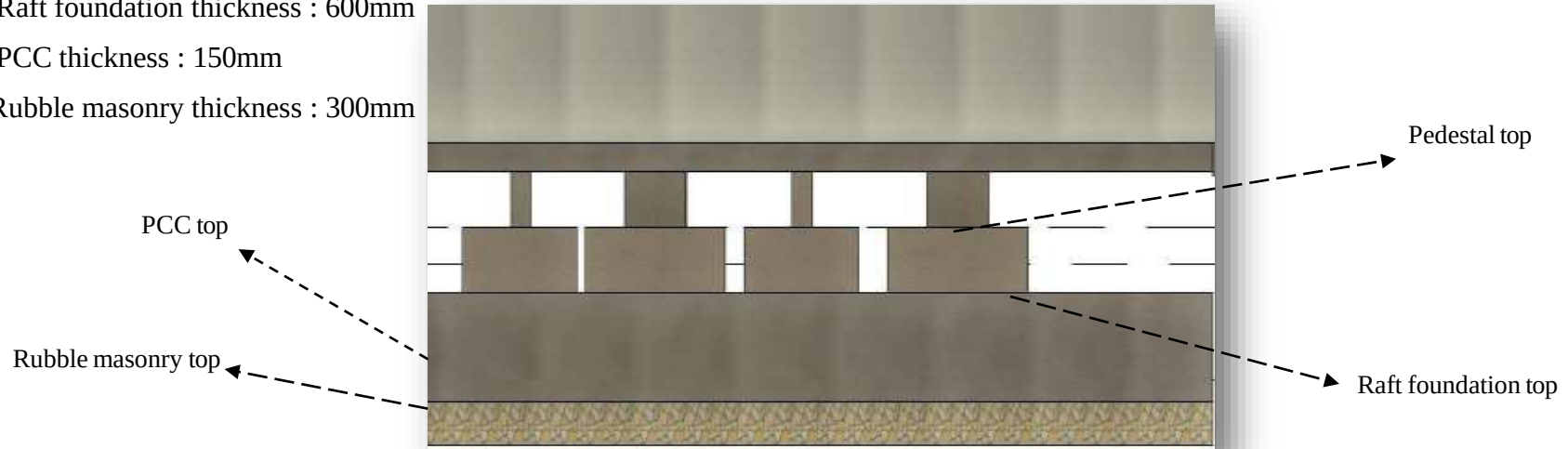
3BHK Flat



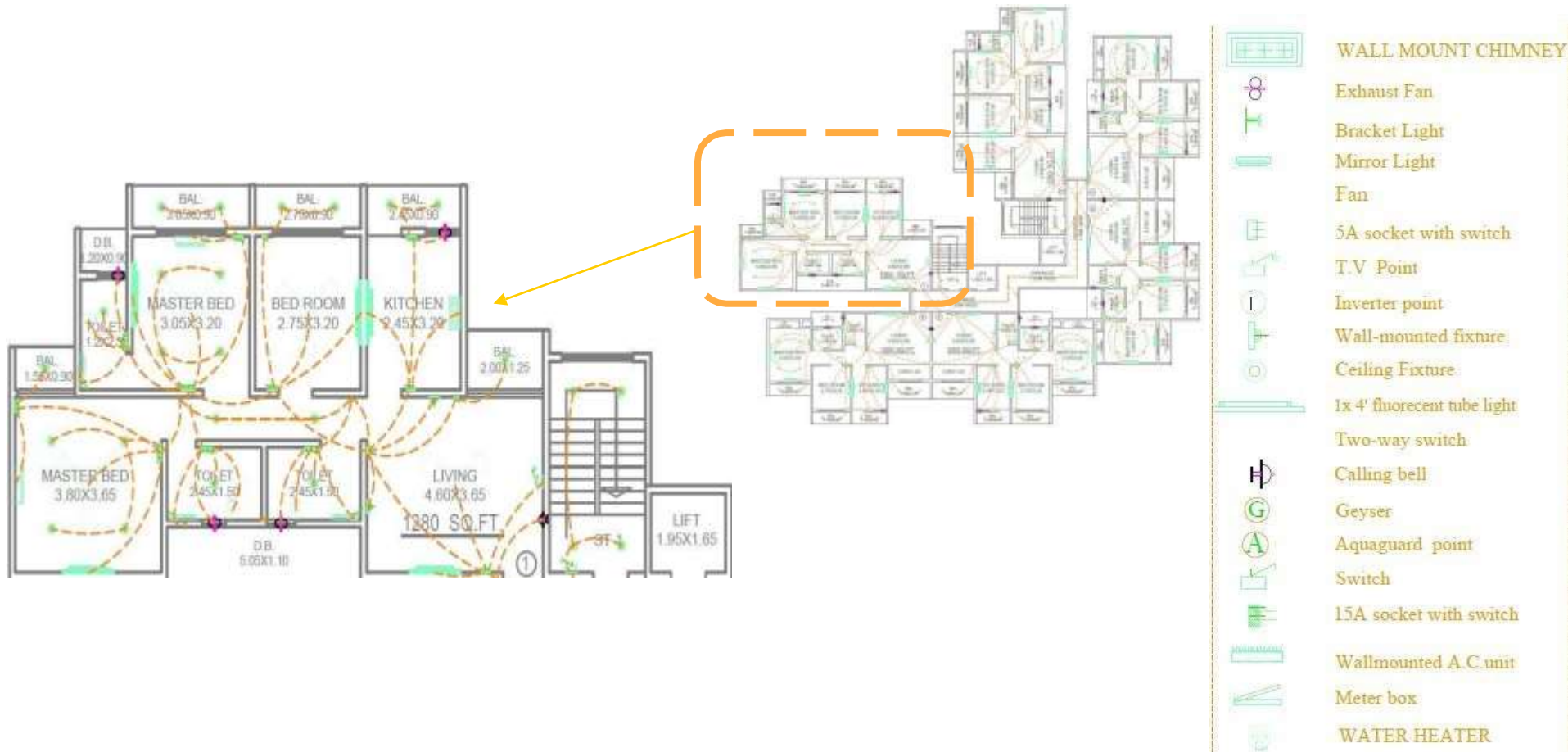
2BHK Flat

STRUCTURAL DETAILS

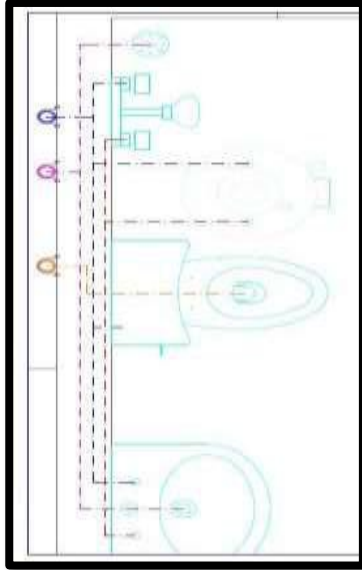
- Foundation type : Raft Foundation.
- Column size considered : 1. 230mm X 700mm
2. 230mm X 450mm
- Beam size considered : 230mm X 400mm
- Pedestal thickness : 450mm
- Raft foundation thickness : 600mm
- PCC thickness : 150mm
- Rubble masonry thickness : 300mm



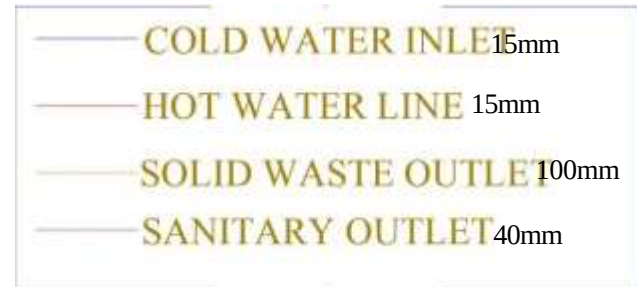
ELECTRICAL LAYOUT FOR 3BHK FLAT



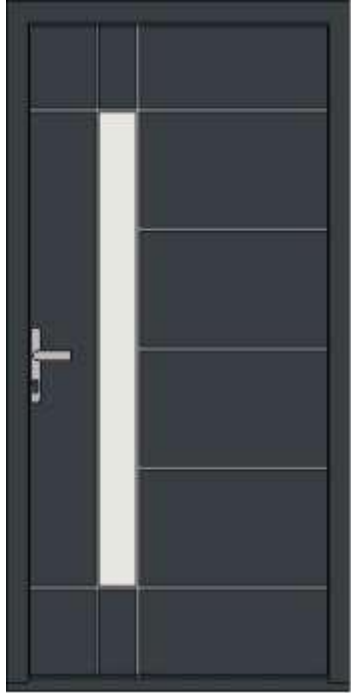
PLUMBING LAYOUT



1.2 x 2.3m washroom



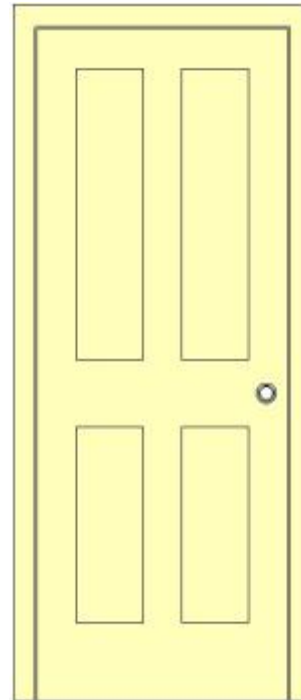
DOOR DETAILS



Type : Teak wood
Size : 1 x 2.1m
Usage : Entrance



Type : Wood
Size : 0.9 x 2.1m
Usage : Bed room

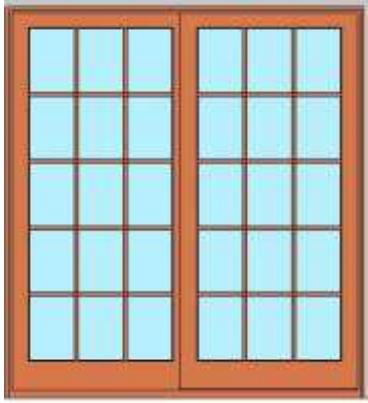


Type : 4 Panel wood
Size : 0.76 x 2.1m
Usage : Kitchen balcony



Type : Single flush door
Size : 0.76 x 2.1m
Usage : Bathroom

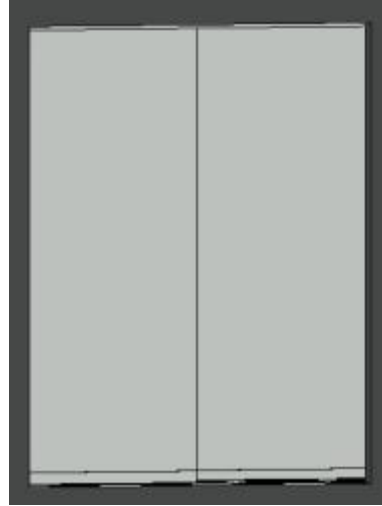
WINDOW DETAILS



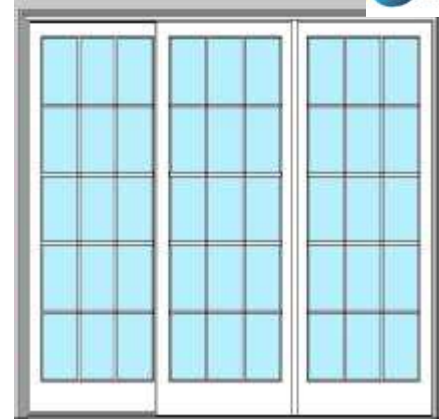
Type : 2 Panel French door
Size : 1.5 x 2.1m
Usage : Balcony



Type : Aluminum with louvers
Size : 0.6 x 0.8m
Usage : Bathroom



Type : Aluminum sliding
Size : 1 x 1.2m
Usage : Bed room



Type : 3 Panel French door
Size : 2.0 x 2.1m
Usage : Balcony



Type : Aluminum fixed
Size : 1.5 x 1.6m
Usage : Staircase

OHT CALCULATIONS

- Total people for entire building = 268
- Considering 135 LPCD as standard from UDPFI guidelines
- Water required $268 \times 135 = \mathbf{36180 \text{ liters}}$
- Considering 1 day storage = $1 \times 36180 = 36180 \text{ liters} = 36.18 \text{ cum}$
- One water tank capacity = 10000 liters
- No of water tank required = $36180/10000 = 3.6 \sim \mathbf{4 \text{ no's}}$

SUMP TANK CALCULATIONS

- Total daily water consumption = **36180 liters**
- Considering the capacity of sump tank as **3 times** the consumption
- Capacity of sump tank required = **100000 liters**
- Dimensions considered = **10m x 5m x 2m**



DIMENSIONS :

- Dia of tank : 2.0m
- Height of tank : 3.65m
- Man hole size : 0.51m

3D VIEWS FROM REVIT



3D RENDERED IMAGES



















BOQ SUMMARY

MYSORE PROJECT G+9					
NAME OF ITEM	QUANTITY	UNIT	RATE	PER	AMOUNT
SITE CLEARANCE	12220	Sqft	₹ 7.40	Sqft	₹ 90,428.00
SURVEYING	1	One day	₹ 5,000.00	One day	₹ 5,000.00
EXCAVATION	5058.08	cum	₹ 412.32	cum	₹ 2,088,582.39
SUFFICIENT WATER AND LIGHTING ARRANGEMENT	1	Lumpsum	₹ 200,000.00	lumpsum	₹ 200,000.00
SOILING	126.452	cum	₹ 1,550.00	cum	₹ 196,000.80
VIBRATING ROLLER COMPACTION	2	per day	₹ 8,000.00	per day	₹ 16,000.00
ANTI-TERMITE TREATMENT	632	sqm	₹ 256.13	sqm	₹ 161,874.16
CENTERING AND SHUTTERING	1385	sqm	₹ 307.35	sqm	₹ 426,510.75
TOTAL					₹ 3,184,395.90
CONCRETE WORK					
PCC WORK (M - 15)	126.452	cum	₹ 3,400.00	cum	₹ 429,936.80
FOOTING CONCRETE WORK (M-40)	1334	cum	₹ 6,000.00	cum	₹ 8,004,000.00
CONCRETE - M30 (FLOOR SLAB)	2636	cum	₹ 5,000.00	cum	₹ 13,180,000.00
CONCRETE - M35 (STAIRCASE)	513	cum	₹ 5,500.00	cum	₹ 2,821,500.00
CONCRETE M35 (COLUMN)	344	cum	₹ 5,500.00	cum	₹ 1,892,000.00
TOTAL					₹ 26,327,436.80
STEEL QUANTITY					
STEEL QUANTITY	159632	MT	₹ 76.00	MT	₹ 12,136,532.00
TOTAL					₹ 12,136,532.00
BRICK WORK					
EXTERNAL PLASTERING WORK	76254	sqft	₹ 115.00	sqft	₹ 8,769,210.00
INTERNAL PLASTERING WORK	68254	sqft	₹ 115.00	sqft	₹ 7,849,210.00
EXTERNAL PAINTING WORK	76254	sqft	₹ 53.50	sqft	₹ 4,079,589.00
INTERNAL PAINTING WORK	68254	sqft	₹ 40.00	sqft	₹ 2,730,160.00
PUTTY	144508	sqft	₹ 40.00	sqft	₹ 5,780,320.00
WATERPROOFING WORK	2564	sqft	₹ 200.00	sqft	₹ 512,800.00
FALSE CEILING WORK	28631	sqft	₹ 120.00	sqft	₹ 3,435,720.00
BRICK WORK IN SUPER STRUCTURE	2033.38	cum	₹ 5,470.00	cum	₹ 11,122,588.60
TOTAL					₹ 44,038,597.60

TILES WORK					
TILE WORK (VITRIFIED TILES)+LAYING COST	60489.44	sqft	₹ 70.00	sqft	₹ 4,234,260.80
BATHROOM TILES ANTI SKID (300x300mm)	480.56	sqft	₹ 45.00	sqft	₹ 21,625.20
GRANITE TILES FOR STAIRCASE	3398.43	sqft	₹ 400.00	sqft	₹ 1,359,372.00
TOTAL					₹ 5,615,258.00
DOOR WORK					
WOOD WORK (MAIN DOOR) 1x2.1m	72	no.s	₹ 9,000.00	no.s	₹ 648,000.00
WOOD WORK (BEDROOM) 0.9x2.1m	840	no.s	₹ 6,500.00	no.s	₹ 5,460,000.00
WOOD WORK (TOILET) 0.76x2.1m	140	no.s	₹ 4,400.00	no.s	₹ 616,000.00
4 PANEL WOOD FOR BALCONY AND KITCHEN	80	sqft	₹ 340.00	sqft	₹ 27,200.00
2 PANEL FRENCH DOOR 1.5X2.1m FOR BALCONY	1525.23	sqft	₹ 500.00	sqft	₹ 762,615.00
3 PANEL FRENCH DOOR 1.5X2.1M FOR BALCONY	2203.11	sqft	₹ 650.00	sqft	₹ 1,432,021.50
TOTAL					₹ 8,945,836.50
WINDOW WORK					
ALLUMINIUM FIXED WINDOWS OF SIZE 1.5X1.6m in staircase	2324.16	sqft	₹ 180.00	sqft	₹ 418,348.80
ALLUMINIUM WITH LOUVERS FOR BATHROOM 0.6X0.8m	650.76	sqft	₹ 400.00	sqft	₹ 260,304.00
ALUMINIUM SLIDING WINDOW	64585	sqft	₹ 500.00	sqft	₹ 32,292,500.00
RAILING	7170	ft	₹ 1,000.00	per ft	₹ 7,170,000.00
TOTAL					₹ 40,141,152.80

BOQ SUMMARY

PLUMBING ITEMS				
Description	Qty. (m)	Unit	Rate	Amount
Providing and supplying in standard lengths ISI mark rigid				
100mm(Working Pressure 6 Kg./Sq.cm)	1652	mmt	₹ 200.00	₹ 330,400.00
50mm(Working Pressure 6 Kg./Sq.cm)	1645	mmt	₹ 69.00	₹ 113,505.00
40mm(Working Pressure 6 Kg./Sq.cm)	1025	mmt	₹ 46.00	₹ 47,150.00
20mm(Working Pressure 6 Kg./Sq.cm)	523	mmt	₹ 52.00	₹ 27,196.00
15mm(Working Pressure 4 Kg./Sq.cm)	425	mmt	₹ 39.00	₹ 16,575.00
Providing GI Pipes for fire extinguishing	4536	mmt	₹ 144.00	₹ 653,184.00
Ceramic Floor Mounted European Water Closet/Western Toilet Commode/EVC S Trap with Normal Seat Cover- White & Premium Normal Flush Flush Tank Combo with Jet Spray	144	per piece	₹ 9,000.00	₹ 1,296,000.00
Floor Trap	144	per piece	₹ 150.00	₹ 21,600.00
White Ceramic Thin Rim Pedestal Wash Basin with Tap	144	per piece	₹ 10,450.00	₹ 1,504,800.00
ALTON LEO10050 Brass 3" High Flow Diverter Full Set With Spout And 5-Function Overhead Shower, Chrome Finish	152	per piece	₹ 5,739.00	₹ 861,448.00
Flush valve	144	per piece	₹ 350.00	₹ 50,400.00
Jaguar Nickel Bib Cock with Wall Flange Tap	144	per piece	₹ 1,300.00	₹ 187,200.00
Bent PVC pipes 15mm dia	137	each	₹ 18.00	₹ 2,466.00
bent PVC pipes 20mm	30	each	₹ 20.00	₹ 600.00
bent PVC pipes 40mm	57	each	₹ 32.00	₹ 1,824.00
bent PVC pipes 50mm	47	each	₹ 35.00	₹ 1,645.00
bent PVC pipes 100mm	129	each	₹ 41.00	₹ 5,289.00
T shaped plumbing pvc pipes 15mm	78	each	₹ 52.00	₹ 4,056.00
T shaped plumbing pvc pipes 20mm	32	each	₹ 60.00	₹ 1,920.00
T shaped plumbing pvc pipes 40mm	24	each	₹ 72.00	₹ 1,728.00
T shaped plumbing pvc pipes 50mm	77	each	₹ 85.00	₹ 6,545.00
T shaped plumbing pvc pipes 100mm	41	each	₹ 90.00	₹ 3,690.00
Total				₹ 5,159,221.00

ELECTRICAL FITTINGS AND LIGHTING FIXTURES				
DESCRIPTION ITEM	UNIT	NO.S	RATE	TOTAL RATE
Ceiling fan, 1400 mm sweep, decorative type, suitable for single phase, 230 V, 50Hz, A.C. supply operation complete with accessories such as 15" long down rod, canopy, electronic stop type fan regulator (similar to Anchor Roms), interconnecting wiring using 2/0 core X 0.75 sq.mm PVC/PVC flexible copper conductor wire etc.	EACH	200	₹ 3,300.00	₹ 660,000.00
Exhaust fan, domestic type, 300 mm (12") dia, 1300 RPM (approx) suitable for single phase, 230 V, 50Hz, A.C. supply operation complete with sturdy metal body and blades, metal canopy, interconnecting wiring/cord and necessary mounting arrangement using 10mm (approx) thick marine plywood / wooden plates as required.	EACH	214	₹ 1,540.00	₹ 329,560.00
Fluorescent tube lighting, compact fluorescent (CFL) tube, electronic type (200 mm) approx) long, ceiling / wall mounting with 2 X 36W/4 W fluorescent tubes pre-wired with electronic ballasts, connector, side holders and interconnecting wiring by using 2/0 core X 0.75 sq.mm. PVC/PVC flexible copper conductor wire including mounting arrangement viz standard down rods, ball & sockets, etc as required (includes "Transformer")	EACH	1000	₹ 700.00	₹ 700,000.00
(A) 1 way switch, 16/20 A, 230V	EACH	6590	₹ 82.00	₹ 539,000.00
(B) 2 way switch, 16/20 A, 230V	EACH	185	₹ 85.00	₹ 15,825.00
(C) Universal socket, 6pin, 10/20A, 230V	EACH	2645	₹ 75.00	₹ 198,375.00
(D) Universal switch socket, 6pin, 10/20A, 230V (combined)	EACH	1952	₹ 128.00	₹ 250,000.00
(E) D.P. Switch, surface type, ballistic modified case having porcelain body, reversible face with screw indicator, 32A, 230V, A.C. supply with ISI mark	EACH	145	₹ 700.00	₹ 101,500.00
15 Ton Air conditioning SYSTEM	EACH	220	₹ 35,000.00	₹ 7,700,000.00
CEILING LIGHT	EACH	1645	₹ 650.00	₹ 1,069,250.00
Lift	EACH	2	₹ 2,400,000.00	₹ 4,800,000.00
Transformer (10 KVA TO 4500 KVA)	EACH	2	₹ 480,000.00	₹ 960,000.00
Generator (250 KVA)	EACH	2	₹ 1,384,000.00	₹ 2,768,000.00
Electrical conduit	mtr	10254	₹ 66.00	₹ 676,764.00
Water Purifier (SL)	EACH	210	₹ 10,000.00	₹ 2,100,000.00
Solar panels	EACH	8	₹ 25,000.00	₹ 200,000.00
Pump of SHP	EACH	10	₹ 16,000.00	₹ 160,000.00
Street Light HAVELLS 100W	EACH	25	₹ 5,500.00	₹ 1,37,500.00
CCTV	EACH	265	₹ 2,000.00	₹ 530,000.00
TOTAL				₹ 25,816,730.00

BOQ SUMMARY

Miselleneous work					
Landscaping considers 0.6% of the total cost	I	Lumpsum	₹ 2,145,162.00	Lumpsum	2,145,162.00
Supplying And fixing Fire Protection System With All Assossaries,	I	R/O	₹ 5,500,000.00		5,500,000.00
					7,645,162.00
Cleaning work					
cleaning Of All the Floors As well As basements From inside As Well AS	100	Labours	₹ 500.00		₹ 50,000.00
Grand Total					₹ 179,121,382.60
CONSULTANT CHARGES	Unit Rate				ESTIMATED COST
MEP CONSULTANT	5.00%				₹ 8,956,069.13
PROJECT MANAGEMENT CONSULTANT	5.00%				₹ 8,956,069.13
MISCALLENIOUS	1.00%				₹ 1,791,213.83
TOTAL					₹ 19,703,352.09
TOTAL COST EXCLUDING CONTINGENCY					₹ 198,824,734.69
TOTAL COST INCLUDING CONTINGENCY	5%				₹ 208,765,971.42
TAX(GST)	12%				₹ 233,817,888.00
GRAND TOTAL COST					₹ 233,817,888.00

SKYLOUNGE MULUND ELITE



SR No	CONTENT	SR No.	CONTENT
1	Project details & Location	9	ROI Study of the Project
2	BEP Document	10	MS project Gantt chart
3	Amenities of Flat	11	Power BI reports
4	Views of Building	12	6D Analysis
5	Rendered Images	13	Tender & Contract Documents
6	Plumbing Details	14	Project Charter
7	Electrical Details	15	Rendered Video
8	4D & 5D Simulation		



PROJECT INFORMATION

Project Location :	Mumbai, Mulund, Maharashtra
Site Area	39683.74sqft.
Total Built-up-area	50323.42sq.m
No.of floors	42
Height of Building	145.33m
Floor to floor Height	3.36m
Width of Building	22.86m
Length of Building	77.01m
Total Carpet area	32000sq.m



The ELITE Mulund

BIM EXECUTION PLAN (BEP) AGREEMENT

By signature below, this BIM Execution Plan, dated 2022-10-23 is herewith approved and will be incorporated as an amendment to the Design Professional Agreement, as a change order to the Contractor Agreement, and as an amendment to other separate consulting and commissioning agent agreements as they may apply to the list of parties co-signing this document.

The BEP shall be updated and amended at each major project phase deliverable, as new key parties or entities are brought on board the project and incorporated into the BIM Execution Plan.

Owner Representatives:

Director

Add. Director : Shaishita Afreen

Project Details:

Sr. No.	Title	Details
1)	Project Name	G+42 SKY LAUNGE & DUPLEX
2)	Project Owner	E-Construct
3)	Project Location	Elite Mulund, Mumbai

Project Team Contacts:

Roles	Name	Company	Phone No.	E-Mail ID
BIM Coordinator	Milan M C	E-Construct	8770019030	madikerimilan17@gmail.com
BIM Modeler	Kalpesh Ahire	E-Construct	9557993292	kalpeshahire1999@gmail.com
MEP & Electrical Engineer	Mujahid Sakbarkar	E-Construct	9073139247	mujahid@gmail.com
Quantity Surveyor	Aklima Khattun	E-Construct	8770019030	aklima@gmail.com

PROJECT INFORMATION

Project Name:	G+42 SKY LAUNGE & DUPLEX
Project Number:	0001
Project Address:	Elite Mulund, Mumbai
Project Description:	RESIDENTIAL

PROJECT PHASES / MILESTONES

Project Phase / Milestone	Estimated Start Date	Estimated Completion Date	Project Stakeholders Involved
Programming/ Pre-Design Phase	19/05/2022	04/06/2022	Owner, Design Professional, Consulting engineers
Schematic Design Phase	05/06/2022	31/08/2022	Owner, Design Professional, Consulting engineers, Contractor
Preliminary Design Phase	19/09/2022	27/09/2022	Owner, Design Professional, Consulting engineers, Contractor, Commissioning agent
Construction Documents Phase (Refer to schedule for early bid packages)	01/10/2022	30/10/2022	Owner, Design Professional, Consulting engineers, Contractor, Commissioning agent



AMENITIES



GYM



BASKET BALL COURT



SOLAR PANELS



PARKING AREA



SWIMMING POOL



CHILDRENS
PLAY AREA



COMMUNITY HALL



JACUZZI

Typical Plan View



Room Tag Typical Plan View

By Room Name Legend

Balcony	Dressing	lift	Passage
balcony	duct	Living Room 1	passage
balcony 2	Kitchen	Living Room 2	Room
Bedroom	kitchen	Lobby	smoke vent
Bedroom 2	Kitchen 2	Master bedroom	staircase
dinning	kitchen 2	Master Bedroom 1	wc

wc2
wc 2





Typical 2BHK Interior work



Sky lounge area

Duplex floors

- Distance between OHT Tank and Terrace is 2.90m
- Distance between service floor 1.80m
- Distance between two floor is 3.36m

3 Level podium parking

3D Elevation view



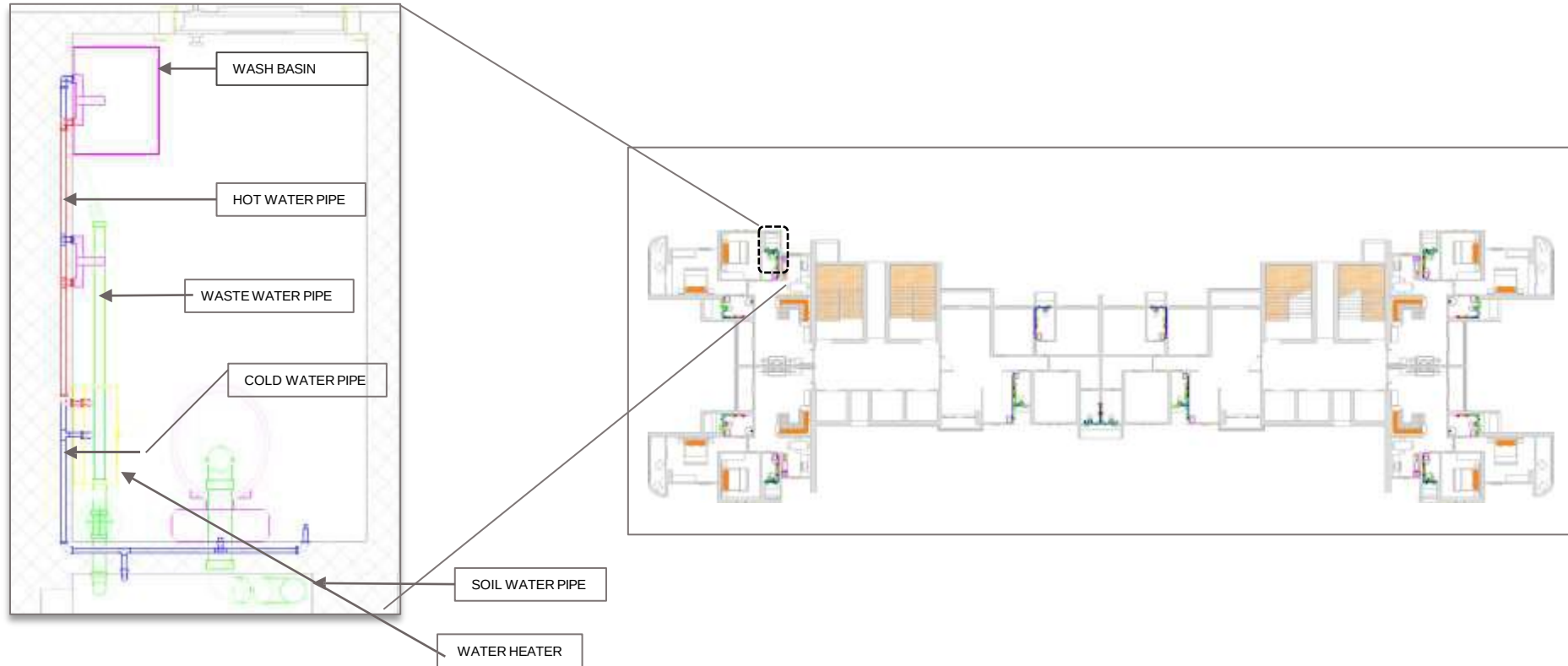




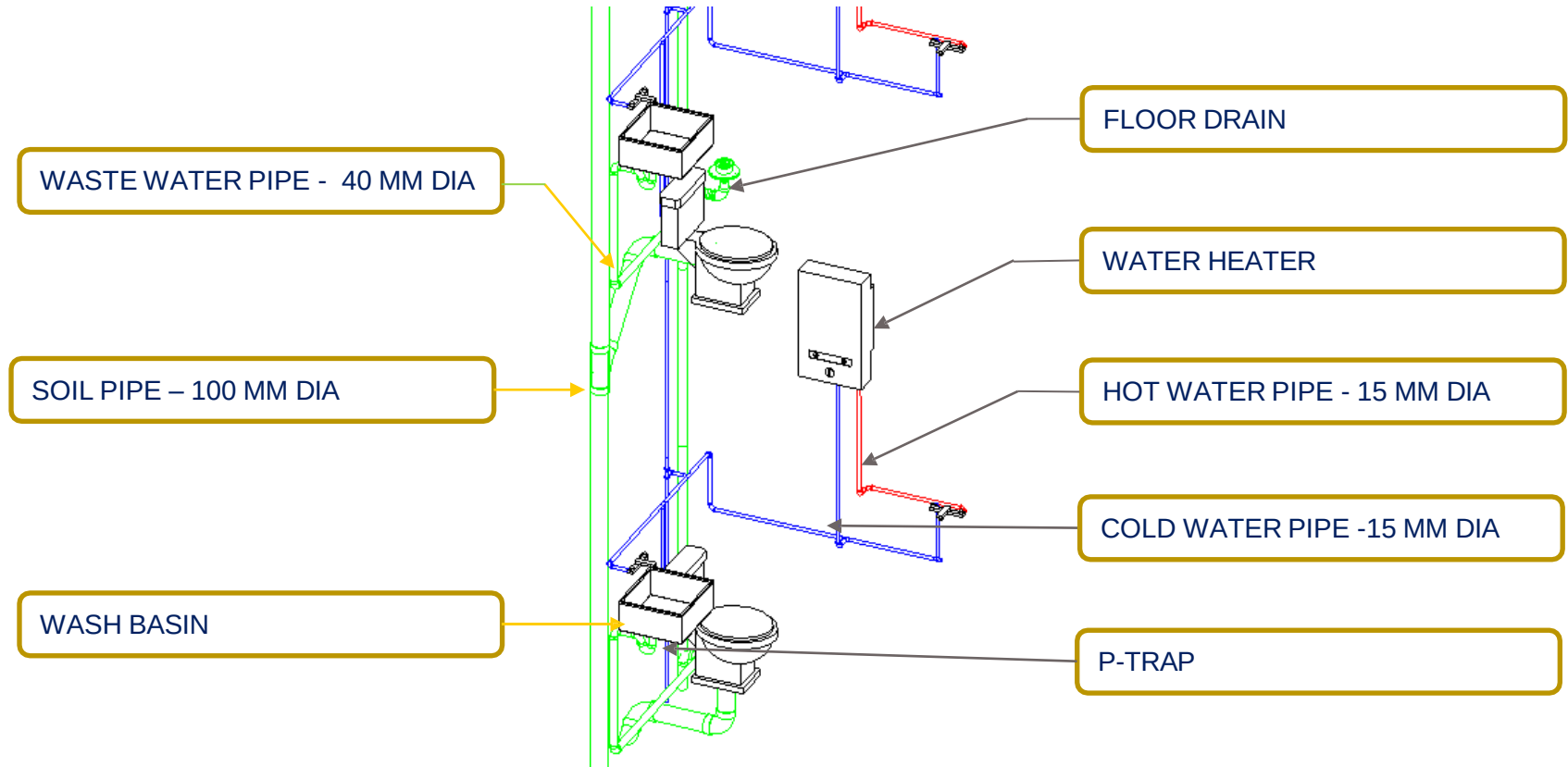




2D PLUMBING DESIGN



2D PLUMBING DESIGN



UNDERGROUND WATER TANK CALCULATION

Total no. of persons = 988

Considering 135 LPCD Water
consumption for residential building =
133380 L

Considering maximum storage for 3 days
= **400140 L**

SLOPE

Waste water pipe = 1:100 (1%)

Soil pipe = 1:100 (1%)

PIPE USED

PVC

PLUMBING FITTINGS AND FIXTURES



P-TRAP



TAP (STAINLESS STEEL)



WATER HEATER



PVC PIPES



FLOOR DRAIN



Y-ELBOW



ELBOW JOINT



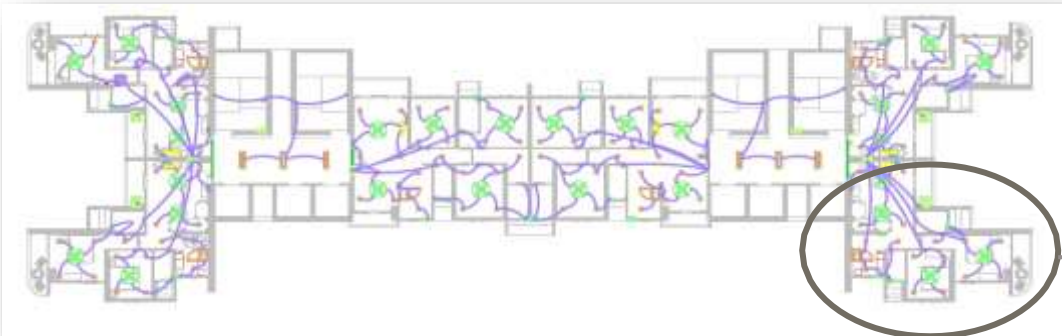
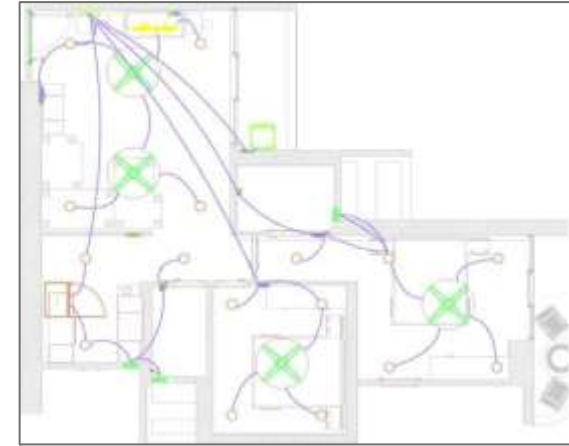
WASH BASIN



T-JOINT

ELECTRICAL LAYOUT- TYPICAL FLOOR (2BHK)

ELECTRICAL DEVICES	NUMBER
ROUND CEILING LIGHT	110
LINEAR BOX CEILING LIGHT	6
WACHING MACHINE	4
T.V UNIT	6
LED BULB	24
REFRIGERATOR	6
EXHAUST FAN	20



Electrical Load Calculation

Total Load = 440.820 KW

Considering 20% extra load = $440.820 + 88.164 = 529 \text{ KW}$

Transformer Load Calculation

Maximum Demand

= Total Load X Diversity Factor = $529 \times 0.6 = 317 \text{ KW}$

Load requirement in KVA

= Expected demand / Power factor = $317 / 0.9 = 352 \text{ KVA}$

Considering 80% Loading Efficiency = $352 / 0.8 = 440 \text{ KVA}$

ELECTRICAL DEVICES AND FIXTURES



CEILING LIGHT (ROUND)

LOAD =10 W



T.V. UNIT

LOAD =60 W



SWITCHES



CEILING LIGHT (LINEAR)

LOAD =20 W



CEILING LIGHT (ROUND)

LOAD =70 W



EXHAUST FAN

LOAD =40 W



RECEPTACLES



LED BULB

LOAD =10 W



REFRIGERATOR

LOAD =160 W

4D CLASH DETECTION

CLASH REPORT BETWEEN ARCHITECTURE MODEL AND PLUMBING MODEL

Image	Clash Name	Status	Distance	Description	Date Found	Clash Point	Item 1				Item 2			
							Item ID	Layer	Item Name	Item Type	Item ID	Layer	Item Name	Item Type
	Clash1	New	-0.227	Hard	2022/5/25 19:42	x:-32.586, y:5.330, z:59.873	Element ID: 712084	20th floor	Metal Deck	Solid	Element ID: 1805457	21th floor	Polyvinyl Chloride - Rigid	Solid
	Clash2	New	-0.227	Hard	2022/5/25 19:42	x:-32.586, y:5.330, z:59.873	Element ID: 945243	20th floor	Metal Deck	Solid	Element ID: 1805457	21th floor	Polyvinyl Chloride - Rigid	Solid
	Clash3	New	-0.220	Hard	2022/5/25 19:42	x:-28.790, y:-10.253, z:125.513	Element ID: 831897	40 th floor	Metal Deck	Solid	Element ID: 1840046	39th floor	Polyvinyl Chloride - Rigid	Solid
	Clash4	New	-0.204	Hard	2022/5/25 19:42	x:28.352, y:-10.199, z:122.364	Element ID: 833519	39th floor	Concrete, Cast-in-Place - C15	Solid	Element ID: 1839152	39th floor	Polyvinyl Chloride - Rigid	Solid
	Clash5	New	-0.199	Hard	2022/5/25 19:42	x:-28.788, y:-10.260, z:130.673	Element ID: 900622	42th LDuplex	Metal Deck	Solid	Element ID: 1840046	39th floor	Polyvinyl Chloride - Rigid	Solid

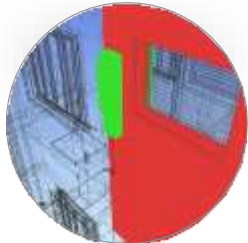
TOTAL CLASHES
=12668

TOLERANCE
LIMIT=0.001 M

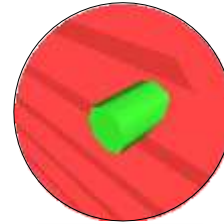
DETECTED CLASHES



Clash between PVC pipe & Metal Deck



Clash Between Shear wall and Water Heater



Clash Between glass and PVC pipe



Clash Between Concrete floor & PVC pipe

After doing multiple iterations the clashes has been rectified and reduced to zero.

SR.No	NAME OF WORK	COST (.)
1.	Site cleaning & preliminary works	5,58,67,684/-
2.	Concreting work	24,56,54,119/-
3.	Fe-500 TMT Rebar	23,03,78,288/-
4.	Mason Work	5,77,22,784/-
5.	Tile Work	9,65,17,180/-
6.	Door and Window works	4,68,75,488/-
7.	Plastering work	12,35,09,736/-
8.	False ceiling	6,26,36,112/-
9.	Painting work	13,1743,718/-
10.	Plumbing and Electric work	10,51,52,742/-
11.	Water proofing work	6,22,315/-
12.	Landscape work	6,30,91,645/-
	Total Cost (A)	122,33,71,813/-

SR.No	ADDITIONAL CHARGES	COST (Rs.)
1.	Add 5% Overhead Charges	6,16,96,854/-
2.	Add 5% General Consultant charge	6,16,96,854/-
3.	Add 5% PM charges	6,16,96,854/-
4.	Add 1% Water Charges	1,05,15,274/-
5.	Contractors Profit 10%	12,33,93,708/-
6.	Add 5% Contingency Charges	7,71,21,067/-
7.	Adding 18% GST	29,15,17,637/-
	Total Cost (B)	59,04,65,702/-
	Total Cost (A)+(B)	RS.191,10,60,064/-
	Total land Cost	191,10,60,064/-

Total cost including land and Construction Cost =

382,21,20,128/-



Microsoft Excel
Worksheet

REVENUE STUDY FOR TYPICAL FLOOR

Type of floors	Area Sqft.	No of flats	No. of stories	Rate per Sqft.	Total Cost
Typical floors					
2 BHK	881.13	4	24	20123/-	170,21,73,983/-
3 BHK	1277.212	2	24	24256/-	148,70,42,605/-
Total					318,92,16,588/-

REVENUE STUDY OF REFUGE AREA

Type of floors	Area Sqft.	No of flats	No. of stories	Rate per Sqft.	Total Cost
Refuge area					
2 BHK	881.13	1	4	20,123/-	7,09,23,915/-
	976.04	1	4	20,123/-	7,85,63,411/-
3 BHK	1367.81	1	4	24,256/-	13,27,10,397/-
	1277.21	1	4	24,256/-	12,39,20,023/-
Total					40,61,17,746/-

REVENUE STUDY FOR 33RD AND 34TH FLOOR

Type of floor	Area (Sqft.)	No. of flats	No. of stories	Rate per Sqft.	Total Cost
33 rd & 34 th floor					
4 BHK	1642.62	1	1	37,580/-	6,17,29,659/-
5BHK	1861.91	1	1	40,153/-	7,47,61,272/-
Double storey	1751.83	1	1	41,582/-	7,28,44,595/-
	3934.93	2	1	42,285/-	33,27,77,030/-
Total					64,21,12,556/-

REVENUE STUDY FOR 35,37 AND 39TH FLOOR

Type of floors	Area Sqft.	No of flats	No. of stories	Rate per Sqft.	Total Cost
35,37 & 39 th floor					
3 BHK	1952.07	2	3	47,869/-	56,06,61,833/-
5 BHK	1277.21	2	3	41,590/-	31,87,14,983/-
Total					87,93,76,816/-

REVENUE STUDY FOR 36th AND 41-42 DULEX FLOOR

Type of floor	Area (Sqft.)	No. of flats	No. of stories	Rate per Sqft.	Total Cost
36th & 41-42 duplex floor					
3 BHK	976.03	2	1	25,685/-	5,01,38,661/-
	1277.21	2	1	32,358/-	8,26,55,922/-
Double storey	1277.21	4	1	32,450/-	16,57,81,858/-
	5807.27	2	1	52,485/-	60,95,89,131/-
Total					90,81,65,572/-

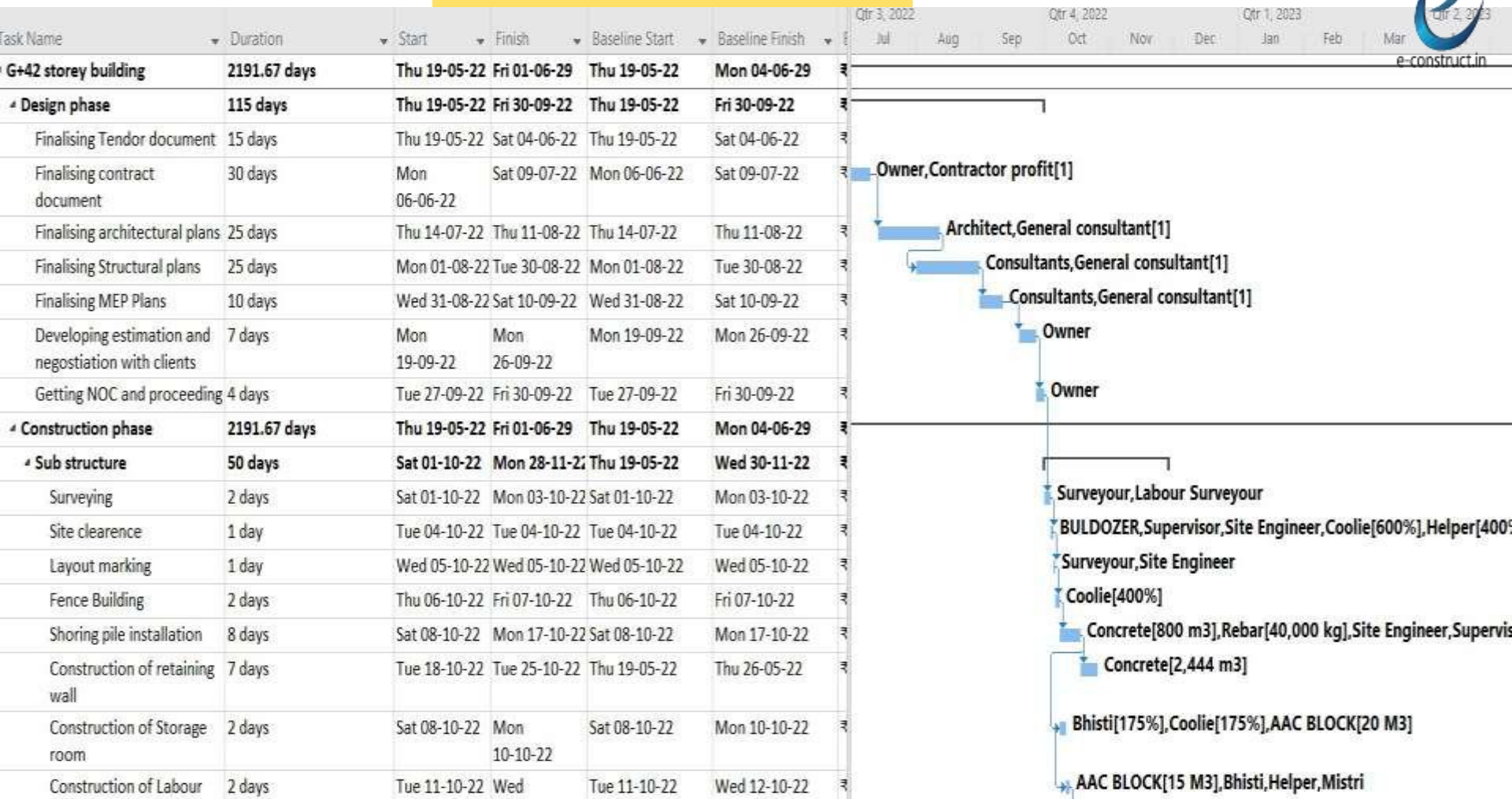
Total Revenue generted

Total Cost of Investment in G+42 Storey	382,21,20,128/-
Total revenue generation from all the flats	611,34,35,518/-
Total profit owner earned	229,13,15,390/-



Microsoft Excel
Worksheet

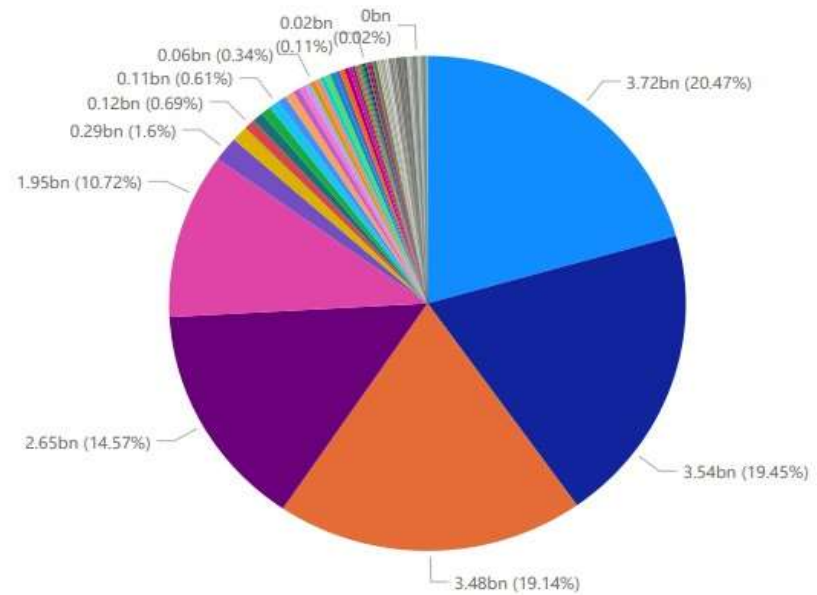
MS PROJECT GANTT CHART



Name	Cost
G+42 storey building	3,72,06,66,880.12
Construction phase	3,53,55,76,317.35
Super Structure	3,47,96,23,320.42
OUTER MISCELLANEOUS WORKS	2,64,90,25,903.09
Land price	1,94,92,81,266.00
GST of 18%	29,15,14,637.00
Design phase	18,50,90,562.77
Internal plastering work	12,49,80,090.00
Finalising contract document	12,33,93,708.77
Architectural works	11,37,41,044.00
Tile Work	11,15,65,553.68
Concreting for slab and beam	10,23,44,979.00
PAINTING WORK	9,99,84,072.00
Contingency charges	7,71,21,067.00
Reinforcement for Shear wall and Column	6,97,49,640.00
Reinforcement for slab and beam	6,46,87,883.15
False ceiling	6,26,36,160.00
OUTER PLASTERING WORK	6,17,54,868.00
Over head charges	6,16,96,854.38
Project Management Consultant	6,16,96,854.38
Casting of Shear wall and Column	5,97,62,797.00
Sub structure	5,59,52,996.93
Landscape work	5,50,29,064.33
PAINTING WORK ON OUTER AREA	4,94,03,894.40
outer road work	4,00,00,000.00
outer boundry wall	3,00,00,000.00
Casting of Raft footing	2,67,15,838.50
Finalising architectural plans	2,05,65,618.00
Finalising MEP Plans	2,05,65,618.00
Finalising Structural plans	2,05,65,618.00
Reinforcement for staircase upto landing	2,05,26,404.00
Swimming pool work	2,00,00,000.00
Total	18,17,72,00,278.13

Cost by Name

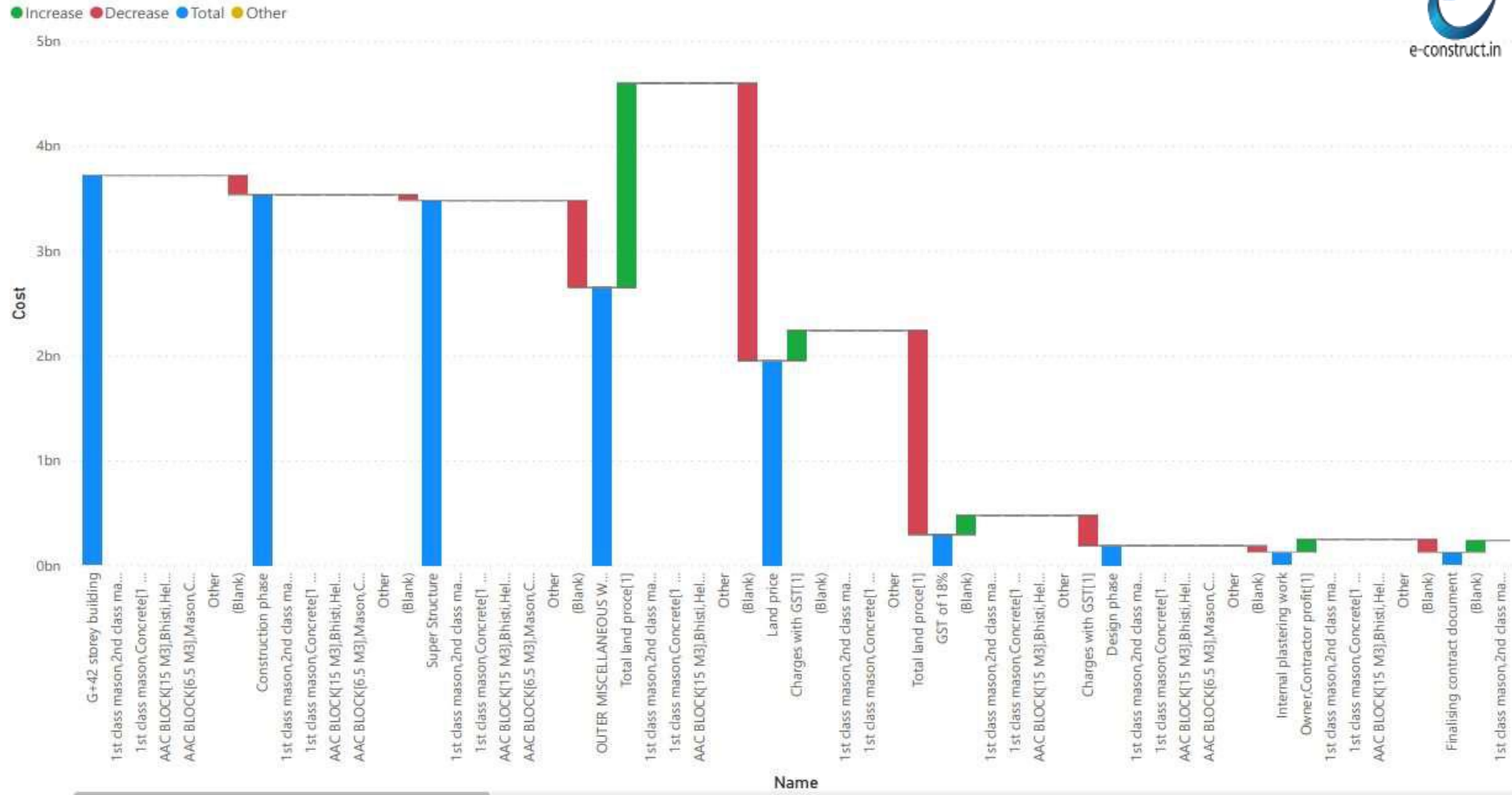
Pie Chart



- Name**
- G+42 storey building
 - Construction phase
 - Super Structure
 - OUTER MISCELLANEOUS WORKS
 - Land price
 - GST of 18%
 - Design phase
 - Internal plastering work
 - Finalising contract document
 - Architectural works
 - Tile Work
 - Concreting for slab and beam
 - PAINTING WORK
 - Contingency charges
 - Reinforcement for Shear wall and Col...
 - Reinforcement for slab and beam
 - False ceiling
 - OUTER PLASTERING WORK
 - Over head charges
 - Project Management Consultant
 - Casting of Shear wall and Column
 - Sub structure
 - Landscape work
 - PAINTING WORK ON OUTER AREA
 - outer road work

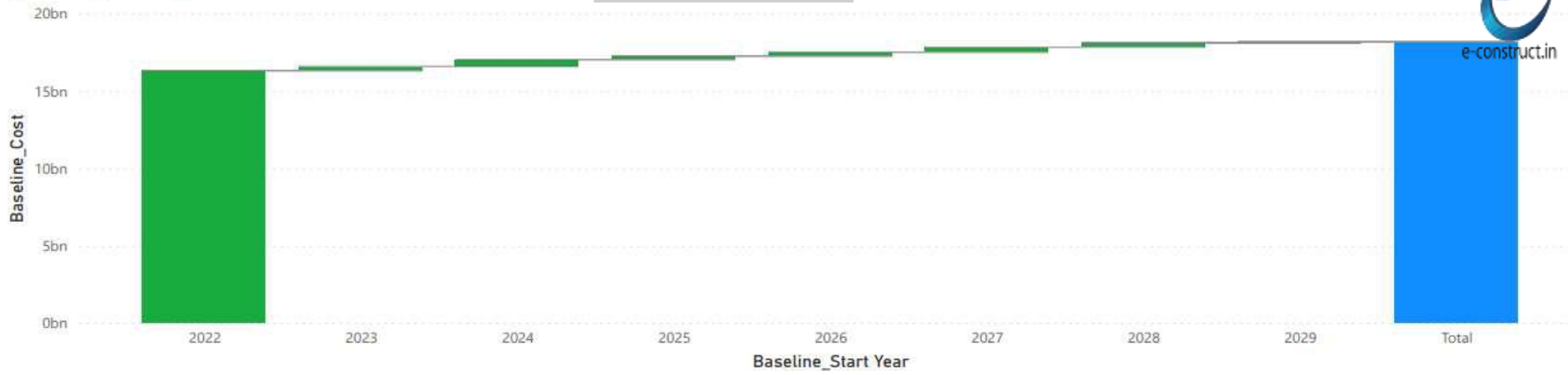
Water fall Chart

Cost by Name and Resource_Names

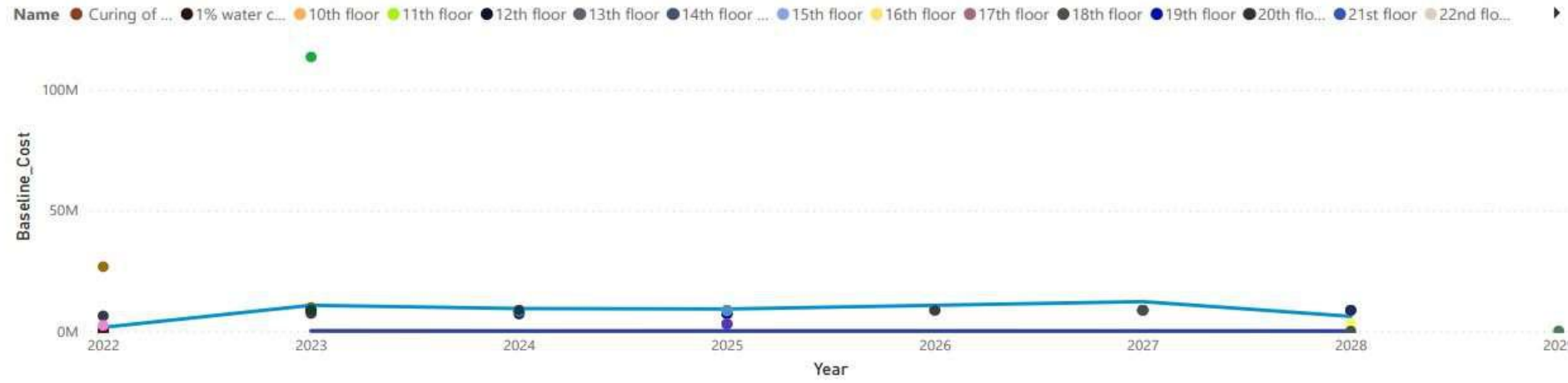


Line Chart

Baseline_Cost by Year



Baseline_Cost by Year and Name



CASH FLOW

Actual Cost

₹ 0.00

Baseline Cost

₹

3,720,666,88

0.12

Remaining Cost

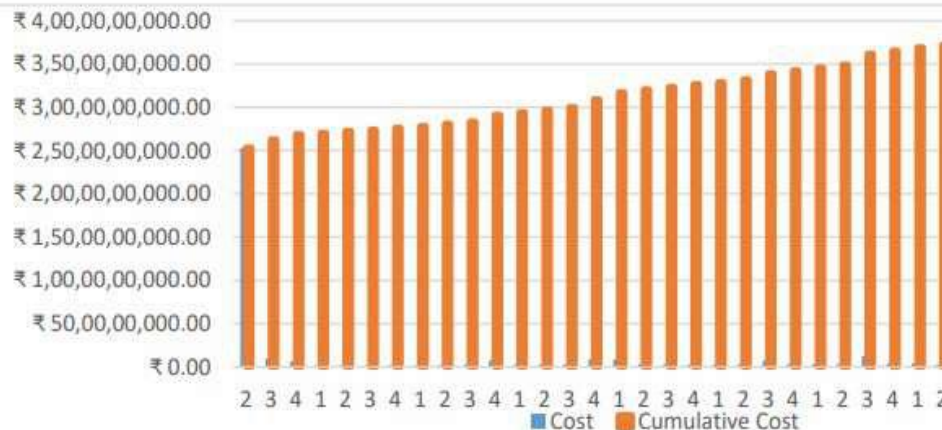
₹

3,720,666,88

0.12

Cost Variance

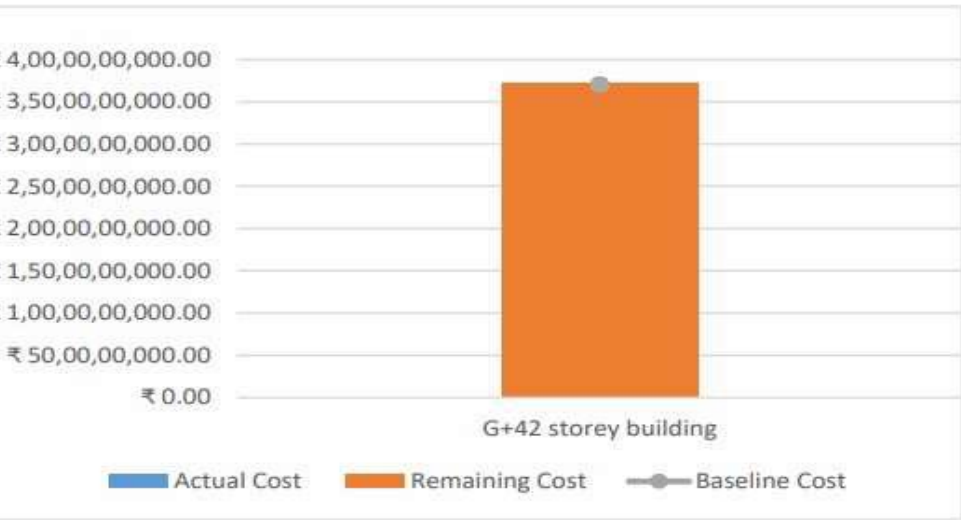
₹ 0.00



TASK COST OVERVIEW

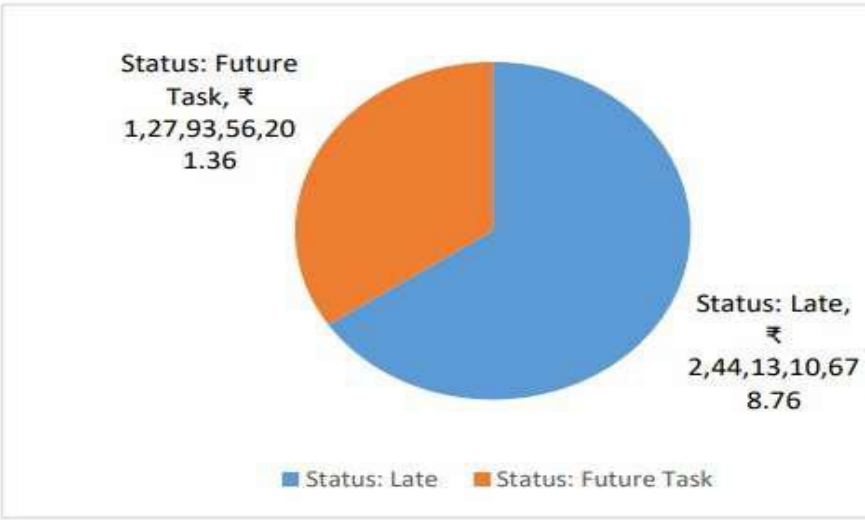
COST STATUS

Cost status for top-level tasks.



COST DISTRIBUTION

How costs are spread out amongst tasks based on their status.



COST DETAILS

Cost details for all top-level tasks.

Name	Fixed Cost	Actual Cost	Remaining Cost	Cost	Baseline Cost	Cost Variance
G+42 storey building	₹ 0.00	₹ 0.00	₹ 3,720,666,880.12	₹ 3,720,666,880.12	₹ 3,720,666,880.12	₹ 0.00

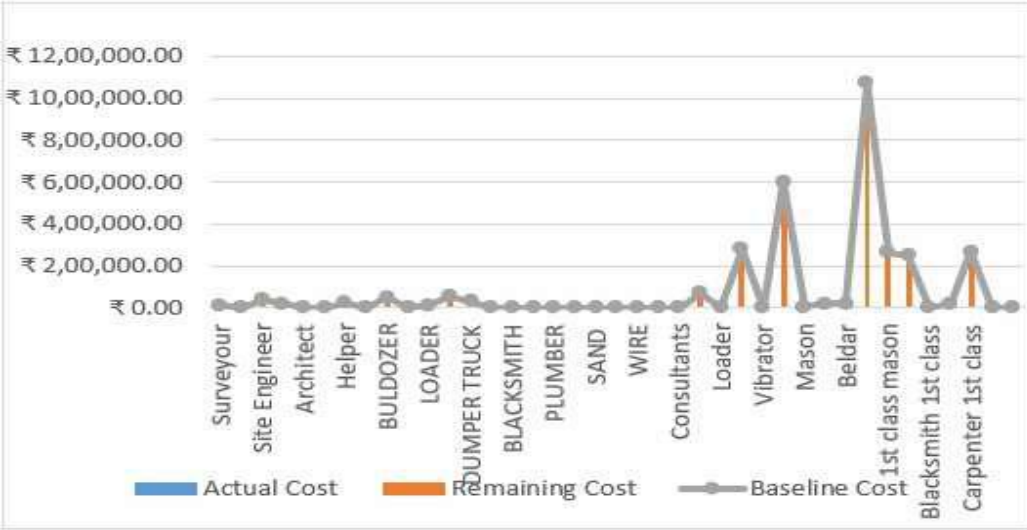
RESOURCE COST OVERVIEW



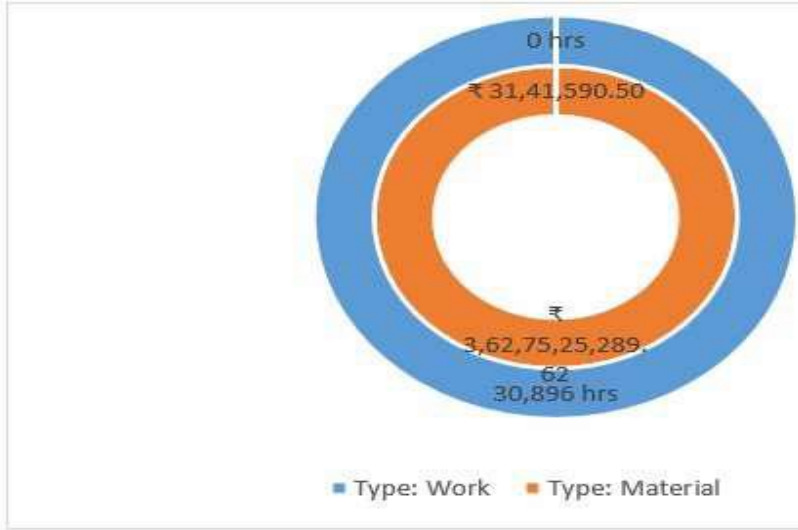
How costs are spread out amongst different resource types

COST STATUS

Cost status for work resources.



COST DISTRIBUTION



COST DETAILS

Cost details for all work resources.

Name	Actual Work	Actual Cost	Standard Rate
Surveyour	0 hrs	₹ 0.00	₹ 5,000.00/day
Labour Surveyour	0 hrs	₹ 0.00	₹ 500.00/day
Site Engineer	0 hrs	₹ 0.00	₹ 1,800.00/day
Supervisor	0 hrs	₹ 0.00	₹ 1,000.00/day

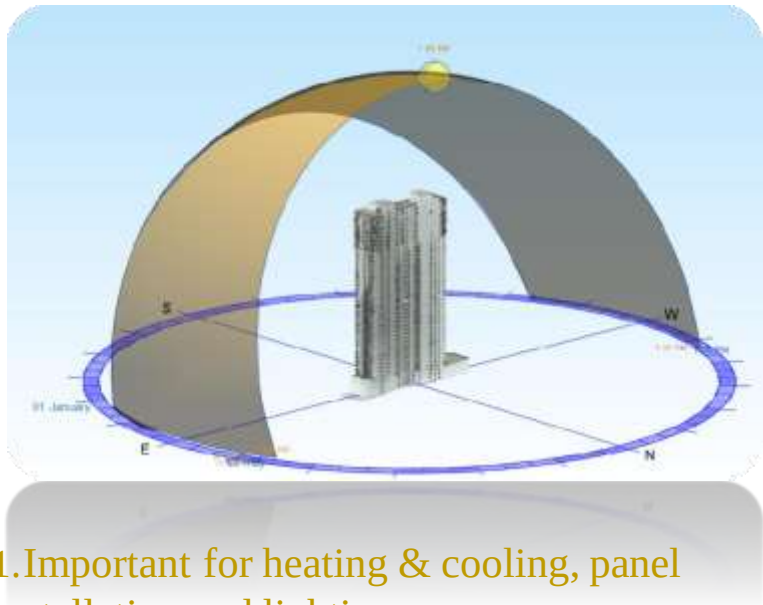
6D ANALYSIS

- 6D dimension of BIM that adds sustainability information to the geometric model of the building.
- Working in Revit makes it simple to find data about the building being modeled, such as its life cycle energy use or its renewable energy potential.

Benefits of 6D BIM

- Reduced energy consumption in the long run
- Faster and more accurate decision making related to component installation during the design process
- Detailed analysis and impact of a decision on economic and operational aspects over the entire lifecycle
- Better operational management of the building or structure after handover

6D Sun path



1. Important for heating & cooling, panel installation and lighting.

2. Sun's path for 1 year.

6D Lighting Analysis

Lighting Analysis - Results Summary

Custom Analysis

For all Floor areas in the Analysis

Total Both - 22% Passing

59% either time below threshold

17% either time above threshold

9:00 am - 32% Passing

May 26

GHI: 764, DNI: 771, DHI: 92

52% below threshold

14% above threshold w/o shades

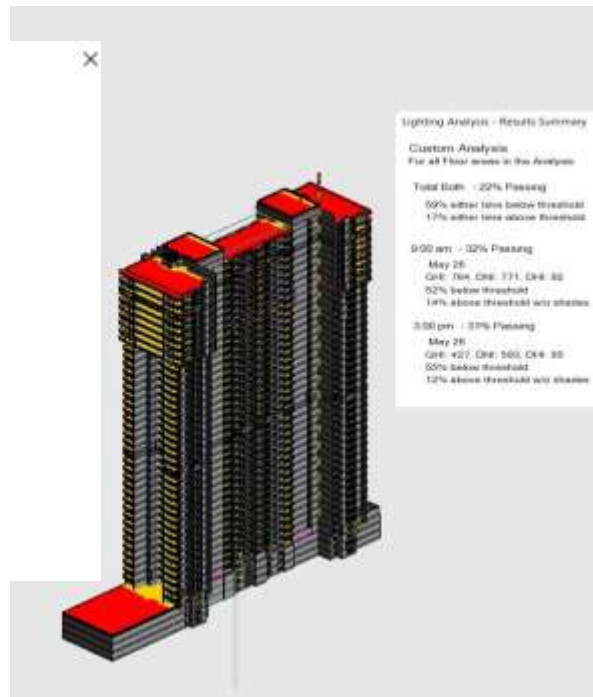
3:00 pm - 31% Passing

May 26

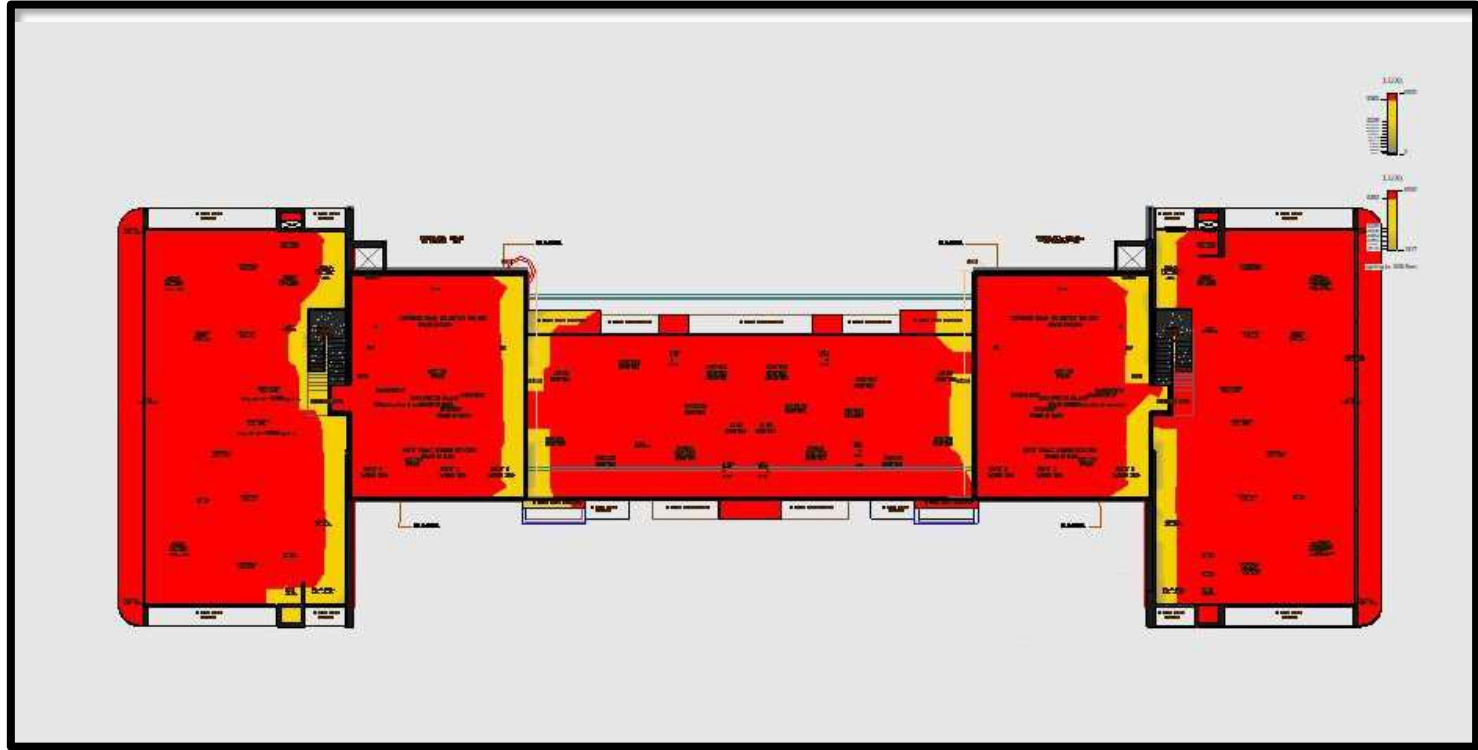
GHI: 427, DNI: 593, DHI: 85

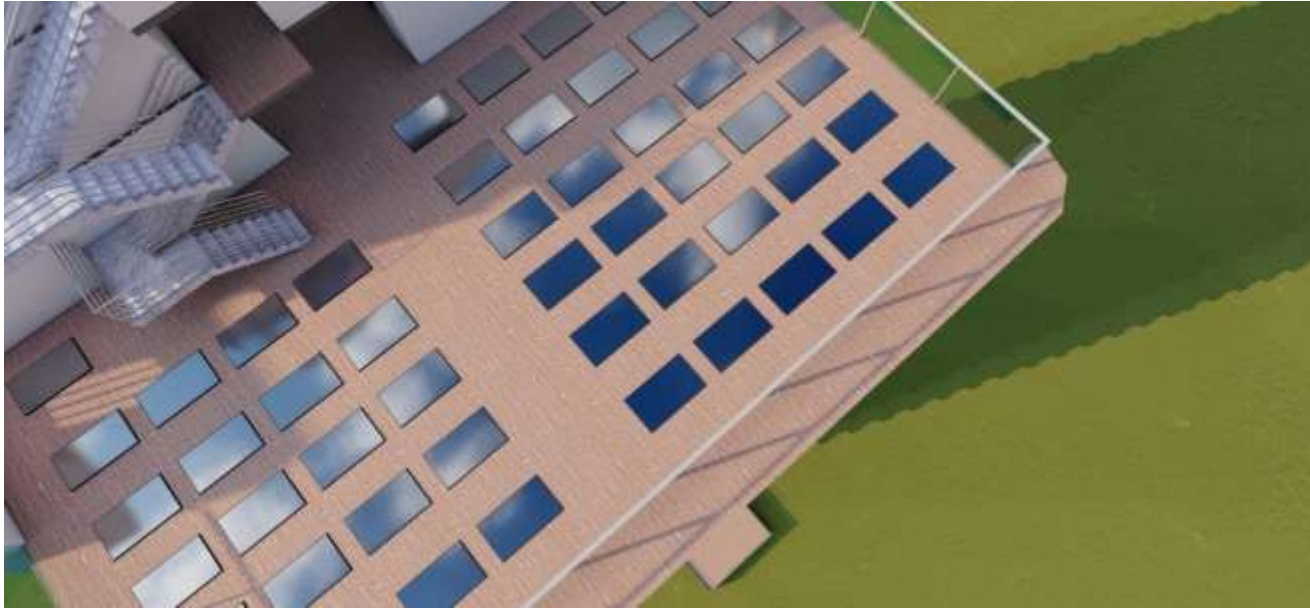
55% below threshold

12% above threshold w/o shades



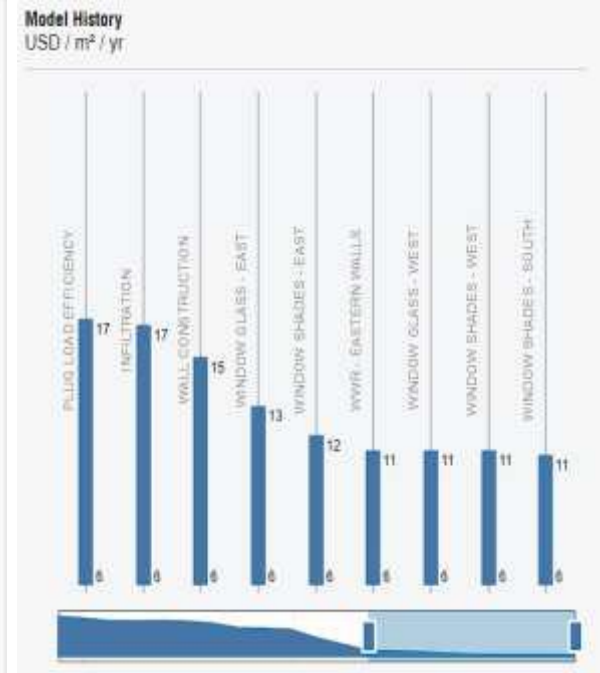
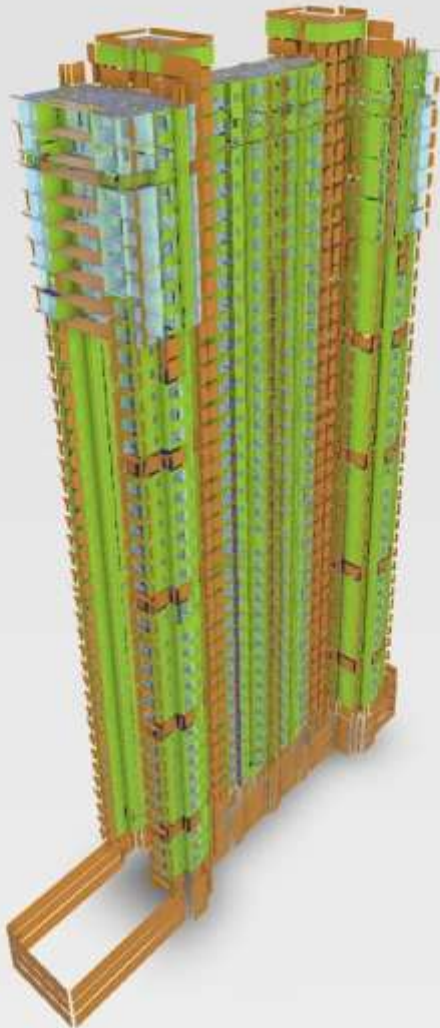
PLAN VIEW OF 6D SOLAR ANALYSIS





Solar panels at a higher altitude will receive more solar radiation as compared to the ground level, resulting in more generation of electricity. The availability of these full solar radiations allows for the formation of a more efficient PV system than ground-mounted PV systems.

6D Energy analysis



7D Facility Management

- It helps in improving the quality of service delivery during the entire lifecycle of a project.
- The dimension is used to track important asset data such as its status, maintenance/operation manuals, warranty information, technical specifications, etc. to be used at a future stage.

7D BIM BENIFITS

- 7D BIM helps to monitor the management of the facility or asset right from the design stage to the demolition stage.
- It makes the changing of building parts and general repair of a building/project throughout its entire cycle a very easy task.
- Pinpoint to the contractor, the maintenance process that must be followed.

COBie SPREADSHEETS

Name	CreatedBy	CreatedOn	Category	Description	AssetType	Manufacturer	ModelNumber	ManufacturerParts	ManufacturerParts	ManufacturerLabor	ManufacturerLabor	ManufacturerUnit	ExtSystem	ExtObject
Doors_Aluprof	<No>	2022-05-26T18:34:55	23-17 11 00	Aluprof	n/a	ALUPROF S	MB-86 ST	n/a	n/a	n/a	n/a	n/a	Autodesk	IfcDoorType
Doors_Aluprof	<No>	2022-05-26T18:34:55	23-17 11 00	Aluprof	n/a	ALUPROF S	MB-86 ST	n/a	n/a	n/a	n/a	n/a	Autodesk	IfcDoorType
Doors_Aluprof	<No>	2022-05-26T18:34:55	23-17 11 00	Aluprof	n/a	ALUPROF S	MB-86 ST	n/a	n/a	n/a	n/a	n/a	Autodesk	IfcDoorType

Name	CreatedBy	CreatedOn	Category	SheetName	RowName	Value	Unit	ExtSystem	ExtObject
Active leaf	<None>	2022-05-2	Approved	Type	Doors_Al	0.9	m	Autodesk	Autodesk
Air permea	<None>	2022-05-2	Approved	Type	Doors_Al	up to Clas	n/a	Autodesk	Autodesk
Analytic C	<None>	2022-05-2	Approved	Type	Doors_Al	<None>	n/a	Autodesk	Autodesk
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Constructi	<None>	2022-05-2	Approved	Type	Doors_Al	n/a	n/a	Autodesk	Autodesk
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GlazingMa	<None>	2022-05-2	Approved	Type	Doors_Al	Glass AL	n/a	Autodesk	Autodesk
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This COBie spreadsheet is an example file that comes with the COBie Extension 1.0
Individual worksheets are organized by project phase as shown below

Contents

People and Companies

Contents

Project, Site, and Facility

Vertical levels and exterior areas

Spaces

Sets of spaces sharing a specific attribute

Types of equipment, products, and materials

Contents

Individually named or schedule items

Sets of components providing a service

Constituents for Types, Components and others

Logical connections between components

Economic, Environmental and Social Impacts at various stages in the life cycle

Contents

NOTE: Submittals and approvals added on 'Documents' worksheet

NOTE: Manufacturer and model added on 'Type' worksheet

NOTE: Serial and tag added on 'Component' worksheet

Contents

Onsite and replacement parts

Required materials, tools, and training

PM, Safety, and other job plans

Assembly-Assembly Type	Category-Connection	Category-Coordinate	Category-Document	Category-Element	
Excluded	Control	Point	Certificates	21-01 00 00 Substructure	
Fixed	Flow	Line-end-one	Client Requirements	21-01 10 Foundations	
Included	Return	Line-end-two	Closeout Submittals	21-01 10 10 Standard Foundations	
Layer	Supply	Box-lowerleft	Contract Drawings	21-01 10 10 10 Wall Foundations	
Mix	Structural	Box-upperright	Contract Drawings	21-01 10 10 30 Column Foundations	
Optional			Contract Modifications	21-01 10 10 90 Standard Foundation Supplementary Components	
			Contract Specifications	21-01 10 20 Special Foundations	
			Design Data	21-01 10 20 10 Driven Piles	
			Design Review Comment	21-01 10 20 15 Bored Piles	
			Manufacturer Field Reports	21-01 10 20 20 Caissons	
			Manufacturer Instructions	21-01 10 20 30 Special Foundation Walls	
			Operation and Maintenance	21-01 10 20 40 Foundation Anchors	
			Preconstruction Submittals	21-01 10 20 50 Underpinning	
			Product Data	21-01 10 20 60 Raft Foundations	
			Punch List Items	21-01 10 20 70 Pile Caps	
			Request for Information	21-01 10 20 80 Grade Beams	
			Requests for Information	21-01 20 Subgrade Enclosures	
			Samples	21-01 20 10 Walls for Subgrade Enclosures	
			Shop Drawings	21-01 20 10 10 Subgrade Enclosure Wall Construction	
			Specifications	21-01 20 10 20 Subgrade Enclosure Wall Interior Skin	
			Test Reports	21-01 20 10 90 Subgrade Enclosure Wall Supplementary Components	
				21-01 40 Slabs-On-Grade	
				21-01 40 10 Standard Slabs-on-Grade	
				21-01 40 20 Structural Slabs-on-Grade	
				21-01 40 30 Slab Trenches	
				21-01 40 40 Pits and Bases	

TENDER DOCUMENT

TENDER DOCUMENT
FOR
CONSTRUCTION OF G+42
SKY LOUNGE AND DUPLEX
AT
MULUND, MUMBAI, MAHARASHTRA



Microsoft Edge
PDF Document

Notice Inviting E-Tender

NIT No: Tender/001/2022

Date: 10.2022

E-Tender-notice

1. Name of Work:

Tenders are invited on the website, the eligible bidders having Digital Signature Certificate (DSC) issued from any agency authorized by Controller of Certifying Authority(CCA), Govt. of India and which can be traced upto the chain of trust to the Root Certificate of CCA, for the following work.

Name of work	Construction of Sky Lounge Tower
Total Estimated cost	382,11,29,128/-
Earnest Money	76442402.56
Period of completion	6 years

Tender Inviting Authority	Contact Person(s)/Tender Dealing Officer(s)
Econstruct Design And Build Private LTD Venkatdhari Heights, 2nd Floor Parapanna Aghahara Main Road, Bangalore - 560068	Kalpesh Ahire, (General Manager), Bangalore, Karnataka, Contact No: (+91)948973484.

2. Time Schedule of Activities:

Sl. No.	Particulars	Time Schedule
a.	Tender e-Publication date	Date & Time as mentioned in the website: e-construct.in
b.	Document download/Sale Start Date	
c.	Document download & Bid submission End Date	
d.	Bid Submission Start Date	
e.	Start Date for seeking Clarification on-line	
f.	Last Date for seeking Clarification on-line	
h.	Date of Pre-Bid Meeting (If required)	
i.	Bid Opening Date	

CONTRACT DOCUMENT

CONTRACT DOCUMENT

FOR
CONSTRUCTION OF G+42 SKY LOUNGE

AT
MULUND, MUMBAI, MAHARASHTRA



MAY-2022

Econstruct Design And Build Private LTD

CONTRACT DOCUMENT

FOR
CONSTRUCTION OF G+42 SKY LOUNGE

AT
MULUND, MUMBAI, MAHARASHTRA

CONTRACT NUMBER: 01/2022-2023 CONTRACT

BETWEEN

Millenium Group Builders and
Developers

&

E-Construct Design & Build Pvt. Ltd



Microsoft Edge
PDF Document

INDEX

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8.	GENERAL TERMS AND CONDITIONS	31-44
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11.	UNIT PRICE CODE	58-64
12.	BILL OF MATERIALS (BOM)	65-69

Thank You